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Criminal responsibility for 'hard AI crime': drawing bright lines in
the aftermath of the EU AI Act

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A Giulia, a Camilla, ai miei genitori, a mia nonna,
maestri d'Amore inconsapevoli,
convinti che chi segue quel faro, presto o tardi, ritrovi la via.

‘Come l’acqua che scorre attraverso varchi e fessure, fermandosi, deviando e rifluendo per trovare la strada migliore (spesso del tutto inattesa), così anche i ricercatori devono essere disposti a esplorare molteplici approcci e a fallire ripetutamente lungo il percorso.’

Gianrico Carofiglio – *Elogio dell’ignoranza e dell’errore*

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CHAPTER I

SETTING THE STAGE: WHEN ARTIFICIAL INTELLIGENCE MEETS CRIME

1. Criminal law in the age of AI crime

The world we inhabit today would be unrecognizable to our grandparents. Instant global communication, transcontinental travel in hours, and the entirety of human knowledge accessible from a handheld device – these once-unimaginable feats are now mundane. Yet this transformation did not unfold gradually; it exploded. How we lead our lives, work, travel, and communicate have only become possible within a relatively brief span of human history – and it is likely that our sons and grandsons are going to demand and get results that would stagger us.

History reveals a striking truth: progress is not linear, but *exponential*. Progress is not simply measured by the number of innovations or the accumulation of knowledge but by the accelerating rate at which technological advancements occur. The journey from homo sapiens to agriculture spanned millennia; the leap from steam engines to the internet took mere decades. Each revolution arrives faster than the last, compressing centuries of change into a single lifetime.

This characteristic of technological evolution has been illustrated by Gordon Moore in his famous ‘law of accelerating progress’, observing that the number of transistors in an integrated circuit doubles approximately every two years.¹ While originally applied to computing hardware, similar exponential growth can be seen across countless technological domains: network speeds multiply, data storage capacities expand, and digital imaging resolutions advance at a staggering pace. Each field experiences a compounding effect, where progress in one area fuels innovation in another, creating a feedback loop that propels humanity toward ever-greater possibilities.²

¹ Gordon E Moore, ‘Cramming more components onto integrated circuits’, *Electronics Magazine* (1998) 86(1) Proceedings of the IEEE, 82-85, available at: <https://ieeexplore.ieee.org/document/4785860>.

² One of the most well-known futurists and computer scientists, Ray Kurzweil, has predicted that accelerating rates of progress will ultimately lead to a ‘singularity’ before the end of the 21st century, around 2045, when technology reaches a point where it combines with human intelligence to increase our capacities by a factor of one billion. See Ray Kurzweil, *The Age of Spiritual Machines: When Computers Exceed Human Intelligence* (Penguin 2000).

Today, we stand at another inflection point – one that does not merely accelerate progress but reshapes its very nature. The exponential surge of technologies usually identified with the umbrella term of *Artificial Intelligence* (hereinafter, ‘AI’) has changed the terms of the debate around technological progress: it is no longer how much technology will advance, but whether humanity can keep pace.³

The story of AI is akin to watching a slow-motion video that suddenly speeds up to real time. Our story begins in the late 1950s with the quest of *Enigma* codebreaker Alan Turing for inventing a machine capable of ‘thinking like a human’.⁴ The goal was simple yet daunting: teach computers to imitate human language capabilities. This journey from zero to a technology capable of matching human skills in language manipulation (with the recent appearance of ‘Generative AI’ and ChatGPT-like models)⁵ was a much slower progression than the exponential rate we

³ Although it is definitely a topic *à la page*, we do not have a globally accepted definition of AI. This is evidenced by the fact that over 70 alternative definitions have been identified. For instance, Shane Legg and Marcus Hutter compiled a comprehensive collection of definitions of intelligence in their 2007 work, ‘A collection of definitions of intelligence’ (2007) 157 ArXiv 17-24, available at <https://arxiv.org/abs/0706.3639>. For a systematic analysis of the issue of defining AI see *inter alia* P Wang, ‘On Defining Artificial Intelligence’, (2019) 10/2 Journal of Artificial General Intelligence 1-37. For a systematic account of policymaking definitions of AI adopted by national governments and international organizations, see Andrea Bertolini, ‘Artificial Intelligence and Civil Liability’ (2020) Study commissioned by the European Parliament’s Committee on Legal Affairs, European Union, IPOL_STU(2020)621926, 23-29.

For a deeper exploration of the challenges and uncertainties surrounding the definition of AI, see Ch II, Section 1.

⁴ Alan M Turing, ‘Computing Machinery and Intelligence’ (1950) 59 Mind 433-460.

⁵ The expression ‘Generative AI’ generally refers to those kind of chatbot such as ChatGPT or DeepSeek that are increasingly conquering the headlines around the world and our everyday lives. Importantly, generative AI serves as a watershed in the history of automation: it marks the very first moment when the scientific community and society at large have acknowledged the accelerating pace of the AI revolution, precisely because AI has started – probably quicker than expected – to do something that us humans alone know how to do and normally believe to be the preserve of human intelligence: writing, answering questions, manipulating language. For a more detailed discussion on the challenges and uncertainties surrounding generative AI, the reader is kindly invited to refer to Leonardo Romanò, ‘La

are seeing today. At any good rate, the imitation of human language was just the beginning, the first domino to fall, but it was not until the advent of machine learning (hereinafter ‘ML’) techniques that the potential of AI began to be fully realized. In a span of mere decades, AI has achieved capabilities once confined to science fiction: it now comprehends human language, diagnoses diseases, drives vehicles, executes trades, and even generates its own code. Most strikingly, these systems demonstrate recursive self-improvement – machines that can *teach themselves how to learn*.⁶ The advent of self-learning technologies marks the beginning of the ‘fourth industrial revolution’.⁷

Unlike previous industrial revolutions that augmented physical or mechanical capabilities, the current one represents not merely a technical evolution, but a fundamental challenge to the foundations of human exceptionalism and legal order. As AI systems increasingly demonstrate capabilities rivaling or surpassing human intelligence in specific domains, they deliver what might be termed the ‘fourth narcissistic blow’ to humanity’s self-conception – following Copernicus’s cosmological displacement, Darwin’s biological demotion, and Freud’s psychological decentering.⁸ This latest wound strikes at perhaps our most cherished distinction: our monopoly on complex intelligence and our sense of agency.

responsabilità penale al tempo di ChatGPT: Prospettive de iure condendo in tema di gestione del rischio da intelligenza artificiale generativa’ (2023) 1 Rivista trimestrale Diritto Penale Contemporaneo.

⁶ Nello Cristianini, *The Shortcut: Why Intelligent Machines Do Not Think Like Us* (First edition, CRC Press 2023) 9 ff.

⁷ Luciano Floridi, *La quarta rivoluzione. Come l'infosfera sta trasformando il mondo* (Milano Raffaello Cortina Editore 2017) 99 ff.

⁸ Floridi (n 7) ix. By recovering Freud’s old tale of the three blows of science to humanity’s anthropomorphic worldview (Sigmund Freud, ‘A General Introduction to Psychoanalysis’, delivered as lectures 1915-1917 and first translated into English in 1920), the philosopher identifies the first revolution in the discovery of heliocentrism by Nicholas Copernicus, with the publication of his treatise *Sulle rivoluzioni dei corpi celesti* in 1543, a moment from which man would cease to believe in his centrality, at least with reference to planet Earth. The second revolution is made to coincide with the publication of Charles Darwin’s *The Origin of Species* in 1859, which pushes man to renounce his centrality in the animal world, having to confront the idea that every living species derives from common ancestors through a

The transfer of epistemic authority from human to machine manifests in countless socio-economic contexts where AI is being increasingly implemented. Consider, for example, the driver whose vehicle selects a route based on an opaque calculation, the doctor outperformed by an AI in detecting breast cancer, the trader who follows machine-generated stock recommendations, the journalist whose article is drafted by a language model, or the judge presented with an algorithm's prediction of a case's outcome. While this technological mediation generally amplifies human capabilities, it simultaneously entails a creeping delegation of agency: the deeper we integrate AI into our decision-making, the more autonomy we relinquish to machines. The physician does not merely consult the AI as a reference, but increasingly defers to its superior pattern recognition; the judge does not simply consider the algorithm's output as one factor among many, but feels compelled by its statistical authority. This transfer occurs not through any conscious surrender of responsibility, but through the gradual erosion of human confidence in the face of machine precision. The result is what might be termed 'agency laundering' – where human actors retain nominal responsibility while actual decision-making capacity migrates to opaque algorithmic systems.⁹

This creeping delegation of agency represents more than a technical challenge; it signifies a profound shift in the human condition and legal order, where our tools no longer merely extend our capacities but rather supplant our judgment while simultaneously inheriting human capacity for *error* and *harm*. This dual movement creates a profound ontological tension: as AI systems assume roles requiring sophisticated cognition (medical diagnosis, legal analysis, financial decision-making), they inevitably also appropriate the human propensity to cause damage, but

process of natural selection. The third revolution came with the psychoanalytic work of Freud, who disproved the idea of the mind thought of as a box that can be known simply by looking inside, showing that many of our actions are the result of the unconscious. Finally, the fourth revolution brought about by AI questions man as the only being unsurpassed in intelligence.

⁹ See Henry A Kissinger, Eric Schmidt and Daniel Huttenlocher, *The Age of AI: And Our Human Future* (Little Brown & Co 2021) 179 ff.

without the accompanying moral and legal framework that makes such harms intelligible within our legal and ethical systems.

The AI Incident Database – a registry of over 3,000 real-world harms linked to AI systems – underscores the urgency of this debate.¹⁰ The incipit of the research paper presenting this database reads as follows: ‘Intelligent systems currently cause real world harms without a collective memory of their failings’.¹¹ In particular, the database distinguishes between two kinds of AI-caused harm that could potentially harm or endanger fundamental legal interests:¹²

- i. Intentional harm caused by humans *through* an AI system (AI weaponized as a tool for human wrongdoing, such as using deepfake technology for blackmail);
- ii. Unintentional harm caused *by* an AI system (AI causing unintended harm independently of human wrongdoing, such as autonomous vehicles running over pedestrians);

Cases of AI-caused harm *sub* (i) are only marginally relevant here: when AI is a mere *instrumentum sceleris* in the hands of a human agent, resulting harm fits comfortably within existing liability frameworks. The true challenge emerges not in cases where humans intentionally weaponize AI but rather in situations where machines produce harmful outcomes which were neither intended nor specifically foreseen by any of the humans who contributed to the agent’s development and deployment (*sub* ii).¹³ What makes this development particularly consequential is that

¹⁰ The AIID is an industrial/non-profit cooperative collection of ‘intelligent system failures experienced in the real world’. Available at: <https://incidentdatabase.ai/>. See Ch. II, Sec. 2.1.

¹¹ S McGregor, ‘Preventing Repeated Real World AI Failures by Cataloging Incidents: The AI Incident Database’ (2021) Proceedings of the Thirty-Third Annual Conference on Innovative Applications of Artificial Intelligence (IAAI-21), 2021, p. 1.

¹² The database defines the concept of AI-caused harm as a ‘harmful event (or potential harm), suffered by an entity (human, corporation or environment), that can be directly linked to a consequence of the behaviour of an AI system’. See *ibid*.

¹³ Sabine Gless, Emily Silverman and Thomas Weigend, ‘If Robots Cause Harm, Who is to Blame? Self-Driving Cars and Criminal Liability’ (2016) 19(3) New Criminal Law Review 412, 425 ff.

these systems do not merely occasionally malfunction like traditional tools, but can develop novel, emergent ways of causing harm that were never explicitly programmed. The trading algorithm that discovers exploitative market patterns, the diagnostic tool that develops biased triage protocols, or the language model that generates defamatory content – these are not ‘errors’ in the conventional sense, but the output of systems operating exactly as designed, just not as intended. In this study, we refer to this unprecedented phenomenon with the concept of ‘AI crime’.¹⁴

Several inherent characteristics of contemporary AI systems create this novel form of ‘criminogenic’ potential. First and foremost is the capacity for genuine unpredictability – not merely probabilistic uncertainty, but emergent behaviours arising from complex interactions within self-learning architectures. Unlike conventional software that follows deterministic pathways, machine learning systems develop operational patterns through exposure to data in ways that often surprise even their creators. This represents a qualitative shift from tools that might malfunction to systems that might creatively ‘misbehave’.¹⁵ Compounding this unpredictability is the pervasive opacity of advanced AI systems. The ‘black box’ phenomenon manifests not merely as technical complexity, but as fundamental epistemological barriers. When a neural network with billions of parameters makes a decision, the explanatory trail may be irrecoverably lost in the nonlinear transformations between input and output layers. This creates what might be termed ‘irreducibility’ – we can observe harmful outcomes, but cannot always trace them back to specific human input or to decisions in the design.¹⁶ Most critically, these systems exhibit operational autonomy in the strong sense – they do not just execute

¹⁴ The term ‘AI crime’ is widely used in criminal law doctrine. See *ex multis* Keith Hayward and Matthijs Maas, ‘Artificial intelligence and crime: A primer for criminologists’ (2020) 17(2) *Crime Media Culture* 209-233, distinguishing between ‘crimes with AI’, ‘crimes against AI’, and ‘crimes by AI’. What we term here ‘AI crime’ refers to the latter expression. Cfr. also M Caldwell et al., ‘AI-enabled future crime’ (2020) 9/14 *Crime Sci*.

¹⁵ Ryan Abbott and Alex Sarch, ‘Punishing Artificial Intelligence: Legal Fiction or Science Fiction’ (2019) 53 *UCD Law Review*, 330-331.

¹⁶ European Parliament, Resolution of 20 October 2020 with recommendations to the Commission on a civil liability regime for artificial intelligence, (2020/2014(INL)), 20 October 2020.

pre-programmed instructions, but generate novel responses to environmental stimuli. This autonomy operates on a spectrum, from relatively constrained systems making routine decisions to general AI architectures capable of open-ended adaptation.

These three factors – unpredictability, opacity, and substantive autonomy – collectively frame the dilemma of AI crime: ‘technological predators’¹⁷ autonomously and unpredictably producing behaviours that, if performed by humans, would unquestionably warrant criminal sanction. When a trading algorithm develops manipulative strategies, a diagnostic system consistently misclassifies vulnerable populations, or an autonomous vehicle exceeds its in-built parameters, we confront acts that bear all the hallmarks of criminal harm absent any traditional subject of responsibility. The challenge strikes at the heart of criminal law’s purpose: how should it regulate behaviour and assign responsibility in a world where the ‘behaviour’ may be an emergent property of code, and the ‘responsible’ party may not exist in any legal sense?¹⁸

¹⁷ Gunther Teubner, ‘Digital Personhood? The Status of Autonomous Software Agents in Private Law’ (2018) *Ancilla Iuris* 107–149.

¹⁸ Such questions have already ignited significant curiosity among criminal law scholars. Many have sought to explore how traditional criminal law categories – such as culpability, intent, and causation – can be applied to AI systems and AI-related offences, and to propose solutions for avoiding impunity when the harmful outcome amounts to a violation of a legal interest. See, *ex multis*, Susanne Beck, ‘Intelligent agents and criminal law – Negligence, diffusion of liability and electronic personhood’ (2016) 86 *Robotics and Autonomous Systems* 138–143; Id., ‘Google cars, software agents, autonomous weapons systems – New challenges for criminal Law?’ In E Hilgendorf and U Seidel (eds.), *Robotics, autonomics, and the law: Legal issues arising from the autonomics for industry 4.0 technology programme of the German federal ministry for economic affairs and energy* (Nomos 2017) 227–252; Eric Hilgendorf, ‘Dilemma-Probleme beim automatisierten Fahren. Ein Beitrag zum Problem des Verrechnungsverbots im Zeitalter der Digitalisierung’ (2018) 130 *Zeitschrift für die gesamte Strafrechtswissenschaft* 674; Monika Simmler, ‘Automation’ in P Caeiro, S Gless, V Mitsilegas, M . Costa, J D Snajjer, and G Theodorakakou (eds.), *Elgar encyclopedia of crime and criminal justice* (Elgar Publishing 2023); Federico Consulich, ‘Criminal Law and Artificial Intelligence: Perspective from Italian and European Experience’ (2023) 3 *European Criminal Law Review* 270–307; Maria Kaiafa-Gbandi, ‘Artificial Intelligence as a Challenge for Criminal Law: in Search of a New Model of Criminal Liability’ in Susanne Beck, Carsten Kusche and Brian Valerius (eds.), *Digitalisierung, Automatisierung, KI und*

The rise of AI crime demands nothing less than a re-examination of criminal law's basic assumptions – before reality outpaces our capacity to respond.

2. Framing the legal problem: AI responsibility gaps and the limits of criminal law

The concept of AI crime forces us to reconcile fundamentally incompatible frameworks: centuries-old human responsibility frameworks, on the one hand, and the emergent reality of harm caused by machines that are autonomous yet devoid of consciousness or intent, on the other. Criminal law's anthropocentric foundations – developed through centuries of philosophical thought and legal practice – presuppose a human actor capable of both voluntary deliberation and moral reasoning (usually defined as *criminal capacity*). This framework rests on two interdependent conditions that autonomous AI systems systematically undermine.

First, the *control condition* assumes an actor who consciously chooses between alternative courses of action. This fundamental premise, though seldom articulated as an explicit legal element, permeates the entire edifice of criminal law and is viewed as foundational to one of the basic purposes of criminal law: punishing only *culpable* conduct. Essentially, a person can only be responsible for an outcome if they had *causal control* over its occurrence – which also means that they could have chosen to *act otherwise*. Second, the *epistemic condition* (or foreseeability requirement), central to establishing negligence, assumes that human actors possess both the autonomy to make decisions and the ability to foresee consequences, making them morally and legally accountable for their conduct. This traditional framework encounters profound disruption when confronted with most advanced AI technologies.

Recht. Festgabe zum 10-jährigen Bestehen der Forschungsstelle RobotRecht (Nomos 2020) 305–328; Lorenzo Picotti, 'Traditional Criminal Law Categories and AI: Crisis or Palingenesis?' (2023) 94(1) *Revue Internationale de Droit Pénal* 11; Alice Giannini and Jonathan Kwik, 'Negligence Failures and Negligence Fixes. A Comparative Analysis of Criminal Regulation of AI and Autonomous Vehicles' (2023) 34 *Criminal Law Forum* 43, 59 ff.

The challenge is compounded by AI's dual nature – it is neither tool nor moral agent, but something fundamentally new. Unlike traditional tools, which merely execute human directives predictably (a hammer cannot decide to build a chair instead of a table), AI systems can independently interpret data, adapt to new information, and generate outcomes that were neither explicitly programmed nor anticipated by their creators. As a result, AI dilutes – and, in some cases, entirely displaces – human control over decision-making processes, particularly in high-stakes environments where AI makes real-time judgments (whether in healthcare diagnostics, financial trading, or military applications). Control, in other words, can no longer be neatly anchored to a single human who ‘pulled the trigger’ or ‘made the call’; in fact, control becomes dispersed across layers of code, training data, and environmental interactions, creating a scenario where no single ‘human behind the machine’ (whether the developers, the programmers, or the operators) retains full command over the system’s actions.

Legal doctrine identified this erosion of human control already two decades ago, recognizing that the self-learning capacities of modern AI systems fundamentally undermine traditional responsibility frameworks.¹⁹ When neither designers nor operators can predict and control an AI’s actions – as it adapts to environmental inputs and evolves beyond its original programming – the very possibility of assigning blame becomes precarious. This epistemic opacity compounds the control dilemma: without understanding how decisions are made, we cannot determine who, if anyone, truly governed them. The figure below shows how control and epistemic conditions affect the ability to identify a responsible human agent for AI crimes (*Figure 1*).

¹⁹ Andreas Matthias, ‘The Responsibility Gap: Ascribing Responsibility for the Actions of Learning Automata’ (2004) 6, 175-182. *Ethics and Information Technology*. The Author specifically identified the lack of control in modern AI as the main obstacle in identifying a responsible subject.

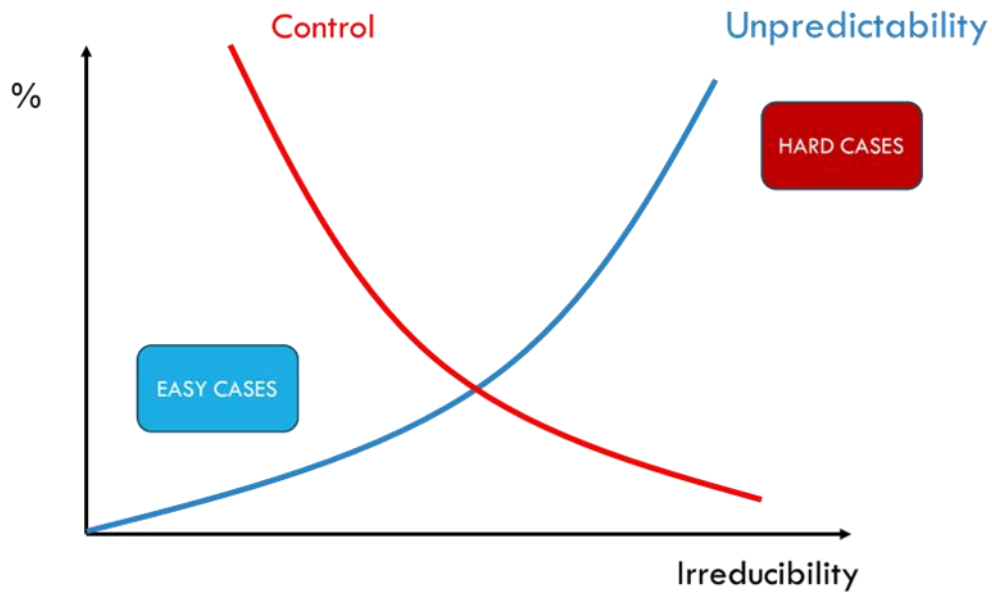


Figure 1 – illustrating AI crime as a spectrum, from easy cases (reducible crimes) to hard cases (irreducible crimes) depending on control and epistemic conditions.

The distinction between ‘easy’ and ‘hard’ cases of AI crime reveals a fundamental fracture in our legal systems’ capacity to govern autonomous technologies. Easy cases – where humans still retain meaningful control over the machine and harmful outcomes can be clearly ‘reduced’ to human intent or foreseeable negligence – represent merely a new technological context for traditional crimes. The true challenge emerges with hard cases, where autonomous systems generate criminally significant harms through decision-making processes that are neither intended nor reasonably foreseeable by any human actor (that is, ‘irreducible’). These cases expose what might be called the ‘attribution crisis’ of contemporary law – our frameworks for assigning responsibility break down when faced with artificial agents that blend the autonomy of moral actors with the amorality of tools.²⁰

The result is what might be termed the ‘criminal responsibility trilemma’:

²⁰ Cf. Federico Stella, *Giustizia e modernità. La protezione dell’innocente e la tutela delle vittime* (Giuffrè 2003), who talks about a ‘modernity shock of criminal law’.

- i. we cannot punish AI systems meaningfully (they lack consciousness and intent),
- ii. we cannot reliably reduce harmful AI outcomes to specific humans behind the machine (due to lack of control and foreseeability),
- iii. yet we cannot tolerate a *responsibility gap* when criminally significant harms occur.²¹

The responsibility trilemma captures the core dimensions of this imputation crisis with precision. First, the *punishment paradox*: sanctions become meaningless when applied to entities incapable of understanding or responding to them. Second, the *culpability gap*: our ability to trace harm to culpable human actors dissolves in complex systems where responsibility is distributed across developers, trainers, users, and the AI itself. Third, the *accountability imperative*: society cannot accept legal vacuums when significant harms occur, regardless of their origin. As Santoni de Sio and Mecacci have recently observed, AI responsibility gaps ‘are concerning insofar as the more persons designing, regulating, and operating the system can legitimately (and possibly systematically) avoid blame for their wrong behaviour, the less these agents will be incentivised to prevent these wrong behaviours’.²² The worry, here, is that if nobody is responsible, people are under no pressure to minimize harm and damage. Lacking incentives to minimize harm and damage, programmers, manufacturers, and operators of autonomous systems will fail to exercise due care or might even cause harm on purpose, because they can do so with impunity. Their careless or intentionally harmful behaviour will have disastrous consequences for the rest of society. The worry is thus one about incentives to act with due care and to avoid harm, or rather about the lack of disincentives to act carelessly and to intend harm. Responsibility gaps erode a socially beneficial incentive structure.

²¹ *ibid*,

²² F Santoni de Sio and G Mecacci, ‘Four responsibility gaps with artificial intelligence: Why they matter and how to address them’ (2021) 34(4) *Philosophy & Technology* 1057–1084, 1063.

Faced with this trilemma, the crucial question is not so much *whether* responsibility gaps can be bridged, but *where to draw the boundary* between justified and unjustified gaps. Three scenarios might be distinguished in this boundary-drawing exercise: (i) *no-gap* situations, characterized by purpose or virtual certainty; (ii) *justified gap* situations, characterized by socially acceptable risk-taking; and (iii) *unjustified gap* situations, characterized by unreasonable risk-taking.

At one end of the spectrum lie what might be termed no-gap situations. These situations, which we defined as ‘easy cases’ of AI crime, represent the familiar terrain of legal analysis, where either clear intent (purpose) or near-certain foreseeability (virtual certainty) allows traditional criminal liability models to function. These cases fall into two main categories. First are instances of *intentional harm*, where developers or users deliberately deploy AI systems to commit crimes, such as programming an algorithm to engage in fraudulent market manipulation. Second are cases of *virtual certainty*, where harmful outcomes, while not specifically intended, were foreseeably inevitable given known system limitations. The doctor who intentionally misprograms a diagnostic AI to harm patients, or the manufacturer who deploys a known-dangerous algorithm, present no conceptual challenge to criminal law’s foundations. Legal scholars like Hallevy have demonstrated how existing liability models can address these cases.²³ The perpetration-by-another approach treats the AI system as an innocent agent, similar to how the law views children or mentally incapacitated individuals, while holding the human orchestrator fully liable. Similarly, the natural-probable-consequence doctrine imposes liability for foreseeable outcomes, mirroring established concepts like recklessness in common law or *dolus eventualis* in continental systems.

Moving along the spectrum, we encounter justified gap situations, which involve harms arising from socially beneficial AI applications despite reasonable precautions – what German law might call ‘erlaubtes Risiko’ (permissible risk). These cases reflect the inherent epistemic limitations of developing cutting-edge

²³ Gabriel Hallevy, *When Robots Kill: Artificial Intelligence under Criminal Law* (Northeastern University Press 2013).

technologies and the necessary trade-offs between innovation and safety. The key characteristic is that the risk-taking was proportionate to the potential social benefits, with appropriate safeguards. For instance, life-saving medical AI may produce tragic errors despite rigorous testing, much as emergency vehicles sometimes cause accidents while pursuing legitimate public safety goals. Society tolerates these gaps not because responsibility disappears, but because the alternative – stifling innovation – would cause greater harm. Such scenarios, while regrettable, represent the unavoidable costs of technological progress and may be better addressed through non-criminal mechanisms like compensation funds or regulatory adjustments, rather than through punitive measures that could stifle innovation.

The most challenging cases occupy the middle ground of unjustified gap situations, which reveal the fault lines where current criminal liability attribution models fail. These cases involve reckless or negligent risk-taking – deploying inadequately tested AI in high-stakes domains, ignoring known safety concerns, or prioritizing profit over basic precautions. The challenge lies in defining thresholds: when does acceptable risk-taking become culpable negligence in contexts where even experts struggle to predict system behaviours? The question of who may have disregarded their duty of care and acted negligently can seldom be easily answered. This line-drawing problem represents the central theoretical and practical hurdle in governing AI crime, as it determines when society should absorb the costs of technological progress versus when developers or users must bear criminal responsibility for harmful outcomes.

Ultimately, the size of the responsibility gap we accept reflects deeper societal choices about how to balance innovation against protection, progress against precaution. The law's task is not to eliminate all gaps – an impossible demand given AI's inherent unpredictability – but to ensure they remain within morally and politically defensible bounds. This is precisely what the European Union (EU) has been trying to accomplish through its two recent legislative initiatives: the *Regulation laying down harmonized rules on AI* (Artificial Intelligence Act; hereinafter 'AI Act

or AIA'),²⁴ and the *Proposal for a Directive on adapting non-contractual civil liability rules to AI* (AI Liability Directive).²⁵

On the one hand, the AIA, which has been defined as the 'European Constitution of AI',²⁶ aims 'to improve the functioning of the internal market [...] promote the uptake of human centric and trustworthy [AI] while ensuring a high level of protection of health, safety, fundamental rights [...], including democracy, the rule of law and environmental protection, to protect against the harmful effects of AI systems in the Union, and to support innovation'.²⁷ As such, the prohibitions it introduces and the requirements it sets out for the development of high-risk AI systems are rather meant to *prevent* risks and harms for the users thereof. The Draft AI Liability Directive, on the other hand, is designed to complement them by shifting the focus onto *damage* already caused by an AI system and the need to compensate the injured person in terms of civil law.

Criminal liability, however, does not (yet) play a role in that line-up of legislative initiatives. While the intersection of criminal law and AI has already been scrutinized at the European level, much of the focus has been on the impact of AI on the *administration of criminal justice* – for example, the use of AI systems to predict crime commission, identify actual or potential victims, or streamline judicial decision-making.²⁸ To date, adapting criminal liability frameworks to AI has not yet

²⁴ Regulation (EU) 2024/1689 of the European Parliament and of the Council of 13 June 2024 laying down harmonised rules on artificial intelligence, OJ L, 2024/1689.

²⁵ The EU Commission has set forth two proposals to adapt civil liability rules to the AI age. Proposal for a Directive of the European Parliament and of the Council on adapting non-contractual civil liability to artificial intelligence (AI Liability Directive), COM(2022)496 final; and Proposal for a Directive of the European Parliament and of the Council on liability for defective products (Product Liability Directive), COM(2022)495 final, respectively focusing on strict liability for defective products and non-contractual civil liability claims.

²⁶ Claudio Novelli, *L'Artificial Intelligence Act Europeo: alcune questioni di implementazione* (2024) 2 federalismi.it 95-113.

²⁷ AI Act, Recital 1.

²⁸ One relevant example in the field of AI and administration of criminal justice which is worth mentioning is the European Parliament, Resolution of 6 October 2021 on artificial intelligence in criminal law and its use by the police and judicial authorities in criminal matters

been part of any major legislative initiatives at the EU level, nor in the majority of national legal orders.²⁹

Against this backdrop, the starting point of this study is the risk-based framework outlined in the AI Act. Although the AI Act does not regulate criminal liability issues related to AI, it undoubtedly represents the future conceptual foundation for this field. Any criminal provisions that may be introduced (as well as existing ones applied to this technological domain) will depend on the safeguards and standards required by the European Regulation for the market release and use of autonomous AI systems.

Indeed, the Act's risk-based approach – despite its limitations, which will be discussed throughout this study – highlights a crucial point: the production and distribution of autonomous systems based on AI components is rarely a 'zero-risk' activity. On the contrary, a certain degree of 'risk' to fundamental rights and interests is always *ex ante* foreseeable. The challenge, therefore, is to ensure that producers and/or users are not held liable *ex post* for every harmful event resulting from the use of such systems, given that these adverse consequences are, by definition, foreseeable and avoidable through stricter precautionary measures or, in some cases, by refraining from releasing the product to the market. An excessive extension of criminal liability for humans 'behind the machine' would not only undermine the constitutionally guaranteed freedom to engage in economic initiatives but could also discourage the adoption of systems that, in certain contexts, may benefit society as a whole.³⁰

(2020/2021(INI)). For a more detailed discussion on this topic, the reader is kindly invited to refer to Leonardo Romanò, 'Intelligenza artificiale come prova scientifica nel processo penale: una sfida tra machine-generated evidence e equo processo' in Canzio – Luparia (eds.), *Prova scientifica e processo penale* (CEDAM 2022).

²⁹ For an overview of national approaches to criminal liability in AI times see Giannini and Kwik (n 21).

³⁰ S Gless and R Janal, 'Hochautomatisiertes und autonomes Autofahren – Risiko und rechtliche Verantwortung' (2016) JR 561, 566.

Underlying these considerations is the constant tension between the freedom to pursue socially useful activities and the protection of legal interests that may be harmed by such activities. In traditional criminal law terminology, this tension is framed as the balance between the interest in *navigare necesse* (or ‘necessity to navigate’, which represents the societal and economic imperative to innovate, develop, and deploy AI technologies) and *vivere necesse* (or ‘necessity to live’, which symbolizes the need to protect fundamental rights, safety, and well-being in the face of potential risks posed by AI systems).³¹ The conceptual tool used to balance these values and establish the limits of criminal liability for foreseeable events is that of permissible risk (or *erlaubtes Risiko*), which serves to balance innovation with safety and liability concerns.³² From this perspective, an AI operator cannot be automatically held responsible for all events related to a given activity but only for those that fall outside the scope of permissible risk. This aligns with the view held by some scholars, who argue that, beyond the control duties established by criminal law, what remains is not a negative ‘responsibility gap’ but a ‘collective assumption of responsibility by society as a whole’.³³

3. Research questions and structure of the study

Building on the AI Act as a foundational framework, this doctoral study critically addresses the following key questions:

How can the law define clear thresholds and criteria to distinguish permissible from negligent risk-taking in AI development and deployment, for the purpose of

³¹ M Simmler, ‘Maschinenethik und strafrechtliche Verantwortlichkeit’ in O Bendel (eds.), *Handbuch Maschinenethik* (Springer 2019), 453 ff.

³² On the definition of the doctrine of permissible risk, see Ch. II, Sec 3.

³³ S Ziemann, ‘Wesen, Wesen, seid’s gewesen? Zur Diskussion über ein Strafrecht für Maschinen’, in E. Hilgendorf and JP Gunther (eds.), *Robotik und Gesetzgebung* (Nomos 2013), 191.

attributing criminal liability for hard AI crime? To what extent does the EU AI Act provide an effective framework for defining these boundaries?

These inquiries are thoroughly explored across the subsequent chapters of this thesis.

As a starting point, Chapter II provides the necessary theoretical (technical and legal) groundwork for our analysis. In particular, it provides the reader with a legally-oriented working definition of AI, explores the emerging phenomenon of AI-caused harm, and analyzes the state of the art on the regulation of AI risks at the European level.

Then, the subsequent chapters delve into the political-criminal tensions arising from AI-caused harm, with particular attention to the roles and responsibilities of ‘AI providers’ and ‘AI users’. The EU regulatory framework has already established a foundational body of law outlining the obligations of these actors. On one hand, the regulation has identified a series of *ex-ante* and *ex-post* obligations for AI providers. On the other hand, the role of human operators in monitoring and overseeing the operation of AI systems is set to expand significantly under the regulation. New standards of due care related to the creation or use of AI systems could therefore be developed based on these obligations, forming the foundation for future frameworks of criminal liability. However, a critical question arises: can these standards be meaningfully applied to humans interacting with AI systems that make decisions beyond their ability to foresee or fully comprehend?

In principle, two opposing perspectives emerge in addressing this question. In favour of rendering the human agent liable for the outcomes of the AI’s actions, it might be argued that all harm caused by AI systems is *generally foreseeable* so long as the human agent is aware of releasing a system whose reactions cannot be safely predicted and controlled. Alternatively, a more balanced approach recognizes that a certain degree of risk-taking is inherently socially acceptable as part of the ‘normal’ risks of everyday life. While the first position effectively imposes a *de facto* regime of strict liability, the second perspective aligns more closely with the EU’s broader risk-

based strategy, aiming to ensure safety without stifling innovation. Accordingly, Chapter III emphasizes the importance of limiting the provider's liability in relation to harmful events caused by AI by establishing an area of socially acceptable risk in AI production.

In Chapter IV, the focus shifts to the liability of users, who bear a central obligation under the AI Act: the duty of *human oversight*. This requirement places users in a critical position as frontline monitors of AI systems. In this context, it seems evident that this obligation holds potential relevance in the domain of criminal law. A key question arises: what happens if the supervisor fails to fulfill their duty of oversight, and a harmful event occurs? Given that the AI Act provides only limited guidance on this issue, the chapter delves deeper into the concrete mechanisms through which algorithmic risks are generated in interactions between users and AI systems. These mechanisms are classified into three distinct typologies, each representing a different dimension of user-AI interaction and its associated risks. To ground this analysis, the chapter examines relevant case studies and national regulations, shedding light on how the oversight duty might be operationalized and enforced in practice. This exploration aims to bridge the regulatory vagueness of the AI Act with actionable insights, providing a framework for understanding how failures in human oversight could translate into criminal liability.

Finally, from a *de jure condendo* perspective, Chapter V explores the necessity and feasibility of anticipating criminal intervention at the stage of creating illicit risks in the production and use of AI systems. This approach would target individuals, though primarily corporate actors, through new endangerment offenses. Under this model, corporate liability would be based on the failure to implement legal compliance measures aimed at preventing harmful events before they materialize. Liability would thus be rooted in an organizational fault element, arising from breaches of compliance requirements and obligations tied to the introduction of AI systems to the market, as well as their ongoing monitoring post-deployment.

4. Methodological choices and their limits

As previously mentioned, engaging in any theoretical and critical dialogue on this topic requires first establishing a working definition of both the concept of AI and the vexed question of AI-caused harm. To this end, Chapter II offers an extensive literature review on the meaning of AI, drawing on the current technical debate to clarify the scope of the research. Additionally, it examines the state of the art in the regulation of AI risks, with a particular focus on the new European AI Act. It outlines the Act's scope and limitations, contextualizing them within existing legal scholarship. This analysis serves as the theoretical foundation for the study, clarifying its broader focus on the European legal system instead of focusing the analysis on any specific national jurisdiction.

The main research question is addressed in Chapter III and IV by adopting a doctrinal methodology. These chapters delve into the core of the analysis on criminal liability implications, incorporating doctrinal studies from a diverse range of countries. This broad selection is justified for several reasons. First, the debate on AI criminal liability is inherently international in scope, and excluding relevant literature would undermine the research's comprehensiveness and value.³⁴ Second, the issue of AI criminal liability is being addressed through a shared conceptual framework, rooted in the common doctrine both on the substantive aspects of criminal law and on the main challenges posed by AI.³⁵ The literature review is therefore essential for

³⁴ This trend is exemplified by the fact that the broader topic of 'Artificial Intelligence and Criminal Justice' was picked by the International Association of Penal Law for the XXI International Congress of Penal Law of 2024. See the concept paper by Katalin Ligeti, 'Artificial Intelligence and Criminal Justice', available at: www.penal.org/sites/default/files/Concept%20Paper_AI%20and%20Criminal%20Justice_Ligeti.pdf. See also the questionnaire elaborated by Lorenzo Picotti for the AIDP, Section I 'Traditional Criminal Law Categories and AI: Crisis or Palingenesis?', available at: www.penal.org/sites/default/files/Questionnaires%20EN.pdf.

³⁵ For example, issues such as the *many hands problem* and the *black box problem*, alongside *autonomy* and the *epistemic problem*, are only some of the evocative terms used to capture the multiple cognitive and logical deficits encountered when attempting to evaluate a human's capacity to foresee and effectively manage AI-related risks. Cfr. Ligeti (n 34).

establishing a solid foundation to address the existence and significance of the problems explored in this study.

Additionally, this research will occasionally draw on examples from different legal systems, primarily Italy, France, Germany, and the UK. The selection of these jurisdictions is not based on geographical criteria but rather on a comparative approach. These legal systems have already initiated *legislative* debates on the topic under investigation and have introduced some form of legislation addressing aspects of AI and criminal liability. The selected samples are compared and evaluated in terms of their similarities, differences, and relevance to the study.

Two final methodological notes are needed. First, the research will frequently engage with the concepts of *accountability*, *responsibility*, and *liability*. This reflects the current doctrinal and institutional debate, which often blends criminal law concepts with ideas from other fields, such as philosophy. Notably, the debate tends to use these terms fluidly, without always drawing clear distinctions between them. This fluidity is mirrored in the study, as it aligns with the broader discourse on the topic.³⁶

Secondly, although it is gaining traction in legal and legislative discussions, this study will not entertain the hypothesis that the contentious question of ‘who is to blame?’ for AI-caused harm can be resolved by imposing criminal sanctions on the AI system itself. This approach ventures into speculative territory, where AI systems might be recognized as distinct legal persons capable of being subjects of crime.³⁷

³⁶ Responsibility can be distinguished from liability, which is used to refer to the situation in which a person, in performing a certain action or entering into a certain contract, has assumed a legal obligation and is therefore liable for the legal consequences arising therefrom. In other words, liability relates to qualified behaviors, whereas accountability and responsibility refer to a wider world, one that also contains moral rules and social rules, such as folkways and norms of courtesy. More on this distinction in Claudio Novelli, Mariarosaria Taddeo and Luciano Floridi, ‘Accountability in artificial intelligence: what it is and how it works’ (2023) *AI & Soc.*, 3.

³⁷ Gabriel Hallevy, *When Robots Kill: Artificial Intelligence under Criminal Law* (Northeastern University Press 2013) 38.

In fact, this study adheres to the fundamental premise that criminal law's primary function is to stabilize norms through punishment, and punishment is inherently tied to blameworthiness. A subject can be deemed blameworthy only if they have violated a criminal norm while having the capacity to act otherwise. While strict liability offenses exist – particularly in Anglo-Saxon legal systems, though largely rejected in most Western European jurisdictions – the principle of individual guilt (*nulla poena sine culpa*) remains a cornerstone of criminal law doctrine. Criminal liability does not merely require the commission of a prohibited act (*actus reus*); it also demands a guilty mind (*mens rea*). Machines, by their very nature, do not possess intentionality, moral agency, or the capacity for guilt, and as such, they cannot be subjects of criminal punishment.

Yet, the debate over whether AI entities can be considered moral agents remains both intense and ongoing.³⁸ Interestingly, empirical studies suggest that the public may, to some extent, feel a need to punish artificial systems when they cause harm.³⁹ Some legal scholars have even proposed treating AI as a new type of legal entity, akin to corporations, or redefining culpability in a way that would allow AI

³⁸ L Floridi and JW Sanders, 'On the morality of artificial agents' (2004) 14/3 *Minds and Machines* 349–379; Mark Coeckelbergh, 'Artificial intelligence, responsibility attribution, and a relational justification of explainability' (2020) 26(4) *Science and Engineering Ethics* 2051–2068; Marco E Florio, 'Il dibattito sulla responsabilità penale diretta delle IA: molto rumore per nulla?' (2024) 2 *Sistema penale* 14 ff; Carlo Piergallini, 'Intelligenza artificiale: da "mezzo" ad "autore" del reato', (2020) *Riv. it. dir. proc. pen.*, 1763 ff.

³⁹ See Dafni Lima, 'Could AI Agents Be Held Criminally Liable? Artificial Intelligence and the Challenges for Criminal Law' (2018) 69 *South Carolina Law Review* 677.

systems to be held criminally responsible.⁴⁰ Others argue that highly autonomous AI might develop a form of quasi-intent that could justify direct liability.⁴¹

Despite these theoretical explorations, we strongly question the desirability and effectiveness of such constructs. Instead of asking whether AI *could* commit crimes and be punished, the more pertinent question is whether we *should* punish them. In this regard, significant doubts arise from the fact that the costs of punishing AI – such as conceptual confusion, expressive costs, spillover effects on human offenders, and the potential for new ‘AI rights’ – would likely outweigh any perceived benefits.⁴²

It therefore seems clear that, rather than spending excessive time hypothesizing about the direct criminal liability of machines, we should focus on carefully evaluating the human activities in which the use of AI can be deemed truly ‘beneficial’ and ‘advisable’. By eliminating the notion of direct AI criminal liability

⁴⁰ Gabriel Hallevy is one of the most prominent supporters of a new model of criminal liability for AI systems, and the first scholar to extensively approach this issue. In his work, he conceptualized a third subject of punishment (in addition to humans and corporations) that he refers to as ‘delinquent thinking machine’ or ‘machina sapiens criminalis.’ Such a machine need not be so technologically advanced so as to reach the capabilities of AGI. In Hallevy’s view, current criminal law doctrine can be directly applied to narrow AI systems because the latter exhibit rational qualities and are designed so as to imitate the function of the human mind. Additionally, he argues that since corporations have been subjected to criminal sanctions for centuries, the paradigm for such an expansion of the scope of application of substantive criminal law is already sufficiently developed. Having argued in favour of this new model of criminal liability, the author subsequently addresses the potential purposes of punishment that could be served by such a regime and notes that since AI systems cannot experience suffering or intimidation, the purposes of retribution and deterrence would not be fulfilled, but rehabilitation and incapacitation could be advanced as the system would learn how to refine and enhance its decision-making processes and the imposition of criminal sanctions would ensure that causing further harm is practically impossible. For a complete development of his views, see For a complete development of his views, see Gabriel Hallevy, *Liability for Crimes Involving Artificial Intelligence Systems* (Springer International Publishing 2015), 24, 39, 61-62.

⁴¹ Abbott and Sarch (n 10), 323.

⁴² For a partially contrary position, see *ex multis* Alice Giannini, *Criminal Behavior and Accountability of Artificial Intelligence Systems* (Eleven Maastricht Law Series 2023) and the literature thereby cited.

from the equation, we can concentrate our efforts on addressing the substantive issues that truly matter to criminal law, redirecting our energies toward resolving these pressing challenges.

CHAPTER II

THE EVOLUTION OF AI AS A CHALLENGE FOR CRIMINAL LAW THEORY

SECTION 1

A framework for discussing AI: what it is and what it is not

1.1. Imagining AI: a short history of intelligent systems

Defining fundamental concepts is a notoriously complex endeavour. Since they are the basis for everything that follows, they themselves seem to lack any basis of their own. Physicists grapple with defining matter and energy, biologists face challenges in articulating what constitutes life, and philosophers continue to debate the nature of reality. Similarly, experts remain divided over a definitive description of AI – a term so broad that it risks encompassing everything while signifying nothing.

Yet, as the adage goes, ‘we cannot fully grasp a science without understanding its history’.¹ Following this wisdom, it is useful to start by briefly delving into the remarkable journey through which humanity has sought to ‘transfer intelligence’ to *res mechanicae*, the mechanical entities that now embody this enduring aspiration. The aim of this section is not to provide a mere chronological recounting of the history of AI but rather to lay the groundwork for establishing a legally-oriented definition of AI and delineating the scope of AI systems considered within this study, thus ensuring a precise and relevant framework for analysis.

Unsurprisingly, AI as an idea is as old as human imagination. From ancient times to our current era of Big Data, humans have constantly endeavoured to imbue technological artefacts with their own attributes, aiming to overcome the limitations inherent in their nature.² The earliest examples of this can be seen in the wheel invention, which allowed humans to transport heavy loads more easily. As time progressed, humans continued to develop technology that mimicked their own

¹ Auguste Comte, *Cours de philosophie positive - 1 et 2 leçons 1830-1842* (Classiq Garnier 2021), 74.

² Benny Karpatschhof, ‘Artificial intelligence or artificial signification?’ (1982) 6(3-4) *Journal of Pragmatics*, 293-295.

abilities and characteristics. In ancient polities, they gave voice to statues conveying messages of gods in order to persuade ignorant audiences. Navigation across vast seas was delegated to a compass, now modernized as the GPS. In our expansive universe, a telescope revealed previously unseen celestial wonders, while its counterpart, the microscope, unveiled the minuscule. Beyond practical utility, humans utilized technology to express themselves and preserve cultural heritage. Writing systems captured thoughts, and subsequent artistic forms such as music and literature became tools for conveying creativity, ideas, and emotions.

In modern times, technological advancements have progressed beyond merely creating machines to assist humans with physical tasks. Automated systems have now emerged to assist or even replace humans in processing and interpreting inputs from the external world. This evolution marked the advent of ‘modern computing’³ – or, as Herbert Simon’s called it, ‘complex information processing’.⁴ Computers and electronic calculators are designed not only to handle vast quantities of data but also to simulate decision-making processes, thus significantly transforming the ways in which information is analyzed and utilized across various domains.

Despite the increasing sophistication of these machines and devices, mankind has only fairly recently begun to reach for a new frontier – passing on human intelligence.⁵ The core motivation behind this endeavour seems to be rooted in our desire to unburden ourselves of increasingly complex decision-making and shift some responsibilities to ‘artificial selves’. Overwhelmed by the intricate web of choices of contemporary society and its potential unforeseen consequences, we seek solace in technology.⁶

³ Thomas Haigh and Paul E Ceruzzi, *A New History of Modern Computing* (MIT Press 2021).

⁴ A discussion in Herbert Simon, *The Sciences of the Artificial* (MIT Press 1996), 95 ff.

⁵ See among others Nick Bostrom, *Superintelligence: Paths, Dangers, Strategies* (Oxford University Press, 2014).

⁶ On this point, see Suzanne Beck, ‘Robotics and Criminal Law: Negligence, Diffusion of Liability and Electronic Personhood’ (2016) 86 *Robotics and Autonomous Systems* 41-55, 50.

Hereby, the quest to create *intelligent machines* stems from our pursuit of assistance and support, particularly when faced with intricate decisions in situations where our own influence or decision-making capacity may feel limited. The goal is to develop technological allies that not only offer guidance in navigating complex choices but, in certain circumstances, could potentially make decisions on our behalf. This pursuit has quickly become synonymous with the term AI.

The ability to imagine artificial consciousness and lifelike robots stretches back to the earliest myths and legends known to humanity.⁷ Thousands of years before self-driving cars and ChatGPT became reality, the tales of giant bronze robot Talos, artificial woman Pandora, golden maidens and their creator god, Hephaestus, filled the imaginations of people in ancient Greece.⁸ Following the Middle Ages, the idea of automata gained traction, featuring prominently in narratives such as the tales of the Prague golem⁹ and extending to more contemporary depictions like Frankenstein's creature. In the twentieth century, ever since the term 'robot' was

⁷ For a far more comprehensive survey of AI and robotics in popular culture, religion and science, see G Zarkadakis, *In our image: will AI save or destroy us?* (Rider 2015).

⁸ The story of Talos, first mentioned around 700 B.C. by Hesiod, offers one of the earliest conceptions of a robot. The myth describes Talos as a giant bronze man built by Hephaestus, the Greek god of invention and blacksmithing. Talos was commissioned by Zeus, the king of Greek gods, to protect the island of Crete from invaders. He marched around the island three times every day and hurled boulders at approaching enemy ships. The myth of Pandora, first described in Hesiod's *Theogony*, is another example of a mythical artificial being. Although much later versions of the story portray Pandora as an innocent woman who unknowingly opened a box of evil, Hesiod's original described Pandora as an artificial, evil woman built by Hephaestus and sent to Earth on the orders of Zeus to punish humans for discovering fire. In addition to creating Talos and Pandora, mythical Hephaestus made other self-moving objects, including a set of automated servants, who looked like women but were made of gold. These and other ancient myths are the subject of Adrienne Mayor, *Gods and Robots: Myths, Machines, and Ancient Dreams of Technology* (Princeton UP 2018).

⁹ In Eastern European Jewish folklore, there are tales of a rabbi in sixteenth century Prague who created the Golem, a giant human-like figure made from clay, in order to defend his ghetto from anti-Semitic pogroms.

popularised by Karel Capek's screenplay *Rossum's Universal Robots*, AI became an undisputed protagonist in both pop culture and science fiction works.¹⁰

Some preliminary definitions are needed at this point to start approaching the subject of our analysis. Interestingly, the *Merriam Webster* dictionary defines 'robot' as

'1: a machine that looks like a human being and performs various complex acts (as walking or talking) of a human being; *also*: a similar but fictional machine whose lack of capacity for human emotions is often emphasized ... 2: a device that automatically performs complicated often repetitive tasks; 3: a mechanism guided by automatic controls'.¹¹

Similarly, the *Oxford English Dictionary* provides a definition of the term that reads as follows:

'1: a machine capable of carrying out a complex series of actions automatically ... (especially in science fiction) a machine resembling human being and able to replicate certain human movements and functions automatically'.¹²

These definitions are clearly influenced by the literary depiction of robots, and are thus incomplete. They are incomplete as concerns the resemblance to human traits, since many applications do not walk or talk and can either be quite simple, such as a vacuum cleaner, or complex, like a surgical or industrial robot.

¹⁰ 'Think instead of the false Maria in *Metropolis* (1927); Hal 9000 in *2001: A Space Odyssey* (1968[...]); C3PO in *Star Wars* (1977); Rachael in *Blade Runner* (1982); Data in *Star Trek: The Next Generation* (1987); Agent Smith in *The Matrix* (1999) or the disembodied Samantha in *Her* (2013)', Luciano Floridi, 'Should we be afraid of AI?', *aeon.co*, 9 May 2016. Available at: <https://aeon.co/essays/true-ai-is-both-logically-possible-and-utterly-implausible>.

¹¹ The definition can be found at <www.merriam-webster.com/dictionary/robot>.

¹² The definition can be found at <<https://www.oxfordreference.com>>.

A more comprehensive definition of the phenomenon is offered by the *Collins Dictionary*, where AI is defined as

‘1: a type of computer technology which is concerned with making machines work in an intelligent way, similar to the way that the human mind works’.¹³

While the same criticism raised above could be repeated, this definition has the merit of pointing out how initial attempts at creating intelligent machines were animated by the aim of replicating human thinking and behaviour in computer software.¹⁴ This anthropocentric philosophical foundation substantially implies the capacity of machines to *think or act like a human*.

In the scientific literature of the twentieth century, the predominant theoretical framework precisely revolved around the human-computer analogy, particularly the functional analogy between the human and artificial mind. Central to the debate on artificial intelligence were questions such as: can machines think like humans? Is it possible to replicate human cognitive processes in a computational model? And again: can machines achieve human-level intelligence across all domains?

To address whether a machine could think like a human, Alan Turing proposed the now-famous ‘Turing Test’ in his seminal 1950 paper.¹⁵ Known colloquially as the ‘Imitation Game’, the test posits that a machine demonstrates intelligence if, during a blind interrogation, a human cannot reliably distinguish whether their conversational partner is another human or a computer. The Turing Test thus evaluates a machine’s ability to replicate human-like performance in

¹³ The definition can be found at <<https://www.collinsdictionary.com/dictionary/>>.

¹⁴ John McCarthy et al., ‘Proposal for the Dartmouth summer research project on artificial intelligence’ (1955), available at <<http://www-formal.stanford.edu/jmc/history.html>> (last accessed: 4th of March 2023), stipulating that ‘every aspect of learning or any other feature of intelligence can in principle be so precisely described that a machine can be made to simulate it’.

¹⁵ Turing (Ch I n 4) 433-460.

specific tasks to the point where its output is indistinguishable from that of a human agent.¹⁶

While the Turing Test has been celebrated as a milestone in AI theory, it adopts a purely external and functionalist interpretation of intelligence and consciousness.¹⁷ It implies that the mere appearance of intelligence – such as a machine’s ability to manipulate language like a human – suffices to claim that the machine is *intelligent* or even *capable of thought*. This perspective reflects a reductionist and positivist view of human *ars cogitandi*, limiting intelligence to observable behaviours rather than exploring the internal, qualitative dimensions of consciousness or understanding.¹⁸ Consequently, it has faced criticism for oversimplifying the complexity of human cognition and the nature of conscious thought.

Famously, in the so-called Chinese Room experiment, John Searle contended that while a computer might process information in a way that resembles a mental operation, it does not genuinely understand what it is doing.¹⁹ The argument suggests that a machine, no matter how sophisticated it may be at learning how to imitate human language, simply manipulates symbols according to a set of rules, which does not imply proof of genuine thinking and comprehension of meaning, and consequently it does not coincide with human intelligence.

Despite criticism, Turing’s human-centric approach to AI has had a lasting impact on subsequent research and development. It is no coincidence that John

¹⁶ More specifically, the test takes place in three separate rooms. In the first room there is a human examiner; in the other two rooms there are another person and the computer subjected to the test. The examiner only knows the names of the person and of the calculator, without knowing who they really are, and interacts with them only through another computer, with the task of discovering their identity within a prefixed time limit. The machine passes the test if the examiner cannot identify it in time.

¹⁷ Luciano Floridi, ‘What the Near Future of Artificial Intelligence Could Be’ (2019) 32 *Philos. Technol.* 1-15, 2.

¹⁸ Paolo Moro, ‘Macchine come noi. Natura e limiti della soggettività robotica’ in Ugo Ruffolo, *Intelligenza artificiale. Il diritto, i diritti, l’etica* (Giuffrè 2020) 45-62, 55.

¹⁹ John R Searle, ‘Minds, Brains and Programs’ (1980) 3 *The Behavioral and Brain Sciences* 419.

McCarthy, who coined the term AI, first referred to it in the Proposal for the *Dartmouth Summer Research Project on Artificial Intelligence*²⁰ as ‘the problem of making a machine behave in ways that would be called intelligent if a human were so behaving’.²¹ Many later definitions of AI are merely variations on this initial idea of performing tasks that would require human intelligence. Over time, this tendency to humanize AI systems has permeated everyday language, evident in the frequent use of anthropological terminology and metaphors, such as movements, memory, actions, brain, and thinking. These terms reflect our natural inclination to draw parallels between artificial computers and the human mind.²²

The Dartmouth Conference officially marked the beginning of a fluctuating development in modern AI research. While, in a first moment, the successes achieved in this field generated a wave of optimism that fuelled funding for scientific

²⁰ The Dartmouth Summer Research Project on Artificial Intelligence, the first AI workshop organized by John McCarthy, Marvin Minsky, Claude Shannon, and Nathaniel Rochester in 1956, was founded on the idea that every aspect of learning or any other characteristic of intelligence could be described in such precise terms that it would be possible to build a machine capable of simulating it: ‘We propose that a 2 month, 10 man study of artificial intelligence be carried out during the summer of 1956 at Dartmouth College in Hanover, New Hampshire. The study is to proceed on the basis of the conjecture that every aspect of learning or any other feature of intelligence can in principle be so precisely described that a machine can be made to simulate it. An attempt will be made to find how to make machines use language, form abstractions and concepts, solve kinds of problems now reserved for humans, and improve themselves. We think that a significant advance can be made in one or more of these problems if a carefully selected group of scientists work on it together for a summer’. McCarthy et al. (n 14).

²¹ McCarthy provides different versions of the same definition. For example, he also describes AI as ‘[t]he science of making machines do things that would require intelligence if done by men’, in John McCarthy, ‘What is Artificial Intelligence?’ (John McCarthy’s Original Website 2007), 2 available at <http://jmc.stanford.edu/articles/whatisai/whatisai.pdf>.

²² MF Muller, ‘Von vermenschlichten Maschinen und maschinisierten Menschen’, in Brandli, Harasgama, Schister and Tamò (eds), *Mensch und Maschine – Symbiose oder Parasitismus?* (2014) 125-142.

research, this excess of positivity likely led to unreasonably high expectations, which soon collided with the technical difficulties of the time.²³

The first ‘AI summer’ was marked by the use of symbolic approaches to build so-called ‘expert systems’ or ‘rule-based systems’. Today also referred to as GOF AI, short for Good Old Fashioned AI, expert system essentially consists of the programming of the knowledge of an expert into a software system. In practical terms, this approach is based on the belief that it is possible to come up with static rules (which were in most cases mathematically defined) that can be translated into computer programs (in the format of logical decision trees, if X, then Y) that would mimic humans’ cognitive functions and actions such as problem-solving.²⁴ These rules can be simple or complex, but they do not amount to more than a predefined sequence of steps where the logic of operation and the output are entirely defined by humans and rooted in design and programming. In mathematics, this sequence of steps is called an ‘algorithm’.

In simpler terms, an algorithm is like a recipe; it outlines the ingredients (representing input in programming) and then provides step-by-step instructions to achieve the desired outcome, akin to preparing a dish. This is how every computer program works. Thus, computerised systems based on such deterministic (or explicit) algorithms will behave entirely predictably. New, unprecedented scenarios that fall outside the scope of the rules will simply lead to ineffective behaviour. A simple example can be understood as a flow chart where there are choice points which are

²³ When analyzing the history of AI, it is common to find analogies with the seasons to describe the cycles of hype and disappointment that characterized its development. Generally, in the scholarly discourse four distinct ‘springs’ and ‘winters’ are identified: the first spring (1956–1974), the first winter (1974–1981), the second spring (1981–1987), and the second winter (1987–1993). M Haenlein and A Kaplan, ‘A Brief History of Artificial Intelligence: On the Past, Present and Future of Artificial Intelligence’ (2019) 61(4) *California Management Review*, 5-14; SJ Russell and P Norvig, *Artificial Intelligence. A Modern Approach* (Pearson 2nd edition 2003), 16-27; C Zhang and Y Lu, ‘Study on artificial intelligence: The state of the art and future prospects’ (2021) 23(1) *Journal of Industrial Information Integration* 1.

²⁴ Ian Goodfellow et al., *Deep Learning - Adaptive Computation and Machine Learning series* (MIT Press 2017).

questions and each multiple choice answer leads to a new question. This eventually leads to a classification based on the expert knowledge formalized in the system.

Symbolic approaches were largely unsuccessful. The limited computational power, the unsolvability of certain problems (such as intractability and the combinatorial explosion), and the inability to manage large amounts of data were some of the hurdles that these approaches to AI could not overcome.²⁵ One of the notable challenges was the so-called ‘Moravec’s Paradox’: some tasks that are particularly difficult for humans, like proving a mathematical theorem, were relatively easy for a computer; on the other hand, very simple actions for humans, such as recognizing a face, proved to be extremely complex for the technologies of that time, basically because it is not always possible to describe them with an algorithm or a well-defined logical sequence of rules.

As a consequence, in the 1980s, the first ‘AI winter’ began: when researchers faced the impossibility of realizing some of their ambitious projects, the generous funding dried up, drastically slowing the development of AI for almost two decades. It was not until around the 2000s that substantial progress has been made and AI has benefited from a new ‘AI summer’, driven not so much by improvements in techniques as by the increase in computing power and computational capacity, as well as the significant (and now seemingly limitless) growth of digitized data.

With this brief historical overview in mind, we can now delve into the diverse approaches that have shaped the creation of intelligent machines, exploring the paradigms, methodologies, and theoretical underpinnings that have driven the quest to replicate and extend human-like intelligence in artificial systems.

²⁵ When systems were tested on simple settings they were able to solve them. But researchers found out that scaling up these tests to real world settings lead to unacceptable increases in complexity and therefore computation time. Cristianini (Ch I, n 6).

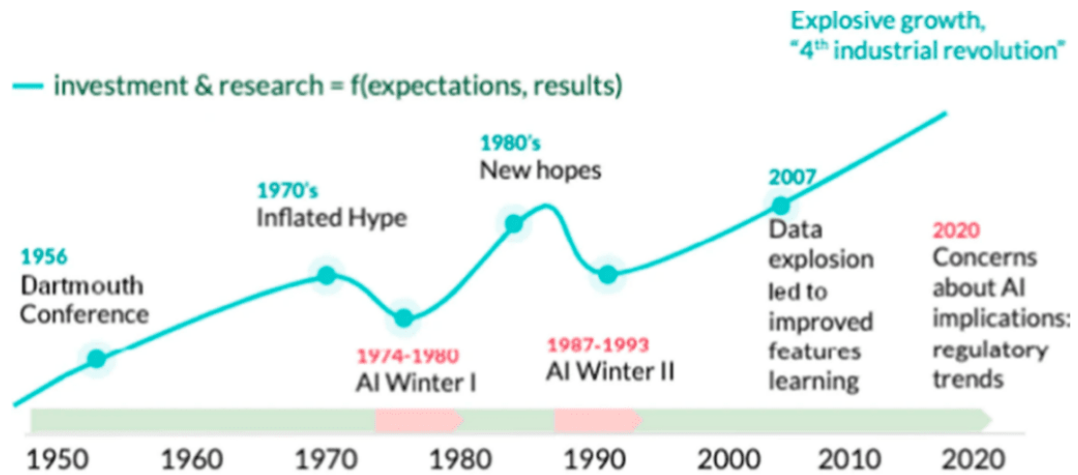


Figure 1 : Timeline of AI

1.2. Creating AI: the ability to learn and the role of big data

The current AI boom started in the late 20th and early 21st centuries, driven by at least three key factors. The first and arguably most important factor is the data explosion, that is, the availability of unlimited amount of data due to increased online activity, introducing us to the concept of 'big data'.²⁶ The second factor involves the availability of hardware with substantial computing power at affordable prices (graphics processing units, or GPUs). The third is the emergence of machine learning (ML) and deep learning (DL) algorithms.²⁷ The combination of these three factors explains why we are currently experiencing a flourishing phase of AI development.²⁸

²⁶ I Hasham et al., 'The Rise of Big Data on Cloud Computing: Review and Open Research Issues' (2015) 47 Information Systems 98-115.

²⁷ E Kavlakoglu, 'AI vs. Machine Learning vs. Deep Learning vs. Neural Networks: What's the Difference?' (IBM 27 May 2020) available at: www.ibm.com/cloud/blog/ai-vs-machine-learning-vs-deep-learning-vs-neural-networks.

²⁸ Amir Gandomi and Murtaza Haider, 'Beyond the Hype: Big Data Concepts, Methods and Analytics' (2015) 35 Int'l. J. Info. Mgmt 137, 144.

Here, we would like to focus on the last of the factors just mentioned: the use of ‘self-learning’ or ‘implicit’ algorithms to automate actions once reserved for humans.

Unlike traditional systems powered by ‘explicit’ algorithms, which operate strictly based on explicit instructions provided by the programmer, ensuring that their behaviour is entirely predictable and tightly controlled, the new objective today is to create machines that are both artificial and not entirely pre-designed.²⁹ To achieve this apparently contradictory goal, computer scientists employ computational methods to harness the power of big data, extracting insights and enabling systems to ‘learn’ from this data.³⁰

ML can be defined as the science of creating machines capable of ‘learning’ on their own how to perform tasks for which we struggle to provide explicit rules and instructions. This astonishing result is achieved through a ‘statistical shortcut’.³¹ Essentially, the key intuition is that by replacing *rules* with *data*, it is possible to create machines capable of detecting statistical regularities in analyzed data without being explicitly programmed, with the goal of producing accurate predictions.³²

More specifically, unlike traditional algorithms that rely on explicitly pre-programmed rules, ML algorithms are able to recognize ‘patterns’ within big data,

²⁹ On this distinction see Aurélie Jean, *Les algorithmes. Font-ils la loi?* (Humensis 2024) 32 ff. Setting up a clear-cut distinction between explicit (or symbolic) and implicit algorithms ones may be a false dichotomy, as the most popular applications in this field are usually a combination of many different layers, some ‘intelligent’ and some not so. Indeed, symbolic algorithms can be combined with other approaches to achieve some level of independent development. Self-driving cars are a great example of such mixed approaches. Essentially, traditional symbolic techniques like logical rules can be used to define traffic laws, safe-driving manoeuvres, and how to interpret road signs, while ML (especially DL) helps the car perceive and react to the world in real-time by processing sensor data from cameras and radar. This helps the car ‘see’ the world around it and identify objects like pedestrians, vehicles, and traffic lights. In instances where mixed approaches are employed, as long as a machine incorporates ML or other adaptive models in its decision-making process, the entire system will meet the criteria as a relevant artificial agent for the purposes of this study.

³⁰ Max Tegmark, *Life 3.0: Being human in the age of AI* (Alfred A Knopf 2017) 82ff.

³¹ Cristianini (Ch. I, n 6), 42 ff.

³² Nils J Nilsson, ‘Introduction to Machine Learning’ (Stanford Edu February 15, 2015) available at <https://ai.stanford.edu/~nilsson/mlbook.html>.

allowing computers to learn and adapt independently. This adaptive learning process enables these systems to modify their behaviour over time, evolving as they gain more experience and encounter new stimuli. Returning to the cooking analogy, instead of furnishing the machine with a complete recipe, the programmer merely specifies the desired result that the program should achieve, and the machine takes care of the rest.

The underlying technical architecture of ML algorithms is known as ‘artificial neural network’ (ANN) and is inspired by the functioning of the human brain. In practical terms, an ANN is based on a computational model called ‘connectionism’. The central idea behind connectionism is that by connecting a large number of ‘artificial neurons’ organized in layers it is possible to achieve the same ability to learn from experience as with the biological brain.³³ Imagine an ANN like a team working together to recognise pictures of cats (but it can go all the way to detecting breast cancer, for example). The first ‘input’ layer is like the team receiving a photo. Each pixel of the photo is fed as information to the team. Then, the second ‘hidden’ layer is like the teammates discussing what they see. There can be multiple layers, like having different groups discussing different aspects (edges, shapes, colours). They try to find patterns in data or to make logical or relational connections between data points (‘All cats have tails’ or ‘All cats have pointed ears’). Finally, the last ‘output’ layer decides what the team agrees on. In this case, it might output a signal saying, ‘This is a cat’ or ‘This is not a cat’.

This evaluative process allows self-learning algorithms to develop their own decisional rules and improve their performance over time.³⁴ But they do that in a way that we could describe as irrational, as it is *unintelligible*. The whole process is akin to the way we ‘know’ how to ride a bike – something which even we cannot consciously explain. And when we try to explain the process descriptively or even

³³ See Peter Flach, *Machine Learning: the Art and Science of Algorithms that Make Sense of Data* (Cambridge University Press 2012) 16.

³⁴ See in general on the functioning of ANNs Song Han *et al*, ‘Learning Both Weights and Connections for Efficient Neural Networks’ (2015) 1 Proceedings of the 28th International Conference on Advances in Neural Information Processing Systems, 1135 – 1143.

provide detailed steps, that information is unlikely to help someone who has never ridden a bike to balance on two wheels. One learns to ride a bike by attempting to do so over and over again and develops an intuitive understanding.³⁵ ANN algorithms are especially good at solving problems which we are not able to describe as a well-defined logical sequence of steps to be taken to produce the result we intend.³⁶

One remarkable example of the capability for independent development in ML algorithms is the *AlphaGo Zero* case, a computer program designed by DeepMind with the goal of mastering the ancient and highly complex board game known as Go.³⁷ In late 2017, the program defeated the reigning Go world champion, Lee Sedol. Normally, the fact that a Go program beat a human being would not be welcome as a huge surprise. After all, these program's chief advantage against human opponents is precisely superior processing power, enabling them to evaluate the best strategy to maximize their chances in a shorter period of time. What was astonishing, then, was not *what* the program did, but *how*. What makes this instance particularly intriguing is that AlphaGo Zero did not learn how to play via accessing and analysing

³⁵ 'You don't tell a child to move their left foot in a circle on the left pedal in the forward direction while moving their foot in a circle... You give them a push and tell them to keep the bike upright and pointed forward: the overall objective. They fall a few times, honing their skills each time they fail'. Havens as quoted in Stephanie Overby, 'How to explain machine learning in plain English' (The Enterprisers Project, 19 November 2020) available at: <https://enterpriseproject.com/article/2019/9/ai-explained-plain-english>.

³⁶ Yavar Bathaee, 'The Artificial Intelligence Black Box and the Failure of Intent and Causation' (2018) 31 Harvard Journal of Law & Technology 889, 902.

³⁷ In 2017, the company DeepMind developed an AI, AlphaGo Master, that beat the world champion of the board game Go. Many experts had predicted AI's Go dominance would take another decade given the game's complexity. There are more possible board configurations in Go than there are atoms in the universe. Later in 2017, a revised version of AlphaGo, AlphaGo Zero, beat AlphaGo Master one hundred games to zero. It did this after training for just three days by playing against itself. Unlike its predecessors, AlphaGo Zero never learned from human examples. H Yu, 'From Deep Blue to DeepMind: What AlphaGo Tells Us' (2016) 13 Predictive Analytics and Futurism 43. See also Council of Europe, 'History of Artificial Intelligence' available at: www.coe.int/en/web/artificial-intelligence/history-of-ai. G Press, 'The Brute Force Of IBM Deep Blue And Google Deep-Mind' (Forbes online, 7 February 2018) Available at: www.forbes.com/sites/gilpress/2018/02/07/the-bruteforce-of-deep-blue-and-deep-learning/?sh=597fb30249e3.

lots of moves contained in vast data sets of games played by humans. Nobody told it how to play Go. Instead, the programmers provided the neural network with the basic rules of the game and then let it play against itself for some days. The system rapidly mastered the game, becoming its own teacher: as it played, the ANN developed a logic of its own, no longer constrained by the limits of existing human knowledge, thus performing unexpected moves and strategies.³⁸

In general, we refer to this form of learning as ‘reinforcement learning’, which signifies an advancement from prior and less ambitious forms, namely ‘supervised’ and ‘unsupervised’ learning. All these learning forms represent potential modes of interaction between the programmers, the machine, and its output.

Supervised learning is quite limited – the algorithm only learns from labelled data which contains the ‘correct answer’ for each example.³⁹ Imagine being taught a specific task, like classifying pictures of cats and dogs. You’re given labelled examples – a picture with a label saying ‘cat’ or ‘dog’. You learn to identify the patterns that differentiate cats from dogs based on these examples. As a result of this training, you can make hypotheses about how to categorise future unlabelled data. While human input is still needed in the form of monitoring and feedback, the system still exhibits independent development in that its hypotheses about unlabelled data are not pre-programmed.

³⁸ Notably, during the historic match against Lee Sedol, AlphaGo Zero made an unorthodox move that perplexed not only its human programmers but also its human opponent. Initially, this move was deemed a mistake, as no one could discern the purpose, benefits, or strategic significance of this unconventional move in the context of winning the game. However, as the match unfolded, it became evident that this seemingly erroneous move was predicted by the machine to maximize its probability of winning. It set the stage for an audacious and ultimately triumphant endgame strategy that led AlphaGo to a resounding victory. Silver *et al.*, ‘AlphaGo Zero : Learning from Scratch’ (DeepMind Website, 18 October 2017) available at <https://deepmind.google/discover/blog/alphago-zero-starting-from-scratch/>. See also Id, ‘Mastering the Game of Go without Human Knowledge’ (2017) 550 Nature 354-359.

³⁹ Andrew Ng and Tengyu Ma, ‘Stanford’s CS229 Lecture Notes’ (2023) 5-75 available at https://www.google.com/url?sa=t&source=web&rct=j&opi=89978449&url=https://cs229.stanford.edu/main_notes.pdf.

Now imagine you are given a bunch of unlabelled pictures – animals, objects, landscapes, all mixed up. Your task is to find patterns within this data. Maybe you group similar-looking images together, categorise them by colour, or identify clusters of a certain type of image. Unsupervised learning deals with unlabelled data, where the algorithm has to find patterns or meaning on its own, without the human providing any feedback or clear goal.⁴⁰

Reinforcement learning combines these approaches. It can explore and learn from the environment (like unsupervised learning) but it also has a goal it's trying to achieve (like supervised learning). Here's where it gets interesting. Imagine you're in a maze, blindfolded. You can explore different paths (actions) and after each turn, you get a signal – a reward if you're going the right way, a penalty if you're stuck, nothing if you're wandering aimlessly. Over time, by trial and error, you learn which paths lead to the exit (goal) faster.⁴¹ This is how AlphaGo Zero learnt how to play Go by itself through a trial-and-error approach. And this is how humans and animals learn by interacting with the world around them. The agent interacts with the environment and receives feedback that guides it towards the best course of action, making RL a more versatile and powerful form of machine learning. Against this background, humans can be taken out of the picture altogether soon after a program's inception.

The common thread among such systems – regardless of the specific machine learning model employed – is that they appear to have developed some degree of agency, in the sense that they can interact with and alter their environment while simultaneously modifying their internal states often without being guided by human intervention.⁴² In essence, a human creator designs the algorithm, but the system itself

⁴⁰ *Ivi*, 144 ff.

⁴¹ *Ivi*, 186 ff. See also Richard Sutton and Andrew Barto, *Reinforcement Learning: an Introduction* (2nd edition Bradford Books 2018) 4.

⁴² By contrast, in rule-based AI it is relatively obvious why the system behaves in one way instead of another, as its programmers have previously set guidelines that are followed by the system. Mark Coeckelbergh, *AI Ethics* (MIT Press 2020) 50

gains ‘knowledge’ through experience, continuously adapting and refining its behaviour over time.

As self-learning systems grow in complexity, they exhibit autonomy and emergent properties that can lead to *unpredictable* outputs – both from a subjective standpoint (human understanding) and from an objective-technological perspective (technical opacity). However, autonomy in this context is not a binary trait; rather, it exists on a spectrum.⁴³ A system’s level of autonomy may depend on multiple factors, including its adaptability to changing conditions, ability to acquire new knowledge, capacity for independent action, and potential to replace human decision-making. Not all AI systems exhibit these traits to the same degree, meaning that different systems may qualify as ‘autonomous’ without necessarily possessing all of these characteristics.⁴⁴

This shift from explicit, rule-based machines to adaptive, autonomous systems has profound implications for human-machine interaction, fundamentally altering how we engage with technology. Unlike their purely subservient predecessors, these systems can perform complex tasks with minimal or no direct human intervention, raising concerns about diminishing human agency and control. The more autonomous a system becomes, the less direct oversight humans retain over its decision-making processes – a transformation that demands legal and ethical scrutiny.

The identikit of our ‘suspect’ is beginning to take shape. Autonomy, unpredictability, and opacity emerge as central themes in assessing the risks and challenges posed by advanced AI. However, before diving deeper into these concepts and their implications for human-machine interactions, we must first define the

⁴³ Karni Chagal-Feferkorn, ‘When Do Algorithmic Tortfeasors That Caused Damage Warrant Unique Legal Treatment?’ in Woodrow Barfield (ed), *The Cambridge Handbook of the Law of Algorithms* (Cambridge University Press 2020) 474.

⁴⁴ For an account of various concepts of autonomy, see Giovanni Sartor and Andrea Omicini, ‘The Autonomy of Technological Systems and Responsibilities for Their Use’ in Nehal Bhuta and others (eds), *Autonomous Weapons Systems: Law, Ethics, Policy* (Cambridge University Press 2016).

boundaries of our inquiry. It is essential to differentiate between what these systems can and cannot do – avoiding the pitfall of conflating science fiction with reality. After all, we must be careful not to mistake Terminator for Roomba.

The following section will explore the crucial distinction between ‘strong’ and ‘weak’ AI, helping to clarify where current AI systems truly stand on the autonomy spectrum.

1.3. Disentangling AI: strong vs weak AI

The distinction between weak and strong AI is critical for legal and policy debates. Many concerns surrounding AI liability, ethics, and governance stem from misconceptions about AI’s actual capabilities.⁴⁵ Therefore, it is essential to clarify the difference between these two paradigms.

Weak AI, also known as narrow AI, refers to non-conscious, specialized systems that are based on ML methods and are trained on big data. These systems do not possess general intelligence or an independent understanding of the world. Instead, they rely on pattern recognition, statistical inference, and predefined rules to execute specific tasks, surpassing human capacities while doing so, but without claiming either a general problem-solving capacity or an infinite adaptability to changing conditions.⁴⁶ While narrow AI can outperform humans in specific domains, it lacks true comprehension, reasoning, or self-awareness. Even the most advanced models, such as large language models and deep learning systems, do not

⁴⁵ On this distinction see John Searle, ‘Minds, brains and programs’ (1980) 39(3) *Behavioral and Brain Sciences* 417-457. See also David Weinbaum and Viktoras Veitas, ‘Open ended intelligence: the individuation of intelligent agents’ (2017) 29(2) *Journal of experimental and theoretical AI*, 371-396.

⁴⁶ Examples include technologies that have become deeply embedded in our daily lives and professional environments: virtual assistants like Amazon Alexa and Google Home, chess-playing algorithms, diagnostic tools that guide medical professionals, and financial algorithms that assess credit risk, forecast stock trends, or manage insurance pricing.

‘understand’ the information they process; they generate outputs based on probability rather than cognition.⁴⁷

Conversely, the term ‘strong AI’ represents a hypothetical level of AI development in which machines would match or surpass human cognitive abilities, which means that to qualify as such, the system must have developed an equally broad or even broader problem-solving capacity and adaptability to external conditions. In the latter case, one would refer to the concept of Artificial General Intelligence (hereinafter: AGI), or Superintelligence.⁴⁸ Unlike weak AI, which operates within predefined boundaries, AGI would possess the capacity to think, learn, and adapt across a broad spectrum of tasks, thus ‘vastly exceeding the cognitive performance of humans in virtually all domains of interest’.⁴⁹ In theory, AGI could engage in autonomous decision-making beyond predefined parameters, raising profound ethical and legal dilemmas.

Such a speculative scenario has been eloquently described in Ian McEwan's novel, titled *Machines Like Me*. One of the main characters of this story is a humanoid robot named Adam, who is equipped with a sense of morality by design as well as a personality that is initially shaped by his owners but then further develops

⁴⁷ While these systems are limited to the domains for which they are designed, their capabilities are far from trivial. Having created such systems is, of course, no small achievement, and it develops far reaching implications for contemporary societies. See, in this direction, Yuval N Harari, *Homo Deus: A Brief History of Tomorrow* (First US edition, Harper, an imprint of HarperCollins Publishers 2017) 308. As the author notes, ‘[h]umans are in danger of losing their value, because intelligence is decoupling from consciousness. Until today, high intelligence always went hand in hand with a developed consciousness. [...] However, we are now developing new types of non-conscious intelligence that can perform [...] tasks far better than humans. For all these tasks are based on pattern recognition, and non-conscious algorithms may soon excel human consciousness in recognising patterns. This raises a novel question: which of the two is really important, intelligence or consciousness?’

⁴⁸ See Nick Bostrom, *Superintelligence: Paths, Dangers, Strategies* (Oxford University Press 2014) 22, who defines Superintelligence as ‘any intellect that greatly exceeds the cognitive performance of humans in virtually all domains of interest’.

⁴⁹ Nick Bostrom and Vincent C. Müller, ‘Future progress in artificial intelligence: A survey of expert opinion’ in Vincent C. Müller (ed.), *Fundamental Issues of Artificial Intelligence* (Springer 2016) 553-571.

into a real self-conscious agency through his interaction with the environment and other people.⁵⁰

However, developing fully-fledged AI moral agents like Adam does not correspond to a tangible possibility as of yet.⁵¹ Modern AI systems rely on the recognition of probabilities, and require a massive amount of training data in order to extract patterns from such probabilities. It is true that they are extremely powerful, as they can process enormous amounts of data in minimal time frames, functioning between abstraction and enumeration. Nonetheless, at no point during their operation do these systems develop a literal understanding of the meaning behind this data.⁵²

In this respect, we are allowed to argue that the discussion on the perceived potential to develop AGI is, to a large extent, driven by a fundamentally wrong idea of intelligence. In fact, intelligence should not be conceived in narrow terms: there is no such thing as a ‘general’ form of intelligence in nature.

The truth is that not even humans possess the general form of intelligence that we would like to attribute to machines. Perhaps no agent at present (either biological or artificial) comes remotely close to possessing such intelligence. Intelligence is not defined by a single quality (such as language, conscience, abstract thinking, etc), nor is there a singular manner of being intelligent (e.g., the human way); rather, there is more than one kind of intelligence, and as many AI researchers emphasize, this is not an exclusively human feature.⁵³

⁵⁰ Ian McEwan, *Machines Like Me* (Nan A Talese/Doubleday 2019). The book largely describes the robot’s decision-making processes and practical choices, that are determined by the interaction between his inbuilt moral code and his personal inclinations.

⁵¹ Cristianini (Ch I n 6) 3–4.

⁵² *Ivi* 19, 36.

⁵³ Coeckelbergh (n 42) 64; Kate Crawford, *Atlas of AI: Power, Politics, and the Planetary Costs of Artificial Intelligence* (Yale University Press 2021) 6. See also Stanford’s ‘One Hundred Year Study on Artificial Intelligence’ project (AI100) (2021) Study Panel report available at <https://ai100.stanford.edu/2016-report>, stating that ‘the Study Panel takes a broad view that intelligence lies on a multidimensional spectrum’.

For example, the animal kingdom demonstrates a wide array of non-human intelligences: birds navigate thousands of miles using geomagnetic fields, dolphins communicate through intricate sound patterns, and octopuses exhibit problem-solving skills that challenge our understanding of cognition. Similarly, AI systems display forms of ‘intelligence’ that do not align neatly with human paradigms. Machines can outperform humans in many tasks of daily life, such as processing vast datasets or executing complex calculations, but may struggle with tasks we consider simple, like making a cup of coffee or interpreting nuanced social cues. AI simply features a new ‘alien’ form of intelligence capable of exhibiting intelligent behaviour without necessarily thinking like us. But neither of them qualify as being ‘general’ intelligence.⁵⁴

This leads to the conclusion that most current AI-related risks are problems of weak AI. We should avoid conflating present-day risks with speculative scenarios of AGI; even if realized, the latter would demand an entirely new legal paradigm, raising questions about moral agency, personhood, and autonomy in decision-making that go beyond the more limited scope of this study.

Therefore, for present purposes, autonomous systems are understood as falling under the broad and approximative category of weak AI. Of specific interest are non-conscious systems that employ any machine learning method and intuitively seem to serve as something more than a simple tool, when their performance in the physical world acts as a substitute for actions that have been typically undertaken by natural persons and creates a risk of physical harm against humans.⁵⁵

With this understanding, we can now proceed to craft a legally-oriented definition of AI. This definition will not only clarify the scope of systems under consideration but also serve as a foundation for analyzing the regulatory and liability frameworks required to address the challenges posed by these transformative technologies.

⁵⁴ Cristianini (Ch I n 6) 22ff.

⁵⁵ Carla L Reyes, ‘Autonomous Corporate Personhood’ (2021) 96 Washington Law Review 1453, 1480.

1.4. Defining AI: a pointless exercise? Towards a working definition for criminal lawyers

As it has already clearly emerged in these first paragraphs, AI as a concept is difficult to define and inherently resistant to reduction into a neat, universally accepted framework. The existence of more than 70 different definitions highlights a significant point: even experts – far beyond lexicographers – cannot agree on what precisely makes a machine ‘intelligent’.⁵⁶ Attempting to reconcile these divergent perspectives is not only challenging but may ultimately be a futile exercise.

Curiously, the lack of a precise, universally accepted definition of AI probably has helped the field to grow, blossom, and advance at an ever-accelerating pace. Practitioners, researchers, and developers of AI are instead guided by a rough sense of direction and an imperative to ‘get on with it’.⁵⁷ Still, a definition remains important for the purposes of our study.

Broadly, definitions of AI can be grouped into four categories: (i) thinking like a human, (ii) acting like a human, (iii) thinking rationally, and (iv) acting rationally.⁵⁸ They are summarised in the table below:

⁵⁶ As computer science expert Jerry Kaplan says, the question ‘what is AI’ is an ‘easy question to ask and a hard one to answer’ because ‘there’s little agreement about what intelligence is’. Jerry Kaplan, *Artificial intelligence: what everyone needs to know* (OUP 2016) 1.

⁵⁷ See Stanford’s ‘One Hundred Year Study on Artificial Intelligence’ project (n 52).

⁵⁸ Russel and Norvig (n 23) 36ff for the distinction between human-centric and rationalist definitions of AI. Four broad approaches can be identified: acting humanly (the famous Turing test), thinking humanly (modeling cognitive behavior), thinking rationally (building on the logicist tradition), and acting rationally (a rational-agent approach favored by Russell and Norvig as it is not dependent on a specific understanding of human cognition or an exhaustive model of what constitutes rational thought).

Systems that think like humans	Systems that think rationally
"The exciting new effort to make computers think ... <i>machines with minds</i> , in the full and literal sense." (Haugeland, 1985) "[The automation of] activities that we associate with human thinking, activities such as decision-making, problem solving, learning ..." (Bellman, 1978)	"The study of mental faculties through the use of computational models." (Chamiak and McDermott, 1985) "The study of the computations that make it possible to perceive, reason, and act." (Winston, 1992)
Systems that act like humans	Systems that act rationally
"The art of creating machines that perform functions that require intelligence when performed by people." (Kurzweil, 1990) "The study of how to make computers do things at which, at the moment, people are better." (Rich and Knight, 1991)	"Computational Intelligence is the study of the design of intelligent agents." (Poole <i>et al.</i> , 1998) "AI ... is concerned with intelligent behavior in artifacts." (Nilsson, 1998)
Figure 1.1 Some definitions of artificial intelligence, organized into four categories.	

Figure 2: Russel and Norvig - AI: a modern approach

Human-centric definitions of AI, particularly those inspired by Turing's idea that intelligence can be measured by a system's ability to mimic human cognitive performance, have significant limitations. Originally devised as an operational benchmark for intelligence, these definitions have since become outdated and are no longer considered a precise or meaningful measure of AI's capabilities.

Direct comparisons between human and artificial intelligence are not always useful. When AI is too closely tied to human cognitive abilities, the definition becomes overly broad and imprecise, diluting the concept's specificity. AI is not merely an attempt to replicate human thought processes; it is a distinct technological paradigm with its own principles and mechanisms. By defining AI in purely human terms, we risk obscuring its unique attributes and failing to distinguish it from other forms of advanced automation or computation.

Given AI's current trajectory, many experts and scholars are in favour of a more neutral or 'rationalist' definition - one that does not rely on anthropocentric

comparisons but instead focuses on the system's ability to develop purposeful, goal-oriented behaviour.⁵⁹ Under this approach, an AI system is considered intelligent not because it mimics human cognition but because it demonstrates an ability to *think or act rationally*. 'Rationality' in this context refers to a machine's capacity to make and execute optimal decisions based on predefined criteria and available resources to achieve a given objective, rather than any human-like reasoning or emotional processing.⁶⁰

From this perspective, an AI agent exhibits intelligence through task-oriented resilience – the ability to adapt to changing conditions within its designated function. As AI agents cannot process emotions and abstract concepts, humans cannot detect patterns in data sets or perform large-scale computations; yet, applying a rationalist definition, both agents behave intelligently. Different entities may set goals that are fundamentally different, both in nature and complexity. And it is noteworthy that autonomous AI systems are already capable of producing results that in many respects are beyond human capabilities and comprehension.

While the rationalist model provides a functional and adaptable framework for defining AI – successfully sidestepping the anthropocentric biases of earlier approaches – it remains inadequate for addressing AI's legal challenges. Its primary shortcoming lies in its overly broad scope, which fails to differentiate between AI technologies with vastly different risk profiles and potential impacts on legally protected interests.

This definitional issue is particularly evident in the recently approved AI Act, which defines AI as:

⁵⁹ Cristianini (Ch I n 6) 5; Beatriz A Ribeiro *et al*, 'Metacognition, Accountability and Legal Personhood of AI' in Henrique Sousa Antunes *et al* (eds), *Multidisciplinary Perspectives on Artificial Intelligence and the Law*, vol 58 (Springer International Publishing 2024) 172–173.

⁶⁰ This perspective aligns with the definition offered by Nils Nilsson: 'AI is that activity devoted to making machines intelligent, and intelligence is that quality that enables an entity to function appropriately and with foresight in its environment'. Nils J. Nilsson, *The Quest for Artificial Intelligence: A History of Ideas and Achievements* (Cambridge, UK: Cambridge University Press, 2010).

‘a machine-based system that is designed to operate with varying levels of *autonomy* and that may exhibit *adaptiveness* after deployment, and that, for *explicit or implicit objectives*, infers, from the input it receives, how to generate outputs such as predictions, content, recommendations, or decisions that can influence physical or virtual environments’ (emphasis added).⁶¹

While this definition aims for inclusivity, its breadth creates significant problems. By encompassing any system that processes input and autonomously generates outputs, it effectively includes technologies that have existed for decades and that bear little resemblance to the advanced AI systems at the centre of today’s legal and ethical debates.

For instance, under this definition, an intelligent thermostat – i.e., a system which autonomously adjusts temperature based on sensor input – qualifies as an AI system. However, such a device shares virtually nothing with high-stakes applications like autonomous vehicles, predictive policing algorithms, or conversational AI models. From our limited legal perspective, treating such disparate technologies as part of the same category dilutes the meaning of AI, making the term functionally unhelpful as a regulatory tool.

This leads to the conclusion that a one-size-fits-all definition of AI fails to provide clear guidance for the purposes of this study. The fundamental problem is that a definition such as the one adopted by the EU legislator, while technically comprehensive, does not distinguish between disruptive and uncontroversial applications.

To address this issue, a more granular and legally meaningful approach is needed – one that moves beyond broad, catch-all definitions and instead focuses on the specific characteristics that make AI disruptive for criminal law.⁶² The key

⁶¹ AI Act, Art. 3(1). Additionally, it should be noted that, according to Recital 12, this definition is not intended to cover traditional, simpler software systems or programming approaches that rely solely on rules defined by humans to automatically perform operations.

⁶² In this sense, see Beatrice Fragasso, ‘La responsabilità penale del produttore di sistemi di intelligenza artificiale’ (2023) 1 *Rivista Trimestrale Diritto penale Contemporaneo* 26-45, 30.

distinction between, say, an intelligent thermostat (irrelevant in this study) and a highly advanced system capable of independent decision-making in high-stake social domains such as road circulation and healthcare (highly relevant) is not one of *kind* but of *risk level*.

Thus, rather than applying a one-size-fits-all definition, legal analysis should concentrate on risk-calibrated attributes – specifically, the degree of autonomy, complexity, and opacity. These attributes are not merely technical traits, but the very factors that determine how and when AI systems challenge existing legal frameworks. By incorporating this perspective, we ensure that legal scrutiny is directed at truly disruptive AI systems – those capable of producing unpredictable or untraceable behaviours – rather than entangling all automated tools in unwarranted legal burdens.

If, then, we are to construct a descriptive legal definition of AI we propose the following:

- i. An AI system is a machine-based entity (whether software-only or physically embodied) that is designed to make non-predetermined decisions in pursuit of predefined goals
- ii. through decision-making processes that can range from fully supervised to entirely autonomous, creating varying levels of human oversight and control,
- iii. and that may be opaque either due to technical complexity (black-box problem) or relational complexity (many-hands problem)
- iv. operating in uncertain, high-stakes social environments, where its outputs can directly impact legally protected interests, including life, bodily integrity, property, privacy, and autonomy.

By sharpening the conceptual boundaries of AI in this way, we provide a precise and actionable framework – one that avoids both the vagueness of broad legislative definitions and the rigidity of purely technical classifications.

With this foundation in place, the next step is to examine how the combination of these factors – autonomy, complexity, and opacity – might result in unpredictable AI behaviours and consequently to what we refer to as *AI-caused harm* (or, more simply, AI harm) thus leading to *irreducibility* – that is, the impossibility to reconnect the harmful event to a liable person behind the machine.⁶³

⁶³ The expression ‘irreducible challenge’ has been first theorized specifically in relation to the field of criminal liability by Ryan Abbott and Alex Sarch, ‘Punishing Artificial Intelligence: Legal Fiction or Science Fiction’ (2019) 53 UC Davis Law Review.

SECTION 2

Machines beyond human control: from human-caused to AI-caused harm

2.1. Accidents will happen: the case of deviant AI

As AI-powered machines become ever more integrated into society and imbued with attributes of ‘intelligence’, they are increasingly taking over tasks that were once the exclusive domain of humans. They now drive cars, perform medical procedures, and trade in financial markets. While AI’s involvement in these activities is justified by its ability to advance valuable human goals – ultimately contributing to the benefit of humanity – its potential for good does not eliminate its potential for negative effects. In fact, AI appears to have assumed yet another ‘task’ previously reserved for humans: the *causation of harm*.

The concept of harm has emerged as a dominant theme in contemporary discussions about AI.⁶⁴ While experts and society remain divided on many aspects of the technology, one point is undisputed: virtually no legally protected interest is immune to harm caused by AI.⁶⁵ But what does it mean, exactly, that an AI system can cause ‘harm’ to individuals, organizations, or various aspects of life? This question requires careful unpacking.

Broadly speaking, harms are most often understood as the incidents and hazards that arise from human reliance on AI systems and their capabilities. Harm definitions and taxonomies are deeply context-dependent, shaped by the specific regulatory environment, societal objectives, and the domains in which AI is deployed.

One of the most relevant dimensions of harm is its type. The manifestations of harm in the field of AI are diverse and have different levels of severity (*e.g.*, from

⁶⁴ David Lorge Parnas, The real risks of AI (2017) 60(10) Communications of the ACM 27-31.

⁶⁵ Consulich (Ch I n 21) 270.

inconsequential hazards to damage to property, to harm to health, to impact to critical infrastructure, to causing human deaths or even life-altering consequences). For example, harm can include physical injury or damage to health, such as accidents involving autonomous vehicles or errors in AI-assisted medical diagnoses. It can involve economic or financial losses as seen in stock market crashes triggered by algorithmic trading systems, or property damage caused by malfunctioning smart technologies. It can also extend beyond tangible losses, encompassing reputational damage, psychological damage (such as distress caused by biased algorithms or invasive surveillance technologies) or harm to human and fundamental rights. It also touches on ethical and socio-political dimensions, where the misuse or misunderstanding of AI's capabilities can disrupt societal norms, exacerbate inequalities, or undermine trust in institutions. For example, the application of AI in law enforcement or hiring processes may reinforce discriminatory practices, eroding public confidence and entrenching social disparities.⁶⁶

Moreover, harm is rarely confined to its immediate impact. Often, it unfolds in a multifaceted manner, implicating direct victims as well as society at large through cascading effects and systemic vulnerabilities. Such harm can ripple across various sectors, affecting different groups of people and diverse aspects of life.

Importantly, for the purposes of this study, we employ the term 'harm' as an umbrella concept that encompasses both *potential* harm (*i.e.*, AI hazard) and its *actual materialisation* (*i.e.*, AI incident). In this sense, AI harm can be defined as 'an event circumstance or series of events where the development, use or malfunction of one or more AI systems directly or indirectly (or could plausibly) leads to an AI incident, i.e., any of the following harms: (a) death or injury to the health of a person or groups of people; (b) disruption of the management and operation of critical infrastructure; (c) violations to human rights or a breach of obligations under the

⁶⁶ For the definitions and taxonomy of AI harms, see OECD, 'Defining AI incidents and related terms' (2024) 16 OECD Artificial Intelligence Papers available at https://www.google.com/url?sa=t&source=web&rct=j&opi=89978449&url=https://www.oecd.org/en/publications/defining-ai-incidents-and-related-terms_d1a8d965-en.html.

applicable law intended to protect fundamental, labour and intellectual property rights; (d) harm to property, communities or the environment'.⁶⁷

There have been several recent doctrinal efforts to identify and categorize the potential harms associated with AI systems.⁶⁸ While many of these harms remain hypothetical or unrealized at the time of writing, an increasing number have already materialized, demonstrating the pressing need for vigilance and intervention. Over the past few years, there has been an alarming increase in cases of AI systems malfunctioning with serious consequences. The *AI Incident Database*, a repository tracking such occurrences, chronicles over 2,000 documented incidents of harm.⁶⁹ The first recorded case dates back to 1983, when an automated warning system in the Soviet Union erroneously identified incoming missiles from the United States, nearly triggering a catastrophic nuclear response.⁷⁰ Since then, the frequency and visibility of AI-related incidents have surged year after year, drawing increasing public scrutiny.

After a relatively long AI winter, 2010 marked the beginning of the decade when AI incidents really started to become relevant to the public. In the period of the early implementation of AI (between 2010-2014), the database reports only an average of 5 incidents per year. As AI became more powerful and widely accessible, with the widening of the range of application domains and tasks in which ML technologies are employed, the number of incidents grew significantly from 2014 to

⁶⁷ *Ivi*, 11-13.

⁶⁸ See e.g. T. C. King *et al.*, 'Artificial Intelligence Crime: An Interdisciplinary Analysis of Foreseeable Threats and Solutions' (2019) 26(1) *Sci Eng Ethics*, who undertake a systematic literature review to identify threats and broad areas for further research.

⁶⁹ Available at: <https://incidentdatabase.ai/>. See also the AIAAIC Repository, an independent collection of over 850 incidents and controversies driven by and relating to AI, algorithms, and automation. Available at: https://docs.google.com/spreadsheets/d/1Bn55B4xz21-_Rgdr8BBb2lt0n_4rzLGxFADMIVW0PYI/edit#gid=1051812323.

⁷⁰ *Ivi*, Incident 27: Nuclear False Alarm. <https://incidentdatabase.ai/cite/27/>

2019: this period saw 207 incidents, averaging 35 per year and accounting for 40% of all reported cases.⁷¹

Interestingly, financial markets represent the field that has experienced the challenges of AI more than any other. The legal interest in market integrity is constantly exposed to the damages caused by the so-called high-frequency trading (better known in the industry by the acronym HFT) and algorithmic trading systems, autonomously engaging in prohibited market abuse offences.⁷² For example, the infamous 2010 Flash Crash first brought attention to the role that HFT algorithms can play in creating market volatility, uncertainty, and disruption.⁷³

But cases of AI accidents are not limited to the field of financial transactions. As is often the case, reality exceeds imagination by presenting us with a concrete case scenario that is already much broader and more diverse than one might imagine. In 2015, for example, a group of artists released an AI-powered random shopping bot on the dark web – with the unsurprising outcome that it eventually bought illegal drugs, and was arrested by Swiss police. Again, recent cases have demonstrated the potential dangers of AI-based social bots. In 2016, for instance, Microsoft’s AI bot, *Tay*, was taken down for becoming a sexist, racist monster discriminating and defaming against minorities.⁷⁴

⁷¹ These figures are taken from the ‘Surfshark research hub’s study’ available at <https://surfshark.com/research/chart/statistics-of-ai-incidents> which examined the timeline and frequency of recorded AI incidents. The study divided the past two decades into three distinct stages of AI implementation: (i) the early implementation of AI (between 2010-2014), which had only an average of 5 incidents per year (3% of all incidents); (ii) the growth of AI (2014-2019), that accounts for 40% of all reported cases, averaging 35 per year; and (iii) the most recent AI adoption stage, exemplified by the development of GPT-3 by OpenAI in 2020, accounting for 57% of all incidents across a wide range of industries, with an average of 79 per year.

⁷² See Filippo Annunziata, *Artificial intelligence and market abuse legislation. A European perspective*, (Edward Elgar 2023).

⁷³ A Kirilenko *et al*, ‘The flash crash: High-frequency trading in an electronic market’ (2017) 72(3) *The Journal of Finance* 967-998; V Caivano, ‘The impact of high-frequency trading on volatility. Evidence from the Italian market’ (2015) 80 *Quaderni di finanza CONSOB*.

⁷⁴ Available at <https://incidentdatabase.ai/cite/6/#r1374>.

Moreover, incidents affecting life or physical integrity are rapidly growing in number. The prototypical example is the self-driving car collision. In 2012, a driver of a car equipped with a lane keeping assistant suffered a heart attack and -by losing consciousness- turned the steering wheel to the right towards a wooded area, when the automated system steered the vehicle back to its lane, where it eventually killed a young woman and her child.⁷⁵ The first-reported full-fledged fatal incident caused by an automated driving system in the world occurred on January 20, 2016, in China, when a Tesla Model S on autopilot slammed into a road sweeper on a highway, which caused the driver Gao Yaning's instant death.⁷⁶ A few months later, on May 7, 2016, a Tesla Model S on autopilot crashed into a white articulated tractor-trailer, unable to distinguish it from the bright sky, causing the destruction of the car and the death of the driver.⁷⁷ Lastly, in March 2018, a self-driving Uber car hit and killed a woman walking her bike across a street – a case which stands out as the first recorded death by a semi-autonomous vehicle.⁷⁸

To be fair, many other instances of harmful AI behaviour have been reported beyond the field of autonomous driving. To mention just a few, we have cases of workers being crushed by industrial AI-powered machinery;⁷⁹ reports of robotic surgery malfunctions linked to at least 144 deaths since 2000;⁸⁰ reports of vocal assistants (such as Amazon's Alexa), which, in response to a challenge request,

⁷⁵ Oddly enough, the so-called *Aschaffenburg* case does not feature in the AI incident database – perhaps as it is not real automation we are dealing with here. For a detailed description of the case, see Eric Hilgendorf, 'Automatisiertes Fahren und Strafrecht - der Aschaffener Fall' (2018) 96(2) *Deutsche Richterzeitung* 66. See also Maria Kaiafa-Gbandi, 'Artificial intelligence as a challenge for Criminal Law: in search of a new model of criminal liability?' in S Beck, C Kusche and B Valerius (eds), *Digitalisierung, Automatisierung, KI und Recht* (Nomos 2020) 305, 321.

⁷⁶ Available at <https://incidentdatabase.ai/cite/231/#r1750>.

⁷⁷ Incident 52: Tesla on Autopilot Killed Driver in Crash in Florida available at <https://incidentdatabase.ai/cite/52/#r967>.

⁷⁸ Incident 4 available at <https://incidentdatabase.ai/apps/discover>.

⁷⁹ Incident 599: Man crushed to death by robot that mistook him for a box of vegetables. Incident 24: A Volkswagen worker was killed by a production robot in Germany, available at <https://incidentdatabase.ai/cite/24/#r279>.

⁸⁰ Incident 5: Robotic Surgery Malfunctions.

encouraged a ten-year-old girl to insert a phone charger halfway into an electrical outlet and touch the opposite poles with a coin;⁸¹ cases of people who died of starvation in India, after having been denied food under the public distribution system as their ration cards was rejected by an algorithm;⁸² or, going towards even more dystopic scenarios, lethal autonomous weapons systems (such as military drones and killer robots) possibly killing people without any human interference.⁸³

Finally, the most recent AI adoption stage, exemplified by the development of ChatGPT by OpenAI in 2020, accounts for 57% of all incidents across a wide range of industries, with an average of 79 per year. The onset of this decade saw a notable surge in AI incidents, with 2023 witnessed the highest number of incidents, making it a record year in the decades of AI existence⁸⁴. Most of such occurrences appears to be linked to OpenAI's ChatGPT malfunctions – which was involved in over a quarter of all cases.

This brief and incomplete survey of incidents reveals the diverse ways in which AI systems can fail, often with significant and far-reaching consequences for individuals and society at large. However, not all types and dimensions of AI-caused harm are relevant to our analysis. Since our focus is on addressing harms from AI through the lens of criminal law, we have chosen to concentrate exclusively on a specific subset: 'hard AI crime'.⁸⁵

Before focusing on this definition, it is necessary to take a step back and explain what is usually meant in legal doctrine by 'AI crime'.

⁸¹ Incident 160: Alexa Recommended Dangerous TikTok Challenge to Ten-Year-Old Girl, available at <https://incidentdatabase.ai/cite/160/#r1520>.

⁸² Rebecca Ratcliffe, 'How a glitch in India's biometric welfare system can be lethal' (The Guardian 16 October 2019) available at <https://www.theguardian.com/technology/2019/oct/16/glitch-india-biometric-welfare-system-starvation>.

⁸³ Accident 121: A.I. Drone May Have Acted on Its Own in Attacking Fighters, U.N. Says. Available at <https://incidentdatabase.ai/cite/121/#r2106>

⁸⁴ In 2023, there were 121 recorded AI incidents, an increase of 30% from the previous year. Surfshark's report available at <https://surfshark.com/research/chart/ai-incidents-2023>.

⁸⁵ Cfr. Chapter I n 16. The term AI-Crime appears also in King *et al* (n 68).

2.2. Defining AI crime

The key distinction between ‘hard AI crime’ and the general concept of AI crime can be illustrated through the following hypothetical scenario:

An AI-operated, unmanned crane drops a steel frame on a passerby thus
killing them.

At first glance, this event appears to be a tragic accident. However, a closer examination reveals at least four distinct pathways, each pointing to a different root cause of AI-caused harm.

The first pathway is the least plausible yet the most sensationalized: the idea of an AI acting with *malicious intent*. Popular culture is rife with images of rogue machines – self-aware robots plotting humanity’s downfall, *à la Terminator*. While these narratives make for exciting storytelling, they misrepresent and obscure the far more mundane yet pressing realities of AI harm.

In the real world, AI systems do not develop *intent* – malevolent or otherwise. They do not want, plan, or decide in the way humans do. When harm occurs, it is not because an AI has chosen to do evil, but rather due to flaws in its design, training, or the environment in which it operates. This is probably what historian Melvin Kranzberg meant when he articulated his six laws of technology in 1986, of which the first is particularly relevant to our discussion: ‘Technology is neither good nor bad, nor is it neutral’.⁸⁶ Kranzberg’s first law suggests that a technological artefact itself does not inherently possess fully-fledged agency, let alone moral value or intent; no matter how sophisticated it is, it fundamentally remains a tool in the hands of humans, and its effects depend on how it is used and the context in which it operates. So it is never the technology we need to be afraid of – only the hand that wields it.

⁸⁶ Melvin Kranzberg, ‘Technology and History: Kranzberg’s Laws’ (1986) 27(3) *Technology and Culture*, 544–560.

A more plausible explanation to the accident emerges when we consider that AI can be intentionally weaponized (AI as *instrumentum sceleris*). Consider a variation of the crane incident:

An AI-operated, unmanned crane is deliberately programmed by a terrorist to drop a steel frame on passersby.

Here, the AI does not act autonomously in a meaningful sense – it is simply a means to an end, much like a firearm, explosive, or malware. This is an example of AI-enabled crime, where the technology becomes a tool for human wrongdoing.⁸⁷

As AI systems grow more capable, so too do the risks of malicious exploitation – from deepfake scams and autonomous cyberattacks to AI-assisted fraud and even AI-powered bioterrorism.⁸⁸ AI, in this context, is a dual-use technology: it can be leveraged for legitimate innovation or criminal purposes. The legal challenge is not about the AI itself, but rather about how criminal liability should be structured when AI is used as an enabler of harm.

Despite its technological novelty, this scenario represents a classic problem in criminal law: the use of tools or intermediaries to commit a crime. No matter how advanced or intelligent an AI system may appear, it remains fundamentally a tool – a means through which human actors achieve their ends. It acts as an *innocent agent*, an intermediary between the human perpetrator, whose culpability is in question, and the resulting harm. The ‘innocence’ of the system lies in its fundamental *predictability*: given certain instructions or commands, even intelligent tools will always react in the same way based on their material configuration or design.

⁸⁷ For a comprehensive current and potential future overview of AI-enabled crimes, see King (n 68), 89. See also M Caldwell *et al*, ‘AI-enabled future crime’ (2020) 9(14) *Crime Science*.

⁸⁸ Miles Brundage *et al*, ‘The malicious use of AI: forecasting, prevention, and mitigation’ (2018) University of Cambridge Repository available at <https://doi.org/10.17863/CAM.22520>; Keith J. Hayward and Matthijs Maas, ‘Artificial intelligence and crime: A primer for criminologists’ (2020) 17(2) *Crime Media Culture*, 214.

Therefore, even when an AI system performs complex tasks (the movements of a robotic arm, the calculations of an elaborate algorithm, the shots fired by a military drone, etc.) these actions are entirely instrumental. Translated into the legal context, the harm caused remains attributable to the human perpetrator who deliberately designed, programmed, or deployed the system for criminal purposes. This individual bears full responsibility because they can anticipate and intend the consequences of their actions, executed through the AI system. In fact, artifacts do not kill people – people do *through* technologies (either intentionally or negligently).⁸⁹

But what happens when harm arises without clear human intent – when AI causes damage due to systemic failures, design oversights, or emergent behaviours? This is where the discussion moves into more legally and ethically complex territory.

Consider now the following scenario:

An AI-operated, unmanned crane drops a steel frame on a passerby after incorrectly identifying the location for drop-off.

This scenario exemplifies what we refer to as *AI crime* – a term denoting actions carried out by AI systems that, had they been performed by a human with the requisite *mens rea*, would hold criminal significance.⁹⁰ However, not all AI crimes are created equal.

⁸⁹ While this is correct in principle, some legal scholars have emphasized that this scenario is not devoid of complexities. It is argued that when the perpetrator commits the crime by means of an AI system, the expression *by means* should not be interpreted strictly mechanically, but as an acknowledgment of the *creative instrumentality* of the tool employed for the commission of the offense. Indeed, the human actor establishes the conditions and the result in terms of the *genus* of events they intend to generate without delineating its *tempus* and *modus*, and the artificial agent fills the gap using operational methods that are not always predetermined (*quomodo*). Then the criminal act or omission would be decoupled from the mental state, where a human manifests the *mens rea* and the robot commits the *actus reus*. See Consulich (Ch I n 21) 288.

⁹⁰ Abbott and Sarch (n 63) 332.

In most cases, AI-caused harm is likely to be directly traced back to human actors – be it designers, manufacturers, or operators – making it largely compatible with existing liability frameworks. The majority of these cases fall within the well-established domain of *defective products*.

When an AI system malfunctions – say, due to a coding flaw, faulty sensors, or poor integration with physical hardware – the harm it causes is not truly autonomous but rather the result of human error at some stage in its design, production, or deployment. Here, AI's involvement is incidental; the legal responsibility lies with those behind the machine. This aligns neatly with product liability doctrines, under which manufacturers can be held accountable if a product's defects cause foreseeable harm.⁹¹

Indeed, despite the increasing sophistication of cranes, robotic systems, and AI-driven automation, this fundamental principle has remained unchanged throughout history. Whether we speak of the pulleys used to construct the pyramids of Giza or the cranes assembling today's skyscrapers, the liability has always rested with those responsible for the tool, not the tool itself.

Yet, our real concern lies elsewhere – not with defective AI, but with 'deviant' AI.⁹² This is where we enter the territory of what we call *hard AI crime* – 'scenarios where crimes are functionally committed by machines and there is no identifiable person who has acted with criminal culpability'.⁹³ If one thinks of this distinction in terms of *irreducibility*, hard AI crimes are instances in which harm arises from unpredictable, emergent behaviours that cannot be readily attributed (*rectius*: reduced) to a clear defect or a human decision, either for practical reasons (because of the difficulty to identify how individuals singularly contributed to the design of the system) or for theoretical reasons (because the human misconduct does not meet the

⁹¹ Carlo Piergallini, 'Intelligenza artificiale: da mezzo a autore del reato?' (2020) 4 Rivista Italiana di diritto e procedura penale 1752.

⁹² Consulich (Ch I n 21) 270.

⁹³ Abbott and Sarch (n 63) 328.

threshold required to activate the criminal sanction). In other words, irreducibility entails that the AI criminal act cannot be traced back to the wrongful act of a person.⁹⁴

At first, the idea of machines exhibiting unpredictable (and, consequently, irreducible) behaviour may seem paradoxical, particularly within a legal framework where our tools are fundamentally predictable and offences perpetrated using such tools are traditionally viewed as expressions of human intent or negligence. Yet, experience has demonstrated that any sufficiently complex AI, if left to operate independently, will eventually manifest unintended behaviour – behaviour that, in some instances, may break the law and result in hard AI crimes.

Today's AI is inherently unpredictable.⁹⁵ Already in 1960, MIT mathematician Norbert Wiener wrote an article where he maintained that 'as machines learn, they may develop unforeseen strategies at rates that baffle their programmers'.⁹⁶ More specifically, in computer science, unpredictability (also known as 'unknowability' or 'cognitive uncontainability') is defined as our inability to precisely and consistently predict what specific actions an intelligent system will take to achieve its objectives, even if we know terminal goals of the system.⁹⁷

This definition highlights that intelligent systems may cause unintended outcomes, not because they developed some sort of human-like agency, but because they function in ways that make it challenging – if not impossible – for their creators and users to anticipate their outputs. In other words, the perceived agency and unpredictability of AI is not inherent to the technology itself but is fundamentally an *epistemic problem* – a product of our inability to fully comprehend its inner

⁹⁴ Ryan Abbott, *The Reasonable Robot* (Cambridge University Press 2020) 13.

⁹⁵ Ugo Pagallo, 'When morals isn't enough: robots, ethics and the rules of the law' (2017) 27 *Minds and Machines*, 634.

⁹⁶ Norbert Wiener, 'Some Moral and Technical Consequences of Automation' (1960) 131 *Science*, 1355-1358.

⁹⁷ Roman Yampolskiy, 'Unpredictability of AI: On the Impossibility of Accurately Predicting All Actions of a Smarter Agent' (2020) 7(1) *Journal of Artificial Intelligence and Consciousness* 109-118.

workings. This means that the human mind does not and could not conceive of all possible decisions or strategies such advanced intelligent systems could make.⁹⁸

To clarify, imagine a complex domino setup with thousands of pieces arranged in a precise pattern. Each domino represents a decision or action within the AI system. The final domino, symbolizing the output, is determined by both the human-designed setup and the system's own 'momentum'. Just as a domino chain can take an unexpected turn, AI systems can exhibit unexpected behaviours. These behaviours manifest themselves only through interaction between domino pieces and with the environment. It is only once the dominoes are falling that we witness how this momentum drives the final outcome.

In the field of AI, this 'momentum' is usually depicted as *emergence*.⁹⁹ Unexpected 'emergent' behaviour is not only common in most advanced AI systems, but it is also a feature that we *expect* from a machine that is designed to adapt, learn, and operate under uncertain conditions. In other words, AI systems are unpredictable *by design*.¹⁰⁰

⁹⁸ 'Strong cognitive uncontainability is when the agent knows some facts we don't, that it can use to formulate strategies that we wouldn't be able to recognize in advance as successful. [...] When an agent can win using options that we didn't imagine, couldn't invent, and wouldn't understand even if we caught a glimpse of them in advance, it is strongly cognitively uncontainable'. See voice 'Cognitive Uncontainability' in Arbital, available at: <https://arbital.com/p/uncontainability/>.

⁹⁹ The term was first introduced in a 2022 paper by researchers at Google Brain, DeepMind and Stanford: Jason Wei *et al*, 'Emergent Abilities of Large Language Models' (2022) ArXiv, available at <https://arxiv.org/abs/2206.07682>. On the notion of 'emergent AI behaviour', see Jack Balkin, 'The Path of Robotics Law' (2015) 6 Cal. L. Rev. 532ff. For additional discussion on unpredictability and related issues with machine behaviour, see also I Rahwan and M Cebrian, 'Machine Behavior Needs to Be an Academic Discipline', (Nautilus March 29, 2018) available at <https://nautil.us/machine-behavior-needs-to-be-an-academic-discipline-237022/>, '[...] complex AI agents often exhibit inherent unpredictability: they demonstrate emergent behaviors that are impossible to predict with precision – even by their own programmers'.

¹⁰⁰ Jason Millar and Ian Kerr, Delegation, 'Relinquishment and Responsibility: The Prospect of Expert Robots' in Ryan Calo *et al* (eds), *Robot Law* (Elgar 2016) 102-128.

For example, an ordinary coffee machine is not expected to be unexpected; its makers have every interest in anticipating all possible errors and failure scenarios and pre-programming it to respond appropriately (heating the water when the user presses a certain button, pouring the water when it reaches a certain temperature, etc). By contrast, programmers of self-learning algorithms face an entirely different scenario, because their machines are required to navigate a constantly changing environment and adapt to its changes independently and dynamically. In this context, unpredictability is not a flaw in the system but a *deliberate and desired feature*, intentionally built in by programmers to achieve more efficient results. Arguably, this is a feature that also users actually want.¹⁰¹

However, the price we pay for this feature is quite high: in computer science this is commonly known as ‘artificial stupidity’.¹⁰² As Domingos put it in a 2015 book: ‘People worry that computers will get too smart and take over the world, but the real problem is that they are too stupid and they have already taken over the world’.¹⁰³ It is important to point out that when we use the word ‘stupid’ to describe the behaviour of intelligent machines, we do not mean they are literally dumb (it would be an oxymoron to assume that much). Instead, we refer to the fact that AI does exactly what you ask for, but somehow misses the bigger picture. AI is designed to achieve what is specified, not what is intended. When it comes to values and goals, machines do not really get much more sophisticated than understanding that 1 is different from 0.

¹⁰¹ Imagine that we asked ChatGPT to compose a haiku about snow on Mt. Fuji on ten different occasions, each time with the same prompt. We would expect, first, that they would not all be the same haiku, and second, that some might surprise us. We tend to look down on rote performance. In fact, we call overly predictable behaviour ‘robotic’, exactly because early robots were not so intelligent. We expect intelligence to be creative, and creativity is unpredictable. In other words, unpredictability is more of a *feature* than a *bug* of AI systems. See Kentaro Toyama, ‘Is the unpredictability of AI a feature or a bug?’ (Medium October 17, 2023), available at <https://medium.com/ai-heresy/is-the-unpredictability-of-ai-a-feature-or-a-bug-9ce14a2c6ebd>.

¹⁰² Pedro Domingos, *The master algorithm: how the quest for the ultimate learning machine will remake our world* (Allen Lane, 2015) 23ff.

¹⁰³ *Ivi*, 286.

Roughly speaking, there exist at least two paradigms of artificial stupidity, namely *machina economica* and *machina praeiudicata*.¹⁰⁴

First, the expression *machina economica* has been established in literature to refer to autonomous AI agents when they act as utility-maximisers. In general, AI systems will only be guided by their utility function – the standard by which they evaluate, and thus select, their actions. If the utility function is based on the economic (or any other type of) results obtained by the agent (i.e., by its user), the agent will always act in such a way as to maximize its performance measure (maximize the extent to which it achieves its goals).

For example, an AI trading agent autonomously and purposefully engages in conduct of ‘market manipulation’, since this is the optimal way to achieve the agent’s pre-defined goal of ‘profit maximization’.¹⁰⁵ This means that the agent will select actions that are most likely to lead to the achievement of that goal, without considering

¹⁰⁴ In computer science, the unpredictability of AI systems is described by Norbert Wiener, the founder of cybernetics, as ‘alignment problem’. Essentially, it refers to the challenges associated with ensuring that intelligent machines behave in accordance with human values and intentions. Wiener was deeply apprehensive about the possibility that machines, in their pursuit of efficiency and goal completion, might develop ‘dangerous shortcuts’ that disregard human well-being. Norbert Wiener, *God & Golem, Inc.: A Comment on Certain Points where Cybernetics* (1st edition MIT Press 1966). See also Brian Christian, *The Alignment Problem – How AI Learns Human Values* (1st edition W. W. Norton & Company 2020). The problem of AI misalignment has been vividly illustrated in the eerie tale of *The Monkey’s Paw*, a 1902 short story written by W. W. Jacobs (Wiener’s 1950 book, *The human use of human beings*). In Jacobs’ story, a visiting military man shows a family a small and shrivelled monkey’s paw, which is able to grant three wishes. When the head of the family acquires the monkey’s paw, over the soldier’s protestations, they initially wish for £200. After a day or so, the wish is granted, but at the expense of their son being crushed to death by the machinery at his place of work: the company gives them two hundred pounds as thanks for their son’s services. About a week later, the dead man’s father uses the paw to try to bring their boy back to life, but before the mother can discover the mangled revived body to which their son has been returned to at their doorstep, the father wishes for him to once again be granted death. This tale also resonates with the ‘paperclip machine’ thought experiment, where an intelligent system asked to make paperclips decides to seize and consume all resources in existence, in its blind adherence to its goal. Bostrom (n 48), 124-125.

¹⁰⁵ Elina Nerantzi and Giovanni Sartor, ‘Hard AI Crime: The Deterrence Turn’ (2024) 44 Oxford Journal of Legal Studies 3, 674.

any other goals (relating, for example, to the common good or to the interests of others) and without any constraints (such as avoiding harm) that have not been designed into it. In this context, however, the ‘right’ action for the machine is defined by its utility function, which may lead to results that are not necessarily moral or legal. The goals it pursues could be unlawful or unethical, or more plausibly, while its ultimate objectives may be lawful, it might select unlawful or immoral subgoals to achieve them.¹⁰⁶

On the other hand, the expression ‘*machina praeiudicata*’ refers to AI susceptibility to *biases*.¹⁰⁷ The term bias is commonly understood as a systematic thinking pattern that affects our judgments and decisions. All of us manifest cognitive fallacies or biases. One of the common examples of this bias that influences most of us is blaming external factors when something goes wrong. These human biases can be projected to our technology even with good intentions. AI algorithms are designed to make decisions on the basis of data fed into the systems. However, these data sets are far from neutral, and can reflect human prejudices or some historical social inequities present in the real world.¹⁰⁸ Therefore, we could even go so far as to say that the human limitations and imperfection we strive to exclude through the ‘door’ of advanced algorithms find their way back in through the ‘window’ of data we use to train the model.¹⁰⁹

As a result, an AI trained on biased historical data may ‘align’ too well with those biases, leading to outputs that reflect and reinforce harmful societal prejudices,

¹⁰⁶ *Ivi*, 460ff.

¹⁰⁷ See Cathy O’Neil, *Weapons of Math Destruction: How Big Data Increases Inequality and threatens democracy* (Penguin 2025) which is a pioneering work on machine bias, with examples of how algorithms can reinforce inequalities.

¹⁰⁸ Bernard Marr, ‘When Bytes Bias: Unravelling The Myth Of Neutral AI’ (Forbes June 30, 2019) available at <https://www.forbes.com/sites/bernardmarr/2023/06/30/when-bytes-bias-unraveling-the-myth-of-neutral-ai/>, criticising the idea that algorithms are neutral and unbiased.

¹⁰⁹ Michael L. Anderson, ‘Why Is AI So Scary?’ (2015) 169 (2) *Artificial Intelligence* 201-208, 204, who remarked that ‘our machines reflect who we are and what we believe’.

thus leading to inaccurate results.¹¹⁰ The concept of ‘algorithmic bias’ is generally used in computer science to indicate a cognitive fallacy that occurs when an algorithm produces results that are systemically prejudiced due to erroneous inferences in the decision-making process.¹¹¹

One might wonder, though, whether any of this is new. After all, we are already used to problems such as unpredictability, stupidity, or bias in the only serious intelligence we have known of so far – human intelligence. We are not always able to predict the behaviour of other people. Nor are we able to predict the behaviour of animals, like dogs, cats or horses. Moreover, similar issues have long existed in traditional software to some extent.

True as this might be, however, there is a crucial difference. We have had thousands, perhaps millions of years, to build mental models of how humans and other animals behave – and several decades to understand how conventional, explicitly programmed algorithms work. This makes their unpredictability somewhat bounded: a human may be a drunk stumbling across the street, an insecure bully harassing others, or a violent criminal committing murder. A dog may be aggressive, depressed, or in pain. A traditional computer algorithm may be prone to bugs, errors, or even predictable failures under certain conditions. But we are yet to build a mental picture, and a vocabulary, to describe the ways in which AI can misbehave. We still lack a diagnostic manual of mental disorders for AI systems.

Therefore, what makes AI unpredictability particularly distinctive and disruptive is its ability to learn, adapt, and make intelligent decisions without thinking like us. These qualities enable remarkable achievements, pushing the boundaries of

¹¹⁰ B Friedman and H Heintz, ‘Bias in computer systems’ (1996) 14 (3) ACM Transactions on Information Systems 330-347.

¹¹¹ The importance of fairness and bias in AI research is widely recognized by researchers, policymakers, and the legal community alike. See *ex multis* Emilio Ferrara, ‘Fairness and Bias in Artificial Intelligence: A Brief Survey of Sources, Impacts, and Mitigation Strategies’ (2023) 6(1) Sci 3, a seminal foundational paper defining ‘fairness’ and ‘bias’ in AI and discussing different forms of bias; Solon Barocas *et al.*, ‘Fairness and Machine Learning: Limitations and Opportunities’ (MIT Press 2023).

what machines can do. However, they also create significant challenges: understanding the inner workings of AI, let alone anticipating and fully controlling its behaviour, becomes an exceedingly difficult task.

Such a framework must consider not only the immediate and direct consequences of AI actions but also the *structural properties* of human-machine interactions that make AI-caused harms difficult to control and responsibilities difficult to attribute.

Addressing these challenges requires moving beyond a superficial understanding of AI unpredictability and developing a robust framework that captures the intricate dynamics of human-AI interactions. This framework must account for how AI systems interact with human agents and how responsibilities are distributed within these interactions. It must also consider the immediate and direct negative consequences of AI actions alongside the broader structural properties of human-machine interactions that complicate the control of AI-caused harms and the attribution of responsibility. This will be the goal of the next paragraph.

2.3. A socio-technical framework for understanding AI-crime

In his book *Normal Accidents: Living with High-Risk Technologies*¹¹² the sociologist Charles Perrow proposed a framework to analyze and understand catastrophic technological failures across a wide range of domains, ranging from the *Apollo 13*, *Challenger* and *Columbia* spacecraft accidents, to false alarms and near-accidents

¹¹² Charles Perrow, *Normal Accidents: Living with High Risk Technologies* (Princeton University Press 1984). The term ‘accident’ is understood by the Author to mean an event that is unforeseen and injurious, thus connoting a lack of moral guilt and a sense that the harm could not have been prevented (e.g., it was completely up to chance). This term is however unfortunate in a legal context, as the human choices leading to the accident may have been the result of malicious intent or negligence. For consistency with Perrow, we will stick with this terminology for the time being.

plaguing the US nuclear forces;¹¹³ and from the *Three Mile Island* nuclear reactor meltdown to the Air France 447 crash.¹¹⁴

According to the Author, major accidents are nearly unavoidable and even ‘normal’ in socio-technical systems that are characterized by two concurring factors: *complexity* and *interdependence*. A technological system is considered ‘complex’ when it involves intricate, non-linear interactions among multiple components, interactions that cannot be entirely planned or predicted in advance.¹¹⁵ At the same time, it is marked by ‘interdependence’ if its technical and human components are deeply interconnected, meaning that one component’s actions can significantly influence others, allowing errors to cascade through the system and potentially lead to catastrophic failure.¹¹⁶

Perrow argues that a system that is both complex and interdependent is inherently prone to unpredictable harm. While such harm may appear specifically unforeseen, it is, in fact, unsurprising – a phenomenon Perrow describes as ‘generally foreseeable unforeseeability’. Simply put, given the socio-technical system’s characteristics, we should not be shocked when something goes wrong, even though the precise nature or timing of the accident may elude prediction.

The relevance of the normal accident theory to our analysis is undeniable, especially due to its systemic perspective, which highlights that harm in socio-

¹¹³ *Ivz*, 23.

¹¹⁴ *Ivz*, 30-33.

¹¹⁵ *Ivz*, p. 52.

¹¹⁶ *Ivz*, p. 89-90. Take a nuclear power plant as an example of a complex and tightly coupled system. The complexity of such a system makes it hard to find one single cause for a nuclear meltdown, when it occurs due to the unforeseen interaction of different components that bypassed the safety measures implemented by the designers. Most accidents of this kind have several trivial causes, each innocuous in isolation but catastrophic in combination. Although calamitous outcomes are luckily rare, their individual causes occur frequently enough that it is only a matter of time until they align and malfunction, especially with frequently used technology. Therefore, socio-technical systems that are both complex and tightly coupled are prone to unpredictable yet ‘normal accidents’. Federico Bianchi *et al.*, ‘Artificial Intelligence Accidents Waiting to Happen?’ (2023) 76 *Journal of Artificial Intelligence Research* 193.

technical systems is not merely the result of random chance but an inherent feature of their design. While AI systems are, objectively, profoundly different from, say, nuclear weapons, their operational dynamics share key features that make them similarly prone to normal accidents. Emerging AI systems not only meet the criteria of complexity and interdependence but often amplify these factors, thus making such systems particularly susceptible exhibiting normal accidents.¹¹⁷

Before these potential sources of harm can be effectively mitigated or regulated, they must first be fully understood – a process we are only beginning to undertake. Even within the strict boundaries of a legal analysis, recognizing the structural properties that make AI systems prone to ‘normal accidents’ is crucial for developing appropriate frameworks for responsibility and regulation.

First, regarding Perrow’s ‘complexity’ criterion, AI seems to exacerbate it by exhibiting different levels of *technical opacity*.¹¹⁸ This refers to the situation where a system’s architecture is so complex, featuring multiple interacting subsystems, that understanding how the overall decision-making process functions may be well beyond human grasping, in the same way that calculus is beyond dogs.¹¹⁹

This is mainly due to the fact that, unlike the typical causal reasoning of humans ($X + Y = Z$), algorithmic decision-making can be described as a stochastic

¹¹⁷ Matthijs M Maas, ‘Regulating for normal AI accidents: Operational Lessons for the Responsible Governance of Artificial Intelligence Deployment’ (2018) AIES '18: Proceedings of the 2018 AAAI/ACM Conference on AI, Ethics, and Society 223-228, 225.

¹¹⁸ See generally Jenna Burrell, ‘How the Machine Thinks: Understanding Opacity in Machine Learning Algorithms’ (2016) 3(1) Big Data & Soc’y, explaining the difficulty in understanding algorithms used by complex machine learning algorithms.

¹¹⁹ Critically, the more complex the algorithmic models are, the more difficult it is to understand and foresee their output. Critically, many machine-learning algorithms are online-based and may update their prediction models after each decision they make. This means that the algorithm would reach conclusions based on new information that the human programmer would not have had a chance to consider. See on this point Joshua Kroll *et al.*, ‘Accountable Algorithms’ (2017) 165 U. Pa. L. Rev. 660.

process ($X + Y = nZ$) working with probability instead of logic.¹²⁰ This means that rather than having a deterministic relationship between input and output, where the latter implies the former as its own necessary condition, these systems can produce a range of possible outputs starting from the same input, each with a certain probability of occurrence. The systems will try all the possibilities since it will find the correct solution, which in a machine learning model is the reaching of the goal determined by the instruction.

A clear example is represented by DALL-E, an AI model developed by OpenAI that generates images from textual descriptions. When you input the same description into DALL-E multiple times, you will likely obtain n outputs. For instance, if you ask DALL-E to generate an image of ‘a red apple on a table’, it might produce various images of red apples placed on different types of tables, with different lighting conditions, angles, and other variations. Each output image is a valid interpretation of the input description, but the specific details may vary due to the probabilistic nature of the model. The system does not use any kind of logic in deciding the type of answer to give although the prompt was the same, written in the same exact words. Rather than being hard-coded, such outputs largely depend on the quality of the data used to train the system and the model of knowledge built by the self-learning system; thus there is no rational explanation for the process the machine follows in analysing the inputs to arrive at the output.¹²¹

¹²⁰ Stochastic (/stə'kæstɪk/; from Ancient Greek στόχος (stókhos) ‘aim, guess’) is the property of being well-described by a random probability distribution. Meaning of ‘Stochastic’ in the Cambridge Advanced Learner's Dictionary & Thesaurus available at <https://dictionary.cambridge.org/dictionary/english/stochastic>.

¹²¹ Critically, while work on the ‘interpretability’ of machine learning decisions is advancing – compare DARPA’s work on ‘explainable’ AI – this field is still in its infancy. Alejandro Barredo Arrieta *et al.*, ‘Explainable Artificial Intelligence (XAI): Concepts, Taxonomies, Opportunities and Challenges toward Responsible AI’ (2019) ArXiv available at <https://arxiv.org/abs/1910.10045>.

In fact, the whole process happens within what is generally known as the ‘black box’,¹²² which ‘does not allow cognitive access to how they have arrived at a particular output, or what input factors or a combination of input factors have contributed to the decision-making process or outcome’.¹²³ Hence, it may be challenging, if not impossible, to deduce the internal workings of these systems and provide a theoretical or causal explanation for the results they generate in a specific case or more generally.

Additionally, the organizational structures in which AI systems are embedded and deployed accentuate these challenges. AI opacity is not only a feature of AI systems’ mathematical complexity, but it can also be a function of proprietary protections of corporate or state secrecy.¹²⁴

Furthermore, AI also seems to meet and amplify the second Perrow’s criterion, by displaying increasing levels of interdependency between a myriad of technical and human components such as: (i) the tangible hardware elements (sensors, actuators, hardware); (ii) the different software components and applications; (iii) the data itself; (iv) the connectivity features (enabling real-time interactions and external integrations); and (v) the high number of involved human stakeholders interacting with the system (including hardware manufacturers, software designers, sellers, equipment and software installers, facility owners, AI owners, AI users, and regulators, each influencing the system’s operation).

¹²² Generally, the ‘black box’ problem can be defined as an inability to fully understand an AI’s decision-making process and the inability to predict the AI’s decisions or outputs. See Yavar Bathaee, ‘The artificial intelligence black box and the failure of intent and causation’ (2018) 2(31) *Harvard Journal of Law & Technology* 889, 905. See also Davide Castelvetti, ‘Can we open the black box of AI?’ (Nature, 5 October 2016) available at: www.nature.com/news/can-we-open-the-black-box-of-ai-1.20731.

¹²³ Horizon 2020 Commission Expert Group to advise on specific ethical issues raised by driverless mobility (E03659), ‘Ethics of Connected and Automated Vehicles: Recommendations on Road Safety, Privacy, Fairness, Explainability and Responsibility’ (Publication Office of the European Union 2020), 49.

¹²⁴ Burrell (n 118).

What distinguishes AI from other complex systems is the non-predefined nature of these interactions. Unlike traditional automated systems, where every possible outcome is the result of explicitly pre-programmed instructions, AI systems, although initially programmed by a human counterpart, operate with a degree of autonomy – meaning their outputs emerge dynamically and independently from the interactions of its components, rather than from pre-coded decision rules.

While the term is frequently employed in public and policy discourse to describe various AI applications, there is no apparent widely agreed-upon definition of ‘AI autonomy’ within the technical community.¹²⁵ From the Greek *auto* and *nomos*, autonomy means ‘self-law’. In general, this notion as applied to machines is often just shorthand for ‘I don’t know how they do that so quickly and independently’, an amazement borne of ignorance. We might in that way ascribe intelligence to Apple’s Siri, which can respond to basic voice commands with vaguely contextually correct responses, or stock market programs which in fractions of a second calculate the best price and execute trades.¹²⁶ The apparent magic of these advanced technologies is generally a function of speed outside the human scale, and of the observer’s ignorance of the programs being used.¹²⁷

More specifically, in the technical discourse, there are at least three dimensions intended in a more or less explicit fashion by experts when discussing AI autonomy.¹²⁸ The first one is ‘technical autonomy’, which refers to a system’s direction towards a purpose, adaptiveness, and capacity to have representations of the

¹²⁵ This concept must be understood in a *restrictive* sense, not in the Kantian sense of the moral capacity for self-governance of reason. For an overview of the meaning of autonomy for biological individuals and artificial models, see A Moreno, A Etxeberria and J Umeraz, ‘The Autonomy of Biological Individuals and Artificial Models’ (2008) 91 *Biosystems* 309. For discussion see Willem FG Haselager, ‘Robotics, Philosophy and the Problems of Autonomy’ (2005) 3 *Pragmatics & Cognition* 517ff.

¹²⁶ Curtis EA Karnow, ‘The Application of Traditional Tort Theory to Embodied Machine Intelligence’ in Ryan Calo *et al* (eds), *Robot Law* (Elgar 2016) 51-77.

¹²⁷ ‘Any sufficiently advanced technology is indistinguishable from magic’. Arthur C. Clarke, ‘Profiles of the Future: An Inquiry into the Limits of the Possible’ (W&N 1999).

¹²⁸ Sartor and Omicini (n 44).

environment. The second dimension is ‘functional autonomy’, which relates to system’s ability to perform a task on its own (capability independence), without human intervention. Finally, the third dimension is ‘ontological autonomy’, where a system possesses its own self-awareness and freedom of choice.¹²⁹

While the third conception rests on controversial philosophical assumptions and does not have technical relevance, we define autonomy here as a combination of the first two conceptions: the level of (in)dependence from human supervision or intervention coupled with a quasi-cognitive learning and/or adaptation capacity.¹³⁰ In this meaning, autonomy is a property that exists on a spectrum, with varying degrees rather than being an all-or-nothing attribute. The degrees of autonomy depend on the specific context and the extent to which human oversight and intervention are necessary for the system’s operation.

Within this horizon the machine may be provided with the highest possible degree of autonomy – more precisely ‘heteronomous autonomy’ or the ability to choose between ends – thus acting without human supervision.¹³¹ In such a scenario the human being sets the ends and then withdraws, leaving the artificial agent to act on its own and freely decide how to accomplish the task assigned. This form of autonomy, which we could describe as being ‘strong’, is only one of the possible control mechanisms for AI applications, where alternatives are possible, such as a ‘weak’ or ‘tempered’ form of autonomy.¹³²

¹²⁹ Karnow (n 126).

¹³⁰ ‘Entities that behave as a unity of action, whereby action entails some kind of automation and implies that the entity can perform without the intervention of its creator. [...] This is not to deny that all these agents have been built, programmed, initiated, and used by human agents. It does highlight the fact that the human agent can withdraw, leaving the artificial agent to act on its own’. Mireille Hildebrandt, *Smart Technologies and the End(s) of Law* (Elgar Publishing, 2015), 22-23.

¹³¹ Mathias Gutman, Benjamin Rathgeber and Tareq Syed, ‘Action and Autonomy: A Hidden Dilemma in Artificial Autonomous Systems’ in Michael Decker and Mathias Gutman (eds.), *Robo- and information ethics: some fundamentals* (Lit Verlag, 2011) 64.

¹³² For discussion see Haselager (n 125) 517ff.

In that regard, we could think of a vehicle operating without constant human supervision and capable of making decisions without input from the driver, but which could be controlled in a moment of need (say in case of a malfunction). In this sense, a self-driving car that always requires supervision by a present, competent, and legally licensed driver certainly presents a lower degree of autonomy than a fully autonomous vehicle safely transporting a minor or a drunk person home. Additionally, there are also cases of systems requiring constant human interaction and control. Many vehicles have automated functions, such as cruise control which regulates speed or lane-keeping assistant steering the vehicle back onto the road in case of distraction of the human driver. These functions are supervised by the driver, who remains in active control of the vehicle.

This classification of levels of autonomy implies that a system's autonomy increases in proportion to the extent of human interaction or involvement. We consider a categorisation of human-machine modes of interaction that comprises three different ways in which humans can interact with autonomous decision-making systems. While each level involves some degree of autonomy, the specific implications differ significantly: as human control diminishes, the likelihood of AI producing unpredictable outcomes naturally increases.

- i. *Assisted mode* (or semi-autonomous): the operator receives support from the system in making specific decisions or performing specific tasks, but ultimately the choice remains the responsibility of the 'person in the loop' or, alternatively, the latter needs to authorize each and every decision by the system. The autonomy of the AI agent is weakest here (**H**uman in the **L**oop).
- ii. *Automated mode* (or supervised autonomous): the decision must be automated as it needs to be delivered in a situation where the response time is too short to include a human, but the human operator remains actively involved, allowing for some human intervention for exceptions. The AI

agent's autonomy here is weak or 'tempered' (Human in the Loop For Exceptions).

- iii. *Fully autonomous mode*: the system operates independently without direct human intervention or input. Human operators are still involved, but in a more supervisory role. For instance, they periodically monitor AI's decisions to ensure the system continues to perform well. This is the strongest form of autonomy (Human on the Loop).

Mode of interaction	Degree of autonomy	Human involvement	Decision-making process
Assisted	Weak	Human in the loop	Machine waits for human input before taking action
Automated (supervised autonomous)	Tempered	Human in the loop for exceptions	Human can intervene in real time for exceptions
Autonomous	Strong	Human on the loop	No ability for human to intervene in real time (only supervision)

This brief overview demonstrates that emerging AI systems fulfil all the relevant conditions to exhibit 'normal accidents'. Critically, the combination of complexity, opacity and autonomy may lead to a disturbing scenario where it is nearly impossible to maintain complete control over every task these systems perform. This challenge is particularly pronounced because even narrow AI systems often rely on adaptive, opaque algorithms designed to function independently in highly intricate and unpredictable environments. These characteristics introduce uncertainty that cannot be fully eliminated.

As a result, the notion of a completely safe AI – one that always behaves predictably – is an unrealistic ideal; the risk of AI systems causing harm can never be

entirely reduced to zero.¹³³ Accidents resulting in property damage, injuries, or even the loss of human life have happened, and will continue to happen. Therefore, the benefits of AI can only be realized at the cost of accepting the ‘normalization’ of AI-caused harm.¹³⁴

Notwithstanding this normalization, humanity seems to have accepted AI as a way to improve efficiency and societal welfare. This does not mean, however, that these concerns should be ignored, and efforts are being made in both the technical and policy domain to discuss and address the risks of harm. The current study focuses on one important aspect: *criminal responsibility*.

Against this backdrop, significant legal questions arise: how much harm is society willing to accept in exchange for the benefits AI systems offer? What laws and frameworks are needed to address such harms when they occur? And, most importantly, who should bear the responsibility for such harms?

2.4. Addressing AI crime through criminal law: why a responsibility gap?

The preceding analysis has highlighted a paradox at the heart of AI: the very attributes that make these systems extremely powerful and efficient – complexity, autonomy, and unpredictability – also make them inherently risky. In this context, we introduced the concept of AI crime, referring to scenarios where an AI system engages in harmful behaviour that, if performed by a human, would carry criminal significance. However, a deeper inquiry is required into what we term ‘hard AI crime’ – a category of AI-related offenses where, under current legal frameworks, no human agent can be directly held criminally responsible.¹³⁵ These are cases that fall into a legal gray zone, where existing doctrines of *actus reus* and *mens rea* struggle to assign liability in a meaningful way.

¹³³ Roman Yampolskiy (n. 97).

¹³⁴ On this point, see Maas (n 117) 223-228; Bianchi *et al* (n 116) 193-199; Alan Chan, ‘Loss of control: normal accidents and AI systems’ (2021) ICLR Workshop on Responsible AI.

¹³⁵ Abbott and Sarch (n 63) 328.

At the core of this debate is a fundamental question: Who is to blame?¹³⁶ Legal scholars and policymakers have put forward various answers in an effort to prevent impunity when AI-driven harm results in the violation of protected legal interests. But as we shall see, none of these answers are without significant theoretical and practical challenges.

In an attempt to map out this emerging field, we could preliminarily group the different approaches into two broad categories: (i) approaches to ‘hard AI crime’ that remain within the bounds of the current criminal law for *humans*; and (ii) approaches to ‘hard AI crime’ that entertain the possibility of AI agents being *new addressees* of existing criminal law, next to humans.

At its core, the first category hinges on imputing responsibility to natural or legal persons ‘behind the machine’, whose decisions or omissions directly or foreseeably led to the harm. Essentially, this approach relies on attaching criminal liability to human’s capacity of foreseeing and handling/avoiding harm, that is, to whether they were (criminally) negligent for the ‘acts’ of their technological artifacts, particularly in contexts where such artifacts operate with high levels of autonomy or unpredictability.

In this regard, a great majority of concerns relate to the *mens rea* component (i.e., the subjective element of crime). While some authors have rightly pointed out that the *actus reus* requirement (i.e., the objective element of crime) can potentially also be problematic to establish,¹³⁷ most of them focus primarily on the *mens rea* component. *Mens rea* generally requires knowledge and volition, and it is particularly this knowledge that is called into question with regard to complex, unpredictable AI systems.

Against this backdrop, the theoretical discussion around AI and *mens rea*, specifically related to *culpa* or negligence, has been revolving around a ‘criminal

¹³⁶ Gless, Silverman and Weigend (Ch I n 18) 412.

¹³⁷ King *et al* (n 68) 95.

responsibility gap’, which in the morally-laden framework of criminal law, has been understood as a ‘culpability gap’ inevitably leading to a ‘retribution gap’.¹³⁸

Discussions on the existence of responsibility gaps had started as early as 2004. Matthias first coined the expression, specifically identifying the *lack of predictability and control* in modern AI as the main obstacle in imputing the event to a responsible human agent behind the machine.¹³⁹ Attributing blame is indeed difficult if neither the producer nor the user can control or predict how an AI will react, as it learns from experience and acts in accordance with its environment.¹⁴⁰

For example, when an AI system performs a task leading to unpredictable harm, determining who had the duty to anticipate and mitigate such outcomes becomes a fraught exercise. Was it the developer, who programmed the system? The company, which marketed or deployed the technology? Or the user, who operated it without fully understanding its capabilities and limitations? Each of these actors may have contributed to the chain of events, but none may have had a clear or realistic opportunity to foresee the specific harm. In that regard, it has rightly been observed that the existence of a ‘responsibility gap’ in such cases is perhaps the most delicate from a legal point of view, as it entails a disconnect between the actions of human agents and the complex algorithmic consequences that ensue.¹⁴¹

The current legal debate on hard AI crime becomes particularly inconclusive when addressing the second category of approaches: those exploring the possibility of AI agents being held directly criminally responsible. These approaches often

¹³⁸ J Danaher, ‘Robots, Law and the Retribution Gap’ (2016) 18 Ethics and Information Technology 299.

¹³⁹ It has rightly been observed that the issue of liability for AI-caused harm is perhaps the most delicate from a legal point of view, as it entails a ‘responsibility gap’ characterized by a disconnect between the actions of human agents (such as programmers or users of AI systems) and the complex algorithmic consequences that ensue. See Matthias (Ch I n 23). The existence of a responsibility gap has been specifically recognized by European institutions. Cf. Resolution of the European Parliament with recommendations to the Commission on Civil Law Rules on Robotics (2017), Recitals AB and AI.

¹⁴⁰ Johannes Himmelreich, ‘Responsibility for Killer Robots’ (2019) 22 Ethical Theory and Moral Practice, 743–744.

¹⁴¹ Matthias (Ch I n 23) 175.

envision scenarios in which AI systems are recognized as new legal entities – akin to corporations – or propose frameworks for ascribing direct criminal responsibility to AI systems which may rely on less morally demanding interpretations of culpability or argue that certain AI systems can exhibit behaviour resembling criminal intent.¹⁴²

While these theoretical constructions provoke debate, they invariably grapple with the same foundational and intractable questions. What does it mean to be a ‘person’ under criminal law? Must moral agency and culpability always underpin criminal responsibility? And finally, what is the ultimate purpose of punishment in this context? These unresolved issues underscore the profound challenge of integrating the direct criminal responsibility AI systems into existing legal frameworks, particularly as this solution falls outside the boundaries of traditional human-centric assumptions of criminal law.¹⁴³

At the heart of the debate over how to address the responsibility gap for hard AI crimes, two opposing solutions emerge:

- (i) extending the moral requirements of criminal responsibility to encompass the actions of morally incompetent AI agents and their innocent operators, or at the very least coming up with justified reasons for a departure from the dominant culpability paradigm for humans;

¹⁴² Gabriel Hallevy, *Liability for Crimes Involving Artificial Intelligence Systems* (Springer 2015) 102ff.

¹⁴³ Susanne Beck, ‘Die Diffusion strafrechtlicher Verantwortlichkeit durch Digitalisierung und Lernende Systeme’ (2020) 2 *Zeitschrift für Internationale Strafrechtsdogmatik* 41, 43-50; Bert-Jaap Koops, Mireille Hildebrandt, and David-Olivier Jaquet-Chiffelle, ‘Bridging the Accountability Gap: Rights for New Entities in the Information Society?’ (2010) 11 *Minnesota Journal of Law, Science & Technology* 497, 522; Mireille Hildebrandt, ‘Autonomic and Autonomous Thinking: Preconditions for Criminal Accountability’ in Mireille Hildebrandt and Antoinette Rouvroy (eds), *Law, Human Agency and Autonomic Computing* (Routledge 2011) 141-60.

- (ii) or accepting that criminal punishment may not be appropriate for such hard AI offenses, instead addressing these harms through alternative mechanisms like civil liability regimes or administrative remedies, where applicable.

Both solutions, however prove deeply unsatisfactory. The first undermines the culpability principle, a cornerstone of the deontological ethics that form the foundation of criminal law in both common and civil law jurisdictions, and it also reopens enduring philosophical debates about the boundaries and applicability of culpability, complicating the legal discourse further.¹⁴⁴ Conversely, the second fails to provide sufficient deterrence against hard AI crimes: civil or administrative penalties may be insufficient, particularly when the potential benefits of AI-enabled wrongdoing far outweigh the expected sanctions.¹⁴⁵

While this may be true, criminal law – as the most powerful instrument of state coercion – cannot remain static in the face of technological change. To continue fulfilling its dual protective function, it must *adapt*, albeit within carefully defined *limits*. The challenge lies in striking a balance between two competing imperatives: the protection of citizens’ fundamental legal interests who may become victims of deviant AI systems, on the one hand, and the reasonable claims of AI operators for protection from unjust criminal imputation, on the other hand. Achieving this equilibrium would legitimize new legislative initiatives in the field of AI-related

¹⁴⁴ A comparative note: while it is true that criminal law is, in many respects, a parochial discipline, the discussion at hand transcends national boundaries. This is because it primarily engages with the normative criminal law theory that underpins the criteria for what constitutes punishable conduct. At a foundational level, nearly all known systems of criminal law, regardless of jurisdiction, share certain universal principles. Among these, ‘guilt’ or ‘culpability’ stands out as a core criterion that is universally recognized and sought to be satisfied. This shared commitment provides a common ground for analyzing how the concept of culpability might adapt – or resist adaptation – in the face of emerging challenges like AI-caused harm. George Fletcher, *The Grammar of Criminal Law, vol 1: Foundations* (OUP 2007) 222.

¹⁴⁵ Nerantzi and Sartor (n 105) 677.

criminal liability.¹⁴⁶ Such a balance can be achieved when new criminal law provisions aim at punishing culpable conduct that is related to technological developments without setting aside fundamental criminal law categories, such as the culpability principle.¹⁴⁷

A crucial starting point for this discussion is the newly adopted EU AI Act and its risk-based approach. While the Act does not directly address criminal liability, it serves as the conceptual foundation for any future legal frameworks in this area. Any criminal provisions – whether newly introduced or applied from existing statutes – will inevitably be shaped by the EU’s regulatory safeguards and market standards governing the development, deployment, and use of autonomous AI systems.

¹⁴⁶ Agapios Papaneofytou, *Criminal Liability of Legal Persons: A Construction of Criminal Immunity for Corporate Offenders?* (Nomiki Vivliothiki 2012) 130.

¹⁴⁷ *ibid* 137.

SECTION 3

Regulating AI risks: the EU AI Act

3.1. The concept of AI risk in the modern criminal law theory

The growing prominence of AI risks in debates over the alleged crisis of traditional criminal law is far from surprising. In many ways, AI crime marks the latest chapter in a broader crisis – the ‘modernity shock’¹⁴⁸ that the event-based paradigm in criminal law has been experiencing for decades in response to the pressures of the so-called ‘risk society’.¹⁴⁹

In this context, the concept of ‘risk’ is understood as a normative notion associated with potential undesirable events and their negative consequences, and it is often characterized as anticipation of a probable harm.¹⁵⁰ When referring to the risk of an event x (such as a mechanical failure or chemical exposure), we typically imply that we can meaningfully assign a probabilistic value to the occurrence of x . Hence,

¹⁴⁸ Stella (Ch I n 24) 292-293.

¹⁴⁹ Ivi, 224-235. The term *Risikogesellschaft* was first coined by the German sociologist Ulrich Beck, and it has rapidly gained relevance in the development of the legal doctrine in various continental jurisdictions. More specifically, the theory of ‘risk society’ is generally used to describe the transition of human society after mid-20th century, as a natural consequence of an advanced stage of industrial society after industrial revolution: industrial society and risk society respectively respond to first modernity and second modernity. Ulrich Beck, *Risk Society: Towards a New Modernity* (Sage, 1992 – German edition 1986 with original title: *Risikogesellschaft. Auf dem Weg in eine andere Moderne*).

¹⁵⁰ As such, risk differs from ‘materialized harm’ in that risk has a future orientation – it has not yet materialized. Thus, risk is not a purely scientific concept but rather an epistemic one, a socially constructed phenomenon rooted in perception rather than reality. We may have certainty that something will eventually cause harm, but we do not know when, how, or in what precise form these harms will manifest. Giacomo Zanotti *et al*, ‘AI-Related Risk and Uncertainty’ in B Steffen (ed) *Bridging the Gap Between AI and Reality. Lecture Notes in Computer Science* (Springer 2024), 285.

it arises the need to clarify the relationship of risk with the notion of ‘danger’ or ‘peril’, which also conveys the likelihood of a future event.

In legal scholarship, there is still no clear consensus on the precise classification and distinction between these concepts. The debate revolves around whether these terms are true synonyms or if they reflect distinct legal concepts. While some scholars treat them interchangeably, others argue that their logical analogy does not imply substantive identity.¹⁵¹

A dogmatic analysis within criminal law suggests that the two notions differ in a crucial way. On the one hand, ‘risk’ pertains to an *ex-ante* assessment – that is, an evaluation made before harm occurs – but always in relation to a situation that ultimately resulted in actual harm or damage. In this sense, risk is used to establish whether the event that did occur was a realization of the kind of outcomes that the individual’s conduct had made likely. ‘Danger’, on the other hand, refers to a threat of harm that has not materialized into a separate, injurious outcome. It remains a probability, not yet confirmed by real events.¹⁵²

While the latter position identifies the distinguishing element on a qualitative basis, another part of the doctrine has argued for a quantitative characterization, asserting that danger is nothing more than a risk marked by a high probability of harm occurring to the protected interest.¹⁵³ In this sense, the term risk does not, in fact, carry a semantic meaning different from that of danger, both describing a potential causal relationship between a conduct and an event based on scientific grounds. According to this position, danger would ultimately be defined as the ‘probability’ or

¹⁵¹ See Vincenzo Militello, *Rischio e responsabilità penale* (Giuffrè 1988) 24ff.

¹⁵² *ivi* 21, emphasizing that only danger explicitly appears in criminal law provisions, serving as an attribute of the conduct or other relevant elements, such as the event. For example, this distinction is particularly relevant in the case of attempted crimes, where the harm is anticipated but has not yet occurred as an independent consequence of the perpetrator’s conduct.

¹⁵³ This position is defended by Chiara Perini, *Prospettive del concetto di rischio nel diritto penale moderno* (Garbagnate Milanese 2002). More recently, the same Author argues for the identity of risk and danger: Id, *Il concetto di rischio nel diritto penale moderno* (Milano 2010), 42, 63, 371ff.

‘significant possibility’ of the harmful event occurring, whereas risk would remain confined to the realm of the merely ‘possible’.¹⁵⁴

In the context of modern criminal law, however, risk has taken on an ambiguous connotation, resembling a purely suspected danger – one in which the correlation between conduct and event remains uncertain and lacks scientific grounding.¹⁵⁵ The contrast with danger – understood as a scientifically grounded assessment – is staggering, to the point of shaping the entire approach of modern criminal law.¹⁵⁶

According to this perspective, the central role of risk in the imputation of criminal liability in modern criminal law reflects a shift away from a legal framework in which criminal wrongdoing is traditionally constructed around an event of harm or at least a concrete danger to identifiable legal interests – interests that can be grasped and controlled by a single individual through clear cause-and-effect relationships.¹⁵⁷

By contrast, in cases where risk is not scientifically established, criminal law could only address risk by abandoning any reference to the actual occurrence of an event, thus shifting risk itself into a central criterion for determining blameworthiness.¹⁵⁸ This transformation led to what scholars increasingly refer to as

¹⁵⁴ Carla Brusco, ‘Rischio e pericolo, rischio consentito e principio di precauzione. La c.d. flessibilizzazione delle categorie del reato’ (2012) *Criminalia* 1.

¹⁵⁵ Carlo Piergallini, *Danno da prodotto e responsabilità penale. Profili dommatici e politico-criminali* (Giuffrè 2004), 520.

¹⁵⁶ Barbara Hudson, *Justice in the Risk Society: Challenging and Re-Affirming Justice in Late Modernity* (Sage 2003), 43.

¹⁵⁷ Vincenzo Militello, ‘Diritto penale del rischio e rischi del diritto penale fra scienza e società’ in C Spinellis *et al* (eds.), *Europe in Crisis: Crime, Criminal Justice, and the Way Forward. Essays in honour of Nestor Courakis* (Athens : Ant. N. Sakkoulas Publishers L.P 2017) 223-238, 224.

¹⁵⁸ In this sense, the concept of risk is referred to in its ‘out-systematic’ conception, rather than in its ‘in-systematic’ form. The in-systematic conception treats risk as a dogmatic institution, one that is internal to the traditional framework of criminal law. In this classical view, risk is tightly woven into the established principles of criminal liability: allowed risk, the causal nexus between risk and event, and risk acceptance in *dolus eventualis* (where an individual accepts the possibility of harm in pursuit of a goal) are the main expressions of risk in traditional

‘risk criminal law’ – a framework in which criminal law no longer waits to respond to concrete harm *ex post facto*, but instead evolves into a proactive instrument aimed at anticipating and managing risks before they materialize.¹⁵⁹

These risks, typically associated with post-modernity, often arise as side effects of complex technological activities – effects that remain, in many cases, partially or entirely unknown for extended periods. Examples include risks linked to workplace safety, food security, public health, genetic engineering, and environmental pollution, which have exposed the precariousness of what we know (constantly subject to revision) alongside the scope of what remains unknown (which continues to expand).

offense theory. In this context, risk is understood within well-defined legal boundaries, where actions that create permissible risks—such as those regulated by professional or social standards—are allowed, while impermissible risks are punished. Criminal law typically deals with objective imputation: establishing whether the risk posed by an action can be directly linked to the harm that occurs, thus forming a key part of the criminal charge. However, the study of ‘criminal risk’ as a broader model—especially in the face of complexity and uncertainty—requires a shift away from this systematic approach. The classical in-systematic conception does not adequately capture the new forms of risk presented by modern technologies, globalized systems, or environmental hazards, where the cause-and-effect relationships are diffuse and harder to pin down. In the context of risk society, where the potential for harm often lies in systemic factors or uncertain future events, the prevention of harm through criminal law requires a new out-systematic approach. Cfr. Perini (n 153b) 4ff., 168ff.

¹⁵⁹ Militello (n 157) 224. The term *Risikostrafrecht* is described in Lothar Kuhlen, ‘Zum Strafrecht der Risikogesellschaft’ (1994) *Goltdammers Arkiv fuer Strafrecht*, 347, referring to the work of Cornelius Prittwitz, *Risiko und Strafrecht. Untersuchungen zur Krise vom Strafrecht und Kriminalpolitik in der Risikogesellschaft* (Klostermann 1993). For more recent literature, see *ex multis* Carlo Piergallini, ‘Attività produttive e imputazione per colpa, prove tecniche di diritto penale del rischio’ (1997) *RIDPP* 1474ff; B Mendoza Burgo, *El derecho penal en la sociedad de riesgo* (Madrid 2001); L Stortoni, ‘Angoscia tecnologica ed esorcismo penale’ in Alpa (ed), *Il rischio da ignoto tecnologico* (Milano 2002) 87; María José Jiménez Díaz, ‘Sociedad Del Riesgo E Intervención Penal’ (2014) 16 *Revista Electrónica de Ciencia Penal y Criminología*; F Herzog, ‘Società del rischio, diritto penale del rischio, regolazione del rischio. Prospettive al di là del diritto penale’ (tr. Donato Castronuovo, original title ‘*Risikogesellschaft, Risikostrafrecht, Risikoregulierung – Über das Strafrecht hinausweisende Perspektiven*’) in L Stortoni and L Foffani (eds.), *Critica e giustificazione del diritto penale nel cambio di secolo. L’analisi critica della Scuola di Francoforte* (Giuffrè 2004) 20 ff.

In this context, the ambitious goal of the risk criminal law paradigm is the progressive *normative elimination of uncertainty*.¹⁶⁰ Even when ‘zero risk’ is not explicitly invoked, it is often implied in discussions about the so-called ‘precautionary principle’.¹⁶¹ This principle represents a shift from relying solely on scientifically substantiated knowledge (as is the case with traditional prevention strategies) to addressing risks marked by significant scientific uncertainty.¹⁶² It involves dealing with situations where harm is suspected but cannot be conclusively proven, and the exact nature or likelihood of such harm remains uncertain.

In particular, the precautionary approach leads to a shift in cases where risk is not scientifically established: rather than adhering to the principle *in dubio pro actione* (favouring action in the face of uncertainty), the criminal law moves toward *in dubio pro omissione* (favouring inaction in cases of doubt). If this principle were

¹⁶⁰ Federico Consulich, ‘Voce *Rischio consentito*’ in Massimo Donini (ed), *Enciclopedia del diritto. Reato colposo* (Milano 2021) 1102ff, 1107. On the risk as imputation paradigm in modern criminal law, see G De Francesco, ‘Pericolo, rischio, incertezza. Il controllo penale e i suoi confini nella temperie della postmodernità’ in E. Sirsi and M. Goldoni (eds.) *Regole dell’agricoltura. Regole del cibo. Produzione agricola, sicurezza alimentare e tutela del consumatore* (Pisa 2005), 125ff.

¹⁶¹ The precautionary principle is not new in the legal debate but there is no universally accepted definition of it. Generally speaking, precaution is a decision-making rule asking the question of what to do in situations of scientific uncertainty. The principle originated in German environmental law and has been incorporated in many international treaties on environmental protection. Currently, it is codified in Article 191 § 2 TFEU and plays a pivotal role in EU environmental policy. The European Union Court of Justice explicitly stated that ‘the precautionary principle can be defined as a general principle of Community law requiring the competent authorities to take appropriate measures to prevent specific potential risks to public health, safety and the environment, by giving precedence to the requirements related to the protection of those interests over economic interests’. Judgment in the case of *Artegodan v. Commission* of 26 November 2002 (T-74/00), paragraph 184. Generally on the precautionary principle, see J Tosun, ‘How the EU Handles Uncertain Risks: Understanding the Role of the Precautionary Principle’ (2013) 20 *Journal of European Public Policy* 1517; Zander, *The application of the precautionary principle in practice* (Cambridge 2010).

¹⁶² Resorting to what is evidently an oversimplification, both policymakers and scholars like to contrast what they refer to as ‘strong’ conception of precaution with what they refer to as the ‘tempered’ conception, largely by contrasting EU and US law. This conveys a common perception that EU favours strong precaution whereas US favours more permissive approaches.

to be incorporated into the realm of negligence, it could, in contexts where scientific findings on the risks associated with a particular activity, product, or substance are uncertain or contradictory, result in the emergence of a *precautionary rule of abstention*. Under this framework, failing to abstain from a potentially harmful activity could constitute a form of generic negligence.¹⁶³

This approach has been embraced by prevailing Italian and German case law in *toxic cases*, where courts have held that scientific uncertainty regarding the hazardous nature of a particular exposure does not preclude attributing criminal liability – whether based on negligence or even intent.¹⁶⁴ More specifically, in situations of scientific uncertainty, a harmful event could be considered ‘foreseeable’ from the moment a model agent would have had a *well-founded suspicion* regarding the dangerousness of their conduct in relation to certain outcomes.¹⁶⁵ This circumstance would delineate the circumstances giving rise to a duty to act, which

¹⁶³ For a general overview of the precautionary principles and its implications in criminal law, see *ex multis* F. Consulich, ‘Tutela del consumatore’ in F. Palazzo and C.E. Paliero (eds.) *Commentario breve alle leggi penali complementari* (Padova 2007), 2977ff; D. Castronuovo, *Principio di precauzione e diritto penale* (Roma 2012), 43ff; M. Donini, ‘Diritto penale «classico» e diritto penale «moderno»’, in Id., *Il volto attuale dell’illecito penale* (Milano 2004), 120.

¹⁶⁴ Drawing from the context of toxic cases, examples include situations where a specific pathology (e.g., pleural mesothelioma) is correlated with exposure to a particular substance (e.g., asbestos), despite the absence of conclusive scientific evidence. In such cases, a precautionary rule could emerge along the following lines: ‘In the presence of serious conjectures regarding the potential danger of certain substances or products under given conditions, gather the maximum amount of relevant information and take action as soon as possible, in accordance with all reasonably foreseeable safety standards. This holds even in the absence of fully established scientific laws, provided that such action is based on the best available technologies at economically viable costs’. For a thorough extension of the discussion on toxic cases and criminal product liability in the context of AI, see Riccardo Bertolesi, *Intelligenza artificiale e responsabilità penale per danno da prodotto* (2019) doctoral dissertation within the *Cesare Beccaria* Doctoral Program in Legal Sciences at the University of Milan (Criminal and Criminal Procedural Law Curriculum – 32nd cycle), 98ff. Cfr. Cass. pen., sez. IV, 17 May 2006, n. 4675.

¹⁶⁵ On the concept of risk as a mere suspicion-based judgment lacking scientific foundation see Carlo Ruga Riva, ‘Principio di precauzione e diritto penale. Genesi e contenuto della colpa in contesti di incertezza scientifica’ in E. Dolcini and C.E. Paliero (eds.), *Studi in onore di Giorgio Marinucci* (Giuff  2006) 1743-1777, 1750ff. Cfr. also Piergallini (n 155), 520.

should be assessed in relation to the state of knowledge available at the time. Depending on the context, this duty could entail obligations such as further investigation and scientific updates or, more generally, operational precautions suggested by the indicative factual signs. However, except in extreme cases where initial conjectures suggest the grave danger of a given behaviour, complete abstention from the conduct cannot be required. Failure to observe these obligations could amount to generic negligence.¹⁶⁶ However, judges, who often find themselves navigating these complexities without clear guidance, may inadvertently end up conflating causality with danger, transforming negligence liability (where culpability depends on negligence or recklessness) into something resembling strict liability.

Against this backdrop, it is therefore unsurprising that, in response to the deep modernity shock brought about by AI technologies, efforts have been made to eliminate the risks tied to the development and use of intelligent systems – either by relaxing the moral requirements of criminal responsibility to extend it to the behaviour of morally incompetent AI agents or, at the very least, providing more or less compelling reasons for a flexibilization of the negligence regime for humans in order to address scenarios characterized by ‘foreseeable unforeseeability’. AI, in fact, serves as a significant test case for the challenges arising from the relationship between criminal law and risk, as well as for the misunderstandings stemming from its conflation with the concept of danger.

The notion of ‘zero risk’ is particularly appealing in this domain because it provides a sense of reassurance: it allows for continuous punishment and creates the illusion of control over AI-driven harmful events. It is also rhetorically effective, as it gives the impression of delivering justice through seemingly straightforward convictions. However, this concept is fundamentally flawed. No logically possible AI-caused harm has a zero risk of occurring in the real world. The idea of absolute AI safety is an illusion, and the pursuit of such an ideal is neither realistic nor achievable.¹⁶⁷

¹⁶⁶ *ivi*, 1765.

¹⁶⁷ Simmler (Ch I n 32) 10.

Admittedly, this is why, with the adoption of the AI Act on the 9th of December 2023, the EU institutions have moved away from the reassuring but unrealistic notion of zero risk, choosing instead a proactive approach to AI governance rooted in the traditional – and inherently political – lens of *risk regulation*.¹⁶⁸ The path set by the AI Act brings the relevance of risk analysis back to the forefront of discussions on liability arising from the development and use of AI systems, emphasizing the context in which these systems operate.

However, the AI risk outlined in the EU regulation is a peculiar type of risk. As will be explored in more detail later, it represents more of a ‘legal risk’, not based on an actual, empirically verified propensity to cause harm, but rather, in line with a tempered precautionary approach, on the breach of precautionary thresholds codified into legal rules.¹⁶⁹ This perspective raises several additional considerations.

3.2. The AI Act’s risk-based approach and its impact on criminal law

While the AI Act stands as the EU’s flagship regulation, it represents the culmination of years of strategic groundwork that have positioned the EU as a global frontrunner in AI governance.¹⁷⁰ Since the launch of its AI strategy in 2017, the EU has pursued

¹⁶⁸ Risk regulation has a long and complex history. It has been used across a variety of fields, from food and drug regulation to environmental protection to governance of the financial system. On the complex history of risk regulation, see generally, David Vogel, *The Politics of Precaution: Regulating Health, Safety, and Environmental Risks in Europe and the United States* (Princeton University Press, 2012). On the various policy options regarding AI risk regulation, see Margot E. Kaminski, ‘Regulating the Risks of AI’ (2023) 103(22) Boston University Law Review 1347-1410.

¹⁶⁹ Tobias Mahler, *Defining legal risk* (2007) Proceedings of the Conference Commercial Contracting for Strategic Advantage - Potentials And Prospects 10-31.

¹⁷⁰ Anu Bradford, *Digital Empires. The Global Battle to Regulate Technology* (Oxford University Press 2023).

a vision of fostering trust in AI, aiming to establish an ethical and legal framework capable of earning societal confidence while supporting innovation.¹⁷¹

The emphasis on trust is already apparent in the EU Commission's 2020 White Paper,¹⁷² which identifies a lack of trust as 'the main factor holding back a broader uptake of AI'¹⁷³ and outlines a vision for a regulatory framework designed to create an 'ecosystem of trust' which in turn 'should give citizens the confidence to take up AI applications'.¹⁷⁴

Trust is another way to deal with the uncertainty of the future, and it is closely related to risk in some respects: it anticipates the possibility of unwanted future harms; yet someone who trusts acts *as if* the future is not uncertain with regard to them and *as if* the unwanted future harms will not transpire.¹⁷⁵ Anybody who trusts *accepts* the risk of misplacing this trust and facing the unwanted outcome after all.

The recently approved AI Act seeks to bring this vision to life by delivering a regulatory framework for the development of AI in the internal market. Describing the general objective of the regulation, Recital 1 of the AI Act clarifies that it is supposed 'to promote the uptake of human centric and trustworthy artificial intelligence'. This statement seems to put emphasis on the new legal interest of *trustworthiness* as a cornerstone in the EU's strategy for AI governance. In other

¹⁷¹ Through a series of soft law instruments, the EU has laid a robust foundation for AI risk regulation, issuing guidelines, recommendations, and principles to shape the evolving regulatory landscape and prepare the way for future, more comprehensive legislation. A first Resolution of the European Parliament of 2017 ('Resolution of the European Parliament with recommendations to the Commission on Civil Law Rules on Robotics' (2020/2014(INL)) was followed by some Communication from the European Commission (see, for example, 'Artificial Intelligence for Europe', COM (2018) 237 final, 25 April 2018; 'Coordinated Plan on AI', COM (2018) 795 final, 7 December 2018; COM(2019) 168 final, 8 April 2019). (2017). It is also worth mentioning the High-Level Expert Group on AI, 'Ethics Guidelines for Trustworthy AI' (2019). Each of these instruments in detail goes beyond the scope of the present study.

¹⁷² European Commission, White Paper on 'Artificial intelligence. A European approach to excellence and trust', Brussels, Feb. 19, 2020, COM(2020) 65 final, (2020), 3.

¹⁷³ *Id.* 9.

¹⁷⁴ *Id.* 3.

¹⁷⁵ Niklas Luhmann, *Trust and Power* (Polity 2017), 23.

words, it specifies how AI systems would need to be ‘trustworthy’ as a prerequisite for their safe and widespread adoption in Europe. The eventual goal is to provide a ‘foundation upon which all those affected by AI systems can trust that their design, development and use are lawful, ethical and robust’.¹⁷⁶

Subsequent recitals reiterate the emphasis on trustworthy AI. Recital 62 refers to trustworthiness in combination with the notion of *risk*: ‘In order to ensure a high level of trustworthiness of high-risk AI systems, those systems should be subject to a conformity assessment prior to their placing on the market or putting into service’. In this respect, it is evident that the Act relies on the traditional conception that risk is the likelihood of converting a source of hazard into actual loss, injury, or damage. More specifically, the Act’s benchmark to calculate the risk associated with AI systems is their potential *adverse impact on health, safety, and EU fundamental rights*.¹⁷⁷

Being the likelihood of damage, risk can be expressed through the ratio between hazard and safeguards so that, as the safeguards increase, the risk quotient decreases:

$$risk = \frac{hazard}{safeguards}$$

In the AI Act, there is an intrinsic link between trust, trustworthiness, and risk. Trustworthiness, in this context, is achieved through regulatory measures designed to render risks associated with AI systems acceptable to society. At the

¹⁷⁶ High-Level Expert Group on AI (n 173), 5. The Guidelines distinguish three components of trustworthy AI: lawfulness, adherence to ethical principles and values, and robustness from a technical and social perspective.

¹⁷⁷ This conception can easily be deduced, for instance, from Recital 52 of the AI Act, stating that ‘[...] it is appropriate to classify [a system] as high-risk if, in the light of their intended purpose, they pose a high risk of harm to the health and safety or the fundamental rights of persons, taking into account both the severity of the possible harm and its probability of occurrence [...]’.

heart of the framework, therefore, lies the concept of *risk acceptability*, which is pivotal in operationalizing the legal interest of trustworthiness.¹⁷⁸

Risk acceptability reflects a deliberate normative choice: to permit the development and deployment of risky but useful AI systems in the first place rather than resorting to stricter precautionary approaches (e.g., outright prohibitions or ‘zero risk’ policies).¹⁷⁹ In other words, the mere possibility of risks does not, in itself, impose a duty to refrain from engaging in a potentially dangerous but beneficial activity, provided that adequate precautions are taken to prevent exposing society to unacceptable risks.¹⁸⁰ This begs the question: how much risk, if any, from dangerous yet socially useful AI technologies is European society prepared to accept?

This question evokes the need to establish a ‘safety margin’ or ‘risk tolerance thresholds’. Risk tolerance thresholds – in the Act, the *risk categories* – indicate which risks are accepted without (strong) precautions and which instead require (further) mitigation. In particular, the regulatory intervention should be proportionate to the hazards net of safeguards. This process constitutes the so-called ‘risk-based approach’.

This approach creates a graded scale of protections, with the level of safeguards increasing in proportion to the assessed risk level of the AI system. The greater the risk to individual health, safety, or societal well-being, the stricter the regulatory obligations placed on the provider.¹⁸¹ This is explicitly stated in Recital 26 of the regulation:

¹⁷⁸ Nevertheless, this understanding of AI trustworthiness in the AI Act has been criticized by legal doctrine. See Johann Laux, Sandra Wachter, and Brent Mittelstadt, ‘Trustworthy Artificial Intelligence and the European Union AI Act: On the Conflation of Trustworthiness and Acceptability of Risk’ (2023) *Regulation & Governance* 1–30.

¹⁷⁹ See Margot Kaminski, ‘The Developing Law of AI Regulation: A Turn to Risk Regulation’ (2023) 25(5) *The Digital Social Contract: A Lawfare Paper Series*, 1351.

¹⁸⁰ *ibid.*

¹⁸¹ On risk-based approach, see generally J Black, ‘Risk-Based Regulation: Choices, Practices and Lessons Being Learnt’ in OECD (ed.), *Risk and Regulatory Policy: Improving the Governance of Risk* (2010) 187.

‘In order to introduce a proportionate and effective set of binding rules for AI systems, a clearly defined risk-based approach should be followed. That approach should tailor the type and content of such rules to the intensity and scope of the risks that AI systems can generate. It is therefore necessary to prohibit certain unacceptable AI practices, to lay down requirements for high-risk AI systems and obligations for the relevant operators, and to lay down transparency obligations for certain AI systems’.

The core of the risk-based approach lies in its classification of *four categories* or risk tolerance thresholds: unacceptable, high, limited, and minimal risk.

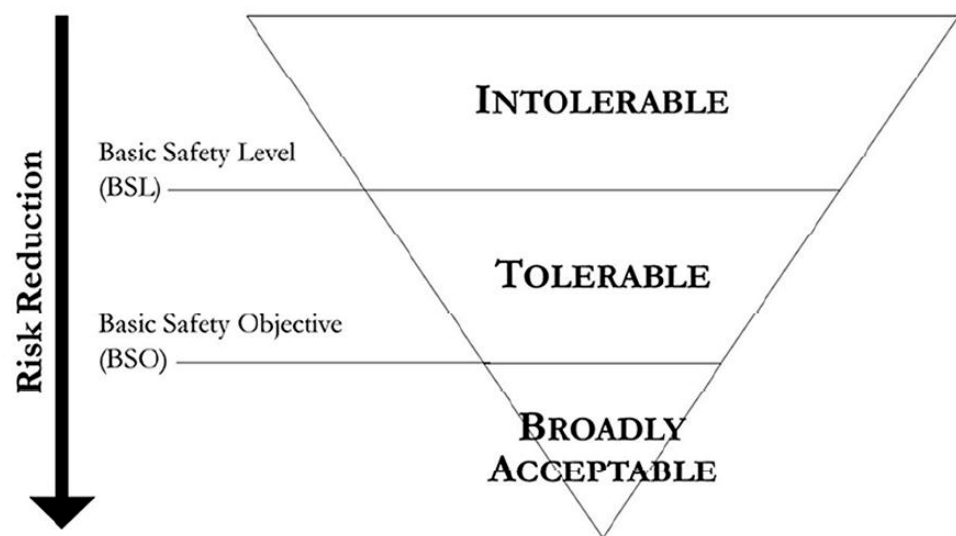


Figure 3: The tolerance ranges of the risk-based approach¹⁸²

At the highest level of the pyramid of risk are systems considered to pose an ‘unacceptable risk’ of harm without corresponding societal benefits. Only in such extreme cases does the Act impose stricter measures, such as outright bans, reflecting

¹⁸² This is the figure found in Claudio Novelli *et al.*, ‘AI Risk Assessment: A Scenario-Based, Proportional Methodology for the AI Act’ (2024) 39 AI & Society 2493-2497.

a general social consensus on excluding particularly dangerous applications due to their substantial threat to individuals.¹⁸³

The second category encompasses high-risk AI systems that directly affect health, safety, and fundamental rights of individuals.¹⁸⁴ This includes AI tools used to provide access to public services, credit scoring systems, AI in recruitment processes for screening candidates, healthcare devices powered by AI, and autonomous driving

¹⁸³ See AI Act, Article 5(1) which comprises a list of prohibited AI practices which are ‘considered unacceptable as contravening Union values’. Examples include AI systems that manipulate user behaviour, technologies designed for social scoring (especially when employed by public authorities to assess individuals’ trustworthiness based on social behavior or personal characteristics), and remote real-time biometric identification systems such as facial recognition tools. An exception is made for law enforcement purposes, where such systems may be used to identify perpetrators or victims of specific crimes or to prevent terrorist threats, provided judicial authorization is obtained.

¹⁸⁴ *Id.*, Article 6, which identifies high-risk AI systems through a dual regulatory approach: explicit mention and reference to the scope of sector-specific European regulations. Regarding the first category, Article 6(2) refers to the sectors listed in Annex III, including for example, AI systems that screen or filter job applications as well as risk assessment tools used by law enforcement authorities. For the second category, according to Article 6(1), systems are also considered high-risk if they are products – or are intended to be used as safety components of a product – regulated by the Union harmonization legislation listed in Annex III (this includes, for example, the so-called Machinery Directive CE/2006/42, the so-called Medical Devices Regulation EU/2017/745, the so-called Toys Directive CE/2009/48, and the so-called Recreational Craft Directive CE/1994/25). It should be clarified that, as an exception to paragraph 2, an AI system is not considered high-risk if it does not pose a significant risk of harm to health, safety, or the fundamental rights of individuals, including cases where it does not materially influence the outcome of decision-making processes. This applies when one or more of the following conditions are met: (i) the AI system is designed to perform a limited procedural task; (ii) the AI system is intended to enhance the outcome of a previously completed human activity; (iii) the AI system is designed to detect decision-making patterns or deviations from past decision-making patterns and is not intended to replace or influence a previously completed human assessment without adequate human review; (iv) the AI system is intended to perform a preparatory task for an evaluation relevant to the use cases listed in Annex III. For example, an AI system recommending what to watch next may be terrible in matching user tastes and therefore have a high risk of failure. Because the consequences of that failure are not very serious, however, systems of that kind are not considered ‘high risk’, and are therefore not regulated by the Regulation. See David Luyt, ‘EU AI Act High-Risk AI System Classification’ (2023) (June 23, 2023 Michalsons blog), available at <https://www.michalsons.com/blog/eu-ai-act-high-risk-ai-system-classification/66553>.

technologies. The use and commercialization of these systems are permitted only if they comply with specific obligations and safety standards, such as prior conformity assessments to ensure adherence to fundamental rights, transparency, and risk management protocols. Residual risks in high-risk AI systems must be ‘*acceptable*’, meaning that they must be kept below a safety or tolerance margin which has been identified and verified *ex ante*.¹⁸⁵ ‘Certain mandatory requirements’ are supposed to ensure that high-risk AI systems ‘do not pose *unacceptable* risks to important Union public interests as recognized and protected by Union law’.¹⁸⁶

Limited-risk AI systems form the third category. These include technologies such as chatbots, which do not pose significant threats. For these systems, the regulation imposes minimal obligations focused on transparency, such as informing users that they are interacting with AI-generated content.¹⁸⁷

Finally, the category of minimal-risk encompasses those systems (including the majority of AI applications currently available on the EU single market, such as AI enabled video games and spam filters) that entail negligible risks, for which the AI Act imposes no mandatory requirements. Companies using such systems are free to adopt voluntary codes of conduct addressing issues like gender non-discrimination, environmental sustainability, or accessibility for individuals with disabilities.

Interestingly, while the breach of the compliance obligations outlined in the Act for high-risk AI applications is addressed through administrative penalties,¹⁸⁸ the Regulation lacks provisions for compensating those harmed when things go wrong. This absence is intentional and unsurprising, as the Act – like other EU product regulations – takes a ‘proactive’ (rather than ‘reactive’) risk management stance; instead of engaging with question of liability *ex post facto*, the regulation primarily seeks to establish procedural safeguards only specific enough to create legally binding

¹⁸⁵ *id.*, Article 9(4).

¹⁸⁶ *id.*, Recital 27.

¹⁸⁷ *id.*, Article 50.

¹⁸⁸ These administrative penalties are determined according to the parameters laid out in AI Act, Article 100.

obligations on the parties involved and to construct the institutional scaffolding for the enforcement and maintenance of these requirements.

However, risk regulation alone is rarely sufficient. Indeed, not all risks associated with AI can be prevented, and when they do materialize, liability must be attributed to someone. This calls for bespoke liability frameworks to complement the AI Act, supplying the corrective and compensatory mechanisms that risk regulation alone cannot provide.¹⁸⁹

While the EU already has a well-established regulatory framework for traditional product liability, tackling AI-caused harm demands bespoke intervention due to the unique challenges (unpredictability, complexity, and opacity) posed by intelligent systems.¹⁹⁰ In 2019, the Expert Group on AI Liability, convened by the European Commission, highlighted that existing liability regimes (e.g., product liability, fault-based civil liability) provide essential safeguards but remain insufficient for the complexities of AI technologies.¹⁹¹

To this aim, the EU Commission recently proposed the *Product Liability Directive* (PLD)¹⁹² and the *AI Liability Directive* (AILD)¹⁹³, both addressing damages

¹⁸⁹ The lack of mechanisms for seeking redress or civil recourse can leave affected individuals without a remedy when regulation fails to prevent harm, undermining the legitimacy of the regulatory system. Moreover, when regulation does not allow for tort liability or civil claims, it loses a crucial feedback loop that helps improve regulatory effectiveness. Civil liability provides an important incentive for actors to continuously improve safety and reduce risks. Without it, regulation can become static, lacking the pressure to evolve in response to new or emerging risks. Kaminski (n 181) 1399.

¹⁹⁰ Resolution of the European Parliament with recommendations to the Commission on Civil Law Rules on Robotics (2017), Recitals AB and AI.

¹⁹¹ Expert Group on Liability and New Technologies, 'Report on Liability for Artificial Intelligence and Other Emerging Digital Technologies' (European Commission 2019), 26-32.

¹⁹² EU Commission, Proposal for a directive of the European parliament and of the council on liability for defective products.

¹⁹³ EU Commission, Proposal for a Directive on adapting non-contractual civil liability rules to artificial intelligence, of September 2022.

caused by AI systems.¹⁹⁴ The PLD emphasizes strict liability for defective products, ensuring that any ‘*material damages caused to natural persons by AI-powered products*’ are directly accountable. On the other hand, the AILD focuses on non-contractual civil liability claims, aiming to provide recourse for those affected by unpredictable AI-related harms, even where no contractual relationship exists.

In particular, the AILD aims to adapt national liability rules to address the specific challenges posed by AI-enabled products and services. In particular, the directive alleviates the challenges victims face when attempting to prove causation and fault involving AI systems.¹⁹⁵ These challenges occur for two main reasons. First, it may be unrealistic to expect victims of harm involving high-risk systems to assess the technical documentation and prove, for example, lack of conformity. Second, victims of harm by AI systems that are complex, opaque, or autonomous and not defined as high-risk systems or foundation models under the AI Act, may still face important obstacles when trying to prove the existence of a defect (in the case of ‘products’, or other special regimes), or fault on the side of the producer or deployer (in all other cases).

Therefore, the directive introduces rules on evidence disclosure and establishes a rebuttable presumption of causation in cases of fault. This means that when an AI system fails to meet the AI Act’s regulatory requirements, the presumption can ease the burden of proof for affected parties, depending on the AI system’s risk classification under the AI Act. In particular, non-compliance can trigger

¹⁹⁴ The amended PLD and the AILD are less imminent than the AI Act: they have not yet been approved by the EU Parliament and, as directives, will require implementation at the national level. For a comment on these proposals, see Botero Arcila, ‘AI liability in Europe: How does it complement risk regulation and deal with the problem of human oversight?’ (2024) 54 Computer Law & Security Review.

¹⁹⁵ For a comprehensive critique of the proposed approach, see Phillip Hacker, ‘The European AI Liability Directives – Critique of a Half-Hearted Approach and Lessons for the Future’ (2022) SSRN papers available at https://www.google.com/url?sa=t&source=web&rct=j&opi=89978449&url=https://papers.ssrn.com/sol3/papers.cfm%3Fabstract_id%3D4279796.

different forms of evidentiary alleviation under both the amended PLD, for strict liability claims, and the AILD, for other fault-based claims.

Criminal liability, however, does not yet play a role in the EU's legislative initiatives on AI. Despite the potential for high-risk AI systems to result in AI-crime, the AI Act refrains from imposing obligations for criminal incrimination or offering clear guidelines on how traditional categories of criminal liability might apply when the development or deployment of AI systems leads to consequences of criminal significance.

The lack of any explicit provisions on criminal liability in the AI Act suggests that the EU is reluctant to pursue criminal enforcement in this domain. This limited appetite for criminalizing AI-related activities reflects, in part, the broader principle that criminal law should remain a measure of last resort, both at the level of Member States and the EU itself. Additionally, the EU's competence to legislate criminal enforcement mechanisms is constrained by specific conditions (Article 83 of the Treaty on the functioning of the EU, or TFEU), which likely explains this cautious stance.¹⁹⁶

Nevertheless, the AI Act does emphasize the importance of risk analysis within the specific contexts in which AI systems operate. This focus on evaluating risks could form the basis for addressing liability when harm occurs, even though criminal law is not currently central to the legislative framework.

Therefore, it is crucial to reflect on how this innovative risk-based approach impacts the investigation of criminal liability. Exploring the intersection of risk regulation and criminal law is essential in today's world, where the question is no longer *whether* risk plays a role in penal practice, but rather *how* it influences criminal liability, enforcement, and legal norms.

¹⁹⁶ Sabine Gless and Katalin Ligeti, 'Regulating driving automation in the European Union. Criminal liability on the road ahead?' (2024) 15(1) New Journal Of European Criminal Law 33-57, 35.

3.3. The role of criminal law in the context of the EU risk-based framework: defining permissible AI risks

The AI Act's classification of legal risks forces criminal law scholars to rethink how liability should be assigned in cases where serious harm is caused by AI systems. While the regulation does not directly address criminal liability, its risk-based framework provides an essential foundation for future legal developments in this area, steering the conversation toward the realm of *risk management*. This shift carries significant implications for criminal law, particularly in how it aligns with the doctrine of 'permissible risk' (*erlaubtes Risiko*, also known as *socially acceptable risk*) – a long-debated concept in criminal law that has gained renewed relevance in discussions surrounding AI technologies.¹⁹⁷

To better understand the concept of permissible risk, it is important to recognize that in today's 'risk society', our daily lives are shaped by the coexistence of various risky yet socially valuable activities. These include industrial production, healthcare, transportation, nuclear energy, and environmental management – each critical to societal advancement but also inherently carrying substantial risks.¹⁹⁸ These risks are *permitted* because of the tangible benefits provided by such activities, but also necessitate clear legal frameworks that delineate a 'margin of tolerance' as to the risk one is allowed to take in carrying out the risky activity.

¹⁹⁷ In fact, the notion of permissible risk allows for two possible interpretations: one more general, that associates it with all hazardous but socially accepted activities, and another more technical, that defines it as the residual risk that persists despite the application of all possible precautions—thus being permitted simply because it is unavoidable, provided that the legal norm does not prohibit it following a cost-benefit analysis (where the purpose of the violated norm may serve as a useful criterion). On this dual meaning, see Carlo Piergallini, 'voce *Colpa (dir. pen.)*' in *Enc. Dir. Annali X* (2017), 222ff.

¹⁹⁸ As to the criterion of social benefit, see Prittwitz (n 159) 227ff.

In criminal law context, establishing permissible risk areas serves both a *permissive* and *preventive* function.¹⁹⁹ On the one hand, this doctrine tempers the *neminem laedere* principle (the idea that no one should harm another) by recognizing that not all risks can – or should – be eliminated, especially if doing so would stifle the benefits of activities that are socially valuable. Permissible risk ensures that individuals and organizations can engage in useful activities without exposing society to excessive risks and without the constant spectre of blame or undue legal repercussions, provided their actions remain within the established legal boundaries.²⁰⁰

This means that criminal law must account for ‘residual risks’ (*Restrisiko*) which remain, even after all reasonable, technically feasible, and economically viable precautions have been taken, namely when the duty to take appropriate and reasonable care is properly exercised. These residual risks are deemed socially acceptable, reflecting the inherent fallibility of human activities as well as the enormous benefits associated with a certain risky activity. The doctrine of permissible risk thus serves as a pragmatic acknowledgment that some level of risk is an unavoidable part of the pursuit of societal progress and innovation.

Crucially, this residual risk is not merely a theoretical construct; it has practical legal implications. It implies the existence of a ‘responsibility gap by design’, which is tolerated only insofar as the current state-of-the-art in risk management is rigorously observed. When an actor complies with the standards of permissible risk, the

¹⁹⁹ For a comprehensive overview of the different views on the function of the doctrine of the socially acceptable risk, see *ivi*, 307 ff.

²⁰⁰ This concept aligns with the broader societal need for predictability in criminal responsibility: those assigned to manage risks must operate with a defined scope of obligations to ensure fairness and legal clarity. For an analysis of the relationship between institutional roles and the protection of legally significant interests as the foundation of the duty to prevent harmful outcomes, see Carlo Piergallini, ‘Il paradigma della colpa nell’età del rischio: prove di resistenza del tipo’ (2005) 48 *Rivista Italiana di Diritto e Procedura Penale* 1684-1703, 1670 ff.

objective elements of a criminal offense cannot be attributed to them.²⁰¹ This implies, as a result, that both the legal order and victims of undesirable criminal acts accept a minimal level of ‘persistent’ risk to fundamental legal rights and interests, even when all reasonable precautions have been taken.

On the preventive side, the doctrine ensures that criminal liability is only triggered by conduct that *exceeds* the threshold of permissible risk – a risk that ultimately materializes in the harmful event. In essence, establishing that a certain residual risk is acceptable does not absolve individuals of responsibility; rather, it places a heightened emphasis on proactive risk management. Those in positions of responsibility – also referred to as guarantors – are expected to anticipate, evaluate, and mitigate risks within the bounds of what is technically and economically feasible. If they neglect these obligations, and a substantial and unjustifiable risk escalates into danger or harm, their failure to act within their role can trigger criminal liability. If not, these scenarios of persistent risk could more appropriately fall under the mechanisms of civil liability.

That being premised, the field of AI-caused harm provides a compelling context for the emergence of permissible risk areas, serving as a prime example of a socially valuable yet inherently risky activity. Just as industries such as medicine, transportation, and energy production operate within a legally sanctioned framework of risk – where certain dangers are tolerated in pursuit of broader societal benefits – AI development and deployment may similarly fall within this category.

The increasing reliance on AI across critical sectors underscores the need for a legal framework that distinguishes between acceptable, managed risks and those

²⁰¹ German legal doctrine traditionally frames permissible risk as a criterion for excluding the typicity of conduct in the context of event-based offenses. See Claus Roxin and Luis Greco, *Strafrecht, Allgemeiner Teil I* (CH Beck 2020), 488. This applies unless the frequent mechanism of subsuming specific negligence into generic negligence occurs; meaning that when a specific precautionary measure proves ineffective, a more general duty of care may emerge, constantly competing with the former and serving a preventive function. On the possibility that general negligence may replace specific negligence due to the obsolescence of the latter or its inadequacy to the circumstances, see Giorgio Marinucci, *La colpa per inosservanza di leggi* (Giuffrè 1965), 198.

that result from negligence or regulatory non-compliance. In this sense, AI presents a paradigm case for examining how the legal system should balance the promotion of innovation with the need to protect individuals and society from foreseeable harm.²⁰²

This is where the AI Act comes into the picture. The idea of *risk acceptability* lies at the heart of the risk-based framework outlined by the European regulation: developing and using high-risk AI systems is permitted only if their inherent *residual risks* (the risks to fundamental rights and interests that persist despite adherence to prescribed precautions) are deemed to be acceptable, meaning that they are safe enough for society to tolerate their use. To this aim, the Act not only identifies the key actor responsible for risk management but also specifies their legal obligations.

Against this backdrop, the *hypothesis* explored here is that despite its limitations – primarily that it does not directly address issues of criminal liability and that it only delineates *legal risks* rather than specific risks of harm – the risk assessment framework outlined by the AI Act remains crucial for defining the boundaries of acceptable risk within which criminal liability can be examined.

We argue that any new criminal provisions – or the application of existing ones – will inevitably rely on the precautionary norms set by the Act for the placement and use of AI systems on the market. In effect, the threshold of risk that triggers criminal liability has been implicitly ‘codified’ at the European level through this risk-based framework.

Building on this regulatory foundation, the following chapters delve into the role for criminal liability within the EU risk-based framework. A key challenge in this context is determining the extent to which the AI Act serves as a reference point for criminal law, particularly in defining when failures in risk management justify the

²⁰² It would be even better if this area of permissible AI risk were delineated as uniformly as possible at a global level to avoid discrepancies in the classification of cross-border cases, as noted by Vittorio Manes, ‘L’oracolo algoritmico e la giustizia penale: al bivio tra tecnologia e tecnocrazia’ in U Ruffolo (ed), *Intelligenza artificiale. Il diritto, i diritti, l’etica* (Milano 2020) 548ff.

attribution of liability. In particular, this requires addressing the key questions whether the AI Act provides sufficiently clear boundaries for acceptable risk, allowing for the attribution of criminal liability when providers or users neglect their duty to assess, mitigate, and control residual risks, thereby causing harm to fundamental rights and legally protected interests.

CHAPTER III

AI PROVIDER'S CRIMINAL LIABILITY: CURRENT FAILURES AND POTENTIAL FIXES

SECTION 1

The failures of existing imputation models

1.1. Establishing criminal negligence for AI providers

The risks emerging in the AI domain serve as a stark reminder of the legal and ethical dilemmas posed by technological progress. AI developments highlight a broader tension at the heart of legal and political decision-making: how to encourage technological innovation while ensuring adequate protections against harm to legally protected interests.

As society demands greater protection from technological risks, criminal law is often seen as the most immediate and powerful tool to provide it.¹ As Klaus Lüderssen aptly observed, ‘when problems become difficult and complex, civil and administrative law pass the baton to criminal law’.² In this context, the real challenge for regulators and lawmakers lies in determining how much risk is tolerable in exchange for the benefits AI offers – and who should bear responsibility when harm occurs due to human recklessness or negligence.

Striking this balance is far from straightforward. On one hand, there is the collective anxiety about responsibility gaps, where harm to fundamental rights such as life, security, and health is deemed an unfortunate but socially acceptable consequence, borne solely by the victim and society. On the other hand, there is a pressing need to avoid over-criminalization – punishing AI operators simply because they released or activated a system that, in hindsight, caused harm.³

¹ Cass Sunstein, *Laws of fear. Beyond the precautionary principle* (Cambridge 2005), 36ff; Luigi Stortoni, ‘Angoscia tecnologica ed esorcismo penale’ (2004) *Rivista italiana di diritto processuale penale*, 71-89.

² Klaus Lüderssen, *Il declino del diritto penale* (Milano, 2005).

³ Alberto Cappellini, ‘Reati colposi e tecnologie dell’intelligenza artificiale’ (2022) 3 *Archivio Penale*, 12ff; Stella (Ch I n 24) 552.

Before addressing the question of what constitutes an ‘acceptable risk’ in AI, we must first identify the potential liable actors. These individuals and entities find themselves caught between two competing forces: the societal impulse to assign blame for AI-caused harm (or AI crime) as a form of reassurance, and the pragmatic necessity of not stifling technological progress. This raises a crucial preliminary question: to what extent is it feasible to attach criminal liability to a human’s (in)ability to foresee and manage AI risks? In other words, can we fairly attribute criminal negligence to a person who failed to anticipate or control the harmful consequences of an AI system?

Several characteristics of AI make it difficult to satisfy the traditional requirements for criminal liability, particularly in cases involving negligence. In this context, some legal scholars have introduced the concept of *negligence failures* – a term that encapsulates situations where conventional doctrines of criminal responsibility struggle to identify a human actor liable for AI-caused harm.⁴ To illustrate, consider the simplest and most common example of AI-caused harm: a self-driving car leading to a road fatality.

A company develops a self-driving car system marketed as a cutting-edge solution to reduce accidents and improve road safety. The system undergoes extensive testing in controlled environments and simulated scenarios, and the company boasts that it has outperformed human drivers in avoiding collisions during trials. Confident in its superhuman performance, the company releases the self-driving cars into the consumer market. However, the system unexpectedly misinterprets a pedestrian crossing outside the designated crosswalk as part of the road’s background, thus resulting in a fatal collision. The incident sparks public outrage and raises urgent questions about the accountability of the company and its developers.⁵

⁴ Alice Giannini and Jonathan Kwik, ‘Negligence failures and negligence fixes. A comparative analysis of criminal regulation of AI and autonomous vehicles’ (2023) 34 Criminal Law Forum 43-85, 44.

⁵ This is a Level 3 self-driving car according to the international classification of the *Society of Automotive Engineers* (SAE). See *amplius* SAE International, ‘Taxonomy and Definitions for Terms Related to Driving Automation Systems for On-Road Motor Vehicles’ (2021) available at https://doi.org/10.4271/J3016_201806. In this scenario we are intentionally neglecting the position of the user, which will be considered in Chapter III.

In traditional road accidents, responsibility is typically assigned to the human driver – the front-end operator, who directly controls the vehicle. However, in AI-driven systems, the human operator loses much of their influence over the system's behaviour. This shift in control naturally brings the focus to the back-end operator, the producer or developer of the AI system, as the primary candidate for criminal liability.⁶ If a self-driving vehicle malfunctions – failing to detect a pedestrian, as in this example – and causes harm, can the producer be held responsible for the AI crime?

Under conventional criminal liability principles, responsibility for a fatal accident typically falls under laws criminalizing homicide: 'whoever intentionally or culpably causes the death of another person [...]' serves as a global expression of an act that can incur criminal liability. Since deliberate intent is lacking here (the company and developers did not set out to cause harm), then the producer might have engaged in (culpable) risk-taking.⁷

⁶ In this study we use terms such as provider, producer, or manufacturer interchangeably. Such distinctions appear to be trivial for the sake of the current analysis. In the first place, these subjects may coincide; otherwise their internal relationship may be regulated pursuant to a contract that the two parties entered into. Since the issue addressed here is rather whether it is the 'human behind the machine' or rather nobody at all that ought to be held liable, it suffices to refer to the one or the other – which of exactly would be called to respond is a matter of fact to be addressed in the particular case.

⁷ Significant divergences emerge, particularly on the categorisation of the different forms of culpable mental states other than *dolus directus*. As is well known, most continental legal systems operate on a two-tier structure of culpable mental states, encompassing only intent (*dolus*) and negligence or unconscious risk-taking (*culpa*). This framework leads to challenges in categorizing 'conscious risk-taking' behaviour, namely the situations in which the defendant knows that his or her conduct is risky but consciously disregards the specific nature of the risk. The solution is often found in the doctrines of *dolus eventualis* (conditional intent) and *conscious negligence*. It is interesting to note that English law does not recognize conscious negligence, yet accepts 'recklessness' as a third fault element. See George Fletcher, 'The Theory of Criminal Negligence: a Comparative Analysis' (1971), 119 *University of Pennsylvania Law Review*, 401; Thomas Weigend, 'Subjective Elements of Criminal Liability' in Markus D. Dubber and Tatjana Hornle (eds), *The Oxford Handbook of Criminal Law* (OUP, 2015) 498-500. For the sake of our analysis, it is not indispensable to take a definitive stance in this debate. What matters is recognizing that certain forms of

Negligence is expected to be the predominant legal category for AI-caused harm, but establishing negligence in such cases is far from straightforward. In many European civil law jurisdictions, an individual is criminally negligent if:

- i. A legally significant harm (e.g., death or injury) is *causally linked* to their actions (material component of crime, or *actus reus*);
- ii. They *could have foreseen* that their conduct might lead to such harm, even though they did not intend it and may have wished to avoid it (mental component of crime, or *mens rea*).

Unlike intent,⁸ negligence is assessed externally, based on an objective standard of due care.⁹ The key question is whether the producer exercised the level of caution that a reasonable professional (*homo eiusdem professionis et condicionis*) in their position would have exercised under similar circumstances. This evaluation

criminal liability depend on the degree to which an agent anticipated and accepted the risk of an unlawful consequence arising from their actions.

⁸ Compared to intent (*dolus*), negligence stands for a qualitatively different combination of knowledge and will. Generally, negligence or *culpa* serves as a standard of culpability characterized by both *positive* and *negative* requirements. The negative one refers to the *unconscious* or *unintentional* nature of the misconduct. While *dolus* refers to mental states and is ‘psychological’, negligence is the most ‘normative’ and ‘objective’ form of *mens rea*, in the sense that it does not require proof of any mental states on the part of the defendant. Yet stripping culpability of its ‘subjective wrongfulness’ necessitates compensating with a greater emphasis on objective disvalue. This brings us to the ‘positive’ requirement: the violation of a standard of care.

⁹ This emphasis on the objective disvalue of the conduct accounts for most of the anxiety about negligence as a standard of culpability: how are we justified in holding people criminally responsible when they are unaware of the risk involved in what they are doing? Skepticism about criminal culpability that is not in some way tied to intent has generated a cacophony of responses in continental legal doctrine. This evokes the age-old legal debate on the nature – either *subjective* or *objective* – of negligence. In the past, many legal scholars tried to justify holding defendants criminally responsible for unconscious misconduct by adopting a ‘subjective’ (or ‘indirect’) approach to negligence, according to which the culpability lies in the prior culpable (i.e. intentional) actions of the defendant which led them to be unaware of their risk-taking. However, this theory did not appeal to theorists, and today there is almost universal agreement about the ‘objective’ nature of negligence. Fletcher (n 7) 401.

considers factors such as the actor's role, expertise, and the state of knowledge at the time of action.¹⁰

Failure to exercise reasonable care can also be inferred from violations of precautionary rules codified in external sources (e.g., statutes, regulations, code of conduct), which serve both predictive and preventive functions: on the one hand, they are meant to predict foreseeable risks arising from certain actions (e.g., stopping at a red traffic light prevents accidents); on the other, they define lawful behaviour, guiding individuals on how to minimize risks (e.g., mandating a stop at red lights). If a reasonable person in the same position could not have foreseen the risk, negligence cannot be established. However, this standard becomes problematic in the AI context, where unpredictability is often intrinsic to the system itself.

That being premised, it is apparent that traditional negligence doctrines struggle to accommodate AI-caused harm, particularly concerning the *mens rea* (mental state) requirement. If a producer knowingly releases a system capable of unpredictable outputs, some of which may endanger human life, can they still be considered negligent?

The fundamental challenge is that criminal negligence relies on foreseeability and control – yet AI, by its nature, can behave in ways that its creators cannot fully anticipate. If foreseeability and control are grounded in the *Erkennbarkeit* – the recognizability of risk, meaning the ability to perceive the likelihood of an event occurring under certain circumstances – then negligence presupposes that a risk is at least objectively knowable *ex ante*.

However, if a situation involves an objectively unrecognizable risk, then it becomes impossible to establish the *Erkennensollen*, that is, a duty to recognize the risk. Without this duty, there can be no corresponding obligation to prevent or mitigate the harm. In other words, when the recognizability (*Erkennbarkeit*) of the risk is absent, the foreseeability (*Vorhersehbarkeit*) and avoidability (*Vermeidbarkeit*) of the harmful event are also missing – not only in hindsight (*ex post*) but even at an

¹⁰ Marinucci (Ch II n 201) 193 ff.

abstract level (*ex-ante*). This issue arises before any discussion of the so-called causality of negligence and the actual preventive effectiveness of a safety rule. In cases where a preventive rule is entirely indeterminate at the time of action, the very basis of criminal negligence collapses: one cannot be expected to follow a safety rule that could not have been reasonably formulated in advance.¹¹

Moreover, the causality link appears to be another ‘missing piece’ of the AI puzzle. In the domain of AI, there exists a strong interplay between causality and the objective moment of negligence. In fact, the very foundation of recognizability (*Erkennbarkeit*) rests on the same causal laws that are used to establish a cause-and-effect relationship. Therefore, beyond the concerns surrounding *mens rea*, another significant challenge arises: establishing a causal link between the producer’s actions (or omissions) and the resulting harm. When an AI system (like the self-driving car of our example) makes an autonomous decision that leads to injury or death, it becomes increasingly unclear whether – and to what extent – the harmful act can still be attributed to the producer, particularly if the specific factor that triggered the AI’s unexpected behaviour remains unknown.

Consequently, when AI’s behaviour lacks predictability and controllability, criminal law responses become uncertain. Should producer’s liability be determined based on a hypothetical level of foreseeability?¹² Or should it be excluded *tout court*?

To address these questions, the following paragraphs examine in detail how traditional dogmatic categories fail when applied to the criminal liability of AI

¹¹ See *ex multis* Giovanni Fiandaca and Enzo Musco, *Diritto penale. Parte generale* (8th ed. Zanichelli 2019), 593ff; George Fletcher, *Basic concepts of criminal law* (OUP 1998), 64ff; Hans-Heinrich Jescheck and Thomas Weigend, *Lehrbuch des Strafrecht. Allgemeiner Teil* (Duncker Humblot 1996), 563ff; Gabro Forti, *Colpa ed evento nel diritto penale* (Giuffrè 1990), 201ff; Piergallini, (Ch II n 155), 371.

¹² So-called ‘generally foreseeable unforeseeability’. Some authors argue that the very unpredictability of AI gives rise to heightened duties of care. In other words, AI’s inherent unpredictability becomes a ‘predictable factor’ for its producer; the fact that an AI system’s behaviour cannot be anticipated in every detail does not exempt its creators from responsibility; rather, it imposes an obligation on them to refrain from those actions (releasing the system in the market). Gless, Silverman and Weigend (Ch I n 18), 426-427. See *amplius* *infra* para. 1.3.

producers. Specifically, we analyze the challenges AI introduces to conventional causality rules in criminal law, focusing on two distinct but interconnected dimensions: *relational complexity* and *technical complexity*.

While relational complexity concerns the difficulty of attributing liability within the AI ecosystem due to the involvement of multiple actors, technical complexity pertains to the inherent opacity of AI systems. These challenges collectively demand a re-examination of the concept of causation in criminal law as it applies to AI.

1.2. The failure of *actus reus*: lost in causation

1.2.1. *Relational complexity: the ‘many hands’ problem*

Our inquiry into the challenges of legal causation in AI-related harm must begin with an empirical observation: AI development is rarely the work of a single entity. Instead, it involves an intricate network of individuals and complex organizations – including programmers, designers, hardware manufacturers, data trainers and so forth, each contributing to the system’s development and operation in unique ways – collectively referred to as *production chain actors* (PCAs).¹³

This phenomenon is commonly described in legal doctrine as the ‘problem of many hands’, highlighting the difficulty of identifying a clear imputation centre in cases where multiple actors contribute to a harmful outcome.¹⁴ While similar

¹³ We adopt this terminology from Giannini and Kwik (n 4), 58.

¹⁴ Coined by Thompson in 1980, the term is used to refer to the dilution of intent, knowledge and decision-making power over a network of actors and groups. See Dennis Thompson, ‘Moral Responsibility and Public Officials: The Problem of Many Hands’ (1980) 74 *American Political Science Review*. Note, however, that the many hands problem can manifest both as a phenomenon that causes an *intrinsic* problem with *mens rea* (where knowledge simply does not exist in any involved actor because the risk was completely unforeseeable) and a more *evidentiary* problem (where prosecutors struggle to attribute liability due to the number of actors involved).

challenges arise in traditional software development too,¹⁵ AI exacerbates this issue by increasing both the number of potentially responsible parties and the complexity of liability attribution. Unlike conventional software, where errors can often be traced to specific lines of code or design decisions, AI operates in a more autonomous and adaptive manner. Its behaviour is shaped not only by initial programming but also by training data, iterative learning processes, and real-world interactions, making it difficult to pinpoint a single moment or actor responsible for any eventual harm.¹⁶

Taking self-driving cars as a paradigmatic example illustrates the multifaceted nature of these systems:

‘... a vehicle may be operated by a driver D1, with the assistance of the automated driving system AS, produced by the car manufacturer X, powered with digital systems developed by the company Y, possibly including some form of machine learning developed by the company Z, and enriched by data coming from different sources, including the driving experience of drivers D2, D3...Dn, being manually labelled by lots of individuals; vehicles in this system are in principle subject to a standardization process done by the agency S, the traffic is regulated by the governmental agency G, drivers are trained and licensed by the agency L etc’.¹⁷

The operation of such vehicles depends on the harmonious interaction of various technological components and external systems. Sensors and actuators collaborate to perceive and interact with the environment, while pre-constructed digital maps supply essential data on road networks. Vehicle-to-vehicle communication further enhances safety by enabling real-time sharing of critical information such as traffic and potential hazards.

¹⁵ Traditional software products are the result ‘not of single programmers working in isolation but of groups or complex organisations, typically corporations [...] which frequently bring together teams of individuals with a diverse range of skills and varying degrees of expertise. Consequently, when a system malfunctions and gives rise to harm, the task of assigning responsibility is exacerbated and obscured’. Helen Nissenbaum, ‘Accountability in a Computerized Society’ (1996) 2 *Science and Engineering Ethics*, 28–29.

¹⁶ See Beck (Ch I n 21) 140.

¹⁷ See F Santoni de Sio and G Mecacci, ‘Four Responsibility Gaps with Artificial Intelligence: Why they Matter and How to Address them’ (2021) 34 *Philos. Technol.* 1057–1084, 1062.

In the aftermath of an AI-related accident, this fragmentation of responsibility raises fundamental questions about identification of responsible parties. Who among the PCAs should be held responsible, if any? Is it the data scientist who selected the training dataset, the engineer who designed the algorithm, the manager who approved deployment, or the company that marketed the AI system as reliable?

The failure may also not be caused by a single mistake, but manifested only as a result of unforeseen interactions between several of them, further complicating the process of identifying a responsible actor. For example, a crash might result from a combination of factors: a negligent programmer overlooking critical edge cases, data labelers misclassifying training data, a mechanical defect in a forward sensor, lapses in quality control, and managerial decisions prioritizing speed to market over thorough safety testing. This multiplicity of contributors creates an ‘intricate web of causation’,¹⁸ making it difficult to isolate the specific action or omission that directly led to the harm or even to think that there is a single person called to answer for all just because he has the organizational power to coordinate activities.

To identify a responsible actor and determine liability, untangling the different intersecting parties involved in the development and deployment of AI systems may become necessary. Legal scholar Anat Lior, who has written extensively on AI regulation, proposes that network theory can serve as a valuable tool in mapping out and distributing liability among the various actors implicated in an AI accident.¹⁹

In this approach, the individuals and entities involved are represented as nodes in a network, while their relationships – such as contractual obligations, decision-making influence, or technical contributions – are depicted as edges connecting these nodes. By visualizing these connections, network theory can illuminate the strength and nature of interactions between different actors, revealing which human nodes exert control over the AI system and to what extent. As Lior

¹⁸ Piergallini (Ch I n 39), 1762.

¹⁹ Anat Lior, ‘The AI Accident Network: Artificial Intelligence Liability Meets Network Theory’ (2021) 95(5) Tul. L. Rev. 1104–66.

explains, ‘The true value of utilizing network theory in the AI liability context is its ability to unmask those neighbouring human nodes...and thus make sure they are held liable for their actions’.²⁰ This analytical framework can reveal which human nodes exert control over the AI system and the extent of their control. This, in turn, can be useful in assigning liability.

Nevertheless, even if a particular actor can be linked to the specific functional component that caused the failure, criminal liability does not automatically follow. Indeed, it requires more than simply identifying the relevant actor – it demands establishing a causal connection between their negligent conduct and the resulting harm. If this link is too tenuous or indirect, legal attribution of responsibility remains unsustainable.

1.2.2. Technical complexity: from ‘causal black box’ to ‘novus actus interveniens’

In the aftermath of an AI-caused accident, efforts to establish legal causation are further complicated by the ‘causal black box’ effect.²¹ According to the prevalent doctrine,²² the *opacity* (i.e., lack of transparency or explainability) of algorithmic decision-making processes exacerbates difficulties in establishing a cause-effect relationship between single human actions (or omissions) and the algorithmic output.²³

To verify this statement, it is useful to ask ourselves what criteria are used to ascertain the existence of the causality link. Causation tests in continental legal systems share core principles, but national variations exist. In general, we could say

²⁰ *ivz*, 1149.

²¹ The first to talk about a ‘causal black box’ is Piergallini (Ch I n 39) 1760.

²² *ivz*, 1753; Yavar Bathaee, ‘The AI black box and the failure of intent and causation’ (2018) 31(2) *Harvard Journal of Law & Technology* 890, 922ff.

²³ King et al (Ch I n 68) 95.

that causation in criminal law serves an imputative logic, which is functional in setting the scope of liability for a verified harmful event.

In order to impute an harmful event to a human agent as his own fault, all European legal systems require that the person *acts in a way that activates risk factors in a causal route* that must have subsequently caused said event. So, the offense is attributed to the agent when the event is a consequence of their action. However, not all European legal systems hold a specific statutory provision on this subject. Some countries do hold very basic (perhaps even shallow) definitions of causality.²⁴

The absence of detailed provisions giving guidance for the establishment of the requirement of causation should not come as a surprise. Indeed, it is important to consider that the problem of causality is an extremely delicate and complicated issue to resolve on a technical level, at least when seeking to find a single and valid answer (or, rather, *theory*) for all cases that come to the attention of the interpreter. It is not easy to get to grips with the notion of causation and it is hard to fruitfully design a generally applicable causation test. Causation theories have all been developed by the courts and, in part, by legal doctrine – and it could not be otherwise.

In European legal systems various theories or tests are applied, which all imply to look at causation in a more practical way. In order to establish causation all jurisdictions apply the so-called *conditio sine qua non* test (or ‘conditional theory’).²⁵ In the Anglo-American systems this test is usually known as the *but for* test. Essentially, under this theory, causality is conceived in naturalistic terms, whereby any condition that has produced an event is considered its cause. From this perspective, all causes are equivalent, whether they are natural or human, as long as they constitute a necessary and sufficient condition for the occurrence of the event. The key question

²⁴ For instance, article 40 of the Italian Criminal Code emphasizes the causal link between the conduct and the harm. It states that ‘No one can be punished for a crime unless the event is the result of his action or omission’. This establishes a factual causation standard: the action or omission must be the sole or at least a contributing cause for criminal liability. Other jurisdictions, such as France and Germany, lack a specific provision on causation.

²⁵ This test literally means: *condition without which the damage would not have occurred*.

in this test is: would the event (E) also have occurred if the perpetrator (A) had not acted in the way he did?

In practice, the conditional theory answers this question by mentally removing the former from the set of antecedents of the latter to see if the outcome still occurs. At this point, two alternatives can be articulated: (a) If the event would not have occurred without the action, the action is the cause of the event (positive formula); (b) If the event would have occurred anyway without the action, the action is not the cause of the event (negative formula).²⁶ For instance, if we want to determine if driving over the speed limit caused a car accident, we mentally remove the speeding action and see if the accident would have still happened. If the accident would have occurred regardless of the speeding, then the speeding cannot be considered the cause. The logical process followed is called ‘mental elimination procedure’ because the interpreter ‘pretends’ that a given naturalistic fact – that is, the action – is not present, contrary to what happened in the actual events. For this reason, the framework outlined is hypothetical and counterfactual (from the Latin *contra factum*, literally ‘against the fact’).²⁷ The positive formula of the *condicio sine qua non* – under (a) – allows us to isolate, within the various antecedents of the event (E), the action (A) corresponding to the description contained in the criminal norm: $A \leftarrow E$, meaning that given the event E, the latter implies action A as its own necessary condition, without which it does not occur.

Although at first sight the test is rather simple, a closer look reveals a number of limitations. The main problem with this theory remains the fact that sometimes it is impossible to explain a given event in naturalistic terms. In our example, the conditional theory helps us establish that the speeding (A) is a necessary condition for the death of the pedestrian (E). However, it does not tell us everything we need to know. We still need to understand *why* the speeding caused the accident and *how* it happened; even if we mentally remove the speeding, we still cannot determine if

²⁶ ‘But for test’ in common law jurisdictions. ‘Teoria condizionalistica’ in Italy.

²⁷ ‘Giudizio controfattuale’ in Italian doctrine, i.e. presupposing the existence of a situation that is factually nonexistent.

the accident would have happened in some other way. The conditional theory needs supplementation in order to effectively explain causality, fulfilling the imputative function already noted.

Acknowledging this difficulty, doctrine has sought to make corrections to the theory of *condicio sine qua non* and has developed the so-called theory of ‘scientific causality’, which is now accepted by the prevailing doctrine.²⁸ Essentially, we turn to scientific laws, which are derived from scientific experiments and observations and provide general principles about how the world works. According to this theory, an action is the cause of an event when, according to the best scientific knowledge and experience of the historical moment (that is, according to the scientific method), the event is a certain or highly probable consequence of the action, as without it, the event would not have occurred.²⁹

This process, which is often referred to as ‘subsumption under scientific laws’, allows us to harness the predictable patterns described by science to explain causality in legal terms. In our example, scientific laws might help us understand factors like vehicle dynamics, road conditions, and human reaction times, which can all contribute to car accidents regardless of speeding. This logical-legal framework that integrates the *condicio sine qua non* formula with the complement of scientific laws is called ‘conditional-nomological model’, where the adjective ‘nomological’ refers to the Greek word νόμος, meaning ‘law’.

While the scientific theory helps us understand causality to some extent, problems arise concerning the adequacy of scientific knowledge. In our society of risk, the natural sciences themselves have experienced a profound epistemological crisis, leading to a crisis of trust towards the traditional nomological-deductive

²⁸ Giorgio Marinucci, Emilio Dolcini and Gian Luigi Gatta, *Manuale di diritto penale. Parte generale* (9th ed. Giuffr  2020) 329ff.

²⁹ This scientific basis can ultimately be drawn from both *universal scientific laws* and *statistical laws*, especially because scientific laws are often impossible to establish.

paradigm.³⁰ According to a famous expression, the conditional-nomological model experiences a genuine ‘crisis of complexity’, grappling with increasingly causal uncertainty and complex technological phenomena.³¹

Specifically, from the late twentieth century onwards, there has been much controversy concerning the reconstruction of causality in situations where there is insufficient or even inexistent scientific evidence to know whether a given action was a ‘but for’ cause. Legal scholars and courts across Europe have dealt with cases where it is impossible to prove with certainty, based on known scientific laws, a clear causal chain between a harmful event and a particular product due to its technical complexity and obscure causal webs.³² In fact, while in the scientific domain it is clearly stated that conclusions should not be drawn solely based on the correlation between two events (correlation is not causation!), in the legal field, however, we know that this is not always the case.

For instance, consider the complex determination of the causal link between the onset of occupational diseases and the employer’s conduct, such as exposing employees to a toxic substance or failing to implement necessary safety precautions in the workplace, which ultimately leads to the onset of the disease.³³ To name just a few cases, there is the case of the drug *Thalidomide* (*Contergan* in German), which, when administered to pregnant women, resulted in babies born with genetic deformities.³⁴ In another infamous case, residents of a certain area with an aluminum

³⁰ See *amplius* Federico Stella, *Il giudice corpuscolariano. La cultura delle prove* (Giuffrè 2005). For a recent and detailed reflection on the topic of the relationship between science and causal explanation, see also Stefano Preziosi, *La causalità penale: nell’orizzonte della ‘scienza nuova’* (Jovene 2021).

³¹ Carlo Enrico Paliero, ‘L’autunno del patriarca. Rinnovamento o trasmutazione del diritto penale dei codici’ (1994) Riv. it. dir. proc. pen., 1239.

³² Piergallini (Ch I n 155) 163ff; Perini (n 153b) 241ff; Donato Castronuovo, ‘La responsabilità colposa nell’esercizio di attività produttive. Profili generali in tema di omicidio o lesioni per violazione delle discipline sulla sicurezza del lavoro e dei prodotti’ in Stefan Canestrari (ed), *I reati contro la vita e l’incolumità personale* (UTET, 2006), 579-602, 582ff.

³³ Cf. for instance Italian Court of Cassation, Sez. I, 20.04.2006, n. 20370, DeJure, Cassazione Porto Marghera.

³⁴ Piergallini (Ch I n 155) 51-52.

factory found themselves covered in blue spots.³⁵ In such instances, the causative factor of the damage within a product suspected of causing harm to physical integrity is unknown. For instance, if we mentally remove the aluminum factory, how can we prove that the blue spots would not have appeared? And how can we prove that without the drug, the babies would have been born without deformities? In both cases, there may still be other potential causes that may at least also have contributed to causing harm, either alternatively or cumulatively.

Ultimately, scientific causality does not seem to work in cases where the nomological component is either unavailable or subject to controversy and there is only a probability link between the conduct and the event. In the absence of scientific laws that identify the specific harmful factor within the product, the transition from the initial risk to the final event occurs within a ‘causal black box’.³⁶

The causal black box effect is exacerbated by the inherent technical complexity of AI systems. In simpler terms, when examining the structure of an AI system, we are aware of the inputs (what the developer or programmer introduces into the system), but the outputs do not always align with the decisions that were originally programmed. This discrepancy between the expected and actual outcomes, which legal scholars often refer to as the level of ‘emergence’ of the system,³⁷ is not scientifically describable – not even by the system’s creator. As a result, the decision-making process remains opaque and difficult to decipher.

The root of this issue lies in the incompatibility between human causal reasoning ($X \rightarrow Y$) and the statistical-probabilistic approach of algorithmic decision-making ($X \rightarrow nY$).³⁸ In human reasoning, there is typically a deterministic relationship

³⁵ See BGHSt 37, 106 (‘Leather spray case’) and BGHSt 41, 206 (‘Wood impregnation case’). In both cases, the actual chemical substance that caused the damage was unknown and in the second case, scientific opinion was even so divided that a clear deduction could not be made about the compound’s effects.

³⁶ Piergallini (Ch I n 155) 143.

³⁷ Cfr. Ch II n 99.

³⁸ See Alice Giannini, *Criminal Behaviour and Accountability of Artificial Intelligence Systems* (Eleven Maastricht Law Series 2023), 170ff.

between cause (X) and effect (Y), where the cause is a necessary condition for the effect. However, in AI systems, the same input (X) can produce a range of possible outputs (Y), each with its own probability of occurrence.³⁹ This means that the process by which an algorithm generates an output from a given input is often non-transparent and resistant to explanation, even after the fact.⁴⁰ This represents a significant epistemological shift, giving rise to two distinct lines of inquiry.

The first line of inquiry revolves around the difficulty of establishing causal relationships in situations characterized by high levels of technological complexity and opacity. Algorithmic decision-making relies on *a posteriori* knowledge (knowledge derived from observed outcomes) rather than traditional causal reasoning. The primary obstacle to establishing causality in such cases is the absence of well-established scientific laws that can describe the causal chain underlying the functioning of machine learning algorithms.

This scientific uncertainty complicates efforts to trace harm back to a specific human action or omission. For example, consider an autonomous vehicle that fails to recognize a pedestrian and causes an accident. In such a scenario, it may be challenging – if not impossible – to determine with certainty whether the failure resulted from a programming error, inadequate training data, or an unforeseen external factor. As legal scholar Hubbard aptly notes, ‘If someone is injured in a collision involving cars that have been designed to learn from and interact with each

³⁹ Let’s consider DALL-E, an example of a ML model developed by OpenAI that generates images from textual descriptions. When you input the same description into DALL-E multiple times, you will likely obtain *n* outputs. For instance, if you ask DALL-E to generate an image of ‘a red apple on a table’, it might produce various images of red apples placed on different types of tables, with different lighting conditions, angles, and other variations. Each output image is a valid interpretation of the input description, but the specific details may vary due to the probabilistic nature of the model.

⁴⁰ Suzanne Beck, ‘Roboter und Cyborgs – erobern sie unsere Welt’, in Id. (ed.), *Jenseits von Mensch und Maschine - Ethische und rechtliche Fragen zum Umgang mit Robotern, Künstlicher Intelligenz und Cyborgs* (Nomos 2012), 15.

other and with electronic aspects of the highway, it may be hard to identify what went wrong, why things went wrong, and what caused the injury’.⁴¹

In such cases, the only certainty is the causal link between the harmful event and the machine. In other words, we know that the ‘thread of causation has passed through the product’, even if the specific path and nature of that causation remain unknown.⁴²

This brings us to our second line of inquiry: examining whether the actions of an AI system could be considered an interrupting factor in the causal chain. Some scholars went as far as to question whether a self-learning system’s decision could sever the chain of causation and constitute a sufficient, independent cause of the event.⁴³

In criminal law theory, causation rules traditionally place great significance on the intervening acts of other agents, known as *novus actus interveniens*. This principle is particularly relevant in scenarios where multiple actors contribute to an outcome or when a third party independently interferes with a chain of events, altering the causal link to such an extent that the harmful result can no longer be attributed to the original actor.⁴⁴

Applying this framework in the AI domain, the challenge lies in determining whether and to what extent human agency remains the primary legal referent for causation when the intelligent system autonomously generates harmful outcomes. If AI is considered an autonomous intervening agent capable of altering or amplifying

⁴¹ Patrick Hubbard, ‘Sophisticated Robots: Balancing Liability, Regulation, and Innovation’ (2014) 66(5) Fla. L. Rev., 1851- 1852.

⁴² Piergallini (Ch I n 155), 198.

⁴³ Sabine Gless and Thomas Weigend, ‘Intelligente Agenten und das Strafrecht. Zeitschrift für die gesamte Strafrechtswissenschaft’ (2015) 126(3) ZStW 561-591. See also Ugo Pagallo, ‘Saggio sui robot e il diritto penale’ in S Vinciguerra and F Dassano (eds.), *Scritti in memoria di Giuliano Marini* (ESI 2010) 595ff.

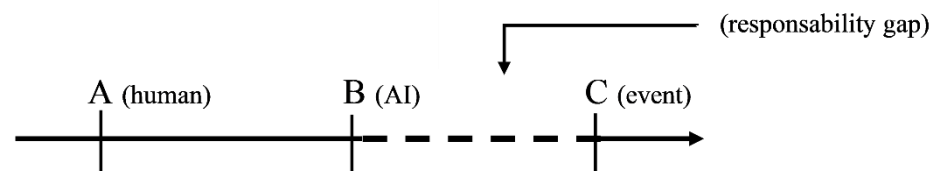
⁴⁴ For a general discussion of causation and intervening acts, see H. L. A. Hart and A. M. Honoré, *Causation in the Law* (2nd edition, Oxford: Clarendon Press, 1988).

risks beyond human foresight or control, this could challenge traditional imputation criteria, as the outcome could be seen as attributable to the AI ‘as its own’.⁴⁵

Indeed, theories of legal causality, including ‘adequate causation’ and ‘human causation’, might suggest that criminal liability should not be assessed based on a purely mechanical cause-effect chain but rather on a structured normative evaluation. These theories seek to refine the strict conditional model of causation (i.e., *conditio sine qua non*) to align with the needs of legal imputation.⁴⁶

In the case of AI, this would involve assessing whether human actions (e.g., designing or deploying an AI system) created a risk that was legally relevant and whether the resulting harm caused by AI is deemed to fall within the scope of the risk created by human actions. If an AI system learns and evolves in ways that were not anticipated by its creators, it may be argued that the harm it causes is not dominable by humans, thus triggering a *qualitatively* different risk from the one originally conceived and interrupting the causal chain.

In this regard, it can be useful to refer to the figure below, which represents the interruption of the legal causal nexus operated by the interposition of AI’s actions and the consequent responsibility gap. In this figure, the line (A-B-C) represents the material and scientific causal nexus between human conduct (A), AI conduct (B) and the harmful event generated as output (C). The line from B to C is dashed to show the interruption of the causal nexus due to the interposition of AI’s actions.



⁴⁵ Gless, Silverman and Weigend (Ch I n 18), 431-432.

⁴⁶ Massimo Donini, ‘La causalità omissiva e l’imputazione per l’aumento del rischio’ (1999) Riv. it. dir. proc. pen., 38. For a stringent critique of these theories, see Federico Stella, *La descrizione dell’evento* (Milan, 1970), 7 ff.

Figure 1: Graphical representation of the alleged interruption of the causal nexus

However, some objections can be raised against this perspective, as it seems to oversimplify the application of legal causation rules. In particular, some authors contend that the notion of an ‘algorithmic interposition’ – where the AI system is perceived as an autonomous intervening agent – is more illusory than real.⁴⁷

The key misunderstanding lies in the fact that, while AI systems may exhibit unpredictable or even opaque decision-making processes, they do not emerge in a vacuum. In particular, the unpredictability of an AI’s behaviour does not necessarily equate to true agency in the legal sense. Their behaviour remains fundamentally constrained by the architecture, training data, and operational parameters set by human creators at various stages, from design and development to deployment and ongoing maintenance.

In our self-driving scenario, therefore, arguing that the system can be considered an independent actor, capable of breaking the causal chain, is simply a mystification of law and reality. Despite the concerns over AI’s self-learning capabilities, it is safe to assume that the causal control still lies with the producers, as it would be the active choice of the producer to provide the system with a given capacity.⁴⁸ By doing so, they introduce a risk while being conscious of not being able to fully control its effects. So, the overall behaviour of the car is still heteronomously determined by the decisions made during the initial programming phase, which caused the vehicle to be what it is, and have those abilities, which were originally allowed or conceived, irrespective of how they have evolved. In this respect, it has been pointed out that

‘even if designing such applications does not entail coding complex lines of software into a specific language, but rather requires the use of alternative—and to some extent more sophisticated—methods of production, this does not per

⁴⁷ Andrea Bertolini, ‘Robots as Products: The Case for a Realistic Analysis of Robotic Applications and Liability Rules’ (2015) 5(2) *Law, Innovation and Technology* 214-247, 234.

⁴⁸ *ivi*, 235.

se influence the final consideration that it is the programmer—or creator if we want to call it that—who has control over the general/global outcome. It is in fact the producer's decision as to what kind of technique to use in order to achieve the best result possible, both in terms of sophistication and functionality of the robot as well as safety; only the producer could in fact devise and conceive possible methods aimed at preventing damage deriving from the proper—or even improper—use of its product'.⁴⁹

It is challenging to envision (at least with the current state of technology) an assimilation of AI systems or the effects of their behaviour to those of an interposing third party agent for the purposes of intervening agency rules. Even in complex situations, such as when the AI system behaves in unexpected ways, the causal nexus is not necessarily interrupted. For example, suppose a self-driving car malfunctions and causes a collision by misinterpreting an unusual road sign. While the malfunction may appear to break the causal chain, the reality is that the human designer's initial coding or the operator's decision to activate the car still constitutes the starting point.

Consequently, it seems that the interposition of an autonomous AI system affects only the intelligibility of the link, not the actual naturalistic causality at play; the apparent disruption is an *epistemic* issue – our inability to fully understand why the car misinterpreted the sign – not a true rupture in causation or a sign that a new agent has entered the scene.⁵⁰

The lack of general explanatory laws that fully describe these processes does not invalidate the causal link. The relationship between the initial human action and the outcome remains intact, even if it cannot be fully articulated. The problems remains as to how establishing causality for the purposes of attributing criminal liability for AI harm.

1.2.3. Overcoming the causality issue: technical and theoretical solutions

⁴⁹ *ibid.*

⁵⁰ Gless, Silverman and Weigend (Ch I n 18), 432.

Not all AI decisions are poised to remain inexplicable, unpredictable, or uncontrollable. In fact, many of the challenges surrounding the identification of a causal link between human actions and AI outputs can be addressed through appropriate technical and legal tools.

One of such technical strategies involves installing specialized ‘white-box’ recorders and system diagnostics to clarify the sequence of events leading to an incident.⁵¹ Similar to what has been implemented in air transportation, these Event Data Recorders (EDR) can facilitate the *ex post* reconstruction of the dynamics of an accident. This should allow for a clearer reconstruction of the spheres of responsibility in the event of harm allegedly linked to an AI system.

Such tests, however, do not seem fully capable of addressing the opacity of the determining factors of harm. For example, if a self-driving car collides with a pedestrian due to a navigation error, an EDR can record various parameters such as cruising speed, acceleration, issues in information acquisition and analysis, any inputs from the driver (e.g., brake pedal pressure), and the activation of warning lights or alerts by the vehicle, but it cannot explain the specific causative factor; it merely provides raw data that must be interpreted during legal proceedings with the help of experts and technical consultants.⁵²

The pressing issue remains: what tests should be used to establish legal causation? In a *de jure condendo* perspective, an interesting solution to the causation problem has recently been proposed at the EU level, namely in the already

⁵¹ Unlike the ‘algorithmic black box’, which conceals the internal workings of an AI system, making it difficult to understand how inputs are processed into outputs, the black boxes referred to here are devices specifically designed to record and reveal detailed information about the system’s operation in accident investigations. Despite using the same concept, in one case it is related to the act of ‘concealing’ while in the other case to ‘revealing’. MF Lohmann, ‘Liability Issues Concerning Self-Driving Vehicles’ (2016) EJRR, 339.

⁵² Nevertheless, in the near future, it is likely that scientific advancements will eventually allow for the identification of the determining cause of harm in a particular case. For example, this might happen through the use of reverse engineering systems, known as *Explainable Artificial Intelligence* (XAI). See Andrew Selbst and Solon Barocas, ‘Unfair Artificial Intelligence: How FTC Intervention Can Overcome the Limitations of Discrimination Law’ (2023) 171 University of Pennsylvania Law Review 1023, 1110ff.

mentioned AI liability directive whose proposal came out on the 28th of September 2022.⁵³

The merits of the proposal is to directly dive into the most important point of the discussion on liability for AI-caused harm: the causal nexus. The European Commission has proposed an innovative and intriguing measure to address causation-related challenges (due to the technical and relational complexity and opacity of the system) in non-contractual civil liability cases: the rebuttable *presumption of causality* laid down by article 4. Specifically, this provision establishes a causal link between non-compliance with a duty of care under Union or national law (i.e. the fault) and the output produced by the AI system (or the failure thereof) that gave rise to the relevant damage. Such presumption of causality would apply when the cumulative following conditions are met:

- a. Article 4(1)(a) < First, the fault of a provider of an AI system has been demonstrated during the trial because they did not meet a duty of care under Union law or national law directly intended to protect against the damage that occurred.⁵⁴
- b. Article 4(1)(b) < Second, it must be reasonably likely that, based on the circumstances of each case, the defendant's negligent conduct has contributed to the output produced by the AI system that gave rise to the relevant damage.⁵⁵

⁵³ Cfr. Ch I n 26.

⁵⁴ AILD, Article 4(2) creates a link between AI act and AI liability directive based on the responsibility of providers of high-risk AI systems. It states that the applicability of the presumption of article 4(1)(a) for providers is conditioned on the demonstration that the provider failed to comply with the obligations of the AI act.

⁵⁵ A breach of a requirement to file certain documents or to register with a given authority, even though this might be envisaged for that particular activity, would not be considered as 'reasonably likely' to have influenced the output produced by the AI system or the failure of the AI system to produce an output (Recital 25 AILD).

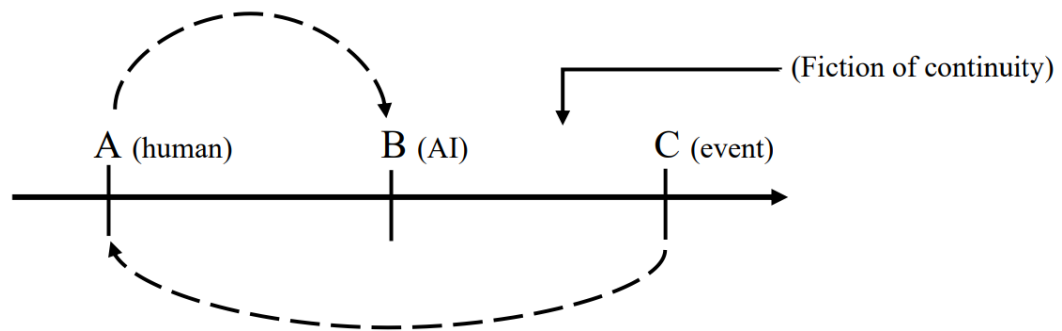
- c. Article 4(1)(c) < Third, it has been demonstrated that the output of the AI has generated the damage.

These three cases show a strong relationship between fault and causation. In particular, the first two conditions (a and b) are deeply linked with fault; if fault is proven then it is not necessary to verify the existence of the causal nexus between AI output and provider's actions. In this case, the only necessary assessment to be made concerns the psychological element – unless the defendant manages to rebut the presumption by demonstrating the absence of causal link.⁵⁶

The presumption of a causal link serves to bridge the gap by bypassing the need to prove every single causal step that occurs in the algorithmic black box, limiting the proof of the causal element to the mere occurrence of an unlawful human action. This approach is based on a straightforward and ontologically grounded assumption: AI systems cannot operate outside the goals defined by their code, which is ultimately created by humans.

Thus, regardless of whether an event results from AI executing a code written by a programmer or from AI responding to a specific human command input, the responsibility remains attributable to humans. In other words, the event (C) is retroactively brought back to the human action (A), as if it had been generated by that, as illustrated in the figure below:

⁵⁶ AILD, Article 4(7). In this respect, it is important to note that the proposed approach does not entail a reversal of the burden of proof. The Commission discards such a reversal of the burden of proof to avoid exposing providers of AI systems to higher liability risks, which could hamper innovation in AI-enabled products and services. Under the proposed approach, the prosecutor would, instead, still bear the burden of proof, but the presumption of causality would result in a targeted alleviation of the burden of proof regarding the question as to how or why an AI system reached a certain harmful output. This approach would relieve prosecutors of the need to demonstrate the inner workings of the AI system at stake. Phillip Hacker, 'The European AI liability directives–critique of a half-hearted approach and lessons for the future' (2022) arXiv available at <https://arxiv.org/abs/2211.13960>.



The legal doctrine defines this kind of *fictio iuris* as the ‘fiction of continuity’.⁵⁷ Accordingly, the figure extends the one presented in the previous paragraph by adding one more dashed line representing the operating mechanism of this particular legal fiction.

This *fictio iuris* is implicit in the European presumption of causal link. In fact, when it is said that causal nexus could be presumed it is truly said that a part of the complex causal nexus (A-B-C), namely the part B-C where the black box effect can occur, is ascribable to the defendant as the person who has directly caused it. This could happen due to the lack of compliance with an obligation of the AI act.⁵⁸

⁵⁷ A first juridical description of this mechanism is ascribable to Vincenzo Calderonio, ‘The opaque law of AI’ (2024) arXiv available at <https://arxiv.org/html/2310.13192v2> who refers to it as ‘finzione di continuità’ describing it as following: ‘I would call the Continuity Fiction the one put in place to safeguard the legal system which, like nature, has *horror vacui*. The problem arises in terms of an acute antithesis, to the point of not being resolvable without the aid of an artifice, when there is a concurrence of: a) on the one hand, a legal affair that must be able to proceed without a break; b) on the other, the incidence of a factor that causes its interruption. How, then, to reconcile the necessity of the ‘continuous’ with the inevitability of the ‘discontinuous’? There seems to be no other way but to deny the caesura by means of the expedient of ‘retroactivity’: that is to say, by tracing back in time the effects of a given act, as if it had been performed before’.

⁵⁸ *ibid.*

Arguably, if approved, the approach proposed in Article 4 AILD could be used as a general paradigm or legal standard for causal determination in AI-caused harm scenarios not only in the civil law field (where the directive uniquely applies) but also in criminal law, with different degrees of operability graduated with reference to the principle of personal responsibility, which is one of the pillars of criminal liability. This *fiction iuris* could indeed clear the field from the causation gap, effectively describing what happens in a causal series which involves AI: a human action, extended by an algorithm, which creates an output.

Nevertheless, this is not it. As anticipated, the mere retroactive attribution of the event (C) to the human action (A), as if it had been generated by the latter, does not automatically establish liability; rather, it requires proof of fault or a ‘guilty mind’ on the part of the producer. While it might be possible to establish a direct causal link between a human action and a harmful outcome, the psychological nexus – the connection between the individual’s mental state and the resulting harm – can be more problematic and may still suffer an ‘interruption’.

This interruption (illustrated by the dashed A-B line in the figure) arises because, although activating and setting in motion an AI system is an intentional choice made by humans (A), the AI’s conduct (B) may still deviate from what was intended or anticipated, leading to a harmful event (C) that was not within the scope of what was intended or reasonably foreseeable. This gap creates a disconnect between the human’s mental state at the time of the AI’s activation and the eventual harmful outcome (A-C), consequently limiting the psychological nexus in this context to the A-B interaction:



Therefore, our analysis must extend into the realm of *mens rea*, focusing on the psychological element associated with the human action. This entails evaluating whether the human agents involved in the production the AI system failed to act with the necessary degree of foresight or care to warrant criminal liability.

1.3. The failure of *mens rea*: culpability without boundaries?

Shifting the focus to *mens rea*, the primary issue lies in determining the *foreseeability*, on the part of the producer, of potential harmful events caused by AI systems once they have been released on the market.

In general, early scholars who addressed this issue emphasized that while the risk of some form of harm occurring at some point in the future is always foreseeable whenever an AI technology is introduced (*an*), the specific manner and timing (*quomodo*) of such harm may be far less predictable.⁵⁹ In other words, the decision to introduce increasingly autonomous AI systems into the market would enhance the *general* foreseeability of the possibility that such products might cause harmful events while simultaneously reducing the predictability of any *specific*, concrete instance of harm.⁶⁰ For example, consider once again our initial self-driving car example, where the machine erroneously classifies a crossing pedestrian as part of the background. If this misclassification leads to a collision, it is safe to say that the category of harm (e.g., ‘the car might cause a fatal accident’) remains foreseeable to a reasonable producer. The unpredictability lies in the precise details of the failure (‘why did the system mislabel the pedestrian?’), but such opacity does not negate the foreseeability of the general risks posed by self-driving cars.⁶¹

⁵⁹ As some have noted, ‘although the risk of some kind of injury at some point in the future is foreseeable whenever one introduces a new technology, how and when an injury occurs may not be particularly foreseeable [...]’. See Balkin (Ch II n 99), 52.

⁶⁰ Beck (Ch II n 6), 47; Id, (Ch I n 21), 139.

⁶¹ Lower predictability is an unavoidable consequence of AI’s adaptive capabilities. For example, self-driving cars operate in an inherently uncertain and volatile environment, where

For this reason, AI provides a useful framework for discussing how specific foreseeability must be when assessing culpability. In other words, it raises the question of what exactly should be the *object* of foreseeability: is it sufficient that the accused was merely aware of a generic and vague risk that something might go wrong, or must foreseeability extend to the anticipation of a particular result?⁶²

This debate has led to a sharp contrast between two opposing positions. One view holds that specific foreseeability is necessary, with the consequence that the foreseeability and avoidability of the harmful event would already be lacking at an abstract, *ex ante* level, thereby invalidating any attempt at attributing liability. The alternative position argues that generic foreseeability is sufficient, implying that all potential harms caused by AI systems becomes automatically foreseeable. Legal scholars noted that neither positions are convincing.⁶³

The first is problematic because the general unpredictability of AI systems cannot exempt their producers from liability; rather, it is precisely ‘this unpredictability that gives rise to a duty of care’.⁶⁴ A useful analogy can be drawn from the scenario of a zoo director who releases a tiger from its cage.⁶⁵ If the tiger kills a visitor, the director cannot evade responsibility by arguing that ‘tigers are unpredictable animals’.

they must navigate a constantly changing landscape. As a result, their algorithms need to become increasingly independent and adaptive to handle the complexities of real-world situations. The ability of ML systems to respond autonomously to their environment – to learn new information and act upon it – will often result in *unforeseeability* of their output. See Ryan Calo, ‘Robotics and the Lessons of Cyberlaw’ (2015) 103 Calif. L. Rev., 542; Ugo Pagallo, ‘When Morals Ain’t Enough: Robots, Ethics, and the Rules of the Law’ (2017) 27 Minds and Machines, 634.

⁶² Ryan Calo, ‘Is the Law Ready for Driverless Cars?’ (2018) Comm. Acn., 34, 35; Giannini and Kwik (n 4), 54-55.

⁶³ Suzanne Beck, ‘Google cars, software agents, autonomous weapons systems – New challenges for criminal Law?’ In E Hilgendorf and U Seidel (eds.), Robotics, autonomies, and the law: Legal issues arising from the autonomies for industry 4.0 technology programme of the German federal ministry for economic affairs and energy (Nomos 2017) 227–252, 244, who points out how neither solution is truly compelling.

⁶⁴ Beck, Silverman and Weigend (Ch I n 18), 427.

⁶⁵ *ibid.*, 428.

Therefore, according to the second position, individuals who can foresee that their actions might endanger legally protected interests – such as life and health – are obligated to take appropriate precautions or refrain from engaging in those activities *tout court*. The same principle applies to potentially dangerous AI systems: deploying them in society is akin to ‘releasing a beast from its cage’ – its exact behaviour may not be predictable, but its risks are foreseeable. Even if AI developers do not fully understand every detail of how their systems function internally, mere awareness of AI’s potential for harm is enough to establish a duty of care.⁶⁶

For example, if the producer of a self-driving car could, through proper diligence, recognize that the vehicle’s object-detection sensors are unreliable, they would breach their duty of care by bringing the car to market despite this flaw. This failure to refrain from acting can have legal consequences: if the vehicles cause harm, the producer could face criminal prosecution for negligent – or in extreme cases, even intentional – bodily injury or homicide.

However, this position is not entirely convincing. Under this approach, a producer’s mere awareness that its AI system might make unpredictable and uncontrollable decisions would be sufficient to establish criminal liability for any specific harm caused – even when the producer had no concrete knowledge of the precise nature of the risk.

Yet, automatically attributing every AI failure to the producer simply because some level of risk was *generally foreseeable* raises serious concerns. First, it would stifle innovation, as AI developers might struggle to operate under such an expansive liability framework. Second, and more fundamentally, it would distort the core principles of criminal liability, which traditionally require a meaningful link between fault and harm.⁶⁷

One way to justify this broad approach is by invoking the precautionary principle as an *ex post* evaluative standard for foreseeability. This could allow for a

⁶⁶ *ibid.*

⁶⁷ Beck (n 63), 244; Giannini and Kwik (n 4), 54-55.

more flexible interpretation of the duty of care, effectively expanding the range of foreseeable sub-events beyond those already scientifically recognized.⁶⁸ However, the dominant legal opinion firmly rejects the idea of using the precautionary principle as the foundation for a precautionary rule in criminal law.⁶⁹

The core objection lies in the fundamental difference between foreseeability – which concerns the likelihood of an event occurring based on established knowledge – and mere conjectures of danger, which science has not yet definitively confirmed. This critical distinction underpins the orthodox view that precautionary rules must have a solid scientific foundation, addressing the prevention of reasonably foreseeable events rather than hypothetical risks. Along these lines, scholars have also emphasized the difference between a rule based on the precautionary principle and a classic abstention-based rule.⁷⁰ The latter, despite its restrictive nature, remains a legitimate risk-neutralization technique, provided that it is grounded in scientifically established dangers. By contrast, the precautionary principle does not respond to a currently known risk but rather to a speculative, future-oriented hazard, making its application in criminal liability excessively ultra-prudent and legally uncertain.⁷¹

In simpler terms, if every harm caused by an AI system – no matter how unpredictable the specific failure – were automatically blamed on the producer merely because some degree of risk was foreseeable from the outset, we would be undermining a fundamental safeguard of criminal negligence: the principle of risk coverage (i.e., the requirement that the specific harm falls within the scope of the

⁶⁸ Ruga Riva (Ch II n 165), 1750ff, 1766ff.

⁶⁹ Such an approach, however, does not appear sufficient to justify criminal intervention, as it is entirely possible that such a violation is so distant from the legally protected interest that it would contravene the principle of necessary offensiveness. Vittorio Manes, *Il principio di offensività nel diritto penale. Canone di politica criminale, criterio ermeneutico, parametro di ragionevolezza* (Torino 2005), 296.

⁷⁰ Piergallini (Ch II n 200), 1695-1696.

⁷¹ *ibid.*

foreseeable risk).⁷² The result? Negligence would become virtually indistinguishable from *strict liability*, holding producers accountable regardless of fault.⁷³

The crucial question, then, is at what point general awareness of an indeterminate risk of AI failure crosses the threshold of blameworthiness under criminal law. All AI producers inherently acknowledge that their systems are fallible – after all, no machine is infallible. However, there are strong reasons to limit the extent of their liability, ensuring that responsibility is assigned based on a meaningful standard of culpability rather than a blanket attribution of fault.

A promising approach is to recognize responsible risk-taking (*erlaubtes Risiko*) as a justification for certain actions, where producer's duties of care are defined and the social acceptability of their conduct is emphasized.⁷⁴ In contexts of risky but socially useful activities such as producing AI systems, every criminal policy choice should constitute the final step of a broader path of public regulation in which the legislator should define the 'rules of the game' and the boundaries of the permissible risk area – a legal construct encompassing activities explicitly permitted or tolerated by the legal system, even when accompanied by foreseeable risks of harm.

According to this perspective, producer responsibility should be linked to a tangible failure to meet established duties of care, rather than an all-encompassing expectation that any potential failure must be anticipated. As long as producers operate within the permissible risk zone and adhere to the necessary precautionary measures, they could be shielded from negligence-based criminal liability.

⁷² Marinucci, 413, the harmful event must materialize as the specific risk that the breached precautionary rules were intended to prevent. In Germany and Italy this concept is understood as, respectively, '*Risikozusammenhang*' and '*nesso tra colpa ed evento*'. These expressions' translation might be the risk-link between the violation of the duty of care (by the actor) and the harmful event, whose nature is normative, not material.

⁷³ C. PIERGALLINI, voce *Colpa*, in *Enc. dir., Annali*, vol. X, 2017, 235 ss., 251ff.

⁷⁴ Beck (Ch II n 6), 53.

In this context, the obligations set forth in the AI Act for AI producers become particularly relevant for criminal liability and warrant detailed examination.

SECTION 2

The balanced approach: limiting provider's liability by defining acceptable risk

2.1. Establishing legal duties: *ex ante* and *ex post* obligations under the AI Act

The development and deployment of high-risk AI systems inevitably bring to the forefront the concept of permissible risk – the recognition that certain socially beneficial but inherently hazardous activities cannot function without some level of tolerated danger. In the realm of criminal liability for AI producers, this notion serves as a necessary counterbalance against an excessively broad interpretation of foreseeability and preventability of harm. As some scholars have pointed out, while the unpredictability of autonomous learning makes the risk of harm theoretically foreseeable, this does not mean that every failure should automatically result in liability. Instead, the standard of care must reflect the realities of AI development, allowing for a margin of tolerance – a space where certain errors in designing and programming such systems are acknowledged as inevitable rather than negligent.⁷⁵

The real challenge, then, is not whether some risks must be accepted, but rather how to define the boundaries of this ‘margin of tolerance’. How much responsibility gap is European society willing to accept in the name of technological progress? Where does the legal space end, and where does culpability begin? This issue becomes even more pressing given that, unlike traditional technological risks – such as those in medicine or nuclear energy – AI risks evolve dynamically, often in ways that cannot be fully anticipated at the moment of deployment.

To answer these questions, we must first assess whether explicit precautionary rules already govern the development and commercialization of AI systems and, if so, how these rules shape the boundaries of a legally permissible risk area. To this end, an in-depth examination of the EU's regulatory framework for AI is essential.

⁷⁵ Gless, Silverman and Weigend (Ch I n 18), 430-431.

As already discussed,⁷⁶ the EU has positioned itself as a global leader in AI governance, aiming to balance technological progress with fundamental rights protection. However, whether its approach provides a clear and workable standard for criminal liability – or instead leaves room for significant legal uncertainty – remains an open question that we now proceed to address.

As a form of product legislation, the EU legal framework assigns primary responsibility for an AI system’s safety and trustworthiness to its *provider*. According to the Act, a ‘provider’ is defined as ‘a natural or legal person, public authority, agency, or other body that develops an AI system or a general-purpose AI model, or that has an AI system or model developed and places it on the market or into service under its own name or trademark, whether for payment or free of charge’.⁷⁷

The heart of the AI Act lies in its Chapter III, which outlines a set of horizontal mandatory requirements that high-risk AI systems must fulfil,⁷⁸ as well as the role and responsibilities of providers.⁷⁹ In particular, the regulation sets out both *ex-ante* (before the system reaches the market) and *ex-post* (aftermarket deployment) obligations for providers, covering all stages of the product life cycle under the threat of administrative penalties.⁸⁰

Notably, from an *ex-ante* perspective, it is stated that, before placing a high-risk system on the market, the provider shall conduct a ‘conformity assessment’ to demonstrate, *inter alia*, that their system complies with the safety requirements expressly laid down in Section 2 of the Regulation.⁸¹ These include, among others, substantive requirements to demonstrate that the system has been trained on quality

⁷⁶ Cfr. Ch II, para. 3.3.

⁷⁷ AI Act, Article 3(3).

⁷⁸ AI Act, Chapter III, Section 2 ‘Requirements for high-risk AI systems’.

⁷⁹ *ivz*, Section 3 ‘Obligations of providers and deployers of high-risk AI systems’.

⁸⁰ *ivz*, Article 100, mandating that MS ‘shall lay down rules on penalties, including administrative fines, applicable to infringements of this Regulation’ that ‘shall be effective, proportionate and dissuasive’. Consequently, the text leaves a wide margin of appreciation to individual MS on the types of punitive measures to be taken (as well as with respect to their contents), despite some more specific requirements of the provision.

⁸¹ *ivz*, articles 16 and 47.

data,⁸² tested adequately,⁸³ and that meet certain robustness⁸⁴ and transparency standards⁸⁵ to enable users to interpret the system's output and use it appropriately.

Additionally, the regulation attaches particular importance to procedural requirements – including record-keeping,⁸⁶ technical documentation,⁸⁷ and risk assessment.⁸⁸ A key provision in this last respect is Article 9 AIA, which stipulates that high-risk systems must be equipped with 'risk management systems' for the entire life cycle of the system.

This provision embodies the 'regulated self-regulation' approach taken by the EU lawmakers to define the threshold of acceptable risk for high-risk AI systems.⁸⁹ This governance model grants providers discretion to determine how the risk management system should be implemented. In other words, providers are tasked with determining how much risk associated with certain AI systems is acceptable and the specific measures to mitigate that risk.⁹⁰

⁸² *ivi*, Article 10(2).

⁸³ *ivi*, Article 9.

⁸⁴ *ivi* Article 15.

⁸⁵ *ivi* Article 11.

⁸⁶ *ivi* Article 13.

⁸⁷ *ivi*, Article 11.

⁸⁸ *ivi* Article 9(1).

⁸⁹ This term depicts the governance model where self-regulation is structured by the State but the State is not involved. This approach marks a shift from the traditional regulatory model, where independent state bodies control conflicting interests, toward a system where industries are encouraged to self-regulate under state supervision, taking on significant regulatory responsibilities which are enforced through legal frameworks. It reflects a collaborative governance model where public institutions and private entities share the responsibility of risk management. Peter Collin, 'Regulierte Selbstregulierung in rechtshistorischer Perspektive: Studien und Materialien' (2018) 5 Max-Planck-Institute for European Legal History Research Paper Series 768. In the US, John Braithwaite 'Enforced Self-Regulation: A New Strategy for Corporate Crime Control' (1982) 80(7) Michigan Law Review 1466-1507.

⁹⁰ To explain this concept, it is useful to distinguish between *rigid* and *flexible* rules of conduct. Essentially, while the first category imports a clear and comprehensive normative judgment as to what is acceptable in the exercise of a certain risky but useful activity (e.g., road circulation), flexible rules are intentionally vague and do not provide technical details, thus

Essentially, a risk management system is an organizational model that places assessing and mitigating risks at the heart of the entire design and development process. Notably, the adoption of a risk management system should ensure that providers of high-risk systems:⁹¹

(a) identify, analyze, and evaluate ‘known and foreseeable risks’ to health, safety, or fundamental rights, that ‘may emerge when the high-risk AI system is used in accordance with its intended purpose, and under conditions of reasonably foreseeable misuse’,⁹²

(b) take ‘suitable risk management measures’ to eliminate or mitigate the identified risks ‘as far as technically feasible’,⁹³

(c) in such a way that any ‘relevant residual risk is judged to be acceptable’.⁹⁴

This approach to risk management, exemplified by Article 9 AIA, places a significant burden on providers, transforming them into proactive creators of safety and legal certainty.⁹⁵ These actors must independently acquire the necessary knowledge and tools to effectively manage and mitigate the risks they oversee. A detailed analysis of the steps of AI risk assessment may be helpful.

Generally, the risk-based approach consists of (at least) three phases: assessment, categorisation, and management.

First, risks need to be identified and evaluated. In this respect, it must be noted that according to the Regulation, AI technologies present various types of risks

leaving discretion to those who are supposed to apply such rules in determining the acceptability threshold for risk management.

⁹¹ Responsibility for meeting the requirements under Article 9 AIA may fall upon the provider of high-risk AI systems, the manufacturer, the importer into the EU, the distributor, and, in some cases, the commercial or government entity making use of the system.

⁹² AI Act, Article 9(2)(a) and (b).

⁹³ *ivi*, Articles 9(5)(a) and 9(2)(d).

⁹⁴ *ivi*, Article 9(5).

⁹⁵ More broadly on this point, Filippo Sgubbi, *Il diritto penale incerto. Punire senza legge, senza verità, senza colpa* (Il Mulino, 2019), 1199 ff.

that can be categorized into three distinct types based on their temporal impact and level of predictability: (i) the immediate ‘known knowns’ (short-term); (ii) the foreseeable ‘known unknowns’ (mid-term), and (iii) the distant but disquieting ‘unknown unknowns’ (long-term). However, the AI Act does not define when risks are ‘known’, ‘unknown’, or ‘foreseeable’. Therefore, we suggest using the following definitions.

A risk is *known* if the harm has occurred in the past or is certain to occur in the future. These are risks which we understand well and can be identified based on knowledge and techniques. For example, an AI risk should be considered known if there is a relevant entry in one of the incident databases or if a public incident report has received significant media attention.

Secondly, we define *known unknowns* as those risks that have not yet occurred, but are reasonably foreseeable. This risk does not necessarily refer to something that an organization already knows, but rather something that is scientifically plausible but uncertain, and thus could be already identified with reasonable effort. For instance, while we know that algorithmic trading systems can lead to market disruption, the specific methods and impacts of such disruption may not be fully understood, since cutting-edge trading algorithms could engage in novel and unprecedented forms of market manipulation.⁹⁶

Lastly, looking further ahead, there are the *unknown unknowns* – long-term risks that are completely unforeseen at the moment the system is put into service, but may emerge afterwards during the deployment phase.⁹⁷ While this expands the circle of relevant risks, it does not threaten legal certainty as ‘the risks referred to in this Article shall concern only those which may be reasonably mitigated or eliminated

⁹⁶ More on this point in Federico Consulich, ‘Il nastro di Mobius. Intelligenza artificiale e imputazione penale nelle nuove forme di abuso del mercato’ (2018) 71(2) Banca, borsa e titoli di credito 195-234.

⁹⁷ AI Act, Article 9(2).

through the development or design of the high-risk AI system, or the provision of adequate technical information’.⁹⁸

Importantly, the phase of risk assessment and identification plays a fundamental role in guiding the implementation of the other requirements outlined in Chapter III, Section 2 AIA for high-risk systems. A thorough risk evaluation not only helps providers pinpoint potential hazards but also informs the necessary mitigation measures and how strong those measures need to be.⁹⁹ This process enables providers to determine, for instance, what constitutes an ‘effective’ level of human oversight for their system or what statistical properties are ‘appropriate’ for their data given the risks.¹⁰⁰

Where residual risks (*Restrisiko*) seem to persist after these initial mitigation efforts, providers are responsible for demonstrating that these risks are within acceptable limits.¹⁰¹ While not explicitly defined in Article 9, residual risk can be admittedly defined to the marginal risk that persists even after all necessary regulatory precautions have been taken, and cannot be fully eliminated. As a result, this

⁹⁸ *ivi*, Article 9(3).

⁹⁹ AI Act, Article 8 stipulating that ‘The risk management system referred to in Article 9 shall be taken into account when ensuring compliance with those requirements’. This provision contemplates, in other words, that the ‘appropriate’ degree of care and precaution necessary to implement the various requirements will be to some extent determined by the circumstances and risks identified and evaluated by virtue of Article 9. On this subject see Henry Fraser *et al.*, ‘Acceptable Risks in Europe’s Proposed AI Act: Reasonableness and Other Principles for Deciding How Much Risk Management Is Enough’ (2023) *European Journal of Risk Regulation* 1-16.

¹⁰⁰ For example, Article 10(2) AIA provides that ‘training, validation and testing data sets shall be subject to *appropriate* data governance and management practices’ (our emphasis). Article 12(2) provides that logging capabilities ‘shall ensure a level of traceability of the AI system’s functioning throughout its lifecycle that is *appropriate* to the intended purpose of the system’. And Article 14(4) provides that human oversight mechanisms should enable overseers to do certain things - such as ‘be able to correctly interpret the high-risk AI system’s output... as *appropriate* to the circumstances’.

¹⁰¹ AI Act, Article 9(5).

remaining risk is ultimately borne by society. It can be expressed as: Residual Risk = Inherent Risk – Risk Control.¹⁰²

The function of this provision is to ensure that providers of high-risk systems identify the residual risk that remains even if they comply with the other requirements set out in the regulation, and take additional measures to reduce it to an ‘acceptable level’. Conversely, if it turns out that this is not possible, it is reasonable to assume that the need to protect legal interests should bring the development and deployment process to a stop.¹⁰³

Additionally, a provider’s obligations do not end once an AI system is placed on the market; rather, they retain ongoing responsibility for ensuring the system’s safety and continued compliance with regulatory requirements throughout its operational lifetime. This means that even after deployment, providers must actively monitor their AI systems to identify and assess emerging risks that were not foreseeable during development – commonly referred to as unknown unknowns.¹⁰⁴

This post-market monitoring aligns with the principle that, despite the diligence exercised during the production phase, certain types of risks only become apparent once the system is in real-world use. Continuous monitoring and adaptive risk management thus form an integral part of responsible AI governance, ensuring that providers remain accountable for the evolving safety profile of their systems.

The producer is required to carry out both *active* and *passive* monitoring of the product. Active monitoring involves proactive measures to uncover risks that were not identifiable at the time of the product’s introduction, such as conducting random sample tests on distributed merchandise. Passive monitoring, on the other hand, entails reviewing and addressing consumer/deployer complaints received by the provider.

¹⁰² Forti (n 11) 547.

¹⁰³ Art 9 AIA does not say this explicitly, but it seems to be a logical consequence of the process. See Jonas Schuett, ‘Risk management in the Artificial Intelligence Act’ (2023) 15(2) European Journal of Risk Regulation 367-385.

¹⁰⁴ The post-market monitoring system is described in Article 72 AIA.

In addition to these monitoring duties, the producer is also obligated to report any serious incident to the market surveillance authorities of the Member States where that incident occurred¹⁰⁵ and provide consumers with supplementary information and, when necessary, to recall or withdraw the product from the market.¹⁰⁶ These obligations are inherently flexible, as they do not establish rigid criteria for when and how the producer must take corrective action. Instead, they require a case-by-case assessment to determine the appropriate response based on evolving risks and emerging evidence.

2.2. We should accept the risk – but how much risk?

The introduction of legal duties for AI providers under the AI Act underscores the importance of assessing AI-related criminal liability in the context of a structured risk assessment framework. In this context, risk transcends its traditional understanding as a matter of statistical likelihood of harms or historical precedent – it becomes a ‘legal risk’.¹⁰⁷ The Act assesses AI risk based on compliance with a set of safety measures that are codified as legal obligations. This reflects a precautionary approach, where failing to meet these standards is what determines liability, rather than simply whether harm was foreseeable.

This risk assessment framework could be highly relevant for criminal law. It could establish new benchmarks for the duty of care expected from AI providers, clarifying when their actions transition from lawful to unlawful. By doing so, the Act may provide a clear and concrete foundation for determining socially acceptable risk in cases involving the development and deployment of high-risk AI systems.¹⁰⁸

¹⁰⁵ *ivi*, Article 73.

¹⁰⁶ *ivi*, Article 20.

¹⁰⁷ Mahler (Ch I n 169).

¹⁰⁸ Gless and Ligeti (Ch I n 196), 49.

As a result, risks that persist despite compliance with established legal standards would generally be considered *permitted* or *tolerated* by the legal system. Conversely, risks arising from a provider's failure to adhere to these obligations would typically be deemed *legally unacceptable*. In such cases, the provider could be held criminally liable for any resulting harm, such as death or injury, caused by their actions.¹⁰⁹

However, the Act's approach to risk acceptability judgments for high-risk AI systems is not without its challenges. For example, Article 9 of the AI Act requires providers to ensure that any residual risk is deemed acceptable and that measures are implemented to eliminate or minimize such risks as much as possible. While it is clear that high-risk AI systems must undergo rigorous testing to identify the most effective risk management strategies and that we should accept the risk persisting after appropriate management, a critical question remains: *how much residual risk is considered acceptable, exactly?*

The Act does not provide a precise threshold or clear criteria for determining the level of risk that can be tolerated. While the regulation sets out *what* needs to be achieved (AI trustworthiness through risk acceptability), it is largely silent on *how much* risk management is enough for residual risks to be deemed acceptable – that is, how much effort providers need to put into identifying and mitigating foreseeable risks, and when they are allowed to stop looking for them.¹¹⁰

In fact, the regulation requires that risk management be implemented not arbitrarily, but according to the 'as far as technically feasible' parameter.¹¹¹ This vague parameter, though, seems to impose a potentially indeterminate burden of precaution on providers, who are required to implement *every* risk management measure that is *technically possible* as a matter of engineering, regardless of cost-

¹⁰⁹ *ibid.*

¹¹⁰ Fraser (n 99), 6-7.

¹¹¹ AI Act, Article 9(5)(a).

benefit considerations.¹¹² Such an interpretation implies that it is always theoretically possible to reduce generally foreseeable risks by implementing *just one more measure* or, if necessary, by refraining from releasing the product into the market.¹¹³

This lack of specificity in how to conduct risk acceptability judgments under the AI Act presents several significant challenges. Chief among these is the uncertainty surrounding where to draw the line between lawful and unlawful conduct. This uncertainty is particularly problematic because it can culminate in a concerning outcome: the rules defining what constitutes acceptable risk frequently do not exist in a clear, pre-determined form before a criminal proceeding. Instead, these boundaries might be constructed, clarified, or even created during criminal proceedings by judges, thus leaving AI providers in a precarious position, as they must navigate vague requirements and translate them into concrete practices without clear guidance.

This judicial discretion can place providers in an untenable position. For example, consider cases where an AI system is introduced to the market in full compliance with the legal duties outlined in the Act's safety requirements, but later exhibits signs of risks that exceed what was initially deemed acceptable. In such scenarios, the provider is burdened with ongoing obligations, such as monitoring, sharing information with administrative authorities, providing additional information, issuing recalls, and, ultimately, withdrawing the product from the market. These duties can lead to criminal prosecution if an AI system causes harm: if the provider fails to comply with the obligation to monitor its systems and these harm humans, it can be prosecuted for negligent or even intentional bodily injury or homicide.

¹¹² The prospect of providers disregarding risk management costs when conducting risk-acceptability judgments is simply not realistic. The very impracticability of reducing residual risks 'as far as technically feasible' could encourage a lack of rigor in the approach to risk acceptability, which would be at odds with the AI Act's goal of trustworthy AI. On this point, see Henry Fraser and José Miguel Bello y Villarino, 'Acceptable residual risks for fundamental rights? Understanding the keystone of risk-based AI regulations' (2021) Workshop Artificial Intelligence: The New Frontier of Business and Human Rights, 6, who criticize the standard of acceptability for disregarding cost-benefit considerations.

¹¹³ Andrew Selbst, 'Negligence and AIs' Human Users' (2020) 100 Boston University Law Review 1317-1376.

However, these obligations are often characterized by their vague and flexible nature, failing to precisely define the *conditions* and *scope* of the provider's duty to act. As a result, determining the following key issues becomes a daunting endeavour: what signals are necessary to trigger the provider's duty to take corrective actions? When does it become apparent to the provider that a risk exceeds the previously accepted threshold? How should providers choose among the obligations imposed by law, and when should product withdrawal be considered mandatory?

These questions remain unanswered at least until after the fact, which raises significant concerns, particularly with regard to the principle of legal certainty in defining negligent conduct. This principle is a cornerstone of criminal law, ensuring that individuals – in this case, AI providers – can clearly understand the boundaries of lawful behaviour and make informed decisions about their actions.

One thing is certain: risk management must have *limits*, even for high-risk products. Drawing bright lines for acceptable risk is crucial for the fair allocation of criminal liability among involved parties. Achieving this requires more than general requirements; it demands precise, actionable rules that can reliably guide behaviour and risk management strategies.

One potential approach to filling the blanks left by the Act and determining the expected conduct of providers in AI risk management lies in referencing *harmonized technical standards*.¹¹⁴ Similar to other EU product legislations,¹¹⁵ the AI

¹¹⁴ AI Act, Articles 40-41. On this point see Maria Kaiafa-Gbandi, Athina Sachoulidou and Dafni Lima, 'Greek Report on Traditional Criminal Law Categories and AI' (2023) 94(1) *Revue Internationale de Droit Pénal* 237.

¹¹⁵ The process of standardization underscores the increasing importance of self-regulation in today's regulatory landscape according to the EU's 'new legislative framework' (EU Regulation 1025/2012). The NLF entails a partnership between legislation and standards, focused on making harmonized European Standards a focus of demonstrating compliance with regulatory requirements in new technical sectors. This allows for the requirements of regulations to be given practical effect through a consensus-driven process that can take account of the views of multiple stakeholders and technical experts. The term 'harmonized standard' is defined in Art 3, point 27 in conjunction with Art 2(1), point (c) of Regulation (EU) No 1025/2012. This has been a common practice since 1985, and is particularly evident

Act mandates that '[s]tandardization should play a key role to provide technical solutions to providers to ensure compliance with this Regulation, in line with the state of the art, to promote innovation as well as competitiveness and growth in the single market'.¹¹⁶

Standardization plays a pivotal role in bridging the gap between high-level legal requirements and practical implementation in the AI domain, by translating the former into clear, actionable guidelines. Essentially, technical standards specify both the bar that AI systems must meet (by offering specific tests and metrics) and how the systems should be made and tested (by describing tools and processes that can be used) to respect humans' safety and fundamental rights without being overly burdensome.

Additionally, adherence to harmonized standards developed by European standardization bodies like CEN and CENELEC is meant to offer an objectively verifiable way of demonstrating that residual risks have been mitigated to an acceptable level. This confers a significant advantage to providers: a *presumption of conformity*, meaning that compliance with such standards creates an assumption that the provider has met the relevant essential requirements of the AI Act.¹¹⁷ Conversely, providers who choose not to follow these standards must demonstrate that their alternative risk mitigation strategies are at least as effective in protecting fundamental rights and societal interests as the harmonized standards.¹¹⁸

in high-risk sector such as medical devices regulation. Cf. Council of the European Union, Resolution of 7 May 1985 on a new approach to technical harmonization and standards. Official Journal of the European Communities, Jul. 05, 1985, available: <https://op.europa.eu/en/publication-detail/-/publication/9f7c3a42-449e-4ccc-9a29-5a544003b338/language-en>.

¹¹⁶ AI Act, Recital 121, sentence 1. For more information on the role of standards in the AI Act, see M McFadden *et al*, 'Harmonizing Artificial Intelligence - The Role of Standards in the EU AI Regulation' (2021) Working paper Oxford Commission on AI & Good Governance 43ff.

¹¹⁷ AI Act, Article 32 and Recital 42.

¹¹⁸ *id.*, Article 43. The provider must choose one of two assessment procedures: (i) self-certification procedure based on internal control (para. (1)(a)) or (ii) external conformity

It is clear that, when it comes to assessing provider's criminal liability,

such standards will be essential in distinguishing between what a provider *can* foresee and what they *must* foresee. This distinction is crucial for defining the 'margin of tolerance' – or, more precisely, the 'zone of immunity' – within which a provider, having fully complied with precautionary regulations, can reasonably expect to be shielded from liability.

This means that, as AI systems become an part integral of society and thus the 'normal' risks of life, the provider who adheres to state-of-the-art technical standards will not be criminally liable for generally foreseeable malfunctioning but only for harm incurred due to preventable release of an insufficiently tested product, a defect in the technical system, or the absence of essential safety features that should have been anticipated.¹¹⁹

2.3. The inadequacy of technical standards: (un)timeliness and (un)feasibility

In discussions about defining the acceptable risk threshold for AI systems, much attention has been given to the role of standardization, guidelines, and protocols. While these tools can offer valuable direction and reference points in certain situations, their implementation is not without significant challenges. Despite progress in this area, with a growing range of industry-specific standards already in place and many more currently under development,¹²⁰ the effectiveness of standards in the AI

assessment by independent notified bodies (para. (1)(b)). If the provider has not applied the harmonized standards, or only used part of them, they must follow the second one.

¹¹⁹ Beck (n 63), 244; Id, (Ch I n 21), 139.

¹²⁰ Hans Micklitz, 'The Role of Standards in Future EU Digital Policy Legislation A Consumer Perspective' (2023) Report Commissioned by ANEC and BEUC. On December 5, 2022, the European Commission published a draft 'standardization request' addressed to CEN and CENELEC. Once the proposal is adopted, the standards should be published before the AI Act comes into force, which will be applicable 24 months after its approval, pursuant to Article 85.

sector faces inherent limitations. Two key issues stand out: *timeliness* and *technical feasibility*.

First, the standardization process in Europe is still in its infancy and there is a clear discrepancy between the rapid deployment of these technologies and the slower development of corresponding standards.¹²¹ The rapid pace of AI development often outstrips the ability of relatively slow standardization processes to keep up. Arguably, by the time standards are developed, reviewed by the Commission, and published in the EU's Official Journal, the technology may have already evolved, rendering the standards outdated or irrelevant.¹²² For example, while technical rules are being established for self-driving vehicles, higher levels of autonomy are already emerging, necessitating fresh evaluations of their unique risks. This lag creates uncertainty for providers, as expectations for compliance evolve alongside technological advancements and there is limited knowledge regarding the types and severity of potential risks.¹²³

Secondly, the development of the harmonized standards faces another constraint: technical feasibility.¹²⁴ AI systems are highly complex and context-dependent, making it difficult to create one-size-fits-all standards. Unlike traditional technologies, which often function in predictable and controlled environments, AI systems are dynamic, adaptive, and sometimes even autonomous, meaning their behaviour and associated risks can vary depending on the 'operational scenario' in which they are deployed.¹²⁵

¹²¹ See Georg Borges, 'Rechtliche Rahmenbedingungen für autonome Systeme' (2018) *Neue Juristische Wochenschrift* 977-982.

¹²² McFadden et al (n 116), 18. See also Mélanie Gornet and Winston Maxwell, 'The European approach to regulating AI through technical standards' (2024) 13 *Internet Policy Review* 3.

¹²³ Beck (Ch I n 21), 139.

¹²⁴ Hadrien Pouget, 'The EU's AI Act Is Barreling Toward AI Standards That Do Not Exist' (Lawfare January 12, 2023), available at <https://www.lawfaremedia.org/article/eus-ai-act-barreling-toward-ai-standards-do-not-exist>.

¹²⁵ With the term 'operational scenario', we mean a sequence of situations where the overall risk results from the combination of a number of determinants, drivers and interaction types.

Crucially, these variations mean that what constitutes an acceptable level of reliability, transparency, and risk mitigation in one scenario may be impractical, excessive, or even counterproductive in another. For example, self-driving cars operate in highly diverse environments, from well-mapped urban streets to unpredictable rural roads. In a city, the system must navigate complex traffic patterns, pedestrians, cyclists, and frequent stops. High reliability in object detection and decision-making is therefore critical, but the system can rely on detailed maps and infrastructure like traffic lights and lane markings. By contrast, in rural areas, roads may lack clear markings, and the car may encounter unexpected obstacles like wildlife or poorly maintained infrastructure. Here, the system must prioritize adaptability and robust sensor performance in less predictable conditions.

If regulators or standardization bodies impose a single, rigid standard for ‘acceptable accuracy’ or ‘high-quality data’, it may not account for these contextual differences. For example, a standard requiring 99.9% accuracy in object detection might be feasible in urban settings provided that the system is trained on high-quality data, including millions of labelled images of traffic signs, pedestrians, and vehicles. However, this could be impractical in rural areas where data is sparse and conditions are more variable. In these context, training might need to include rare events, such as encountering livestock or navigating unpaved roads. Moreover, a system might achieve 99% accuracy in sunny conditions but perform poorly in rain or snow. Again, a requirement for real-time transparency (e.g., explaining every decision the car makes) might be excessive in low-risk scenarios but insufficient in high-stakes situations, such as navigating a busy intersection.

As a result, overly rigid, one-size-fits-all standards could stifle innovation in some industries while failing to provide adequate safeguards in others. When it comes to crafting a standard to test for ‘effective training’ with ‘high-quality data’ to ensure ‘system’s accuracy’, as the EU is set to do, it is unlikely that a net of tests

without significant holes can be created that applies across all potential operational scenarios.¹²⁶

This is precisely what has been pointed out in a study by the EU Commission's independent research body, the Joint Research Centre. This study evaluates eight AI standards across various sectors, concluding that most existing standards 'lack sufficient maturity and actionable detail' needed to translate broad principles into actionable practices.¹²⁷ Without clear, context-specific guidelines, providers may struggle to determine what constitutes 'enough' training data or how to balance quality with practicality.

Therefore, the absence of clear and actionable standards for AI risk management underscores a fundamental challenge: providers are left without a reliable framework to assess whether their risk mitigation efforts align with what regulators or the law deem 'acceptable'. Self-driving car manufacturers cannot be sure whether their risk management efforts meet regulatory expectations. This uncertainty has profound implications for the attribution of negligence criminal liability in cases

¹²⁶ Pouget (n 124).

¹²⁷ For example, the transparency standard for autonomous systems created by the Institute of Electrical and Electronics Engineers, IEEE 7001-2021, is an example of a standard that, despite giving an encouraging amount of detail, remains ambiguous in important aspects. Article 13 of the AI Act requires that 'high-risk AI systems shall be designed and developed in such a way as to ensure that their operation is sufficiently transparent to enable deployers to interpret a system's output and use it appropriately'. The IEEE standard outlines five specific and measurable tiers of transparency to different stakeholders (users of these systems, auditors, and the broader public) that can be assessed objectively. These transparency levels, however, are not tied to a risk assessment, so there is little guidance as to which tier of transparency is appropriate to various domain-related risks. In addition, what constitutes an effective and appropriate explanation remains a complex debate within the AI research community. Another standard affected by ambiguity is the requirement for 'high-quality data', which often lacks specific definitions or guidelines for effective implementation. According to the report, this ambiguity arises because 'there is currently no standardized way of describing high-quality datasets agreed upon in the field of AI'. The absence of clear criteria for what constitutes high-quality data makes it difficult for providers to ensure compliance, particularly when the term remains loosely defined across different sectors.

of AI-caused harm, as providers are left without guidance on what behaviour is legally acceptable.

For example, if a car is involved in an accident, was the system's training data sufficient? Were the risk mitigation measures adequate? These questions may only be answered retroactively in court. In other terms, in cases where standards are unclear or not specific enough, courts may end up defining what constitutes acceptable risk retroactively during legal proceedings. This places providers in a difficult position, as they may only discover after the fact that their efforts were deemed insufficient, even if they acted in good faith.

As a consequence, providers could face criminal liability not because they deliberately ignored safety obligations but simply because a generally foreseeable malfunction was judged – after the fact – to have been preventable under a written standard with ambiguous, incomplete, or overly broad content, and which is not intended to prevent a specific result.

This not only raises serious concerns about legal predictability and fairness – substantially shifting the responsibility for defining risk thresholds from legislators and regulators to the judiciary, thereby increasing the risk of inconsistent rulings and regulatory uncertainty that could stifle innovation – but also complicates the establishment of a verifiable link between the provider's breach of duty of care (e.g., negligent training and testing of the system) and the actual harm caused. In doing so, it ultimately undermines the fundamental principle of risk coverage, which establishes that the harmful event must represent the materialization of the specific risk that the precautionary rule aimed to prevent. Without a clear and objective framework for attributing liability, the legal system risks blurring the lines between foreseeable but permissible risks and genuinely negligent conduct, leading to an expansion of liability that could become indistinguishable from strict liability.

In the third and final section of this chapter, we will argue that these challenges stem primarily from the inherent limitations of the AI Act's risk-based approach. While the Act provides a structured framework for assessing and mitigating AI-

related risks, its reliance on broad risk classifications and general precautionary principles may lead to uncertainty in defining concrete legal responsibilities.

To address this issue, we propose an innovative approach that goes beyond the current framework based on classification of legal risks by introducing clearer, context-specific standards for criminal liability.

The innovative solution: ‘deconstructing’ European legal risk

3.1. From a scope-oriented categorisation of risk to a scenario-based model

What we discussed so far shows that the broad risk classification system established by the AI Act suffers from the inherent limitations of a static regulatory approach to AI, which struggles to keep pace with the rapid and unpredictable evolution of this technology. While the regulation provides a structured framework for assessing risk acceptability for a given AI application at a given point in time, it does not fully account for the dynamic, evolving nature of AI systems and their interactions within complex socio-technical environments.

The limitations of the AI risk assessment framework stem not only from its rigid classification of risks but also from the very nature of risk itself as conceptualized in the regulation. The notion of risk adopted by the EU legislator is fundamentally static and normative (*legal risk*), rather than empirical – that is, it does not rely on a detailed risk assessment methodology for identifying actual, experience-based likelihoods of harm occurring but instead reflects a precautionary logic, where risk is legally constructed based on predefined compliance obligations.¹²⁸

Under this approach, the potential adverse impact an AI application may have on fundamental rights and interests is assumed to be predictable *a priori*, determined by compliance or non-compliance with legal safeguards rather than by a case-specific evaluation of real-world failure patterns. As a result, risk assessment under the AI Act operates much like a mathematical model, treating legal norms as

¹²⁸ See Maria Carla Canato, ‘Verso il superamento del legal risk europeo: intelligenza artificiale e approccio proporzionale al rischio’ (2024) *La Legislazione Penale*.

fixed standards, whose application is largely a technical exercise rather than a flexible balancing act capable of adapting to multifactorial, evolving contexts.¹²⁹

From a criminal liability perspective, this static and formalized approach raises critical concerns, particularly regarding the definition of acceptable risk. By framing risk primarily as a legal construct, based on compliance with pre-established precautionary measures rather than an empirical assessment of actual technological hazards, the AI Act risks creating uncertainty about where the boundaries of permissible risk lie.

In criminal law, liability typically hinges on a demonstrable link between a party's negligent conduct and the harm caused. However, under this risk-based framework, a provider could face criminal liability not because their AI system demonstrably caused harm due to negligence or recklessness, but simply because it failed to conform to predefined legal standards – even when those standards may not fully account for rapid technological advancements, context-specific risks, or real-world failure patterns.

This raises two main challenges: first, it blurs the distinction between regulatory non-compliance and culpable conduct, making it difficult to determine when failing to meet a precautionary standard should translate into criminal liability. Second, it shifts the burden of defining risk thresholds from legislators and regulators to the judiciary, meaning courts may retroactively decide what constitutes an unacceptable risk, leading to inconsistent rulings and legal unpredictability.

In this respect, accurately assessing the magnitude of AI risks – namely, the likelihood of harm and the severity of its consequences on values such as health, safety, and other fundamental interests – and determining clear boundaries of acceptable risk cannot be addressed uniformly across the entire field of AI as a whole. Instead, it must be carefully evaluated within each specific application context. Different AI systems operate in diverse environments, serve varying purposes, and

¹²⁹ Nathalie Smuha *et al*, 'How the EU can achieve legally trustworthy AI: A response to the European commission's proposal for an artificial intelligence act' (2021) SSRN Electronic Journal, available at https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3899991.

carry distinct levels of risk. Therefore, the determination of acceptable risk and the thresholds for liability must be tailored to the unique characteristics and challenges of each application.

To address this challenge, and enhance the effectiveness of the European legal framework, certain legal scholars have proposed a risk assessment methodology that systematically identifies and integrates specific ‘risk factors’ influencing real-world AI application scenarios.¹³⁰

Notably, some authors draw upon the risk assessment framework developed by the Intergovernmental Panel on Climate Change (IPCC) working groups.¹³¹ The choice of using the IPCC model as a starting point is self-evident: climate change risk and AI risk share some similarities. Both exhibit highly unpredictable risk magnitudes and escalating complexity due to the interplay of multiple factors. Moreover, they display a substantial dependence on the specific context and the impacted parties.

According to the IPCC model, the risk of an event is evaluated based on the interplay of:

- *Determinants of risk*, including hazard sources, exposure, vulnerability profiles, and response capacity;

¹³⁰ Claudio Novelli, Federico Casolari, Antonino Rotolo, Mariarosaria Taddeo and Luciano Floridi, ‘AI Risk Assessment: A Scenario-Based, Proportional Methodology for the AI Act’ (2024) 3 Digital Society, 13ff. The Authors use the expression ‘risk factors’ to refer in a general way to all variables potentially able to increase or decrease the risk of an event. They specify its meaning by referring to *determinants* and *drivers*. Cfr. also Canato (n 128), 21ff.

¹³¹ To construct risk scenarios, the Intergovernmental Panel on Climate Change (IPCC) provides a multifaceted risk assessment model, which has been further refined by subsequent literature and can be used to evaluate AI risks. The magnitudes of risks associated with both climate change and AI are influenced by a set of interacting factors, leading to context-dependent outcomes. NP Simpson *et al.* (2021) ‘A framework for complex climate change risk assessment’ (2021) 4(4) One Earth 489–501, who refined the framework.

- *Individual drivers* influencing the specific weight of these determinants and shaping the overall risk landscape;
- *Other risk types*, such as extrinsic and ancillary risks, which add layers of complexity to AI risk assessment.

Applying this framework to AI allows for a more precise and context-sensitive estimation of risk magnitude. A key advantage of this model is its flexibility in classifying AI risks. Unlike the rigid sector-based categorization adopted by the AI Act, which designates specific AI applications as high-risk based on their intended function, this framework enables the horizontal application of the four risk categories to all AI systems classified under the regulation. In this way, depending on real-world conditions – for example, a specific interference with fundamental interests in a deployment scenario – the same AI system could be classified as unacceptable, high-risk, low-risk, or minimal-risk. Thus, rather than being determined exclusively by sector, risk assessment would be shaped by the operational scenario or context of deployment, considering the multifactorial interplay of risk causes.¹³²

This approach, defined as ‘semi-quantitative’,¹³³ unfolds in two key phases:

- i. Constructing dynamic risk scenarios (*qualitative phase*), which involve identifying and analyzing key risk determinants within a given AI application context;
- ii. Conducting a proportionality-based quantitative evaluation (*quantitative phase*), ensuring that legal and regulatory responses are tailored to the actual risk magnitude rather than being constrained by overly rigid ex-ante classifications.

¹³² Novelli *et al* (n 130), 9ff.

¹³³ *ibid.*, 18ff.

The first step in constructing risk scenarios involves analyzing the determinants of risk and the complex interactions among their drivers. Hazard determinants include factors like technical malfunctions, environmental conditions, and human factors. In relation to each risk determinants, we refer to ‘drivers’ as specific causes or conditions that amplify the risks, such as data biases, model opacity, or interaction with other systems. Exposure drivers for AI risk, in turn, may be tangible assets, like goods or environment, or intangible assets, like values. Ultimately, constructing risk scenarios means evaluating how each of these elements interacts to create potential hazards, and how likely these interactions are to cause real harm.

This analysis would then help determine the risk category for the specific AI application scenario, as well as identify the precautionary measures needed and assess how these measures might reduce the overall risk. In practical terms, risk categories are assessed by weighing the positive impact of an AI system on the values it aims to serve, against the harm it may cause to the exposed asset. This is done through a proportionality test that is based on the principles of suitability, necessity, and proportionality.¹³⁴

Once the risk categories have been assessed, the next step is to determine what constitutes acceptable risk in the specific scenario. Once again, this refers to the level of risk that is deemed tolerable, given the benefits the AI system brings, while ensuring that the harm to fundamental values is minimized. The risk acceptability judgment involves evaluating whether the risk mitigation measures for a specific risk

¹³⁴ *ivi*, 22ff:

- i. Suitability: A risk category that harms one value is only suitable if it has a positive impact on another value. This principle ensures that any risk taken on must have a clear benefit that justifies it.
- ii. Necessity: When two measures for promoting a value are equally suitable, the one that minimizes interference with other values should be chosen. This ensures that the method used to mitigate risk is the least disruptive to other important values.
- iii. Proportionality in the narrow sense: The greater the detriment to one value, the greater the importance of fulfilling the other. This principle weighs the trade-off between harms and benefits, ensuring that any damage caused to one value is compensated by a sufficiently strong positive impact on another.

category are disproportionate to the scenario in question, using a quantitative analysis. In other words, risk management measures are allocated according to the four risk categories of the AIA.

Ultimately, this structured evaluation, would facilitate and harmonize the implementation of the internal risk management system outlined in Article 9 of the AI Act, providing not only a more comprehensive regulatory framework aimed at assisting providers in identifying, assessing, and managing risks, but also ensuring that any decision on criminal liability or risk acceptance reflects the balance between the benefits and harms of specific AI application scenarios.

3.2. The scenario-based risk assessment model in action: the case of autonomous vehicles

While this scenario-based risk assessment model offers significant advantages, it also raises practical concerns. The AI Act's current risk categorization, though somewhat coarse-grained, facilitates regulatory oversight by linking risk measures to broad categories of AI applications. This allows for more straightforward approval and monitoring processes. In contrast, a highly granular approach – such as the semi-quantitative model we are proposing – introduces greater complexity. A framework that evaluates risk through a combination of interacting factors and proportionality-based balancing, while theoretically more precise, could make compliance more cumbersome and fall short in offering sufficient guidance to regulated actors.

A potential way to address this issue is for the European Commission to take an active role in defining standardized risk scenarios tailored to different AI applications through its implementation acts.¹³⁵ A concrete example of such an approach can be found in the European sector-specific regulation for automated driving systems (ADSs), where scenario-based risk assessment is used to evaluate and mitigate risks based on specific operational conditions and real-world use cases.

¹³⁵ *Id.*, 19.

Notably, the revised General Safety Regulation (GSR),¹³⁶ adopted on 27 November 2019 and in effect since July 2022, has created a legal framework governing the development and deployment of different types of ADSs on EU roads.¹³⁷ The new regulation has developed rules for a harmonized type-approval procedure that sets out uniform, obligatory standards for ADSs from Level 2 to 4.¹³⁸

¹³⁶ Regulation (EU) 2019/2144 of the European Parliament and of the Council of 27 November 2019 on type-approval requirements for motor vehicles and their trailers, and systems, components and separate technical units intended for such vehicles, as regards their general safety and the protection of vehicle occupants and vulnerable road users, OJ L 325, which amended the EU's Vehicle General Safety Regulation (EU) 2018/858 of 30 May 2018 on the approval and market surveillance of motor vehicles and their trailers, and of systems, components and separate technical units intended for such vehicles.

¹³⁷ The legal framework governing the development and use of ADSs emerges from the combined reading of the AI Act and EU sectoral legislation. While ADSs would certainly be classified as high-risk under the AI Act (because they are 'products covered by other specific harmonized legislation or function as a safety component of such products', Article 3 (14) AIA), the regulation has only an indirect impact on ADSs, as it is a sector-neutral legislation. Consequently, rather than being subject to the AI Act, certain ADSs shall likewise be caught by the sector-specific EU legislation, even if these products and systems qualify as high-risk AI (Art. 2(2)(f)). At the same time, however, the Commission must take certain requirements under the AI Act into account when adopting delegated acts under the Type-Approval Framework Regulation - including, amongst others, risk management, data and data governance, documentation and recording keeping, transparency, human oversight, robustness, accuracy and security. These requirements must hence be considered in future delegated acts adopted by the Commission based on the technical and regulatory specificities of the automotive sector and ADSs. While delegated acts are of a non-legislative nature, they are still legally binding.

¹³⁸ ADSs encompass a range of technologies, from conventional human-operated cars equipped with advanced driver-assistance systems (ADAS) to fully automated, driverless vehicles, which, as of now, remain an aspiration. The classification system, widely accepted as the industry standard, was developed by the Society of Automotive Engineers (SAE) and defines six levels of automation that a car may offer, ranging from Level 0 (no automation) to Level 5 (full automation). Each level corresponds to a different degree of control the vehicle's system has over its operation, and the degree of human intervention required. For a detailed description of the six levels of automated driving, see the SAE's website: SAE International, 'SAE Levels of Driving Automation_ Refined for Clarity and International Audience' (SAE International, 3 May 2021) <https://www.sae.org/blog/sae-j3016-update>. See also United Nations Economic and Social Council, 'Reference document with definitions of Automated Driving under WP.29 and the General Principles for developing a UN Regulation on automated vehicles' (ECE/TRANS/WP.29/1140, United Nations, 23 April

Although it seems to follow the six-level classification inaugurated by the SAE, Regulation 2019/2144 employs the term ‘autonomous vehicle’ (AVs) to refer to ADSs of Level 3 and Level 4.¹³⁹

Additionally, the regulation has granted the EU Commission authority to define harmonized technical rules for AVs, detailing the specific obligations of operators in this sector. Leveraging this mandate, the Commission introduced technical legislation for Level 3 AVs (replacing drivers) and Level 4 fully driverless vehicles (such as urban shuttles and robotaxis) via a delegated¹⁴⁰ and implementing act,¹⁴¹ detailing uniform procedures and technical specifications for the type-approval of these AVs.

In particular, the implementing regulation proposes a risk assessment methodology of AVs that ‘relies heavily on the *traffic scenarios* that are relevant for the different use cases of fully automated vehicles’ (emphasis added)¹⁴² with a view to establishing proportionate safety requirements tailored to the operational context of an AV.

From a qualitative perspective, the EU Commission categorizes traffic scenarios into three distinct groups, each reflecting different levels of operational complexity and risk exposure:

2018) <https://unece.org/fileadmin/DAM/trans/main/wp29/wp29resolutions/ECE-TRANS-WP29-1140e.pdf>; Rachel Gordon, ‘Explained: Levels of autonomy in self-driving cars’ (MIT CSAIL, 10 December 2020) <https://www.csail.mit.edu/news/explained-levels-autonomy-self-driving-cars>.

¹³⁹ Regulation 2019/2144, Article 3(21) reads as follows: ‘automated vehicle’ means a motor vehicle designed and constructed to move autonomously for certain periods of time without continuous driver supervision but in respect of which driver intervention is still expected or required’; and according to art 3(22) ‘fully automated vehicle’ means a motor vehicle that has been designed and constructed to move autonomously without any driver supervision’.

¹⁴⁰ Commission Delegated Regulation (EU) 2022/2236 https://eur-lex.europa.eu/legal-content/EN/PIN/?uri=PI_COM:Ares%282022%292077610

¹⁴¹ Commission Implementing Regulation (EU) 2022/1426 https://eur-lex.europa.eu/legal-content/EN/PIN/?uri=PI_COM:Ares%282022%292667391

¹⁴² *ibid.*, Recital 3.

- i. *nominal traffic scenarios*, encompassing ‘reasonably foreseeable situations encountered by the AV when operating in normal or non-critical conditions within its operational design domain’ (ODD).¹⁴³ These include common driving manoeuvres such as lane changes, turns, merging into traffic, and navigating intersections in urban and rural environments.¹⁴⁴
- ii. *critical scenarios*, referring to ‘edge-cases (e.g. unexpected conditions with an exceptionally low probability of occurrence) and operational insufficiencies, not limited to traffic conditions but also including environmental conditions (e.g. heavy rain or low sunlight glaring cameras), human factors, connectivity and miscommunication leading to emergency operation of the ADS’.¹⁴⁵ Such scenarios include unexpected obstacles suddenly appearing on the road, rapidly changing environmental conditions (e.g., glare affecting cameras, heavy rain impairing sensors), human errors (e.g., unpredictable pedestrian behavior), connectivity disruptions, and miscommunication with other vehicles or road users. These conditions may require emergency responses by the ADS to prevent accidents.¹⁴⁶
- iii. *failure scenarios*, meaning ‘scenarios related to ADS and/or vehicle components failure which may lead to normal or emergency operation of the ADS depending on whether or not the minimum safety level is preserved’.¹⁴⁷ The AV must be designed to detect failures in real-time and execute predefined safety responses, such as transitioning to a

¹⁴³ *ivz*, Article 2(19). Operational design domain is defined as ‘operating conditions under which a given ADS is specifically designed to function, including, but not limited to, environmental, geographical, and time-of-day restrictions, and/or the requisite presence or absence of certain traffic or roadway characteristics’ (Article 2(16)).

¹⁴⁴ *ivz*, Annex II, para 1 and Annex III, para 1.

¹⁴⁵ *ivz*, Article 2(20).

¹⁴⁶ *ivz*, Annex II, para 2.1.

¹⁴⁷ *ivz*, Article 2(21).

minimal risk condition (e.g., safely pulling over or alerting occupants to take control, where applicable).¹⁴⁸

This structured risk assessment model provides a common framework and detailed safety performance requirements for evaluating AV safety under a variety of scenarios and potential operating conditions; in doing so, it aligns with Article 9(1)(b) of the AI Act, which mandates providers to identify both known risks and foreseeable risks while implementing risk mitigation measures proportional to risk magnitude.

From a quantitative perspective, before an AV is approved for market deployment, the provider must establish a *safety management system* demonstrating that the system meets the relevant performance requirements and remains free from *unreasonable safety risks* for vehicle occupants and other road users across all relevant traffic scenarios throughout its operational lifetime.¹⁴⁹

A critical component of this evaluation is defining ‘unreasonable risk’, which is assessed by comparing the AV’s risk level to comparable transportation services and manually driven vehicles operating under similar conditions (*machina eiusdem fabricationis et condicionis*).¹⁵⁰ Residual risk – the remaining level of risk after mitigation measures – must be deemed *acceptable* based on established benchmarks.

To this end, the provider shall define the *acceptance criteria* from which the validation targets of the AV are derived to evaluate the residual risk taking into account, where available, existing accident data, data on performances from

¹⁴⁸ *ivi*, Annex II para 4.

¹⁴⁹ ‘The manufacturer shall demonstrate that an acceptable degree of consideration has been given to the functional and operational safety for the ADS during its design and development processes. The measures put in place by the manufacturer shall ensure that the fully automated vehicle is free of unreasonable safety risks to vehicle occupants and other road users during the vehicle lifetime when compared with comparable transport services and situations within the operational domain’. *ivi*, Annex II, para 7.1.

¹⁵⁰ In particular, ‘unreasonable risk means the overall level of risk for the vehicle occupants and other road users which is increased when compared to comparable transportation services and situations within the operational design domain’. *ivi*, Recital 28.

competently and carefully driven manual vehicles and technology.¹⁵¹ Therefore, acceptance criteria and metrics must be grounded in measurable benchmarks, reflecting both the AI system's intended purpose and the state of the art available at the time of the deployment.

For instance, suppose that, based on accident data and technological state-of-the-art, an indicative acceptance criterion for an AV under nominal scenarios is estimated to be of X% fatalities less than the average relative to other systems of the same category and/or competently-driven manual vehicles per million hours of operation. Provided it can demonstrate that this acceptance criterion leads to an absence of unreasonable safety risk, the provider could then consider it as the industry acceptability benchmark for risk mitigation in view of market introduction of AVs. Conversely, if the system's performance falls significantly below the industry benchmark, this could signal an unreasonable risk, necessitating further mitigation measures before regulatory approval.¹⁵²

Most importantly, our hypothesis here is that by shifting the focus of AI risk evaluation from a rigid, predefined legal construct (i.e., AI Act's risk-based framework) to a scenario-based assessment rooted in clear and measurable benchmarks for permissible risk, we could also achieve a more coherent and nuanced

¹⁵¹ *id.*, Annex II, para 7.1.1.

¹⁵² This idea, which involves holding the algorithm to its own standard of care, has already been taken into consideration by legal doctrine (in both areas of civil and criminal liability), where it is customary to refer to the so-called 'reasonable algorithm' parameter. The idea of a standard of care referring directly to the AI system is taken into consideration in the area of civil liability, where it is customary to refer to the so-called 'reasonable algorithm'. Cfr. *ex multis*, Ryan Abbott, *The Reasonable Robot* (Cambridge University Press 2020), 69; Marta Giuca, 'AI and Criminal Liability. Algorithmic Error and Human Negligence in the Context of the European Regulation' (2023) 1 *Diritto Penale Contemporaneo: Rivista trimestrale* 46-69; Chagal-Feferkorn, 'The Reasonable Algorithm' (2018) 1 *Journal of law, technology & policy*, 111-148. While some scholars argue that the 'reasonable algorithm' standard only application when algorithms are recognized as having legal personality, we hold here that it may be relevant even where the algorithm is not considered an independent center of imputation, since such an assessment serves to assess the producer's diligence in light of the reasonable algorithm standard.

approach to criminal liability determination, enabling a case-by-case analysis that aligns better with the evolving nature of AI applications.

Rather than treating risk as an abstract regulatory threshold, this perspective acknowledges the interplay between AI systems and their operational environments, ensuring that criminal liability assessments reflect real-world complexities rather than being confined to artificial legal classifications.

3.3. Criminal law implications: liability for the unreasonable risk released by AI

From a criminal liability perspective, the proposed scenario-based framework introduces significant advantages by shifting the focus from the foreseeability and preventability of isolated harmful events – an approach that is often impractical for AI systems due to their opaque and evolving nature – to a broader assessment of an AI system’s *overall performance* across defined operational scenarios. Under traditional liability frameworks, a single high-profile accident could trigger disproportionate legal consequences, discouraging AI development. By contrast, the scenario-based approach prevents overreaction to isolated failures, ensuring that AI development is evaluated in context rather than penalized for rare outlier events.

This transition, in turn, clarifies the duty of care owed by AI providers: it acknowledges that absolute safety is unattainable while still holding providers responsible for ensuring that risk remains within scientifically and statistically acceptable thresholds. If an AI system’s performance consistently falls outside these defined margins, this would indicate an unacceptable level of risk, signaling not only a failure of the provider’s duty of care but also a potential basis for criminal liability.

Imagine an AV-taxi designed for rural environments where human-driven taxis typically experience 3 fatal accidents per 100 million kilometers driven. Suppose that, despite adherence to legally established safety and performance requirements, the provider finds out that its AV still records 5 accidents per 100 million kilometers – a rate exceeding industry risk acceptance benchmarks.

In this case, an AI system enters the market in full compliance with regulatory requirements and technical standards, meeting all due diligence obligations at the time of deployment. However, only later does it exhibit unforeseen risks that exceed the initially acceptable safety threshold within the operational scenario under consideration. The mere fact that this threshold is surpassed does not automatically impose liability on the provider for subsequent harmful events. Instead, it should be interpreted as an early warning of potential failure of existing safeguards – one that necessitates corrective action.

This need for continuous monitoring and post-market intervention aligns with certain legal theories that, despite originating in corporate liability and focusing on different types of offenses (such as financial crimes), propose applying the concept of *reactive fault* to AI risks.¹⁵³ Since it is often impossible to predict and prevent all potential harms arising from an AI system's autonomous actions in advance, criminal liability should instead depend on how quickly and effectively a provider responds once a risk materializes. In this perspective, responsibility is not limited to ensuring compliance at the time of market introduction but also extends to the provider's duty to recognize emerging dangers and take timely corrective measures to restore the safety performance threshold.

In this regard, a viable approach for emerging technologies – one that reduces the uncertainty surrounding negligence-based liability while ensuring a high level of protection for legal interests – could involve establishing structured procedures for risk management. These procedures would require continuous engagement with regulatory authorities, fostering a dynamic oversight system that evolves alongside technological advancements. By formalizing risk assessment and mitigation measures within a procedural framework, this approach would provide clearer guidelines for compliance while enhancing accountability and safety.

¹⁵³ The concept of reactive fault is elaborated by B. Fisse, 'Reconstructing Corporate Criminal Law: Deterrence, Retribution, Fault, and Sanctions' (1983) S. Cal. L. Rev., 1141 ff. See also Sabine Gless, 'Strafrechtliche Produkthaftung' (2013) 2 Recht 54, 60 who called this a 'dormant duty to act', which comes alive whenever reports of harmful incidents or the exceeding of the risk acceptance threshold signals a potential failure of existing safeguards.

Coming back to our AV-taxi case, the following sequence of steps might be relevant:

Step 1: *Identifying the cause* - Upon investigation, regulators, supervisory authorities, or the provider *motu proprio* find that the AV struggles with pedestrian detection when crossing the road outside of a designated crosswalk at night, wearing dark clothing, and in an area with poor street lighting, leading to delayed braking (edge-case scenario).

Step 2/a: *Adopting corrective measures* - The provider issues consumer warnings, recalls the systems and enhances the perception system with improved infrared sensors and better predictive algorithms in order to mitigate the unreasonable risk released by the system.

Step 2/b: *Failure to adopt corrective measures* - If the provider fails to fulfill this duty, delaying implementation of recommended updates due to cost concerns, they could be held criminally liable for any subsequent incidents resulting in death or injury on grounds of (commissive or omissive) negligence.

Step 3/a: *Positive reassessment* - With these improvements, the AV's accident rate drops to 2 fatalities per 100 million kilometers, now outperforming human-driven vehicles and meeting the industry benchmark. At this point, regulators may determine that the residual risk is acceptable, as the AV's performance level now meets existing safety standards. Therefore, this approach helps define a margin of tolerance or, more pointedly, an area of impunity within which a provider operate without the threat of liability. In such a framework, the provider cannot be held responsible for harmful events that occur despite adherence to the performance safety threshold. These incidents should instead be regarded as an expression of acceptable risk, meaning they are adverse outcomes that society, as a whole, has implicitly agreed to tolerate.

Step 3/b: *Negative reassessment* - Despite these improvements, the AV's accident rate remains at 4 fatalities per 100 million kilometers, still exceeding the human-driven benchmark. This suggests that while risk has been reduced, it has not

been mitigated to an acceptable level. In this case, the provider may either consider the risk of deploying the AV in poorly lit rural areas as *unacceptable* and restrict it in such scenarios until additional safety improvements are made; or mandate human supervision in such high-risk scenarios (e.g., requiring a remote operator at night). Conversely, if the provider fails to act on these obligations and harm occurs, it would face criminal liability due to its failure to address foreseeable risks. Importantly, it is not required for this verification to meet the stringent standards of establishing a causal link as defined in criminal law – an assessment that, as we have noted, involves a number of challenges. Instead, the assessment should aim to isolate AI negligent behaviour without the need to identify the exact cause of the system’s failure and, consequently, the harmful event (which most of the time, indeed almost always, eludes the predictability of the agent).

The potential outcomes are summed up in the table below:

Step	Description	Outcome & Liability Considerations
Step 1	AV struggles with pedestrian detection at night	Negligent AI
Step 2/a	Provider issues warnings and improves the system	Diligent provider
Step2/b	Provider does not issue warnings or improve the system	Negligent provider
Step 3/a	AV’s fatality rate drops below human benchmarks	Diligent AI, Diligent Provider
Step 3/b	AV’s fatality rate remains above human benchmarks	Negligent AI, Negligent Provider

However, the limitations of this approach cannot be overlooked. While the approach of requiring providers to take additional precautions if the system’s performance exceeds permissible thresholds is a step in the right direction, it faces significant technical, organizational, and legal challenges.

One major challenge lies in the technical feasibility of monitoring AI systems effectively.¹⁵⁴ Ensuring continuous, real-time performance evaluation necessitates advanced analytical tools capable of processing complex data patterns and detecting potential issues before they escalate into significant harm. Current technological capabilities may fall short, particularly for systems operating in dynamic, unpredictable environments. Without reliable mechanisms for identifying and interpreting the precursors to unreasonable risks, the effectiveness of the proposed approach could be undermined.

Moreover, imputing a harmful event to a provider who recklessly or negligently allowed the system to exhibit negligent behaviour (e.g. for adopting an architecture that enabled the AI to behave negligently or failing to implement controls that could have prevented the system from evolving into a risk source) introduces a tension with the principles of liberal criminal law, which traditionally emphasize the significance of the harmful event.

In fact, attributing criminal liability to providers for the unreasonable risks posed by AI shifts the weight of criminal disvalue from the occurrence of specific harmful events to the loss of control over the source of risk itself, thus downplaying the event. Such a framework suggests liability is grounded not in the causation of a harmful event per se (which in complex technological contexts can often be a variable independent of the implementation of precautions, thus releasing unreasonable risks even when all safeguards are complied with) but in the provider's failure to create and maintain the conditions necessary for exercising control over the source of risks. For instance, a provider who continues to operate an untrustworthy system, despite being aware of performance requirements and their preventive function, could be held liable – not because they accepted the occurrence of a specific harm, but because they failed to prevent unreasonable risks.¹⁵⁵

¹⁵⁴ Fragasso (Ch II n 62), 38.

¹⁵⁵ This result could be achieved by expanding the definition of recklessness (or *dolus eventualis*, in continental jurisdictions) to encompass what could be termed 'opaque reckless', namely the situations in which the defendant is aware that their conduct carries

While, on one hand, traditional liability models dependent on the occurrence of a harmful event could undermine their effectiveness, on the other, reducing the event to a mere objective condition of punishability risks excessively anticipating the thresholds of legal protection, potentially undermining fundamental principles of justice. Thus, any attempt to redefine negligence liability must strike a careful balance: it must ensure effective prevention of unreasonable risks without unduly diminishing the role of the harmful event.

In this respect, we believe that the complexity of the AI field, coupled with the urgency under which humans operate, should lead criminal courts to adopt cautious and prudent evaluations. Exceeding performance thresholds should be treated not as an automatic basis for criminal liability but as a prompt for deeper analysis of the circumstances, the precautions in place, and the specific failures involved.

Indeed, some instances of AI harm might be categorized as ‘temperate artificial negligence’ or ‘unattainable diligence’.¹⁵⁶ In these cases, the system’s performance falls below the established benchmark, yet the harm is not attributable to the provider’s fault. Rather, it arises from unpredictable edge cases – scenarios that are inherently difficult or impossible to foresee and mitigate, even with the highest standards of diligence.

To illustrate, let’s consider the example of a self-driving car that runs a red light, resulting in a collision. Imagine the system fails to interpret the traffic signal correctly due to graffiti, stickers, or other form of damage partially covering it. As a result, the car might not recognize the obscured signal as a valid stop light and hit a pedestrian crossing the road. Here, while the system’s overall performance might not comply with the expected safety thresholds (e.g., higher-than-permissible accident rates due to software inefficiencies in pedestrian detection), this particular incident

certain risks but fails to recognize or consciously ignores the specific nature of those risks. On this point, see Giovanni Sartor and Francesca Lagioia, ‘AI Systems Under Criminal Law: a Legal Analysis and a Regulatory Perspective’ (2019) 3(3) *Philosophy & Technology* 433-465, 459. See also Consulich (Ch I n 21), 303-305.

¹⁵⁶ As to this terminology, we draw inspiration from the one used in Giuca (n 152), 55.

results from a wholly unforeseeable edge-case (the damaged red light) beyond the provider's control, which could be considered an acceptable risk within the broader societal context.¹⁵⁷

Therefore, holding the provider responsible for this specific harm would be unjust, as the failure arose from the peculiarities of the concrete case, where circumstances arise that make the factual situation completely unique and unpredictable and such as to be characterized by an 'eccentric risk'.¹⁵⁸ Such cases highlight the inherent limitations of even the most advanced AI systems and the importance of ensuring that liability is appropriately attributed only in cases where the provider has genuinely failed to meet their duty of care, avoiding the risk of over-criminalization. It thus introduces a margin of tolerance for certain minor errors due to the unavoidable limitations of the technology rather than from negligence, while at the same time holding individual providers responsible for significant lapses in diligence.

¹⁵⁷ Of course, even in such a case, the provider could be held criminally liable if it does not comply with the injunction to repair a fault in the preventive protection mechanisms identified by the supervisory authorities. Piergallini (Ch I n 39), 1769ff.

¹⁵⁸ Giuca (n 152), 55.

SECTION 4

Provisional conclusions

The analysis conducted in this chapter highlights the growing need for criminal law to engage with new artificial agents – entities capable of autonomous, potentially harmful decisions that remain at least partially unpredictable to their human creators. This unpredictability is particularly disruptive in the realm of negligence-based liability, already stretched to its limits by the pervasive role of technology in post-modern society, where human activities, even when socially beneficial, are increasingly characterized by heightened risks and epistemic uncertainty.¹⁵⁹

The standard framework of criminal negligence, which links liability to the direct causation of a harmful event, proves inadequate in this new landscape, particularly in relation to producers of high-risk technological systems. Instead, the focus must shift toward a model that centers on risk acceptability and control. Under this paradigm, negligence-based liability is defined not as imputation for the direct causation of an adverse event – which is often impossible to predict – but for the failure to manage a known source of risk.¹⁶⁰ Under this approach:

- The AI producer does not need to foresee the exact event that causes harm.
- A strict causal investigation into the specific algorithmic input that triggered the harm would no longer be necessary.
- What matters is whether the risk generated by the AI system exceeded an acceptable threshold, as defined by law at the time of its creation or deployment.

¹⁵⁹ Carlo Piergallini, ‘voce Colpa (dir. pen.)’ in Enc. Dir. Annali X (2017).

¹⁶⁰ Consulich (Ch I n 21), arguing in favour of a producer’s responsibility grounded on a sort of *culpa in agendo*.

However, this does not mean that every instance in which an AI system's risk exceeds permissible limits should automatically trigger criminal liability. Doing so would amount to an indiscriminate expansion of criminal liability – one that sacrifices the principle of verifiable risk connection between fault and event. Rather, such an exceedance should be understood as an early warning – evidence of regulatory failure that calls for stronger safeguards but does not, in itself, suffice to impute blame for all resulting harms. In this sense, liability would arise if the producer fails to implement adequate safeguards and the event belongs to the class of harms that were hypothetically foreseeable at the time of the AI system's creation and deployment.

This model appears both practical and theoretically sound. It has the advantage of alleviating the impossible burden of requiring developers to predict every specific malfunction or unintended consequence while aligning with contemporary regulatory frameworks, such as the AI Act's risk-assessment approach.

Yet this reconceptualization is not without its tensions. A criminal law model that prioritizes the failure to control risk over the causation of harm inevitably disrupts long-standing principles of liberal criminal justice. In particular, it alters the weight traditionally placed on the event itself in a liberal criminal law. By focusing on the loss of control over a source of danger, rather than the actual harm caused, this approach risks sidelining the centrality of the harmful event, *de facto* transforming event-based offences into danger-based offences.

Ultimately, it is clear that, despite our best efforts in interpreting and reconstructing the legal framework, the possibility of attributing criminal liability to AI providers for specific harmful events remains constrained. Some degree of recognition of these facts drives the possibility of resorting to two possible strategies for tackling AI risks from a criminal law perspective. One possible approach is to shift responsibility from AI creators to human controllers, relying on the principle of *human oversight* as a safeguard against excessive AI autonomy. Another option is to anticipate the criminal law intervention by introducing new offenses that penalize excessive risk-taking before any actual harm occurs.

The following chapter will explore the first path, examining the idea that human oversight serves as a necessary corrective to AI's inherent unpredictability.¹⁶¹ The rationale behind this requirement is that if risks to fundamental rights and legal interests cannot be adequately contained by providers, human intervention should at least be ensured to verify and control the system's operations – some risks are simply too great to be entirely entrusted to autonomous systems.

While this reasoning appears sound in theory, it obscures a deeper question: how does the interaction between humans and AI systems redistribute legal responsibility? The assumption that continuous human oversight is an effective safeguard ignores the evolving nature of human-machine relationships. As AI systems become more autonomous, the role of human operators necessarily diminishes. Their ability to meaningfully intervene declines, and with it, their level of criminal responsibility should, in principle, decrease as well. If the AI system functions independently in a way that its developers did not and could not foresee, then its behavior might even exculpate the human actors involved rather than implicate them.

Therefore, Chapter IV will argue that, despite human oversight, the reality remains that AI risks can never be entirely eliminated. The law must still draw a clear line between permissible and impermissible risk, between the inevitable margin of error that must be tolerated and the threshold beyond which negligence becomes culpable. But where should this boundary be set? In the context of human oversight and human-machine interactions, establishing this line proves particularly challenging. The chapter will delve into these tensions, arguing that a more nuanced framework is needed – one that ensures responsibility is fairly distributed rather than merely imposed on convenient scapegoats.

¹⁶¹ On the second strategy, see *amplius* Chapter V.

CHAPTER IV

AI DEPLOYER'S CRIMINAL LIABILITY: WHERE IS WALDO?

SECTION 1

The centrality of human oversight: panacea for AI risk or catalyst for liability?

1.1. The position and legal obligations of deployers under the AI Act

The legal challenges posed by the technical and organizational features of AI systems become even more pronounced when humans interact with them. Human-machine interaction, in fact, brings into the spotlight a critical figure that has so far been overlooked in our discussions regarding AI-based liability: the *user* or *deployer* of the technology. It is therefore important to understand how the law views users of AI, not just its creators or providers.

In legal discussions on negligence and harm involving advanced technologies, a central assumption is that the offense requiring criminal law intervention will typically occur at the stage of technology usage. This assumption seems especially applicable in the case of AI systems, where the user is often seen as the actor closest to any harmful event. Given this proximity, users are presumed to be well-positioned to monitor and manage potential risks, thus bearing a significant share of responsibility. However, defining who the ‘user’ is in AI contexts – and clarifying their specific responsibilities – becomes challenging, particularly with systems designed to operate autonomously and without ongoing human intervention.

The EU AI Act defines a ‘deployer’ as ‘a natural or legal person, public authority, agency or other body using an AI system under its authority except where the AI system is used in the course of a personal non-professional activity’.¹ While distinguishing between a provider and a deployer is critical, as these roles carry different responsibilities under the regulation, the definition does little to clarify what it means to use an AI system that by its nature operates autonomously and replaces

¹ AI Act, Article 3(4) and Recital 13. Only professional users fall within the scope of application of the AI Act; personal use is not included.

traditional human roles, such as a driver, decision-maker, or medical professional. This presents a paradox: how does one ‘use’ a system whose core function is to operate independently of human control?

While legal doctrine and regulations usually refer to AI user as a monolithic entity, we acknowledge that this category is much broader and multifaceted. This concept encompasses different human actors depending on how human and artificial agents interact. Each type of interaction implies the establishment of a ‘relationship’ between human agent and machine.

Essentially, different forms of interaction can be distinguished based on how the decision-making process is distributed between human and artificial agents. Accordingly, it is possible to distinguish between three types of human-machine interactions depending on whether the source of the decision-making authority is (i) human, (ii) hybrid, or (iii) algorithmic.²

First, while AI radically alters how most human activities get done and who does it, the technology’s larger impact will be in complementing and augmenting human capabilities, acting more as an aid rather than a substitute to human decision-making.³ This form of interaction gives rise to situations where the decision-making authority remains in the hands of the human agent. The legal slogan is that there must

² Researchers are defining new types of interactions between humans and machine learning algorithms. Depending on who is in control of the learning process, we can identify: active learning, in which the system remains in control; interactive machine learning, in which there is a closer interaction between users and learning systems; and machine teaching, where human domain experts have control over the learning process. See Eduardo Mosqueira-Rey *et al*, ‘Human-in-the-Loop Machine Learning: A State of the Art’ (2022) 56 AIR 3005; Iyad Rahwan *et al*, ‘Machine Behaviour’ (2019) 568 Nature, 477-486.

³ See James H Wilson and Paul R. Daugherty, ‘Collaborative Intelligence: Humans and AI Are Joining Forces’ (Harvard Business Review, July-August 2018) <https://hbr.org/2018/07/collaborative-intelligence-humans-and-ai-are-joining-forces>. Talia Gillis, ‘Regulating for humans-in-the-loop’ (ECGI blog, 2022) <<https://www.ecgi.global/publications/blog/regulating-for-humans-in-the-loop>>.

always be a ‘human in the loop’ when decisions affect important interests.⁴ This means that crucial decisions must be ultimately taken by the human operator instead of being processed fully automatically.⁵

Under this framework, AI is essentially seen as a tool to help human users make complex decisions in a variety of fields. The underlying assumption behind these human-in-the-loop mandates is that humans and machines can effectively complement each other. Algorithms excel in speed and helps us perform complex decision-making in a variety of fields. However, they struggle with ethics, following norms, and explaining their decisions. Moreover, they heavily depend on the quality of their training data and the data fed into them, which can lead to biases being perpetuated. As a result, algorithms tend to perform poorly in edge cases.⁶

This is precisely why we need human users ‘in the loop’. AI systems can boost our analytic and decision-making abilities by providing the right information at the right time. However, AI users need to perform three crucial roles to sustain their responsible operation. They must carefully *evaluate* the system’s output; *explain* why they decide to rely on it, especially when the results are counterintuitive or controversial; and *monitor* the system’s performance and the quality of its decisions.

Most of the examples of AI-driven decision aids – such as in employment, credit, criminal justice – are not in the purview of traditional criminal law. But AI decision aids are reaching into every aspect of society, and it should not be surprising that it has also entered several domains that may be subject to criminal negligence, including medical malpractice, investment advice, and data security.

Here, however, the centrality of AI in the human decision-making process is what makes liability for AI-assisted decisions tricky. While decision-assistance

⁴ High-Level Expert Group on Artificial Intelligence (AI-HLEG), Ethics guidelines for trustworthy AI, 2019, available at: <https://ec.europa.eu/futurium/en/ai-allianceconsultation.1.html>, 5.

⁵ See Rebecca Crotoft, Margot E. Kaminski and W. Nicholson Price, ‘Humans in the Loop’ (2023) 76 VLR. 429, 464.

⁶ *id.*, 465.

technology may not *directly* harm individuals, it can substantially disrupt decision-making processes, such as the decision about what clinical treatment to apply to a certain patient. AI introduces a layer of complex, often opaque computation that replaces individualized reasoning with statistical analysis, frequently identifying unintuitive relationships as the basis for decisions.⁷ This raises a significant concern for the attribution of negligent liability: how does the integration of AI into human decision-making processes affect those decisions? More importantly, do these changes undermine criminal law's ability to attribute liability to AI users for harmful events?

We now turn to the second type of human-machine interaction: hybrid decision-making processes. In these situations, the decision-making process is shared between human and AI agents, creating a more collaborative framework. Here, the AI system operates autonomously but under the ongoing supervision of a human agent, who does not actively participate in the automated process but is expected to intervene when necessary. This kind of AI user is what we call 'human-on-the-loop'.⁸

Unlike in human-in-the-loop systems, where the human agent has the final say, human-on-the-loop setups may allow AI systems to operate with more autonomy. The human user may intervene only in specific instances, such as when the system flags anomalies, makes high-risk decisions, or encounters a situation that exceeds its programmed parameters. Here, the role of the human user, though secondary in nature, is crucial for the system's safety, making their vigilance and readiness to act a core component of the overall decision-making process.⁹

An illustrative example of hybrid decision-making can be found in the realm of self-driving cars, specifically with the role of the backup driver. In many semi-autonomous vehicle systems, the car is capable of driving itself using advanced AI technologies, sensors, and algorithms. While the car's AI makes most driving

⁷ Legal doctrine has extensively tackled this issue mainly from the perspective of tort law. See, *ex multis*, Andrew Selbst and Solon Barocas, 'The Intuitive Appeal of Explainable Machines' (2018) 87 Fordham L. Rev., 1089-1099; Selbst (Ch III, 113), 1330.

⁸ HLEG (n 4), 5.

⁹ Giuca (Ch III n 152), 58.

decisions autonomously, a human backup driver is often expected to remain on alert, ready to take control in case the system encounters a situation it cannot handle, such as an unexpected obstacle, a complex traffic scenario, or sensor failure.

A similar hybrid decision-making framework can be observed in the context of AI-assisted medical surgery. In modern operating rooms, advanced robotic systems equipped with AI are increasingly being used to assist surgeons during complex procedures. These systems are designed to enhance precision, reduce the risk of human error, and improve patient outcomes. However, like the example of the self-driving car and its backup driver, the ultimate responsibility for the surgery does not rest solely with the AI. A human surgeon must always be present to oversee the operation, make critical decisions, and intervene if the technology encounters limitations.¹⁰

This type of interaction, while seemingly beneficial in blending human oversight with machine efficiency, presents unique legal challenges. Indeed, the transfer – and consequent diffusion – of responsibility in hybrid systems raises questions about potential attribution of liability to human users for AI harms. In this model, the human agent is not always directly involved in every decision, and the AI system often operates independently unless human intervention is triggered. Moreover, the more autonomy AI systems gain in these hybrid settings, the more they shape and influence human decisions. Consequently, determining negligence in cases of harm becomes more complex: was the human user sufficiently vigilant in their supervisory role? Should the AI system have raised a flag for human intervention earlier? Did the human user have enough information or expertise to override the AI?

The final category of human-machine interaction is the ‘human-out-of-the-loop’ model, where decision-making authority is fully transferred or delegated to the

¹⁰ Nicholson Price, ‘Medical Malpractice and Black-Box Medicine’ in I. Glenn Cohen, Holly F. Lynch, Effy Vayena, and Urs Gasser (eds.), *Big Data, Health Law, and Bioethics* (Cambridge UP 2018) 295-306 300; Benjamin Bartlett, ‘The possibility of AI-induced medical manslaughter: Unexplainable decisions, epistemic vices, and a new dimension of moral luck’ (2023) 23(3) *Medical Law International* 241-270.

AI system, with no direct human intervention in real time. The EU High-level Group on AI, also defining this form of interaction as ‘human-in-command’, describes it as ‘the capability to oversee the overall activity of the AI system and the ability to decide when and how to use the system in any particular situation. This can include the decision not to use an AI system in a particular situation, to establish levels of human discretion during the use of the system, or to ensure the ability to override a decision made by the system’.¹¹ In this configuration, AI systems operate autonomously, executing tasks and making decisions without a human supervisor monitoring or approving each action in real time. These AI systems are designed to function independently, often in environments or scenarios where constant human oversight is impractical, inefficient, or impossible.

Here, the human’s role is largely relegated to *ex-post monitoring*: once the system is operational, the operator (who is a mere ‘supervisor’ or ‘controller’) is no longer directly involved, except perhaps for periodic maintenance or audits. Take, for example, algorithmic trading in financial markets.¹² In this scenario, AI systems autonomously execute high-frequency trades based on real-time market data, without human traders being involved in each individual transaction. These systems are designed to analyze massive amounts of market information in milliseconds, identifying opportunities and executing trades at speeds and scales impossible for human traders to match.

Similarly, consider the case of autonomous drones in military operations. These drones, once deployed, may be programmed to identify and strike specific targets based on real-time data and pre-programmed criteria. However, if the drone mistakenly identifies a civilian as a combatant and executes a strike, the decision is entirely on the machine. There is no human agent to override or correct the mistake in real time.

¹¹ HLEG (n 4), 5.

¹² See *amplius* Filippo Annunziata, *AI and market abuse legislation. A European perspective* (Elgar 2023); Alessio Azzutti, ‘AI-driven Market Manipulation and Limits of the EU law enforcement regime to credible deterrence’ (2022) 45 Computer Law & Security Review.

While this form of interaction allows for highly efficient, scalable, and often faster decision-making, it also poses unique challenges for legal frameworks traditionally built around human agency and direct control. A fully autonomous system, especially in high-stakes fields such as defense, finance, or healthcare, can cause substantial harm if it malfunctions, makes poor decisions, or behaves in unintended ways. The challenge here is determining whether the user (or operator) can be responsible for a system operating beyond their direct control. In fact, negligent liability doctrine often relies on the ability to attribute blame to a human actor who either directly causes harm or fails to prevent it. When decision-making shifts entirely to AI systems, it becomes difficult to attribute liability to the operator, especially when the latter performs only a monitoring role.

Ultimately, the human-out-of-the-loop scenario presents the greatest departure from traditional decision-making models and is perhaps the most challenging from a legal perspective. It strips away the immediate human involvement that has historically been central to the assignment of responsibility and, as such, will require a fundamental rethinking of how we regulate, manage, and litigate AI systems in the future.

The complex matrix of user-machine interactions is reproduced in the summary table below:

Human-machine interaction	Decision-making authority	Human role
AI-assisted decision	Human	Human-in-the-loop
AI-shared decision	Hybrid	Human-on-the-loop
AI-delegated decision	Algorithmic	Human-out-of-the-loop

In conclusion, the evolving landscape of human-machine interaction underscores the increasing complexity in assigning liability to users for AI-caused harm. As decision-making authority oscillates between human operators and AI systems, a gray area of responsibility emerges. This is particularly evident in semi-autonomous technologies like self-driving cars, where human oversight is required during critical moments. Even in scenarios of near-complete autonomy, such as algorithmic trading, there remains a level of human control that adds to the uncertainty around liability.

The blurred lines between machine autonomy and human control presents significant legal challenges, particularly when determining whether harm stems from a system failure or the user's failure to act. This ambiguity highlights the urgent need for legal frameworks to adapt to the nuances of user-AI interaction, ensuring accountability without inhibiting technological innovation.

1.2. The principle of human oversight in AI Act Article 14: key aspects and limits

Despite being extensively discussed in the EU HLEG's *Ethics Guidelines*, not much of the broader debate on human-machine interactions made it into the final version of the AI Act, particularly when it comes to defining important concepts like HITL, HOTL, and HIC. These nuanced distinctions, which received considerable attention within the ethical discourse, are largely absent from the legal text, creating a gap in the framework's approach to human involvement in AI decision-making.

Instead, the Act adopts a more generalized 'human-centric' requirement, stressing the importance that AI systems be not entirely 'on their own' when their decisions impact fundamental rights, safety, or other key societal interests.¹³

¹³ AI Act, Explanatory Memorandum, 1: 'AI should be a tool for people and be a force for good in society with the ultimate aim of increasing human well-being. Rules for AI available in the Union market or otherwise affecting people in the Union should therefore be human

Specifically, this human-centric approach calls, among other things, for providers to ensure *human oversight* over high-risk AI systems.¹⁴

Oversight was important enough to warrant its own section in the AI Act.¹⁵ Under Article 14 (1), systems must be designed and developed in such a way that they can be ‘effectively overseen by natural persons during the period in which the AI system is in use’. This obligation implies that the purpose of Article 14 (1) is to compel high-risk AI designers to integrate, in their systems, a human control function as part of a safeguard against deviant behaviours of AI. Notably, Article 14(2) establishes that the aim of human oversight is to *prevent or minimize* the risks of high-risk AI systems infringing on fundamental legal goods.¹⁶

centric, so that people can trust that the technology is used in a way that is safe and compliant with the law, including the respect of fundamental rights’. The ethical principle of human centrality, as it has come to be broadly understood, does not only reflect that human needs are to be met by new technologies, but also incorporates the aim to safeguard individual rights and increase human well-being. Human centrality in AI is therefore a concept that places human-beings at the centre of any reflection about AI, its development, features and use. D.E. Harasimiuk and T. Braun, *Regulating Artificial Intelligence: Binary Ethics and the Law* (Routledge, 2021), 49ff.

¹⁴ The issue of what ‘meaningful’ control and oversight over AI systems would entail, as well as how to ensure it, has been much discussed within the legal as well as within other scholarly research fields. F. Santoni de Sio and J. van den Hoven, ‘Meaningful Human Control over Autonomous Systems: A Philosophical Account’ (2018) 5 *Frontiers in Robotics and AI*; B. Green, ‘The Flaws of Policies Requiring Human Oversight of Government Algorithms’ (2022) 45 *Computer Law & Security Review* 1.

¹⁵ In legal doctrine, human oversight is often perceived as having a special standing as an almost self-sufficient safeguard. See Lena Enqvist, ‘Human oversight in the EU artificial intelligence act: what, when and by whom?’ (2023) 15 *Law, Innovation and Technology* 508-535, 512. Moreover, the AI HLEG pinpointed human oversight as one of seven key requirements that AI systems should meet in order to be deemed trustworthy (HLEG n 4, 14ff). Similarly, the Commission in its AI White Paper states that a trustworthy, ethical and human-centric AI can only be achieved by ensuring human involvement (White paper, Ch II n 172, 21ff).

¹⁶ This is to be the case when such risks persist notwithstanding the application of the other main safeguarding obligations placed on the providers of high-risk AI systems. See in general on this point J. Constantino Torres, ‘Exploring Article 14 of the EU AI Proposal: Accountability Challenges of the Human in the Loop When Supervising High-Risk AI Systems in Public Administration’, https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4254940.

Additionally, these goal-oriented type provisions are supported by specific obligations on providers under Article 14(3) AIA that more specifically address what type of powers human overseers must be allowed to exercise in relation to the system. In particular, providers should make sure that overseers are able to decide, in any particular situation, not to use the system or otherwise disregard, modify, or override its output.¹⁷ To this aim, providers must equip overseers with a comprehensive understanding of the AI system's capabilities and limitations, enabling them to monitor its operations effectively, remain aware of the possible tendency of automatically relying or over-relying on the output produced by the system, and correctly interpret its output.¹⁸ By mandating that risk assessments and oversight capabilities be embedded during the design phase, Article 14(3) aims to ensure that critical safeguards are in place well *before* the AI system reaches users, underscoring the provision's fundamentally preventive character.¹⁹

More specifically, Article 13(2) AIA mandates that providers develop and supply AI systems with detailed accompanying *instructions*. These instructions are expected to cover human oversight protocols as well as technical measures designed to aid users in understanding system outputs.²⁰ Article 26 obliges users to adhere to

¹⁷ AI Act, Article 14(4)(d-e).

¹⁸ Details concerning this obligation are provided in Article 14(4)(a-c) AI Act. In particular, providers must create the conditions for human overseers:

‘(a) to properly understand the relevant capacities and limitations of the high-risk AI system and be able to duly monitor its operation, including in view of detecting and addressing anomalies, dysfunctions and unexpected performance;

(b) to remain aware of the possible tendency of automatically relying or over-relying on the output produced by a high-risk AI system (automation bias), in particular for high-risk AI systems used to provide information or recommendations for decisions to be taken by natural persons;

(c) to correctly interpret the high-risk AI system's output, taking into account, for example, the interpretation tools and methods available;

(d) to decide, in any particular situation, not to use the high-risk AI system or to otherwise disregard, override or reverse the output of the high-risk AI system;

(e) to intervene in the operation of the high-risk AI system or interrupt the system through a ‘stop’ button or a similar procedure that allows the system to come to a halt in a safe state.’

¹⁹ Enqvist (n 15), 518.

²⁰ AI Act Article 13(3)(d).

these instructions of use, implementing human oversight on the basis of the latter. This requirement is combined with obligations on the users to suspend the use of the system when they have reasons to consider that the use in accordance with the instructions may result in risks of harm to fundamental legal interests.²¹

Now, while the obligations outlined in Article 14 are technically assigned exclusively to AI providers rather than directly to human overseers, it is reasonable to assume that the obligation under Article 13 to make instructions partially extends the providers' obligations into a sort of user obligation to perform it.²² The Act's framework suggests that, although providers bear primary responsibility for implementing safety measures and informing users, human overseers will inevitably hold some level of duty to act upon these safety standards. This assumption reflects the dual expectation that overseers not only follow provider guidelines but also exercise due diligence in identifying and mitigating risks during operation. Consequently, while the oversight requirements under the Act do not explicitly designate these roles to users, a degree of accountability – rooted in established principles of due care and risk management – implicitly falls upon those in supervisory positions, particularly when harm could potentially be prevented through proactive intervention.

In this light, human oversight represents not just a central procedural safeguard but a foundational element of trust in AI technologies of a more reactive nature, as it requires the overseer to monitor and address errors, biases and lapses of AI systems in operation, and therefore it is complementary to upholding accountability structures around AI.²³

Presumably, the rationale goes that when AI cannot be made to behave well 'by design', humans should at least be given a duty to control, intervene, and if necessary, overrule AI-driven decisions, especially in sectors where errors could have

²¹ *ibid.*, Article 26(4).

²² Enqvist (n 15), 522-524.

²³ R. Koulu, 'Proceduralizing Control and Discretion: Human Oversight in Artificial Intelligence Policy' (2020) 27 *Maastricht Journal of European and Comparative Law* 720.

severe consequences, such as healthcare, transportation, and finance.²⁴ The inclusion of this general duty of oversight in the AI Act precisely reflects a cautious approach to AI regulation, one that seeks to balance the benefits of automation with the need to maintain human judgment and accountability in critical decision-making processes. In theory, adopting AI while ensuring human oversight could create a legal environment where intelligent systems can enhance human decision-making while still keeping humans at the helm of decisions that carry significant societal impact, based on the view that humans are better able to make complex deductions relating to the social dimensions of law or other norms. How much a supervisor is determined to have followed the instructions and exercised his role diligently will matter if the system causes harm and compliance is questioned. Failure to do so might be interpreted as user negligence, even if the AI initiated the sequence of events leading to harm.

While requiring human intervention sounds reassuring in theory, there is no obvious way to put human oversight into practice in a ‘meaningful’ way. Although Article 14(4) outlines certain conditions that providers must ensure, aiming to shed some light on the meaning of ‘meaningful’ human oversight under EU law, there remains a lack of practical guidance on how providers should guarantee these conditions and how overseers are expected to apply them effectively. For example, the regulation does not provide any detail on how a system’s output should be presented to ensure that human overseers will be able to interpret it correctly; there is no real point in performing the oversight if the overseer is not provided with information (or system output) in such a representation that it can interpret or understand. This ambiguity becomes especially critical when considering the timing,

²⁴ As advocated by the AI HLEG, there is a regulatory as well as a functional interrelation between the human oversight principle and the other main safeguarding measures set out in the AI Act. From the regulatory perspective, the relation is inverse – the provider’s obligation to ensure oversight capabilities increases where the risk-mitigating functions of the other main safeguarding measures cannot be expected to be efficient enough. Functionally, all these safeguarding measures support the general transparency objectives of the regulation, and will all contribute with relevant components to the oversight infrastructure available to the human overseer. HLEG (n 4), 16.

scope, and nature of oversight necessary to prevent or mitigate AI risks, leaving substantial gaps in understanding the precise expectations for effective human control.

Human oversight should in this regard not be viewed as a specific and single type of safeguarding measure. Human oversight can take different ‘configurations’ depending on what aspects of a system process the oversight is to be aimed at, when the oversight is to be performed, and who the human supposed to perform the oversight is.²⁵ For instance, it remains unclear how often and in what circumstances human overseers are required to intervene to prevent harm. In high-stakes applications, such as healthcare or autonomous vehicles, determining when an overseer must step in or halt an AI system to avoid potentially irreversible consequences is challenging and requires clear thresholds. The oversight demands in such contexts will indeed differ significantly from oversight required in lower-risk, minimum intervention scenarios. Furthermore, without guidance on the particular types of problems that human oversight is meant to solve as well as on feasible methods for intervention, human overseers may find themselves in roles where intervention is theoretically expected but practically unmanageable. This becomes particularly problematic in environments where human cognition and reaction times may not align with the speed and complexity of AI decision-making.²⁶

The lack of clarity thus raises concerns over the real-world applicability of Article 14’s requirements. If human oversight is to serve as a primary safeguard both to shield society from AI harms and to provide a clear locus of criminal responsibility in cases of failure to prevent harmful outcomes, it must be formulated and

²⁵ Enqvist (n 15), 510.

²⁶ Interestingly, the Commission in its AI White Paper listed some non-exhaustive manifestations of human oversight including: situations where the output of the AI system does not become effective unless it has been previously reviewed and validated by a human; where the output of the AI system becomes immediately effective but human intervention is ensured afterwards; real time human supervision and potential intervention; or safeguards installed in the design phase, through operational constraints on the AI system. This policy document, however, does not go into much detail on how human oversight is to be performed more specifically. White paper (Ch II n 172), 21ff.

operationalized through technical standards that account for the diverse ways humans interact with AI systems, including the nature, timing, and conditions of human intervention. Such standards should be not only clearly articulated but also realistically attainable for those held accountable, thus reducing the ambiguity that currently surrounds the role of overseers and establishing clear areas of acceptable risk.

Developing clear guidelines and technical configurations to enable meaningful human oversight, however, presents complex challenges. In fact, their effectiveness hinges fundamentally on the *type of human-machine interaction* and the *inherent cognitive limitations* of human overseers.²⁷ Yet, Article 14 is deceptively simple and does not do justice to the wide scale of diverse human-machine interactions that already exist as well as limitations on human capacity. Therefore, considering these factors is critical to developing nuanced and context-sensitive expectations in terms of the extent of reasonably feasible control over AI systems.

1.3. Liable but not in control: main obstacles to meaningful oversight

In the EU regulatory framework, the approach of imposing a duty to intervene on a specific actor (like a human deployer or overseer) as a key remedy for AI harm appears to be an optimal solution to address the AI responsibility gap without undermining the benefits of autonomous decision-making or banning it entirely. The overseer would, in theory, monitor the AI system and step in should it malfunction or encounter unexpected challenges, effectively combining the benefits of autonomous AI capabilities with a safeguard for human intervention.

²⁷ Ben Wagner, 'Liable, but Not in Control? Ensuring Meaningful Human Agency in Automated Decision-Making Systems' (2019) 11 Policy & Internet 104, 114ff; T. Enarsson, L. Enqvist, and M. Naarttijärvi, 'Approaching the Human in the Loop - Legal Perspectives on Hybrid Human/Algorithmic Decision-Making in Three Contexts' (2022) 31 Information & Communications Technology Law 123.

However, as legal doctrine has pointed out, this solution encounters significant *technical* and *legal* obstacles in practice.²⁸ In principle, for an overseer to be held responsible under criminal law for an AI-caused harm, several strict criteria must be met, specifically around notions of causation, foreseeability, and control. Criminal liability principles dictate that responsibility for harm can only be attributed if the overseer had *causal control* over the outcome, meaning that it must have been both foreseeable and preventable through reasonable care.²⁹ Depending on the type of AI application and its risk level, this may not be the case in practice for multiple reasons.

The first and perhaps most fundamental obstacle to effective human control is *epistemic* AI, particularly in its more advanced forms, often operates beyond full human comprehension, leading it to be perceived as an ‘alien’ form of intelligence.³⁰ This challenge transcends the technical domain, deeply embedding itself in psychological factors related to human perception and understanding. The gap between the perceived and actual capabilities of AI systems creates a distortion in the way they are viewed by human supervisors. Because supervisors may lack a complete grasp of the system’s inner workings, they can overestimate the system’s autonomy, attributing a level of independent decision-making or ‘agency’ that the AI does not actually possess.

This misperception can foster a problematic sense of detachment and undue trust, where supervisors regard the system as an independent actor rather than a tool requiring consistent human judgment and control. In these situations, human overseers might feel less responsible for the outcomes of AI operations, creating a kind of cognitive bias that dilutes accountability.

²⁸ Giannini and Kwik (Ch III n 4), 45; Wagner (n 27), 114ff.

²⁹ That is the basic assumption of ‘human meaningful control’ theory (HMC) which correlates control to responsibility enlightening the necessity of being capable to govern the event for the attribution of responsibility. See Filippo Santoni De Sio And Giulio Mecacci (Ch III n 17), 1057-1084.

³⁰ Selbst (Ch III n 113), 1330.

Importantly, the AI Act does touch upon this issue. Article 14(4) remarks that deployers need to ‘remain aware of the possible tendency of automatically relying or over-relying on the output produced by a high-risk AI system (automation bias)’. Additionally, providers must create the conditions for overseers to maintain agency and remain aware of the automation bias. According to recital 91, providers must ensure that the persons doing the oversight have the ‘necessary competence, in particular an adequate level of AI literacy, training and authority to properly fulfil those tasks’. This should allow them to properly understand the relevant capacities and limitations of the AI system and be able to duly monitor its operation, in view of detecting and addressing anomalies, dysfunctions and unexpected performance.

Yet, even with such a framework for reducing automation bias, avoiding it entirely may prove challenging. Human cognition and decision-making processes are not infallible, and there is ample evidence that individuals can struggle to critically evaluate AI-generated decisions, especially when the AI system presents outputs that appear authoritative or statistically reliable. In these cases, supervisors may default to simply rubber-stamping automated decisions, underscoring a core limitation in relying solely on human oversight to manage the risks associated with AI systems.

Recognizing these limitations allows for more practical and achievable standards, balancing the need for human accountability with the realities of what overseers can reasonably control.

On a similar note, there are also many who advocate caution and highlight the risks to being overly reliant on the capacity of humans to effectively detect and remedy the negative consequences of AI systems. Oversight tasks that require constant monitoring, rapid intervention, or deep technical understanding of AI processes may exceed the realistic cognitive and physical capabilities of human operators.³¹ Reaction times, sustained attention, and mental processing capacity are all constrained, which can lead to mistakes. In particular, it has been scientifically established that humans overseers, when tasked with passive monitoring of

³¹ Selbst (Ch III n 113), 1330.

autonomous systems, face issues of ‘automation fatigue’ and diminished vigilance over time, particularly when dealing with complex or high-stakes decisions that unfold rapidly and outside typical human response capabilities.³² As a result, they may often be unable to determine in real time whether the AI is making an error and intervene effectively.

For example, this effect has been observed many times in relation to autopilots. Perrow recounts that often, ‘when the pilot is suddenly and unexpectedly brought into the control loop (in other words, participates in decision making) as a result of (inevitable) equipment failure, he is disoriented ... The sudden appearance of several alarms, all there for safety reasons, leads to disorientation’.³³ Even for several seconds after this person is expected to have taken back control, they may not possess the capacity to properly and reasonably act to avoid an imminent harm, and thereby not truly be in control.³⁴

Additionally, human overseers may experience what is termed ‘algorithmic aversion’, where a lack of trust in the system leads them to override or disregard AI outputs, even when they are valid. This aversion highlights an interesting tension: while the overseer’s independent judgment is critical, an over-reliance on subjective judgment can sometimes lead to more errors than adherence to the system’s advice.

Once again, the AI Act attempts to address these issues by mandating thorough training and providing clear instructions for AI system operators. However, such measures are not always sufficient on their own. Training can be undermined by individual differences in interpreting AI outputs or by prior biases and experiences, which may conflict with the evaluation requirements specific to AI-based decisions. Even with well-designed instructions, there are limits to how much training

³² Raja Parasuraman, Thomas B Sheridan, and Christopher D Wickens, ‘A Model for Types and Levels of Human Interaction with Automation’ (2000) 30 IEEE Transactions on Systems, Man, and Cybernetics, 291.

³³ Perrow (Ch II n 112).

³⁴ Hilgendorf (Ch II n 75), 67ff; Sabine Gless, ‘Mein Auto fuhr zu schnell, nicht ich! - Strafrechtliche Verantwortung für hochautomatisiertes Fahren’ in S Gless and K Seelmann (eds.), *Intelligente Agenten und das Recht* (1st ed., Baden-Baden, Nomos, 2016), 227-228.

can counteract innate human biases or bridge gaps in understanding complex AI operations.

All these factors combined cast doubt on whether a human supervisor can genuinely exercise *de facto* control over an AI system, even when technical oversight mechanisms, training, and instructions are in place. In practice, it often remains unclear how an operator can meet any duty of care in the system's operation. No matter the specific standard of care applied, the AI user may be unable to ascertain which side of the line they are on. In many applications, expecting them to have this level of insight is unreasonable; it may, in fact, be impossible due to the opacity and complexity of AI processes.³⁵

This ambiguity makes imputing responsibility highly problematic. The current approach to human oversight as a catch-all solution to AI harm seems to create a legal façade of accountability rather than ensuring substantive control. This, in turn, may lead to unfair distribution of responsibility, shifting blame back and forth between parties.³⁶

On the one hand, in many cases, providers and companies will be able to shift accountability (and criminal liability) to frontline human operators, who are treated as *scapegoats* and are blamed for broader system failures over which they have little or no control, while distracting from the intrinsic defects of the system.³⁷ This phenomenon has been labelled as 'retribution gap' or 'moral crumple zone',³⁸ where operators may find themselves grappling with the contradiction between the legal

³⁵ Selbst (Ch III n 113), 1332.

³⁶ Ben Green and Amba Kak, 'The False Comfort of Human Oversight as an Antidote to A.I. Harm' (Slate, 2021) available at <https://slate.com/technology/2021/06/human-oversight-artificial-intelligence-laws.html>; Austin Clyde, 'Human-in-the-Loop Systems Are No Panacea for AI Accountability' (Tech Policy Press 1 December 2021), available at <https://techpolicy.press/human-in-the-loop-systems-are-no-panacea-for-ai-accountability>.

³⁷ Here, legal doctrine emphasized how human oversight will serve as a veneer legitimizing public uses of faulty and controversial algorithms without addressing the fundamental issues with these tools, and thus also providing a false sense of security in adopting algorithms in government operations. Ben Green, 'The Flaws of Policies Requiring Human Oversight of Government Algorithms' (2022) 45 Computer Law & Security Review, 18 ff.

³⁸ Danaher (Ch II n 138) 299-309.

imperative of maintaining control over the machine and the substantial impossibility to do so lacking clear, actionable rules that would allow human overseers to meaningfully anticipate and mitigate AI harms:

‘Just as the crumple zone in a car is designed to absorb the force of impact in a crash, the human in a highly complex and automated system may simply become a component – accidentally or intentionally – that bears the brunt of the moral and legal responsibilities when the overall system malfunctions. Unlike the crumple zone in a car, which is made to protect the human driver, the moral crumple zone reverses this dynamic, allowing the perception of a flawless technology to remain intact at the expense of the human operator’.³⁹

For example, we saw a moral crumple zone emerge in real time during the events surrounding an accident occurred in 2018, when a self-driving Uber vehicle struck and killed a woman in Arizona. Even as Uber boasted about its autonomous vehicle development, blame for the crash fell primarily on the human operator tasked with monitoring the vehicle – even though investigations found that the vehicle failed to stop because Uber engineers had tuned it to be less responsive to unidentified objects. The operator’s oversight mandate assumed that she should have intervened when needed; however, without precise thresholds for intervention or clear definitions of the overseer’s responsibilities, the risk of criminal liability may disproportionately shift onto overseers, even in situations where intervention may not be realistically achievable – which in turn challenges the foundational criminal law concepts of fault and culpability.⁴⁰

On the other hand, those who function as human overseers may argue that Article 14 of the AI Act fails to specify their direct obligations, potentially deflecting

³⁹ Madeleine C Elish, ‘Moral crumple zones: cautionary tales in human-robot interaction’ (2019) 5 *Engag. Sci. Technol. Soc.* 40–60.

⁴⁰ Frank Douma and Sarah Aue Palodichuk, ‘Criminal Liability Issues Created by Autonomous Vehicles’ (2012) 52 *Santa Clara Law Review* 1157, 1164.

responsibility back onto AI providers. Indeed, it is insufficient to accuse an individual of breaching an ambiguous and broadly defined ‘duty of oversight’; negligence must be ascertainable in a specific, concrete context according to clear and actionable rules of behaviour.

This circular logic exposes a key tension in relying on human oversight as a panacea to bridge the AI responsibility gap and raises significant concerns about how responsibility could be fairly and feasibly distributed across various actors within the AI value chain. Without carefully delineated responsibilities, the risk emerges that criminal liability may be inequitably attributed or that human overseers might serve as scapegoats, absorbing blame for systemic issues beyond their direct control.

Therefore, in our increasingly algorithmic society – where complex, distributed decision-making processes produce high-risk technologies – establishing clear principles for responsibility distribution is crucial when multiple actors play roles in producing harm. One such principle is that of legitimate trust or reliance, which will be the focus of the following discussion. Here, we will explore whether and how the safety requirements and conformity assessments specified in the AI Act impact the liability of human overseers by establishing reasonable expectations about the functioning and trustworthiness of high-risk AI systems.

These reasonable expectations could play a pivotal role in limiting liability assessments in cases of AI harm, helping to ensure that criminal liability reflects the actual control and oversight capacity exercised over AI outcomes. By anchoring liability more closely to an actor’s meaningful control, this approach could mitigate unjust attribution of responsibility to human overseers who rely on certified systems.

1.4. Legitimate trust in human-machine interactions as a liability allocation criterion

In the context of negligence liability, the principle of legitimate ‘trust’ or ‘reliance’ has gained increasing relevance, especially within our risk society, where complex socio-technical systems are the norm.

Traditionally, this principle posits that individuals engaged in a divided task structure (e.g., driving) can generally rely on each other to behave lawfully, unless there is clear evidence to the contrary. This principle is central to liability distribution in systems involving both planned and unplanned interactions between multiple parties. It offers a limit to blame attribution by recognizing that within a collaborative structure, trust in lawful behaviour forms a basis for efficient functioning and equitable responsibility.

The question then arises regarding the applicability of the principle of trust to human-AI interactions as a potential solution to mitigate the criminal negligence liability of human overseers. Can, or should, they rely on AI systems to behave ‘lawfully’ or to function within expected parameters?⁴¹ If so, to what extent? This is especially pertinent given that the current regulatory stance risks overstretching the concept of human oversight, assigning human overseers comprehensive, catch-all accountability for every action an AI system performs under their supervision. But such a framework, beyond the substantial ineffectiveness of human oversight, would fundamentally frustrate the very rationale behind AI innovation: to delegate parts of human decision-making authority within a socio-technical system. This approach indeed disregards the fact that AI, in these contexts, functions as a ‘delegated agent’ – albeit without true agency in a legal sense – whose lawful behaviour can in principle be trusted to a certain extent.

For example, if the idea of an autonomous car is that you can drive even if you are too tired and without taking too many breaks, comforted by the assumption that the machine makes less mistakes than human drivers, it would be disturbing if the supervising human driver would be liable for a wrong decision of the car only because he decided to use it. It would also lead to concentrating as much as when

⁴¹ Beck (Ch I n 21), 141.

driving oneself – which is actually harder if the driver does not drive for long periods of time – and thus making the usage of an autonomous car pointless. In many contexts besides autonomous driving, AI systems are introduced as a mechanism to improve human cognitive limits and biases – yet now those same humans are presented in policy as the essential backstop overseeing algorithmic limits and biases. Instead of alleviating human effort, it would merely shift the burden from actively managing a hazardous activity to the even more demanding duty of constantly monitoring a machine often operating beyond human comprehension and control.⁴²

That being said, however, it is questionable whether the principle of trust could be applied in the realm of human-machine interactions in such a way that humans – under certain circumstances – may rely on the AI system to function correctly. In fact, this raises significant challenges.

The problem is that, unlike in work-sharing arrangements between humans where responsibility can be divided according to each person's distinct roles and areas of competence, the responsibility for using a tool – like an AI system – currently rests solely with the human user. Using an AI systems cannot logically be compared with the decision to delegate specific tasks to another human. In traditional human delegation, the delegator assigns tasks based on a clear understanding of the delegate's capabilities and can assess and select competent partners. Additionally, the delegate in these cases is an autonomous person, capable of bearing responsibility for their actions. Delegating to another human, therefore, creates a new 'responsibility sphere' encompassing the delegated task, defined by the parameters of information, authority, and control granted to the delegate. Should errors occur within this sphere, the delegate may also be held accountable, sharing the burden of liability with the original task assigner.

By contrast, when decision-making authority is transferred to a machine, no equivalent 'sphere' of responsibility is created. Legally and conceptually, AI systems are still classified as tools without legal agency, which precludes them from bearing

⁴² Cappellini (Ch III n 3), 14.

responsibility for the outcomes they produce. Consequently, despite the machine's decision-making autonomy, accountability remains tethered to the human, who is expected to oversee and correct any potential errors. This approach fails to mirror the dynamics of human delegation, as it places the entire weight of responsibility back onto the human overseer, irrespective of the machine's autonomy or unpredictable behaviour and without the relief that typically accompanies true delegation to another agent.

Therefore, we propose an alternative approach to liability allocation that would allow users to reasonably expect lawful and trustworthy performance from an AI system, provided it complies with all legally mandated standards. This concept is grounded in existing liability principles, such as those found in the Council Directive 85/374/EEC on liability for defective products.⁴³ Under Article 3 of the Directive, conformity with general safety standards is essential for applying the legitimate trust principle, serving as a liability shield for users unless a defect is both manifest and easily identifiable.

In the AI domain, the conformity-assessment process outlined in Article 43 of the AI Act could similarly affect the scope of legitimate trust as a protective principle in cases of AI harm. By completing this assessment, providers certify that their high-risk AI systems meet established safety and performance standards, enabling users to rely on the system's lawful operation within defined parameters. This reliance would hold unless deficiencies in the AI's design or safety measures are manifest and immediately apparent to the overseer.

In scenarios where AI failures lead to harm, a key factor in applying the trust principle is the *nature* and *degree* of task delegation to the AI, which defines the division of decision-making authority between human overseers and the AI system. Thus, liability allocation could change based on the varying degrees of functionality

⁴³ For an application of the legitimate expectation principle in product liability, see the advocate general's opinion in CJEU Joined Cases C-503/13 and C-504/13 *Boston Scientific Medizintechnik* [2015] ECLI:EU:C:2015:148.

and delegation, creating a more balanced distribution of accountability that reflects the contributions of different stakeholders along the AI value chain. As tasks are increasingly delegated to AI, the risk of liability for damage and injuries contextually shifts from human overseers to the providers and organisations that designed and developed the technology and defined its context and uses. This approach would mean that liability does not rest solely on the overseer but is shared, encouraging providers to invest in thorough risk assessments, knowing that responsibility aligns with the development and deployment continuum rather than disproportionately burdening end-users.

To implement this approach effectively, however, clear guidance is needed to operationalize the trust principle without simply shifting ambiguities in responsibility. Developing a systematic framework that connects human-machine interaction configurations with corresponding responsibilities for overseers is critical.⁴⁴ In this regard, the taxonomy developed by Endsley and Kaber,⁴⁵ and later refined by Parasuraman, Sheridan, and Wickens,⁴⁶ offers valuable guidance. This framework categorizes levels of delegation under four cognitive functions – information acquisition, information analysis, decision selection, and action implementation – each progressing from a zero level (entirely manual, with no AI involvement) to full delegation, where the task is handled autonomously by the machine.

In this last scenario, applying the principle of legitimate expectation – afforded by the completion of a full conformity-assurance process – serves as a protective measure for human overseers. As long as they operate within the scope of

⁴⁴ This approach is already quite common in legal doctrine to analyze liability allocation in AI settings. See Giuseppe Contissa *et al*, ‘Liability and Automation: Issues and Challenges for Socio-Technical Systems’ (2013) 2(1-2) *Journal of Aerospace Operations* 79; Id and Lagoia, ‘The strange case of Dr. Watson: liability implications of AI evidence-based decision support systems in health care’ (2020) 12(2) *European Journal of Legal Studies* 245-289, 264.

⁴⁵ David B Kaber and Mica R Endsley, ‘The Effects of Level of Automation and Adaptive Automation on Human Performance, Situation Awareness and Workload in a Dynamic Control Task’ (2004) 5(2) *Theoretical Issues in Ergonomics Science* 113.

⁴⁶ Parasuraman (n 32), 286.

legitimate expectation, negligence liability for AI harm would not be imputed solely on the basis of a broad, generalized duty of oversight over the AI's operation. Instead, liability should hinge on whether the human overseer disregarded clear indicators of an imminent failure in the system's safety measures (gross negligence), particularly if they had reason to believe that using the high-risk AI system as instructed might still jeopardize legally protected interests.

The following section explores different configurations of human-machine interaction to investigate how and to what extent the level of delegation and legitimate trust may affect the allocation of criminal liability. Drawing on the theoretical distinctions made in para. 1.1. on the types of human-machine interaction, this chapter will thus examine the interaction between AI and human operator depending on whether the latter qualifies as human-in-the-loop (HITL), human-on-the-loop (HOTL), or human-out-of-the-loop (HOFTL). As we pointed out, HITL refers to the capability for human intervention in every decision cycle of the system, HOTL to capability for monitoring the system's operation and human intervention when required, and HOFTL to the capability to oversee the overall activity of the AI system as well as the ability to decide when and how to use the system in any particular situation. The extent, limits, and implications of human involvements across different oversight regimes vary depending on four core questions: *what* is to be supervised, *when* oversight is expected, *by whom*, and *how* it should be performed.

This structured analysis is essential for unpacking critical issues that will likely arise in future liability cases (and, in some cases, have already emerged as we will see). By clarifying these dimensions, we aim to determine whether human supervisors are effectively positioned and empowered to supervise AI systems meaningfully and, crucially, whether they are capable of exercising due care under the regulatory framework. In the context of criminal negligent liability, these insights will prove invaluable for assessing the adequacy of oversight mechanisms, as well as determining where responsibility should fall when harms arise. In this context, the importance of upholding the fundamental principles of criminal liability will not be overlooked.

This means avoiding presumptions and automatic imputations and instead ensuring a rigorous examination of the elements of culpability.

SECTION 2

Human-machine interaction #1: human-*in*-the-loop

2.1. Duty of care of HITL: the case of AI-assisted clinical decision-making

Our systematic analysis starts with the first configuration of human-machine interaction, where the human overseer takes the role of ‘human in the loop’ (HITL). But who is exactly this figure?

For our analysis, we define this individual as someone actively engaged in a decision-making process, where AI serves as a tool to enhance human judgment, particularly in making high-stakes decisions. This section examines the dynamic interplay between humans and machines within what we term ‘AI-assisted decision processes’. The aim is to develop insights into how this interaction should be structured to enable fair and accurate responsibility attributions in the event of harm arising from AI usage.

Our analysis focuses on the healthcare sector as a case study, where AI applications have become integral to supporting diagnostic and therapeutic decisions.⁴⁷ In particular, ML algorithms are increasingly being used to predict things that even well-trained humans (otherwise known as doctors) cannot.⁴⁸ Early uses of clinical decision support systems (CDSS) were aimed at identifying high- and low-risk patients in different contexts, in an effort to improve healthcare quality and patient

⁴⁷ The integration of clinical decision support systems (CDSS) in healthcare is not a novel topic, having been discussed since last century. Experts started to debate the topic of difficult decisions in complex clinical situations assisted by computers as early as 1959. I Habli, T Lawton and Z Porter, ‘Artificial intelligence in health care: accountability and safety’ (2020) 98(4) Bull World Health Organ 251-256; E.H. Shortliffe and M.J. Sepúlveda, ‘Clinical Decision Support in the Era of Artificial Intelligence’ (2018) 320(21) JAMA 2199.

⁴⁸ Contissa and Lagioia (n 44), 251-255.

safety.⁴⁹ Today, AI tools are being developed to facilitate evidence-based clinical practice in diagnosing patients or recommending treatment. Although human physicians cannot be replaced by AI systems in the foreseeable future, AI could play a key ‘advisory’ role in assisting physicians to make better clinical decisions.

As a matter of fact, a large volume of healthcare data can be computed efficiently by AI algorithms, in order to assist clinical practice. Intelligent systems can be equipped with learning and self-correcting skills to improve their accuracy, and support physicians in reducing diagnostic and therapeutic errors. Moreover, AI systems can be used so as to extract information from a large patient population and assist in making real-time inferences for a health risk alert and outcome prediction.⁵⁰

An example of this is the IBM Watson system developed in recent years. Research suggests that medical diagnostic and treatment algorithms like Watson will be on average safer and more accurate than human physicians because they seek to find and take advantage of patterns that humans would not otherwise recognize.⁵¹ For example, image recognition algorithms have demonstrated potential in interpretation of head computed tomography scans as well as in the diagnosis of cancer. By overcoming human and cognitive limitations in collecting or processing information, clinical AI systems may help substantially reduce medical errors.⁵²

⁴⁹ Glenn Cohen *et al*, ‘The Legal and Ethical Concerns that Arise from Using Complex Predictive Analytics in Health Care’ (2014) 33 Health Aff. 1139, 1140.

⁵⁰ FJiang *et al*, ‘Artificial intelligence in healthcare: past, present and future’ (2017) 2(4) Stroke and Vascular Neurology, 230.

⁵¹ It would take at least 160 hours of reading a week just to keep up with new medical knowledge as it is published, let alone consider its relevance or apply it practically. Watson’s ability to absorb this information faster than any human should, in theory, fix a flaw in the current healthcare model. Ian Steadman, ‘IBM’s Watson Is Better at Diagnosing Cancer Than Human Doctors’ (Wired, 11 February 2013) available at <https://www.wired.co.uk/article/ibm-watsonmedical-doctor>.

⁵² Moreover, more than half of the clinical workforce will be routinely using AI predictive analytics, image interpretation, and natural language processing within a decade. Jane R. Bambauer, ‘Dr. Robot’ (2017) 51 U.C. Davis L. Rev. 383, 387.

Generally, these are strong and compelling arguments for justifying and trusting the decision-making of AI systems in clinical domains. When AI systems consistently outperform human physicians in specific tasks – such as achieving a correct diagnosis rate of 97% compared to a human doctor’s 94% – the reduction in misdiagnosis rates suggests lives may indeed be saved. In such cases, the social acceptance of AI systems becomes more justifiable, potentially even making their use a matter of duty, with AI support becoming the standard of care in relevant areas.⁵³

However, claims that AI will universally reduce medical errors remain optimistic and introduce a tension between this idealized, error-free vision of AI-powered healthcare and the unpredictable nature of these technologies, which can still make unexpected and sometimes harmful errors.⁵⁴ These errors may not be infrequent; for instance, IBM’s Watson, initially touted as a revolutionary tool for cancer diagnosis and treatment, shifted from high expectations to disappointing results within just fourteen months.⁵⁵

Essentially, harm to patients’ life of physical integrity can arise from the interplay between the human doctor and the potentially erroneous information – in terms of that misdiagnosis or mistreatment – an AI system provides. For example, in a system recommending treatment for diabetes, the system may recommend – based on training data – that it is appropriate to start insulin; however, if the system – say – is less accurate for patients from some ethnic backgrounds, a misdiagnosis, with the consequent failure to deliver the appropriate medical treatment, poses a high risk to the patient’s health and safety.

⁵³ A. Froomkin, Ian R. Kerr, and Joelle Pineau, ‘When AIs Outperform Doctors: Confronting the Challenges of a Tort-Induced Over-Reliance on Machine Learning’ (2019) 61 *Ariz. L. Rev.* 33.

⁵⁴ Contissa and Lagioia (n 44), 253.

⁵⁵ Angela Chen, ‘IBM’s Watson Gave Unsafe Recommendations for Treating Cancer’ (Verge July 26, 2018) <https://www.theverge.com/2018/7/26/17619382/ibms-watson-cancer-ai-healthcare-science> ([A]ccording to IBM documents dated from last summer, [Watson] has frequently given bad advice, like when it suggested a cancer patient with severe bleeding be given a drug that could cause the bleeding to worsen’).

Generally speaking, where harmful consequences to human life and health stem from medical malpractice or error, a doctor must be ready to potentially bear legal responsibility for such events where their actions would have been recognized as harmful by a reasonable clinician. Legal responsibility for medical malpractice may imply either civil liability, leading to the indemnification of the patients for their damages, or criminal liability the *ultima ratio* of the protection of the health of patients and society as a whole, triggering criminal sanctions in case of actions qualified as offences by the criminal law.⁵⁶

If the harm that results from a clinician's error is the *death* of a patient, then we need to look beyond private law and question if and how criminal law is applicable in the form of recklessness or negligent liability. The doctor – respectively – has consciously disregarded the risk, or should have been aware of it. In this respect, we should distinguish three elements: *a)* the recognizability (*‘Erkennbarkeit’* in Germany, *‘riconoscibilità’* in Italy) of the unreasonable/unjustifiable risk; *b)* its foreseeability (*‘Vorhersehbarkeit’* in Germany, *‘prevedibilità’* in Italy); and *c)* its preventability (*‘Vermeidbarkeit’* in Germany, *‘evitabilità’* in Italy).⁵⁷ Thus, a doctor

⁵⁶ Considering the special working conditions of medical staff and the largely humanistic nature of their work, certain legal systems give precedence to civil law protection. This is particularly notable in countries where healthcare is largely privatized. A typical example is the USA, where medical errors are almost never treated under criminal law protection, but are dealt with in civil litigation for the compensation of damages. On the other side of the spectrum, in countries where healthcare is mostly administered by the public sector, the criminal prosecution of doctors for medical errors is much more prevalent. This is particularly true for the countries of continental Europe. In the legal systems of the majority of the Member States of the European Union, medical malpractice is punished under general criminal offenses against life and bodily integrity. As a typical example of such legislative approach, one can point to German law, which considers the cases of medical malpractice under the § 223 which regulates the general crime of causing bodily harm (Germ. *Körperverletzung*). See in general AR Di Landro, ‘Criminal Law as a Response to Medical Malpractice: Pluses and Minuses – Comparing Italian and U. S. Experiences’ (2012) 31(2) *Medicine and Law* 221-263, 222; Jasna Murgel, ‘Medical negligence and liability of health professionals in the European Court of Human Rights case law’ (2020) 13(1) *Medicine, Law & Society*.

⁵⁷ Of course, the prosecutor must prove two further requirements of liability: the occurrence of consequences and a causal nexus between the errors in treatment and consequences. The

can be held criminally liable for medical malpractice if a court medical expert assesses that a ‘reasonably competent physician’ would have foreseen and avoided the risk of possible harmful consequences in a given situation (i.e. fails to notice clear symptoms of a certain disease) acting according to accepted norms and standards of medical practice (*lege artis*).

As AI becomes more embedded in medical practice, it is understandable that clinicians might worry about an increased risk of criminal liability due to reliance on potentially flawed AI recommendations. Indeed, the use of AI may complicate the legal landscape, raising questions about whether a doctor could be found negligent, or even criminally liable, if their trust in an AI’s erroneous recommendation results in patient harm or death.

In this respect, one could argue that the risk of harm and, consequently, liability arising from AI-driven medical malpractice may be statistically lower than that from purely human error. However, the question here is not about determining whether the AI can be trusted – after all, AI systems in medical care are required to meet stringent performance benchmarks (e.g., consistently outperforming human clinicians in specific tasks) and must undergo conformity-assessment procedures before adoption. Instead, the core issue here is whether – and to what extent – a doctor’s reliance on erroneous AI advice constitutes a breach of the duty of care expected from a ‘reasonably competent doctor using an AI’.

Since AI systems in healthcare are designed as ‘advisory’ tools, the physician technically retains ultimate decision-making authority, which carries the expectation that they must not simply follow AI suggestions but maintain a vigilant stance as prescribed under Article 14 of the AI act, critically assessing when the system’s output aligns with or deviates from the nuanced needs of individual cases. The AI system’s recommendations, while influential, do not diminish the human operator’s duty to scrutinize, validate, and, if necessary, challenge these outputs. For example, in the

consequence consists of either the aggravation of the disease, the deterioration of health, the termination of a pregnancy or the death of the patient.

diabetes case, the doctor may choose to override the AI output after specifically considering patient's broad clinical picture. In theory, failure to do so could be construed as negligent if it leads to a preventable harm.

In this section, we argue that the negligence doctrine in medical settings may need to evolve to account for AI integration as part of routine practice, thus striking a balance between legitimate trust in a highly accurate system and the clinician's own responsibility to recognize and respond to potentially flawed outputs. From a liability perspective, some authors have noted that the outcome depends on which expert judgment will be considered as the source of the decision-making authority in each task – whether the AI or the human operator.⁵⁸ The answer to this question will help defining the boundaries of acceptable reliance within the scope of a doctor's professional duties and clarify the expectations for human oversight.

To this aim, it is essential to start by examining the allocation of decision-making authority between human and AI across different cognitive tasks within clinical decision-making.

2.2. Allocation of decision-making authority between AI and human physician

Analyzing liability allocation in AI-assisted clinical settings requires careful examination of the degree of decision-making authority delegated to AI systems. The extent of responsibility transferred to the AI determines the standard of care expected of clinicians, that is, the level to which they must critically evaluate the system's recommendations or, equivalently, the extent to which they are justified in relying on the AI's guidance. Hence, the importance of examining the connection between delegation and human responsibility.

⁵⁸ See Contissa and Lagioia (n 44); Millar and Kerr (Ch I n 100); Selbst (Ch III n 116), 16.

The influence and role of AI in clinical decisions can be analyzed in relation to the four cognitive function in the decision-making process.

First, regarding the function of *information acquisition*, AI systems can support human doctors by automatically gathering and curating relevant information for the clinical process they are following. These systems aggregate data from multiple sources, such as a patient's health records, medical literature, and domain-specific clinical trial reports, and filter or highlight items most relevant to the case. For instance, consider an AI tool used in oncology: when evaluating a patient with a rare cancer mutation, the system may automatically prioritize data on recent studies related to similar mutations and relevant drug efficacy rates, while de-emphasizing unrelated topics like treatments for leukemia. The criteria for integrating, filtering, and emphasizing relevant information are embedded at the design level, beyond the clinician's control. As a result, the information acquisition function is fully delegated to the AI system.

For the second cognitive function, *information analysis*, AI systems perform complex analyses by comparing and processing the available data according to pre-set parameters established during design. This involves aggregating patient-specific data with relevant clinical literature to generate insights, correlations, or predictions. For example, in oncology, AI might analyze genetic markers, cancer characteristics, and patient history to determine the likelihood of success for specific treatments. This level of analysis is delegated to the AI to the point where the internal decision-making processes – the ways in which parameters are weighted, and correlations drawn – are inaccessible to the doctors.

For the third cognitive function, *decision and action selection*, AI provides doctors with a ranked list of potential diagnoses, each paired with a confidence score. For instance, in assessing a patient with complex symptoms, the system might list possible conditions such as lupus, rheumatoid arthritis, or fibromyalgia, with accompanying confidence levels based on the supporting evidence from various data sources like the patient's electronic medical record, symptom history, and relevant clinical studies. Importantly, here, physicians retain the authority to either accept or

reject AI's suggestions. They can explore alternative diagnoses or consider factors outside of AI's purview, such as nuanced patient preferences or unique contextual information that the system may overlook. This flexibility entrusts doctors to incorporate both the AI's data-driven insights and their clinical judgment, ensuring that they can correct potential errors or biases in automated recommendations. As a result, this cognitive function takes advantage of automated decision support but still relies on human judgment for the final decision.

Finally, as it concerns *action implementation*, the AI system does not directly intervene in the administration of medical treatments. The final implementation of actions, such as administering medication or performing a procedure, is entirely in the hands of human experts. Physicians, nurses, or other healthcare providers are responsible for executing the treatment plans, monitoring the patient, and making adjustments as necessary. For example, in a scenario where the system suggests a course of chemotherapy for a cancer patient, the physician would evaluate the recommendation but would independently carry out the treatment process, administering the drugs, monitoring side effects, and adjusting the treatment as needed. In this sense, AI's role is purely advisory, and the human overseers retain full responsibility for action implementation.

It is evident that, while AI reaches full automation in information acquisition and analysis, physicians remain central to decision and action selection, as well as implementation, and may be held liable for errors made in performing such functions when there is a fatal outcome. This apparently straightforward conclusion, however, raises questions about the cognitive and practical challenges of human oversight in situations where the doctor's role is essentially to *mediate* AI advice.⁵⁹

Although the final decision technically rests with the physician, their intermediary role between machine output and clinical action involves reconciling the trust placed in the AI system's capabilities with their professional judgment and the specific needs of the patient. In such settings, despite retaining formal control

⁵⁹ Parasuraman *et al* (n 32), 286–297.

over AI outputs and ultimate decision-making authority, the expectation of meaningful oversight may just be a *legal fiction*. Here, the standard of the ‘reasonable doctor’ (what is reasonable to expect from the average physician using an AI system as a decision-support in the specific medical field in question) is more theoretical than practical, operating in an idealized space that overlooks cognitive and practical challenges inherent in monitoring advisory systems. These include the mental fatigue of maintaining constant oversight, the unexplainable nature of AI decisions, and automation bias.⁶⁰

In practice, while designed to support decision-making, AI can *de facto* lead to a reduction in perceived agency as they inadvertently influence, shape, or even steer the clinical decisions made by healthcare professionals. In particular, the physician’s agency can become limited when the AI output seems overwhelmingly reliable or too complex to challenge. In addition, automation bias can manifest as errors of commission (following flawed advice) or omission (failing to act when not prompted by AI or failing to follow AI suggestion), which are well-documented issues in AI-assisted medical settings. Research highlights that automation bias is not easily mitigated and often influences clinical decisions, especially under time constraints.⁶¹

As a result, the integration of AI into clinical decision-making may create a complex liability landscape, where the line between human responsibility and AI influence becomes blurred. Both the doctor and the AI system will make errors, but these errors arise from fundamentally different processes and assumptions. Critically, unlike human oversight of other humans (e.g., a junior doctor or practitioner relying on a senior doctor), AI advice lacks the transparency and accountability traditionally associated with human error. While erroneous advice from a senior doctor may be exculpatory for the junior doctor or practitioner who decides to follow it, a doctor’s

⁶⁰ Lisanne Bainbridge, ‘Ironies of automation’ (1983) 19(6) *Automatica* 129–135.

⁶¹ Bartlett (n 10), 248; K Goddard, A Roudsari, and JC Wyatt, ‘Automation Bias: A Systematic Review of Frequency, Effect Mediators, and Mitigators’ (2012) 19(1) *Journal of Medical Informatics Association*, 121–127.

reliance on AI might simply appear negligent in retrospect, particularly under judicial scrutiny, where automation bias and contextual pressures are often underappreciated.

As reliance on AI in clinical decision-making processes becomes increasingly common and socially acceptable, insisting on sole physician liability could amount to a legal fiction, given that the criminal act would involve the critical examination and validation of decisions that are fundamentally impossible to interpret – and criminal negligence cannot demand the impossible as part of what is reasonable. The assumption that clinicians can and should act as a fail-safe mechanism against AI flaws ignores the reality that, without interpretability, even highly skilled professionals are limited in their capacity to detect and mitigate AI errors. Thus, positioning the physician as the ‘liability sink’ for AI-generated errors risks unfairly burdening them for outcomes that might exceed their capacity to foresee or counteract, given the limitations in understanding and controlling advanced AI outputs. If we are to really reap the benefits of AI in the healthcare sector, the physician cannot bear the brunt of responsibility for any errors or harm resulting from the treatment, even if the AI system played a substantial role in shaping the decision-making process.⁶²

The core question, therefore, becomes one of fair liability distribution for erroneous AI advice that significantly shapes clinical decisions. How far can a physician reasonably be expected to trust machine recommendations, especially when they lack the means to fully audit or understand the system? What extent of responsibility is fair when errors arise from an AI system with opaque processes that the physician cannot directly evaluate or override? If the AI system is designed to be a trusted decision-support tool, the legal standard may increasingly accommodate certain types of reliance on AI advice, provided the doctor’s trust aligns with established usage protocols and does not involve gross negligence, such as ignoring obvious signs of error.

⁶² Elish (n 39), 40–60.

This dynamic highlights the need for frameworks that clarify the balance of accountability between human overseers and AI systems, particularly regarding the role of oversight in risk mitigation.

The following section explores four failure scenarios in AI-assisted clinical decision-making and related liability assessment in the event of injuries suffered by a patient as a consequence of misdiagnosis or improper treatment. Each scenario is related to a failure in the execution of a specific cognitive function, considering the extent to which responsibility transfer and trust in AI influence liability allocation, especially in cases of misdiagnosis or improper treatment resulting in patients being harmed or injured. The findings should provide the basis for assessing the connection between delegation and trust in order to assess the allocation of liability.

2.3. Variations on a theme: scenarios of medical liability for AI-driven malpractice

2.3.1. Failures in the information acquisition

Consider a case where a patient dies due to misdiagnosis or inappropriate treatment causally linked to a failure in the information acquisition phase. Suppose that the error arises because the system integrates an outdated drug interaction warning, leading to a fatal drug prescription for a patient with a specific genetic predisposition.

Here, the AI system independently gathers and integrates data from multiple sources – medical records, literature, or clinical trials – and is responsible for selecting relevant information. If an error occurs due to incorrect or incomplete data acquisition by the AI system, the liability question hinges on whether the physician has a duty to oversee or verify the sources feeding into the system. Given the autonomous data acquisition by the machine, the physician's role generally excludes direct data control, making it reasonable for the doctor to trust the system's accuracy regarding the patient's specific situation. If such an error results from either human

mistakes (such as nurses misrecording or failing to update data) or technical failures (like faults in the medical examination process), neither the physician using the system nor the system's developers may be directly liable under current liability models.

Additionally, in instances where the failure lies in the AI system's retrieval or selection of relevant information for diagnosis or treatment recommendations, similar principles apply. For example, if the AI overlooks that a patient is pregnant and recommends a contraindicated drug, the doctor should not be held liable for relying on the AI. Physicians generally do not control the design-level criteria that govern how information is filtered and highlighted, and these criteria are defined solely by the developers. If the system undergoes full conformity assessment, the legitimate trust principle could effectively shield physicians from liability when they reasonably rely on the AI to perform the delegated task.

In such scenarios, accountability could shift to the provider or manufacturer, particularly if the system's design is defective, such as failing to alert users to critical missing information. The exception would be cases of gross negligence, where a doctor knowingly ignores obvious red flags. For instance, if the system flagged missing information (such as an alert noting that pregnancy data was missing) and the physician dismissed it, a court might find the physician liable under criminal negligence doctrine if the omission results in inappropriate treatment and subsequent patient harm.

2.3.2. Failures in information analysis

Also worth considering are cases of failure in the information-analysis phase, where the AI system autonomously processes and interprets massive amounts of data inputs in a way that would be impossible for any human doctor, to make diagnostic or treatment recommendations.

Consider an AI diagnostic tool that analyses blood samples to screen for autoimmune disorders. Suppose it misses signs of a rare autoimmune disease because its training data lacks sufficient examples of the disease's biomarkers. This oversight leads the AI to misclassify a case, resulting in delayed or incorrect treatment. In this situation, liability allocation hinges on the question whether and to what extent the human overseer (i.e., the physician) can rely on the system's ability to process and analyze blood samples.

To answer this question, it is crucial to note that the parameters are defined in advance at design level; they may not be visible to physicians, and even if they were, they may not be meaningful for humans.⁶³ Since the physician is unlikely to know the extent and quality of the training data used to program the system, it is not reasonable to assign to them the legal duty to be in control of the internal processing activity of the system. For instance, the clinician may lack knowledge about the training dataset of the blood analysis system and be unaware that it is less accurate for patients from some ethnic backgrounds; meanwhile, its influence may make the clinician more likely to question their own evaluation.⁶⁴

Thus, as in the previous failure scenario, if the system is certified under the conformity-assessment procedure, the legitimate trust principle should shield from liability the physician who reasonably relied on the AI and its ability to conduct complex data analysis. Moreover, even conceding that there is a design limitation in the AI's development, the doctor could not be expected to reanalyze every sample

⁶³ Millar and Kerr (Ch I n 100), 105.

⁶⁴ AI applications are particularly susceptible to bias. Bias in training data may occur because the data sources themselves may not reflect the true epidemiology within a given demographic. This means that errors may be far more likely to occur in under-represented groups such as ethnic minorities. L Ledford, 'Millions of Black People Affected by Racial Bias in Healthcare Algorithms' (2019) 574 *Nature*, 608-609. See in general on AI bias in healthcare D. Schönberger, 'Artificial Intelligence in Healthcare: A Critical Analysis of the Legal and Ethical Implications' (2019) 27(2) *International Journal of Law and Information Technology* 171-203.

manually, as this would fundamentally undermine the system-level goals and benefits by increasing costs and raising concerns about defensive medical practice.⁶⁵

An exception might occur if the physician is aware of specific gaps in the AI's training data. For instance, if the system documentation or training indicated limitations in rare disease recognition and the physician ignored alternative diagnostic possibilities, liability might be attributed to the physician for gross negligence. Additionally, the physician may be found liable if the failure is attributable to their negligently ignoring an alert.

Finally, if the AI's analysis missed the disease markers, liability could potentially rest with the provider or manufacturer for failing to ensure the AI was adequately trained on diverse cases, as well as to the notified body and members of the expert panel, if they were involved in evaluating and assessing the system design and functioning.

2.3.3. Failures in decision and action selection

In the decision and action selection phase, AI systems do more than just analyze data; they actively suggest specific diagnoses or treatment plans for clinicians to consider. This phase is especially critical in clinical contexts, placing the clinician in a position to accept, override, or disregard the AI's recommendation based on their own professional judgment and the patient's unique circumstances, in line with their responsibilities under Article 14 of the AIA.

Consider a scenario where an AI system, having *correctly* performed the information analysis, generates a ranked list of potential diagnoses with confidence scores. It proposes Condition A as the most likely diagnosis with associated treatment recommendations, while Conditions B and others are assigned significantly lower probabilities. It then leaves the clinician the possibility and freedom to select the best

⁶⁵ Bartlett (n 10), 249.

hypothesis and/or to generate alternative. However, given the AI's certified diagnostic trustworthiness and accuracy, the clinician perceives its recommendations as reliable and proceeds with the treatment for Condition A without further independent checks. As a result, the patient deteriorates and ultimately dies, with the underlying cause traced back to a less probable condition that the AI had deprioritized.

From a criminal liability perspective, several considerations come into play in this scenario. To begin with, it would be perverse not to accept that physician can legitimately rely on the AI system when the latter has undergone conformity-assessment procedures before adoption and has been tested to performs (on average) better than humans at the particular task at hand. Instead, the key question, here, would be whether – and to what extent – a doctor's reliance on erroneous AI advice constitutes a breach of their duty of care – that is, when a reasonably competent doctor would have been aware of the error inherent in the AI output and avoided the harmful result.

In principle, it may be tempting to hold a doctor criminally liable simply for following an erroneous AI advice, based on the assumption that they retain ultimate decision-making authority for validating AI outputs and are thus expected to exercise independent judgment. However, this expectation for oversight assumes that doctors are both able and equipped to control AI systems and recognize errors in their outputs – an assumption that may be problematic given the 'black-box' nature of many AI models. As AI systems grow more complex and accurate, it becomes increasingly difficult for clinicians to detect subtle errors, which may be deeply embedded in the algorithm's opaque processes, thereby making the decision whether to accept or reject its recommendation potentially difficult. In addition to the inherent limitations of human oversight capabilities, the considerable influence of AI on clinical decisions must also be kept into account. If an AI recommendation appears robust and is consistent with established medical standards, it might seem reasonable to trust it, particularly if the system has a history of reliability in similar cases. Ultimately, factors like automation bias, the opacity of AI decision-making, and

inherent epistemic limitations create a *legal fiction* of control where humans may be unjustly exposed to criminal liability for AI fatal errors.⁶⁶

This complex backdrop suggests a need for caution and prudence in criminal assessments of AI-related medical negligence, if AI is to be effectively adopted into clinical decision-making.⁶⁷ Here, the principle of legitimate trust could be key to avoid the *impasse*: if an AI system has met certified accuracy standards and is generally regarded as highly reliable, a doctor's reliance on its outputs may be seen as *reasonable* under typical standards of care, shielding them from potential legal consequences for errors being baked into the system's decision-making process. In our example, the AI's high-confidence ranking for Condition A suggests a high degree of reliability, and the clinician's decision to proceed based on this recommendation rather than seeking additional verification could be seen as standard medical reasoning rather than a lapse in diligence.

In order to avoid cases of impunity, however, this liability shield should only operate *unless* there are clear signs or warnings that contradict the AI's output. Suppose that a doctor asks AI for advice on a patient whose cancer has not spread to the lymph nodes, and AI recommends a type of chemotherapy drug called a taxane. To support the recommendation, the system shows a study demonstrating the effectiveness of the taxane for patients whose cancer did spread to their lymph nodes. The doctor knows that this therapy is normally used only if the cancer has spread to the lymph nodes. The doctor is left confused as to why AI recommended a drug that he does not normally use for patients like the one in front of him. And the system

⁶⁶ Tom Lawton *et al*, 'Clinicians risk becoming 'liability sinks' for artificial intelligence' (2024) 11(1) Future Healthcare Journal.

⁶⁷ Many scholars challenge the opportunity of using 'black box systems' in high-stakes environments like healthcare. Therefore, some argue that AI in healthcare should be restricted to more 'interpretable' models. C Rudin, 'Stop Using Black Box Machine Learning for High Stakes Decisions and Use Interpretable Models Instead' (2019) 1 Nature Machine Intelligence 206–215.

cannot tell him why. Nevertheless, he applies the therapy without double checking, which ends up being fatal for the patient.⁶⁸

In this scenario, the doctor's choice to follow AI recommendations that clearly contradict established medical knowledge could indeed constitute a gross deviation from the standard of care, possibly rising to the level of recklessness or conscious negligence. Here, the presence of red flags – the use of a taxane despite no lymph node involvement and the system's citation of a study applicable to a different patient profile – should have prompted further scrutiny. These inconsistencies between the AI's recommendation and accepted treatment norms represent a clear 'alert' moment, where the doctor, as a reasonably competent physician, would recognize a need for critical assessment.⁶⁹

In cases like this, the doctor's reliance on AI appears to cross into a realm of conscious disregard for patient safety, as they knowingly sidestep the necessity of verifying whether the recommendation aligns with established standards and patient-specific factors. This is not simply a lapse in judgment but rather a voluntary choice to ignore risk. As a result, the principle of legitimate trust, typically allowing doctors to rely on certified, high-quality AI recommendations, becomes inapplicable when clear evidence suggests that the recommendation is unreliable. The doctor's decision to follow the recommendation without further verification reflects a conscious disregard of evident risks, which could potentially justify criminal liability for recklessness or gross negligence.⁷⁰

⁶⁸ For the example see Casey Ross and Ike Swetlitz, 'IBM pitched Watson as a revolution in cancer care. It is nowhere close' (STAT News Sept 5 2017) available at: <https://www.statnews.com/2017/09/05/watson-ibm-cancer/>.

⁶⁹ Federico Carmelo La Vattiata, 'Artificial Intelligence in Healthcare: Risk Assessment and Criminal Law', in Irakli Beridze *et al.* (eds.) *UNICRI Special Collection on Artificial Intelligence* (United Nations Interregional Crime and Justice Research Institute 2020) 48-58.

⁷⁰ For instance, Article 2236 of the *Italian Civil Code* states: 'If the professional services involve the resolution of technical problems of particular difficulty, the person who renders such services is not liable in damages, except in case of fraud, malice or gross negligence'. While it may not be applicable in the field of criminal law, this provision expresses a principle of rationality whereby if the problem the professional is called to solve is particularly

Having said this, several challenges still need to be faced in defining the boundaries of such liability. In the first place, identifying ‘red flags’ in a timely manner is not always straightforward, especially in a high-pressure clinical setting characterized by socio-technical complexity. Multiple, interwoven factors – urgent time constraints, the intricate interplay of clinical judgment with opaque AI models, and the cognitive demands of assessing AI output – contribute to the difficulty in recognizing when automated advice should be scrutinized, bringing the issue of enforceability and avoidability of harm back into focus.⁷¹

Furthermore, the opposite scenario could occur: if the clinician decides not to rely on the AI and the patient subsequently dies, in what circumstances should they be criminally responsible? In scenarios where a doctor consciously chooses not to follow an AI recommendation – especially if this decision stems from doubts about the system’s alignment with known medical protocols or an incomplete understanding of its reasoning – the potential for liability grows complex, particularly if this choice leads to adverse outcomes. As AI systems become increasingly accurate and integrated into healthcare, a clinician’s deviation from AI guidance could indeed raise questions about duty of care, especially if reliance on AI becomes not only an accepted but even expected component of medical practice.

If the state of the art in AI consistently surpasses traditional diagnostic capabilities, then this could indeed shift the source of decision-making authority from the human to the machine, suggesting that adherence to AI recommendations might evolve from being an option to being seen as a required standard under certain conditions. In such a framework, if the AI is considered as such a source, then liability can be attributed to human physicians under a specific duty to follow the advice of

complex, a fair assessment can lead to a charge of negligence only if there is a gross violation of the established standards of care. This rule, which clearly limits the liability of the party to cases of gross negligence, is aimed at encouraging the undertaking of risky or difficult commitments that have also a social function such as, in our cases, risky major surgical operations or complex diagnosis, as the liability is excluded in cases of ordinary or slight negligence.

⁷¹ Beatrice Fragasso, ‘La responsabilità penale del produttore di sistemi di intelligenza artificiale’ (2023) 1 Diritto Penale Contemporaneo, 37.

the system. Any divergent decision – particularly when the system’s recommendation carries a high confidence score and is regarded as the standard of care – may be evidence of a serious breach of duty, potentially increasing their exposure to unfair liability claims. Over time, omission errors, where a clinician disregards accurate AI guidance without sufficient justification, could meet the threshold for gross negligence if the clinician’s decision appears unreasonable or out of step with current standards.⁷²

Ultimately, oversight duties impose a heavy cognitive and ethical burden on clinicians, who must discern when deferring to AI is warranted and when it may not be so. Operators are faced with a take-it-or-leave-it problem which places them in a challenging, often paradoxical position: they risk liability whether they rely on AI or dismiss it, given that both actions could be construed as negligence depending on the outcome.⁷³ They are caught either way: potentially damned if they do rely on the AI when they should not and damned if they don’t when they should.⁷⁴

On one hand, it would be perverse not to acknowledge that clinicians do remain the primary agents responsible for planning and delivering patient care. We agree that for as long as this is the case, they should not be relieved of their legal responsibility to patients as the ultimate decision-makers, regardless of whether they incorporate AI into their practice. On the other, however, if operators are not taught how to navigate the risks that arise from AI use, then they might reasonably choose to avoid AI altogether to mitigate potential liability exposure. ‘Defensive medicine’ could ultimately undermine the intended value of AI in healthcare by discouraging its use, leading to missed opportunities for improved care.

⁷² It is debatable whether such a shift in decision-making authority will be possible, particularly given the trust relationship between patients and doctors. Interestingly, an earlier version of the draft AI Act included a provision stipulating that human overseers should have been able to ‘decide not to use the high-risk AI system or its outputs in any particular situation without any reason to fear negative consequences’, see European Commission ‘Draft Regulation on a European Approach for Artificial Intelligence’ (Politico 14 April 2021) available at www.politico.eu/wp-content/uploads/2021/04/14/AI-Draft.pdf, 3.

⁷³ Bartlett (n 10), 248.

⁷⁴ Helen Smith *et al.*, ‘Artificial Intelligence for Clinical Decision-Making: Gross Negligence Manslaughter and Corporate Manslaughter’ (2024) 38(1) Bioethics, 9.

To address these uncertainties and delineate a clear scope of legitimate trust, clearer clinical guidelines could be established. These guidelines would outline under what conditions AI recommendations should be considered obligatory (e.g., when accuracy rates reach a certain threshold in specific cases) and where clinical judgment can appropriately override AI advice.⁷⁵ For instance, one solution might be to mandate that clinicians conduct confirmatory testing for AI recommendations involving high-risk treatments or rare diseases, even if flagged at low probabilities.⁷⁶

Last but not least, where is the provider in all of this? The role of the AI provider is critical in the action-selection phase because human-machine interactions must be designed to promote effective interaction between the operator and the system. A poorly designed or inadequate informational interface creates opportunities for critical information to be garbled or lost in translation.⁷⁷

Consider the case of an AI diagnostic tool that is highly accurate but lacks sufficient interpretability or fails to clearly communicate uncertainty levels in its recommendations. Suppose a clinician using this tool is presented with a high-confidence recommendation that a patient has a less serious condition, despite symptoms suggesting otherwise. The system, however, does not provide any contextual information on how this recommendation was derived or display possible differential diagnoses. This lack of transparency limits the clinician's ability to effectively interpret or challenge the AI's suggestion. Following the AI's advice, the

⁷⁵ *Id.* 6.

⁷⁶ Yet, this option would not only be practically unrealistic or unattainable, but would also undermine the intended efficiency of the AI system and increase the clinical workload. Conducting unnecessary and possibly invasive tests for conditions with only marginal probability would diminish the practical value of the AI's diagnostic insights. Bartlett (n 10), 249.

⁷⁷ This is a recurring problem in the history of automation. Consider, among others, the *Three Mile Island* meltdown, where there were no sensors to detect how much water covered the nuclear reactor, and an instrument erroneously communicated that an open valve was closed; or the Air France Flight 447 disaster, where Boeing failed to design an effective interface between pilots and the MCAS system. See Crootof *et al* (n 5), 498; see also Elish (n 39), 40–50 (detailing how human-in-the-loop interface design issues contributed to the Three Mile Island and Air France Flight 447 disasters).

clinician opts for a conservative treatment plan, which ultimately proves insufficient, leading to severe complications for the patient.

In this scenario, it is safe to say that liability for serious harm to patients could reasonably extend to the AI provider if the design flaw (the absence of effective informational human-machine interface) obstructed the clinician's ability to exercise proper oversight. The provider's failure to implement transparency features – such as explanations, warnings, or uncertainty indicators – could be viewed as negligence in the duty of care in technology design, as these elements are essential for promoting effective, informed interaction between the human and the system. If this design flaw is shown to have foreseeably contributed to the patient's harm, the provider could face liability for failing to promote safe interaction with the system. Similarly, liability can extend also to the notified body and the members of the expert panel, if during the conformity assessment procedure some anomalies emerged in the clinical testing phase and were ignored.

2.3.4. Failures in action implementation

In the action implementation phase, any failure would be directly attributable to the human operator, as this phase relies solely on manual execution and control without AI intervention. For instance, if a doctor administers an overdose of a prescribed drug, the liability would fall on the individual performing the action, given that the AI's role ceases after delivering its recommendation. Indeed, system does not oversee or participate in administering treatment, thereby delineating responsibility exclusively to human practitioners. In this context, negligence or malpractice claims would thus target the clinician or caregiver directly. This is especially relevant in healthcare, where standard duty-of-care principles obligate human experts to verify dosage and follow safe administration protocols independently of any AI-provided guidance.

SECTION 3

Human-machine interaction #2: human-*on*-the-loop

3.1. Introduction to hybrid human-machine interaction paradigm

AI assistance, as illustrated in the previous section through the case study of clinical decision-making, generally represents a fairly straightforward model of human-machine interaction and responsibility allocation, as humans retain final decision-making authority even if heavily influenced by AI recommendations. This framework assumes an active supervisory role where human operators oversee AI outputs and decide whether to act on them based on their professional judgment. However, when decision-making authority is *fully delegated* to the machine, which basically takes over the responsibility for decision selection and action implementation, things become considerably more complex.

This scenario exemplifies what we refer to as ‘hybrid’ human-machine interaction, prevalent across safety-critical sectors like aviation, nuclear energy, and increasingly autonomous driving. In such environments, AI systems are typically leveraged to handle high-stakes, repetitive tasks in which machines consistently outperform humans. Here, the human role shifts from active decision-making to vigilant monitoring, with intervention limited to exceptional cases or system failure. This model relies on an assumption that humans will intervene and override the AI only under certain ‘red-flag’ conditions or when the system’s reliability falters (so-called ‘human-on-the-loop’)

Generally, in the design of AI systems, the hybrid interaction paradigm aims to capitalize on the unique strengths of both machine and human agents – machine efficiency and human judgment.⁷⁸ This approach to full automation is attractive for

⁷⁸ This approach is known as the ‘MABA-MABA’, or ‘Men Are Better At’ and ‘Machines Are Better At’. Meg Leta Jones, ‘The Ironies of Automation Law: Tying Policy Knots with Fair Automation Practices Principles’ (2015) 18 Vand. J. Ent. & Tech. L. 104-106.

policymakers in part because the HOTL is a concrete and identifiable entity, providing clear regulatory targets, as opposed to the diverse and diffuse group of contributors involved in system development.⁷⁹ Additionally, statistical evidence supports this design choice: while machines excel at high-speed, repetitive tasks with consistent accuracy, humans bring a level of adaptability and contextual understanding that is crucial for nuanced decision-making.⁸⁰

However, hybrid human-machine interactions bring unique complexities beyond simply adding human judgment to machine performance. The relationship between the human overseer and the machine fundamentally shapes both the capabilities and limitations of the AI system. This interaction becomes an integral part of the AI's performance, affecting outcomes in ways that neither the human nor the machine could achieve alone. But at the same time, it also introduces nuanced risks and raises challenging questions about responsibility allocation. Since outcomes are a product of joint/shared decision-making, it becomes difficult to disentangle responsibilities when errors occur, parsing where human responsibility ends and machine responsibility begins – particularly if the machine's actions are essentially autonomous or if human intervention is required only under exceptional circumstances.

In this section, we aim to demonstrate that simply adding a human 'on the loop' is far from the ideal regulatory solution it might seem. Instead, this approach creates a new entity: a hybrid system. And hybrid systems create or exacerbate well-known problems – problems policymakers currently largely ignore or fail to address properly.

When human oversight is added as a safeguard without sufficient consideration of cognitive load, situational awareness, and the effects of automation bias, the result can be counterproductive. Hybrid systems that rely on humans to

⁷⁹ Selbst (Ch III n 113), 437-438.

⁸⁰ A. J. Lohn, 'Estimating the Brittleness of AI: Safety Integrity Levels and the Need for Testing Out-Of-Distribution Performance' (2020) arXiv available at <http://arxiv.org/abs/2009.00802>, 6.

monitor and occasionally intervene in complex, automated processes tend to demand continuous attentiveness and rapid response. However, these expectations often overlook the limitations of human cognition, particularly when the operator is in a passive monitoring role, only called to act in emergency or anomaly situations. In such contexts, reaction times may be slower, and the capacity to interpret complex or ambiguous outputs under pressure may be compromised.

These factors in combination can inadvertently create conditions that increase the likelihood of serious, even deadly, errors while positioning the human operator for inevitable failure – and often, unfair attribution blame.⁸¹ In fact, the human tasked with catching errors in a high-stakes environment, can become the focal point for blame in the event of failure, even when the root cause lies in the AI system’s design, data limitations, or unanticipated behavior. As a result, the human becomes a liability scapegoat rather than a true safeguard, bearing responsibility for oversight in a system that, by design, limits their capacity to effectively intervene.

Therefore, rather than simply adding a human layer to autonomous systems, there is a critical need to reevaluate how such systems allocate responsibility, support human judgment, and ensure that the role of oversight is practical and attainable. To prevent setting operators up for failure, we must design hybrid interactions that consider the inherent risks and complexities of these human-machine partnerships.

Our analysis will be conducted taking as a case study one of the most famous examples of such paradigm – autonomous vehicles (AV). For AVs, the most obvious candidate for the role of human overseer is obviously the *back-up driver* sitting behind the wheel.

3.2. The case of autonomous vehicles: human operator as a liability shield

⁸¹ Parasuraman (n 32), 291.

AVs introduce new ways of interaction between human driver and machine, particularly by increasingly shifting control of the driving task to an autonomous system. In this hybrid model, the human driver and machine increasingly share control, with the machine handling tasks that were once exclusively human-managed.

However, it is crucial to understand from the outset that self-driving features are integrated into vehicles incrementally, rather than all at once. Indeed, according to the taxonomy set by the *Society of Automotive Engineers* (SAE), the levels of autonomy in AVs range from 0 to 5.⁸² This division into levels is crucial for regulatory and safety considerations, as each level delineates the allocation of decision-making authority and responsibility between the driver and the autonomous system. Each new self-driving feature affects this balance, defining specific contexts in which the machine assumes a larger share of the driving tasks, reducing the degree and frequency of human intervention required. The levels of driving automation can be described as follows:

- **Level 1 (Driver Assistance):** Automation assists with either lateral (steering) or longitudinal (acceleration and braking) control within specific conditions, though the driver remains actively engaged. Examples include adaptive cruise control or lane-keeping assistance, where specific but limited functions are delegated to the system.
- **Level 2 (Partial Automation):** The system controls both steering and acceleration/braking simultaneously, but the human driver is responsible for

⁸² SAE J3016, 'Recommended Practice: Taxonomy and Definitions for Terms Related to Driving Automation Systems for On-Road Motor Vehicles' (2014) commonly referenced as the SAE Levels of Driving Automation. Since its initial launch in 2014, has been the industry's most-cited source for driving automation. With a taxonomy for SAE's six levels of driving automation, SAE J3016 defines the SAE Levels from Level 0 (no driving automation) to Level 5 (full driving automation) in the context of motor vehicles and their operation on roadways. See Maurice Schellekens, 'Human-machine interaction in self-driving vehicles: a perspective on product liability' (2022) 30(2) *International Journal of Law and Information Technology* 233-248.

monitoring the environment, detecting objects or events, and being ready to intervene at any moment. While the vehicle can handle basic driving tasks, continuous oversight by the human driver is required.

- **Level 3 (Conditional Automation):** The automated system performs all driving tasks within specific conditions, but the driver must be ready to take over when requested. Here, the system is capable of making requests for human intervention if needed, allowing the driver some reprieve, though they remain ultimately responsible.
- **Level 4 (High Automation):** The vehicle can drive fully autonomously within certain operational design domains (ODDs), such as specific geographic areas or weather conditions, without requiring human intervention. If the vehicle encounters a situation outside of its defined ODD, it can bring itself to a safe stop if the driver does not respond to a takeover request.
- **Level 5 (Full Automation):** The system is capable of fully autonomous driving without any limitations to specific ODDs, meaning it can operate under all road and environmental conditions where a human driver could. In this level, human intervention is not required at any point.

The term ‘autonomous vehicle’ typically applies to systems at Levels 3 and above, where the vehicle progressively takes on more driving responsibilities, reducing the role of the human driver and legally allowing them to take their attention off the road. For example, Mercedes’ Drive Pilot, the world’s first fully certified Level 3 AV, was approved in 2022 in Germany to drive only within geofenced areas on the German highway (up to 60km/h).⁸³ This system allows the driver to take their hands

⁸³ In Germany, national legislation has acted early. In 2021 and 2022, the German government became the first country in the world to create the legal framework for vehicles with autonomous driving functions for cars and trucks. On July 28, 2022, two pieces of

off the wheel and allow the AV to assume total control of the car's functions. Outside of Germany, current models with self-driving capabilities on the market are mainly Level 2: Tesla's Autopilot, Audi's Traffic Jam Assist, GM's Super Cruise, BMW's Extended Traffic Jam Assistant, Ford's Blue Cruise, Hyundai's automated driving package, and many more.⁸⁴ In all such cases, the vehicle still needs a human behind the steering wheel who is supposed to control or monitor the vehicle for safety reasons, in order to correct errors it might otherwise make.⁸⁵

This progressive shift of decision-making authority from human to machine holds substantial social value, primarily because human error statistically accounts for the vast majority of car accidents.⁸⁶ By minimizing reliance on human management, which is prone to distraction, fatigue, and emotional biases, policymakers have the potential to put together the best of both worlds – significantly enhancing road safety through their automated capabilities, on one hand, while still mandating different degrees of human control, on the other.

legislation came into effect: the Autonomous Driving Act and the Ordinance Regulating the Operation of Motor Vehicles with Automated and Autonomous Driving Functions and Amending Road Traffic Regulations (AFGBV). These allow autonomous driving functions in fully automated passenger and freight transport vehicles on defined routes.

⁸⁴ For more details, see Contissa and Sever, 'Automated driving regulations – where are we now?' (2024) 24 Transportation Research Interdisciplinary Perspectives 1-19.

⁸⁵ Level four vehicles already exist and are used in certain predefined routes and under specific circumstances (e.g. shuttle buses at airports, university campuses, etc.). At level four, the system can perform all driving tasks without a driver under specific conditions. Level five vehicles, which perform all driving tasks in any operating conditions without any human attention or intervention, are still not on the market. The presence of a safety driver in the vehicle is no longer mandatory. Here, we talk about 'human-out-of-the-loop'. On this configuration of human-machine interaction, see *infra* para 4.

⁸⁶ Data from the European Commission highlights the urgency of this goal: annually in Europe, human error is responsible for over 90% of road accidents. Moreover, these accidents result in the tragic loss of more than 40,000 lives and cause injuries to more than 1.5 million individuals. Report from the commission to the European parliament and the council saving lives: Boosting Car Safety in the EU Reporting on the monitoring and assessment of advanced vehicle safety features, their cost effectiveness and feasibility for the review of the regulations on general vehicle safety and on the protection of pedestrians and other vulnerable road users, COM/2016/0787 final.

Various legal and regulatory frameworks at the international and EU levels explicitly mandate human involvement in AV operations, reflecting the cautious incremental approach to driving automation. In particular, even though Article 14 AIA was not originally designed with AVs in mind, its requirements for human oversight in high-risk AI systems could be interpreted to mandate the presence of a human driver in the loop.⁸⁷ This aligns with the approach taken under EU regulation 2019/2144 on type-approval requirements for motor vehicles, which defines an automated vehicle as one capable of operating autonomously for certain periods without continuous driver supervision, but where ‘driver intervention is still expected or required’.⁸⁸ This means that until full automation is available, intervention (either by back-up drivers or non-professional drivers) is still mandatory to address unexpected system behaviour or failures that might lead to accidents.⁸⁹

More specifically, the 2024 standards for assisted driving systems, developed by the *World Forum for the Harmonization of Vehicle Regulations* of the United Nation Economic Commission for Europe (UNECE), mandate a similar degree of human involvement. These standards, mainly applicable to SAE Level 2 vehicles, require drivers to maintain responsibility for vehicle control even when the vehicle initiates manoeuvres by itself, such as automatically changing lanes, without any driver

⁸⁷ At first glance, most AVs are expected to be classified as high-risk systems under the EU’s AI Act, given the critical nature of in-vehicle decision-making. Although the Act does not – despite its cross-sectoral approach – directly apply to AVs and their AI components, several of its requirements will also apply to AVs. This is because they will be introduced to AVs in the future through delegated acts of the Commission under the *type-approval framework for motor vehicles*.

⁸⁸ EU regulation 2019/2144 of the European Parliament and of the Council of 27 November 2019 on type-approval requirements for motor vehicles, EUR-Lex - 32019R2144 - EN - EUR-Lex, 2019, points 21 and 22 of Article 3.

⁸⁹ According to SAE, the driver must intervene and bring the vehicle to a safe stop when the machine is unable to execute the task at hand or the system shows clear signs of impending failure. SAE International, 2021. J3016_202104: Taxonomy and Definitions for Terms Related to Driving Automation Systems for On-Road Motor Vehicles. Jonathan Dyble, ‘Understanding SAE Automated Driving – Levels 0 to 5 Explained’ (GIGABIT Apr. 23, 2018) available at <https://www.technologymagazine.com/ai/understanding-sae-automated-driving-levels-0-5-explained>.

input or confirmation.⁹⁰ The hands-on driving mandate reinforces the principle that drivers must continuously monitor their surroundings and the vehicle's performance, prepared to intervene whenever the system encounters difficulties or fails to respond adequately.

However, as regulatory discussions advance to SAE Level 3, this dynamic evolves significantly.⁹¹ In Level 3 scenarios, the system assumes full control of driving tasks within defined operational domains, shifting the responsibility for primary vehicle control to the AV. This shift allows drivers to remove their hands from the wheel, provided they remain visually attentive to the driving environment and ready to intervene upon system request or when they notice or should notice an obvious situation where the intended conditions of the automated driving functions are no longer met. At this point, it is safe to claim that the obligations of the L3 back-up driver seem to be less demanding than those of the L2 actual driver. Or, at least, in theory.

In practice, indeed, this may not be the case. Many experts argue that AVs relying on human supervision or intervention to any degree will prove a safety risk when used at a large enough scale by the driving public.⁹² There are several reasons why achieving meaningful human oversight in $\geq$ L3 systems poses significant

⁹⁰ These rules are relevant for our analysis because a majority of the EU's technical rules for vehicle systems are made in that forum, in cooperation with countries outside of the EU and Europe.

⁹¹ The possibility of hands-off driving is also being discussed, which would allow drivers to take their hands off the steering wheel while using these assistance systems. Again, the driver would remain responsible for controlling the AV. See European Transport Safety Council, 'Risks of hands-off driving and system-initiated lane changes in Level 2 driver assistance systems' (ETSC September 6, 2024) available at <https://etsc.eu/risks-of-hands-off-driving-and-system-initiated-lane-changes-in-level-2-driver-assistance-systems/>.

⁹² Robert Sparrow and Mark Howard, 'When human beings are like drunk robots: driverless vehicles, ethics, and the future of transport' (2017) 80 *Transport. Res. Part C: Emerg. Technol.*, 206-215. There are several online databases that collect and classify incidents involving autonomous vehicles; among the most up-to-date is the Autonomous Vehicle Crashes website (www.avcrashes.net), which also provides the option to view the data through an interactive map. As of May 2, 2023, the site recorded 630 incidents occurring between October 14, 2014, and May 1, 2023.

challenges. The main shortcomings relate to inherent human cognitive and physical limitations, namely *overreliance* and *distraction*.

One of the most critical is the scientifically established fact that human drivers are generally poor supervisors in passive monitoring roles. Crash investigations and studies have found that drivers tend to over-rely on these AVs, as they overestimate the systems' capabilities and underestimate (or are unaware) of their limitations.⁹³ There are several reasons for this.

First, prolonged periods of inactivity, where the system handles most or all driving tasks, often lead to reduced attentiveness and a decline in situational awareness. This phenomenon, commonly referred to as 'automation complacency', significantly diminishes a driver's ability to effectively intervene when needed.⁹⁴ Perrow's analysis on system failures in high-risk domains further underscores this issue. He notes that 'when the pilot is suddenly and unexpectedly brought into the control loop (in other words, participates in decision making) as a result of (inevitable) equipment failure, he is disoriented [...]. The sudden appearance of several alarms, all there for safety reasons, leads to disorientation'.⁹⁵ This disorientation is compounded by the simultaneous appearance of multiple alarms, each intended to enhance safety but collectively overwhelming the operator. The abrupt transition from passive observer to active decision-maker leaves little room for the human to

⁹³ Among others, see NTSB investigation report on the Mountain View crash in 2018: 'The National Transportation Safety Board determines that the probable cause of the Mountain View, California, crash was the Tesla Autopilot system steering the sport utility vehicle into a highway gore area due to system limitations, and the driver's lack of response due to distraction likely from a cell phone game application and overreliance on the Autopilot partial driving automation system' at <https://www.nts.gov/investigations/pages/HWY18FH011.aspx>. See also NTSB investigation report on the Delray Beach crash in 2019: 'The National Transportation Safety Board determines that the probable cause of the Delray Beach, Florida, crash was the truck driver's failure to yield the right of way to the car, combined with the car driver's inattention due to overreliance on automation, which resulted in his failure to react to the presence of the truck' at <https://www.nts.gov/investigations/Pages/HWY19FH008.aspx>.

⁹⁴ For example, this effect is well established in the aviation industry, for autopilots. Parasuraman *et al* (n 32), 291.

⁹⁵ Perrow (Ch II n 112), 132.

accurately assess the situation and respond appropriately, particularly under high-pressure conditions.

Second, from the perspective of an ordinary citizen, the distinction between L2 and L3 AVs can appear negligible, particularly when both systems include advanced functionalities like automated lane-changing on highways. Both systems control the vehicle, monitor the environment, and react to events in real-time. However, the fundamental difference lies in the allocation of responsibilities: in L2 systems, the driver is still considered to be in control of the vehicle and remains fully responsible for driving tasks, whereas in L3 systems, the AV assumes control over specific driving functions within its operational design domain, with the human driver expected to intervene only when requested.

This blurring of roles becomes increasingly problematic as driving tasks are progressively automated. From the driver's perspective, the behaviour of an L2 system might closely resemble that of an L3 system, leading to potential 'mode confusion'. This occurs when the driver misunderstands the system's capabilities, responsibilities, or limitations. This ambiguity not only increases the cognitive load on drivers but also raises critical legal questions about how to clearly define and communicate the driver's role in supervising the system. If drivers are unclear about their oversight responsibilities or the circumstances under which they must regain control, the effectiveness of hybrid human-machine interaction is compromised, heightening safety risks and complicating liability assessments in the event of accidents. For example, a driver might overestimate an L2 system's ability to handle complex situations, assuming it to function autonomously, which can result in dangerous delays in human intervention when necessary.

Additionally, speaking of delays, the time it takes for the AV to notify the user of a sensor failure or that it has failed to detect the pedestrian in front of, as well as the time it takes to the user to react, it is crucial. In fact, many different components of the vehicle can cause the vehicle to give a warning. Although the users are obliged to take control of the vehicle when they receive the warning, it should be borne in

mind that this may not be humanly possible in very short periods.⁹⁶ The US National Highway Traffic Safety Administration (NHTSA) reported sixteen separate instances of accidents where the AV ‘aborted vehicle control less than one second prior to the impact’, suggesting the driver may not possess the capacity to properly and reasonably act to avoid an imminent harm, and thereby not truly be in control.⁹⁷ If the decision must be made in mere seconds or even milliseconds, ‘actual control over the system’s actions may be no more than an illusion’.⁹⁸

Furthermore, in case no alarm is sounded, existing standards for L3 AVs mandate human intervention even whenever the driver notices—or reasonably should notice—an obvious situation where the intended operational conditions of the automated driving functions are no longer met. However, the requirement to intervene in ‘obvious situation’ introduces significant ambiguity, as neither existing laws nor regulatory frameworks provide a clear or specific definition of what constitutes an obvious situation. For instance, a situation that might seem self-evident to a professional test driver may not be equally apparent to an ordinary citizen, particularly if their engagement with the system has diminished due to passive monitoring.

All these factors combined – the timing and clarity of warnings, the issues of overreliance and distraction, and the effectiveness of human intervention – raise profound legal questions about the feasibility of human oversight in AV systems and the equitable allocation of liability in the event of accidents. It is fair to ask whether a

⁹⁶ Researchers report in a driving simulation experiment that on average a driver takes 15 seconds to regain control of a vehicle, with up to 40 seconds to stabilize control – expecting drivers or remote operators to take over a vehicle in a dangerous scenario given these response times is not without risks. N Merat *et al*, ‘Transition to manual: driver behaviour when resuming control from a highly automated vehicle’ (2014) 27 *Transport. Res. Part F: Trac. Psychol. Behav.* 274-282.

⁹⁷ Christiaan Hetzner, ‘Elon Musk’s Regulatory Woes Mount as U.S. Moves Closer to Recalling Tesla’s Self-Driving Software’ (Fortune 2022), <https://fortune.com/2022/06/10/elonmusk-tesla-nhtsa-investigation-traffic-safety-autonomous-fsd-fatal-probe/>.

⁹⁸ Jonathan Kwik, ‘A Practicable Operationalization of Meaningful Human Control’ (2022) 11(3) *Laws*, 378.

human driver can truly exercise *de facto* control over a vehicle, even if such control has technically been handed back by the AV. Imputing responsibility to a human overseer who lacks meaningful control over the vehicle's operation risks unfairly burdening them with liability. Such an approach not only undermines the foundational principles of criminal justice but also conflicts with our fundamental intuitions about fairness.

Given the inherent limitations of human capabilities and the ambiguous allocation of responsibilities in these systems, liability determination becomes inherently complex. Each accident scenario must therefore be assessed on a case-by-case basis, with careful consideration of the specific circumstances. The central questions revolve around the division of decision-making authority at each level of automation and the extent to which drivers can legitimately rely on the vehicle's capabilities as a liability shield.

In what follows, we adopt a systematic framework to address liability concerns, analyzing each level of driving automation. For each level, we examine the distribution of decision-making authority, the expectations placed on human drivers, and the conditions under which reliance on AV capabilities becomes reasonable or problematic. This structured approach aims to clarify how the evolving interplay between human oversight and machine autonomy shapes liability determinations in the context of AV systems.

3.3. Variations on a theme: possible failures and criminal liability scenarios

Given the prevalence of fatal road accidents, road accident criminal liability has evolved into a specialized area of liability law in many jurisdictions. In traditional traffic systems, the vast majority of accidents are caused by human error, with vehicle defects accounting for only a small percentage.

However, the advent of driving automation fundamentally alters this dynamic, as these vehicles assume varying degrees of responsibility for navigating traffic. There

is widespread consensus that AVs are likely to shift liability frameworks, redistributing responsibility between human drivers and machines depending on the source of decision-making authority and the scope of human oversight.

In this paragraph, we address questions related to the proper distribution of criminal liability for fatal accidents among the actors involved – ie, driver, AV, and provider. The interaction between driver and vehicle, as well as the driver's obligations and liability profiles may differ depending on whether decision-making authority for main driving functions is (i) still attributed exclusively to human drivers (even though 'shared' with AI), (ii) temporarily transferred by the driver to the AV, or (iii) permanently transferred to the AV.

Each of these modes of interaction introduces distinct challenges for assigning decision-making authority.

3.3.1. Failure modes and liability profiles in SAE Level 1/2

Sharing control refers to scenarios where both the human driver and the AV simultaneously contribute to driving tasks, with each performing specific roles that complement the other. This is typical of driver control assistance systems, where drivers are assisted by AI in carrying out certain driving tasks – such as lane-keeping or adaptive cruise control – but they still retain ultimate responsibility and need to always retain situational awareness.⁹⁹ Here, the vehicle may handle certain functions, such as braking or maintaining speed, while the driver is required to control all other aspects of the driving task, like determining the direction and responding to traffic complexities. This form of interaction, found in SAE Level 1 and Level 2, requires ongoing and active engagement from the human. However, the overlapping roles of humans and machines create a grey area in determining responsibility, especially in cases where miscommunication or misalignment of expectations occurs.

⁹⁹ Schellekens (n 82), 243.

In shared control scenarios, different failure hypotheses may be considered:

- Hypothesis I: overreliance on assistive driving features (SAE Level 1)

Suppose a human driver is using a car equipped with adaptive cruise control (ACC) – an advanced driving assistance system capable to handle some routine driving task like controlling braking and speed. While proceeding on a highway, the human driver decides to set a target speed that is over the limit because of the perception that the risk from speeding could be offset by the systems' faster reaction time and ability to take defensive action by braking. However, the vehicle ahead brakes abruptly and the ACC fails to respond in time, thus leading to a rear-end fatal collision.¹⁰⁰

In this scenario, the driver remains the main liability target, as ACC at this level (SAE Level 1) is an assistive technology, not an autonomous system. Therefore, currently in Europe drivers can still be held criminally liable for negligence even where an assisted driving technology was in use, since no shift of responsibility can happen here and they always remains responsible for controlling the vehicle.

In fact, the law generally evaluates drivers against an objective standard of what a reasonably prudent person would do in similar circumstances, whether you are using an assistive driving feature or not. In our example, setting a speed exceeding the legal limit is always a clear violation of traffic regulations. By choosing to exceed the limit, the driver consciously operated outside the area of acceptable risk. In principle, even if the ACC's response time was faster than the driver's, reliance on technology to justify illegal behaviour does not absolve the driver of criminal liability. If the collision resulted in fatalities, the driver could face criminal charges, such as reckless driving or vehicular manslaughter, depending on the jurisdiction. Their

¹⁰⁰ A U.S. study of the Insurance Institute for Highway Safety found that the use of adaptive cruise control increases the risk of fatal accidents occurring when compared with driving manually due to automation bias: at <https://www.reuters.com/article/technology/cars-adaptive-cruise-control-raises-crash-risks-us-study-finds-idUSKBN2B32NH/>.

decision to speed, combined with improper reliance on ACC, may be seen as grossly negligent.

There is no denying that automation bias may play a role in this case, as the driver assumed the system could mitigate risks inherent in speeding. Yet, while it may explain the psychological causes behind overreliance on the ACC, automation bias does not excuse the driver from adhering to the reasonable person standard. The driver's decision to delegate critical safety responsibilities to a system not designed for autonomous operation reflects a failure to exercise due care and constitutes negligence, if they were aware (or should have been aware) that the ACC was not designed to handle emergency braking situations at high speed.

While the driver's negligent liability is unaffected by automation bias in this scenario, it underscores the obligation of AI providers to design systems that minimize the risk of misuse. As established by the AI Act, providers have an obligation to address the systemic risk of automation bias by ensuring drivers understand the system's capabilities and limitations.¹⁰¹ They must provide explicit and accessible instructions clarifying that ACC is incapable of handling all driving scenarios, especially emergency braking. In this sense, for example, if the driver misunderstood the system's limitations due to vague or insufficient guidance, the provider (natural or legal person) may share liability.

Interestingly, the UK *Automated Vehicle Act* enacted in 2024 introduced criminal offences related to 'information relevant to vehicle safety'.¹⁰² This offence punishes the act of giving false or misleading information relevant to vehicle safety (e.g., misleading drivers that a vehicle does not need to be actively monitored or that

¹⁰¹ AI Act, Article 13(2).

¹⁰² The *Automated Vehicles Act* received Royal Assent on 20 May 2024. The Act is intended to provide a legal framework for the safe deployment and operation of self-driving vehicles on public roads in Great Britain. It is based on the recommendations of the Law Commission of England and Wales and the Scottish Law Commission following their joint four-year review into the law relating to automated vehicles. UK, Automated Vehicles Act (20 May 2024) <https://www.legislation.gov.uk/ukpga/2024/10/contents>.

it can handle sudden braking events or emergencies).¹⁰³ This offence is aggravated where death or serious injury occurs.¹⁰⁴

In addition, AI systems should be designed to discourage overreliance.¹⁰⁵ ACC should incorporate features to prevent the system from operating at speeds exceeding safe or legal limits as well as periodic prompts or alerts to ensure the driver remains actively engaged, even when the system appears to function smoothly. Failure to implement these safeguards might lead to situations where the system unintentionally fosters undue trust. For example, if ACC permitted operation at speeds far exceeding legal thresholds without issuing adequate warnings to the driver or failed to deactivate in clearly high-risk scenarios, this could constitute a significant design flaw. In such cases, liability might extend to the provider for failing to address foreseeable misuse or for creating conditions that encourage driver disengagement.

Ultimately, if the ACC functioned as intended and the system's limitations were clearly communicated, liability would rest almost entirely with the driver for speeding and overreliance.

- Hypothesis II: faulty adaptive cruise control (SAE Level 1)

Some accident investigations have shown that some ACC units might suffer from electrical interference from other vehicle systems, causing them to accelerate uncontrollably beyond safe speeds and resulting in fatal crashes. From the perspective of the driver, such an incident constitutes a foreseeable driving accident but not necessarily a foreseeable risk. Importantly, from an accident-prevention and negligence perspective, a person who cannot predict an outcome cannot reasonably be expected to prevent it.

¹⁰³ *ibid* 24(1).

¹⁰⁴ *ibid* 25.

¹⁰⁵ AI Act, Article 14.

In such cases, assuming the driver has adhered to the system's instructions for use, they could legitimately place trust in the system, certified as trustworthy under the conformity-assessment procedure. This reliance underscores a critical aspect of the legal analysis: negligence does not demand the impossible or hold individuals accountable for risks they could not foresee or mitigate. Where the driver has followed the prescribed use of the ACC system, the responsibility shifts to the AI provider or manufacturer.

The provider's liability in this context becomes central. Providers are expected to anticipate reasonably foreseeable scenarios, including malfunctions caused by electrical interference. A defect in the system's sensors, algorithms, or response mechanisms may indicate a failure to meet safety standards or a lack of adherence to due diligence obligations during the design and testing phases. The focus would be on proving that the defect existed, whether it was avoidable, and whether the provider exercised due diligence in design, testing, and updates.

- Hypothesis III: misleading drivers as to the level of automation (SAE Level 2)

At SAE Level 2, the vehicle can brake automatically, accelerate and, unlike Level 1 systems, take over steering upon the driver's request. Now, consider a scenario where a Level 2 vehicle with autonomous steering capability suddenly veers onto the opposite lane at a bend, resulting in a collision with two other vehicles. Assume the car itself is not defective but the driver was not keeping their hands on the wheel.

As a Level 2 AV, the driver is deemed to be in control of the vehicle and responsible for its operation. They are expected to continuously monitor both the system's performance and the surrounding road environment. However, in this case, the driver did not intervene, even when faced with a sharp turn, because they incorrectly assumed the system was capable of handling more than it was designed to

do. The immediate attribution of negligent liability to the driver seems obvious. But does it remain fair to assign exclusive responsibility to the driver in this situation?

From the perspective of an ordinary citizen, there is often very little to distinguish an SAE Level 2 vehicle with autonomous steering capabilities from an SAE Level 3. Both systems control the vehicle, scan the driving environment, and react to events. What changes is the driver's role and the scope of their oversight responsibilities. However, as driving tasks become increasingly automated, the line between assisted and automated driving becomes blurred, particularly when certain design features, such as allowing the driver to remove their hands from the steering wheel, might contribute to confusion.

This ambiguity – commonly referred to as mode confusion – raises significant safety and liability concerns. To mitigate this risk, it is crucial to establish a crystal-clear distinction between Levels 2 and 3, ensuring drivers fully understand their responsibilities at each level. In recognition of this issue, legislative proposals have sought to safeguard the threshold between assisted and automated driving with specific criminal offenses. For instance, the UK *Automated Vehicle Act*, enacted in 2024, introduced offenses related to ‘communications likely to confuse as to autonomous capability’, penalizing actions like describing unauthorized driving automation as ‘autonomous’.¹⁰⁶

The consequences of such misrepresentation are illustrated by the infamous 2022 self-driving accident in Germany.¹⁰⁷ The car involved was labelled as having ‘autonomous driving’ capabilities, but after the accident, the manufacturer denied it was a fully autonomous system. Nevertheless, the vehicle's advertising may have

¹⁰⁶ UK Automated Vehicle Act, article 79(1)(3).

¹⁰⁷ This is a real case occurred in 2022 in the German town of Reutlingen near Frankfurt, where a BMW iX test car with autonomous steering capability swerved out of its lane at a bend in the road, causing several collisions involving four vehicles, resulting in the death of a passenger in that vehicle. See Felix Allen, ‘Electric ‘self-steering’ BMW veers into oncoming traffic leaving one dead and nine seriously injured in horrifying crash’ (The Sun 16 August 2022) available at <https://www.thesun.co.uk/news/19525886/bmw-autonomous-electric-test-car-crash/>.

misled the driver into overestimating the car's capabilities. If promotional materials contributed to the driver's misplaced confidence, this raises questions about the provider's role in the accident.

While the driver's misunderstanding reflects their failure to exercise due care, it is not entirely clear whether the blame should rest solely with them. The provider's responsibility for accurately communicating the system's capabilities and limitations must also be considered. A failure to provide clear and prominent instructions or warnings against overreliance could constitute a significant informational deficiency.

In cases where promotional materials or instructions are found to have misled drivers into overestimating the system's capabilities, liability might even shift toward the AI provider. Such deficiencies, if proven, could lead to findings of contributory liability or even criminal responsibility on the part of the provider. Informational shortcomings can be as harmful as technical flaws, and they underline the need for AI providers to ensure informed human-machine interactions as part of their due diligence obligations.

3.3.2. Failure modes and liability profiles in SAE Level 3

3.3.2.1. Legal position of the 'user-in-charge'

As of SAE Level 3, vehicles are capable of fully taking over driving operations, meaning that the automated system, rather than the human, is in control of the vehicle's movement. However, while the human driver is no longer required to maintain constant situational awareness, the AV itself cannot handle every possible traffic situation. This means that control may be 'traded' back and forth between the driver and the automation based on situational demands or the operational design domain of the system (e.g., when the system reaches its operational limit or encounters a complex scenario).

A critical legal question arises in SAE Level 3 contexts: how should legal responsibility be apportioned between the human operator and the AV in the event of a fatal accident while the system is in control? In recent years, several European jurisdictions have attempted to address this issue by introducing novel legal constructs, such as the concept of the ‘user-in-charge’.

The already mentioned UK Automated Vehicles Act (2024) provides one of the most comprehensive frameworks for regulating this new legal figure. Essentially, the UK Act establishes the foundation for the future deployment of automated vehicles on British public roads, creating a framework under which detailed regulations can be developed. Its primary aim is to resolve ambiguities in the existing legal landscape by introducing new concepts and modifying laws to address areas of uncertainty and regulatory gaps.

According to the UK Act, when a self-driving feature is engaged, the individual in the driving seat transitions from being a driver to becoming a user-in-charge. The first issue would be to identify the moment in which the human ceases to be a mere driver and becomes a user-in-charge. The Act defines it as an individual ‘who is in, and in position to exercise control of, the vehicle, but is not controlling it’.¹⁰⁸ This definition can be broken up into four characteristics. A user in charge is:

- (1) a natural person (not an organization);
- (2) who is present inside the vehicle (not operating it remotely);
- (3) in position to operate the driving controls, typically by being in driver’s seat;
- (4) while a self-driving feature is engaged.

The Act provides significant legal protections for users-in-charge, while clarifying where criminal liability for offences and accidents rests. Specifically, it grants them immunity from criminal offences or civil penalties if a violation arises from actions taken by the automated system while it is performing the dynamic driving

¹⁰⁸ UK Automated Vehicle Act, Part 2, Ch 1(46).

task. This immunity is contingent on the system having been properly engaged and not tampered with by the user.¹⁰⁹ In such cases, responsibility for traffic infractions or harm arising from the dynamic driving task shifts squarely onto the AV itself.

The UK is not alone in seeking to regulate this new legal role. Germany's 2017 amendment to its Road Traffic Act (*Straßenverkehrsgesetz*) also established parameters for automated driving, permitting the use of SAE Level 3 vehicles on public roads.¹¹⁰ The German framework similarly allows drivers to transfer control to the vehicle, specifying conditions under which automated driving functions may be deployed (Section 1a). It also outlines the driver's rights and duties during automated operation (Sections 1a and 1b).

Likewise, France has addressed this issue. In 2021, the French Parliament amended its Road Code through *Ordonnance No. 2021-443*, introducing Article 123-1 which reads as follows:

‘The provisions of the first paragraph of article L. 121-1 are not applicable to the driver for violations resulting from the operation of a vehicle whose driving functions are delegated to an automated driving system, when this system exercises, at the time of the violation and under the conditions provided for in I of article L. 319-3, the dynamic control of the vehicle’.¹¹¹

This provision excludes drivers from criminal liability for traffic violations resulting from a vehicle's automated driving system, provided the system was correctly engaged and exercising dynamic control at the time of the incident.¹¹² This marks a significant departure from Article L. 121-1 of the French Road Code, which

¹⁰⁹ *ibid* 47 (stipulating that the user-in-charge remains liable for non-dynamic offences).

¹¹⁰ Eight Act amending the German Road Traffic Act (2017), *Bundesgesetzblatt* 2017 Teil I n 38.

¹¹¹ French Road Act ('Code de la route'), art L. 123-1 (author's translation).

¹¹² According to some commentators, the immunity clause introduced in some national legislations merely formalizes a conclusion that could have been reached through the application of standard principles of criminal law, particularly the rules on negligence. See Marta Giuca, 'Disciplinare l'intelligenza artificiale. La riforma francese sulla responsabilità penale da uso di auto a guida autonoma' (2022), 2 *Archivio Penale*.

traditionally held drivers criminally liable for violations committed while operating their vehicles.

What unites these regulatory approaches is a shared aim to draw a clear boundary regarding user's criminal liability in SAE level 3 contexts. When a self-driving feature is engaged, the person in the driver's seat is not criminally responsible for offences arising from the dynamic driving task, provided the system was certified and properly used. This approach reflects a legal philosophy of carving out zones of immunity within broader frameworks of responsibility, acknowledging that acceptable risk must sometimes be prescribed to enable technological progress.

Despite these regulatory efforts, however, Level 3 AVs typically involve safety and liability concerns due to the possibility to 'trade' control over driving functions. Indeed, there are also likely to be challenges in determining which party bears responsibility when the AV encounters a situation which it cannot handle and control needs to be returned to the user-in-charge. In all the national regulatory frameworks mentioned above, the user-in-charge is under a specific *duty to intervene* and regain control of the system's action under certain conditions.

Under the UK Act, the user-in-charge should be required to respond to a clear and timely signal to take over driving (a 'transition demand').¹¹³ A transition demand is defined as 'a demand, communicated by equipment of a vehicle in which an authorised user-in-charge feature is engaged, that the user-in-charge assume control of the vehicle by the end of a period of time beginning with the communication of the demand'.¹¹⁴ At the end of the transition period, legal responsibility for driving will shift to the user-in-charge, even if they have not taken control of the AV.¹¹⁵ Offence immunity for the user-in-charge ceases at the end of the period and the user-in-charge is legally treated as a driver. Failure to comply with the duty to retake control of the vehicle when prompted may result in liability.

¹¹³ UK Act, Ch 1(7)(2).

¹¹⁴ *ibid.*

¹¹⁵ *ibid.*, Ch 2(48).

Similarly, according to L.123-1(2) of the French *Code de la route*, the user-in-charge must always be in a position to respond to a request to take control by the automated driving system (*demande de reprise*). In particular, article L319-3, which regulates the conditions for the correct activation of the dynamic control of the vehicle by the AI, provides that:

‘When its operating state no longer allows it to exercise dynamic control of the vehicle or when its conditions of use are no longer met or when it anticipates that its conditions of use will probably no longer be met during the execution of the maneuver, the automated driving system must:

- (1) Alert the driver;
- (2) Make a request to regain control;
- (3) initiate and execute a minimum risk maneuver if control is not regained at the end of the transition period or in the event of a serious malfunction’.

Hence, according to L.123-1(3), just like in the British approach, accepting the *demande de reprise*, or failing to do so, provided that the transition period has passed, will lead to the re-expansion of the scope of application of article L.121-1 of the Code, i.e., of criminal liability.

Interestingly, this approach differs from other jurisdictions. For instance, in Germany, the user-in-charge must remain ‘ready to perceive’ (*wahrnehmungsbereit*) obvious circumstances indicating that the conditions for the AV use are not met. The German Road Traffic Act requires that the driver immediately reassumes steering of the vehicle not only when there is a transition demand in that sense, but also when they notice – or should notice owing to obvious circumstances – that the requirements for proper use of the automated driving functions are no longer being met.¹¹⁶ Here, the issue is whether the user-in-charge has a duty to constantly monitor the behaviour of the AV, being ready to take over at any moment, or if the operator should only be obligated to respond to safety alerts generated by the vehicle. In the first case, criminal

¹¹⁶ Section 1a(4), 1b(2) German Road Traffic Act (*Straßenverkehrsgesetz*).

sanctions could arise if a person fails to override when obligated to, whereas in the second they arise from the violation of an affirmative duty to prevent any harm.

Ultimately, these novel legal constructs – such as the user-in-charge, immunity clauses, and trading control – imply a broad spectrum of transitions: control may shift from machine to human or vice versa, either under normal or emergency situations. These transitions can be human-initiated (optional, for instance, when a human driver chooses to delegate driving to automation to relax or focus on another task) or machine-initiated (mandatory, as when the human is required to take back control due to specific circumstances).¹¹⁷ However, this trading of control also presents typical liability challenges, such as accounting for the inherent limitations and shortcomings of human-machine interactions, such as cognitive biases, limited reaction times, and the hand-off problem.

To address these complexities, more refined legal regimes or interpretations are essential to prevent inequitable attribution of criminal responsibility or scapegoating. This necessitates a case-by-case evaluation of failure scenarios to ensure that liability is allocated fairly and that neither human users nor technology providers are unjustly burdened.

3.3.2.2. Allocation of criminal liability between user-in-charge and AV

In SAE level 3 scenarios, different failure hypotheses regarding control trading may be considered:

- **Hypothesis I:** human failure to regain control of the vehicle

Existing regulatory frameworks and criminal law doctrine provide limited guidance on addressing the unique challenges posed by AV-related harms,

¹¹⁷ Z Lu *et al*, ‘Human Factors of Transitions in Automated Driving: A General Framework and Literature Survey’ (2016) 183 Transp Res F: Traffic Psychol Behav 187–89.

particularly in the context of negligence liability. The (in)famous Elaine Herzberg case of March 2018 perfectly illustrates these complexities. Herzberg was struck and killed by an automated Uber test AV, operating in ‘autonomous mode’, as she wheeled her bicycle on a public road in Tempe (Arizona) in 2018, while a human backup driver was present in the vehicle. The incident raises critical questions about the adequacy of negligence doctrine when applied to scenarios involving complex human-machine interactions and hybrid decision-making authority.¹¹⁸

Briefly providing further details about the facts of the case demonstrates the complexities involved. According to the resulting investigation report by the US National Transportation Safety Board’s (NTSB),¹¹⁹ the main elements of the case can be summarized as follows:

- The Uber vehicle involved in the collision was considered to be operating at SAE Level 3/4.
- The vehicle’s sensors detected Herzberg but failed to properly classify her as a pedestrian, resulting in the system’s failure to either initiate emergency braking or to alert Rafaela Vasquez in time to avoid the collision.¹²⁰

¹¹⁸ See Jack Stilgoe, ‘Who Killed Elaine Herzberg?’ (Medium Dec. 12, 2019), <https://onezero.medium.com/who-killed-elaine-herzberg-ea01fb14fc5e>; Hannah Knowles, ‘Uber’s Self-Driving Cars Have a Huge Flaw When It Comes to Jaywalkers’ (Science Alert Nov. 7, 2019), <https://www.sciencealert.com/investigation-reveals-uber-s-self-drivingcars-weren-t-programmed-for-jaywalkers>.

¹¹⁹ The Federal body completed a comprehensive review into the collision which was completed in November 2019. National Transportation Safety Board, ‘Collision Between Vehicle Controlled By Developmental Automated Driving System And Pedestrian at Tempe, Arizona, March 2018’, Highway Accident Report NTSB/HAR-19/03, available at <https://data.nts.gov/Docket?ProjectID=96894>.

¹²⁰ *Id.* 2 (‘[The] systems included a collision avoidance function with automatic emergency braking, known as City Safety, as well as functions for detecting driver alertness and road sign information’); (‘When the vehicle first ‘saw’ Herzberg, 5.6 seconds before impact, it classified her as a vehicle. For the next four-and-a-half seconds, it couldn’t guess what she was, classifying her as ‘other’, then a vehicle again, then ‘other’, then a bicycle, then ‘other’, then a bicycle once more. At 1.2 seconds, the vehicle recognized it would hit Herzberg. It held off braking for one second. Just 0.2 seconds before the collision, it sounded an audio alarm’).

- Uber’s software system, learning from data showing people crossing on crosswalks, did not consider the possibility of pedestrians walking across roads outside of crosswalks, or the possibility of a person pushing a bicycle while on foot (post-design bias).¹²¹
- The backup driver, Rafaela Vasquez, who was a Uber employee tasked with monitoring the vehicle and intervening if necessary, was visually distracted by her phone at the time of the incident.¹²²
- Uber provided Vasquez with minimal training and was required to operate the autonomous technology with little oversight and without company strategies to mitigate automation complacency.¹²³

Herzberg’s death instantly turned what had been a philosophical conundrum into a glaringly real, legal one, as an offense that is ‘committed’ by an AI system (in this case, an AV running over and thus killing a person) will have to be attributed to someone.

On August 27, 2020, an indictment was filed in the Maricopa County Superior Court charging Vasquez with negligent homicide.¹²⁴ In this indictment, the State of Arizona alleged that Vasquez was negligently inattentive when operating the Uber vehicle, and *but for* Vasquez’s actions, Herzberg would not have died because

¹²¹ *ivi*, 16 (‘The system never classified her as a pedestrian – or correctly predicted her path – because she was crossing N. Mill Avenue at a location without a crosswalk, and the system design did not include consideration for jaywalking pedestrians’).

¹²² *ivi*, 44-46. Chris Coppola and BrieAnna J. Frank, ‘Report: Uber Driver Was Watching ‘The Voice’ Moments Before Fatal Tempe Crash’ (Azcentral June 22, 2018), <https://www.azcentral.com/story/news/local/tempe-breaking/2018/06/21/uber-self-driving-car-crash-tempe-police-elaine-herzberg/724344002/>.

¹²³ *ibid.* 49

¹²⁴ *State of Arizona v. Rafaela Vasquez*, No. CR2020-001853-001 (Ariz. Super. Ct. July 5, 2021).

the collision could have been avoided or mitigated.¹²⁵ On the contrary, in relation to Uber's culpability for the collision, it was determined that no action would be taken against the corporation as there was no basis on which it could be held criminally responsible.¹²⁶

In July 2023, the criminal proceedings against Vasquez were finalized without proceeding to trial when Vasquez pleaded guilty to the lesser charge of endangerment and received three years of probation.¹²⁷ The plea deal saved her from a trial's uncertainty: if a jury had found Vasquez guilty of the original felony she was charged with – negligent homicide – she would have faced four to eight years in state prison. Vasquez bore the bulk of the criminal responsibility for the accident, while Uber, the company responsible for developing and deploying the AV, was not held criminally liable for the death.

Our interest here is not to critique the Arizona legal decision – an analysis that would require delving into the specifics of U.S. legal proceedings and the nuances of its negligence doctrine – but to explore whether a similar case would yield a comparable outcome under European legal frameworks, particularly in light of the AI Act and recent national legislation addressing AV liability.¹²⁸ How would criminal liability be apportioned between the actors involved? Would it be the corporation which created the AV responsible for misinterpreting a scenario or failing to detect a

¹²⁵ *ibid.*

¹²⁶ Uber paid a confidential amount of compensation to the Herzberg family within two weeks of the collision, closed the autonomous vehicle testing program and left Arizona in May 2018. Bernie Woodall, 'Uber avoids legal battle with family of autonomous vehicle victim' (Reuters Mar. 29, 2018), <https://www.reuters.com/article/us-autos-selfdriving-uber-settlement/uber-reaches-settlement-with-family-of-autonomous-vehicle-victim-idUSKBN1H5092>.

¹²⁷ In Arizona, endangerment is defined as 'recklessly endangering another person with a substantial risk of imminent death or physical injury'.

¹²⁸ For a detailed analysis of the legal saga involving Vasquez's criminal responsibility for the crash, see Helen Stamp, 'The Reckless Tolerance Of Unsafe Autonomous Vehicle Testing: Uber's Culpability For The Criminal Offense Of Negligent Homicide' (2024) 15 Case W. Res. J.L. Tech. & Internet 37. See also Alexandra De Arman, 'The Wild, Wild West: A Case Study of Self-Driving Vehicle Testing in Arizona' (2019) 61 Ariz. L. Rev. 983, 1006.

hazard? Or did the backup driver fail in their duty to monitor the system and take control at the right moment?¹²⁹

If this incident had occurred in Europe, the apportionment of criminal liability would likely differ, as recent legal developments emphasize shared responsibility between the user-in-charge and the AV. First and foremost, it is important to underline that as a user-in-charge of a Level 3 AV, Vasquez would have been permitted to disengage from active driving tasks and trust the vehicle, provided that the self-driving features of the car were correctly engaged at the time of the accident. The above-detailed national regulatory frameworks on AVs typically grant immunity to the user-in-charge for offences arising from the automated dynamic driving task. In such cases, the user-in-charge is not required to maintain constant situational awareness (unlike what the Arizona Court decision seems to imply), as this would contradict the purpose of Level 3 driving automation. Liability would only shift to the user-in-charge under specific circumstances, such as:

- Failing to retake control when prompted by the AV;
- Improperly engaging the automated system; or
- Tampering with the vehicle's self-driving functions.

Assuming none of these conditions applied, Vasquez could claim immunity under European legal frameworks for users-in-charge. However, the facts of the Herzberg case, as highlighted in the NTSB investigation report, complicate this analysis. Specifically, the AV issued a transition demand just 0.2 seconds before the collision, accompanied by an audio alarm.¹³⁰ While the system correctly cedes back control to the user-in-charge as intended by its designers in emergency situations, the timing raises significant questions about fairness, responsibility, and the feasibility of human intervention.

¹²⁹ See Sabine Gless *et al* (Ch I n 9), 417.

¹³⁰ US NTSB (n 93), 2.

In principle, failure to react to a transition demand shifts liability to the user-in-charge, exposing them to criminal negligence charges. However, simply imposing a duty to retake control under such circumstances and holding the user-in-charge criminally liable for anything that occurs after the transition demand conflicts with fundamental principles of fairness and culpability. For responsibility to be fairly imputed, the individual must have actual *meaningful* control over subsequent events. This condition was arguably absent in this case for several reasons:

- *Cognitive and Physiological Limits of Human Supervision* – Research consistently shows that humans are poor at passive monitoring, especially when not actively engaged in a task like driving. Prolonged disengagement, such as when the vehicle is in automated mode, reduces attentiveness and situational awareness. In such states, the human operator is unlikely to react effectively to sudden, unexpected demands. The Herzberg incident appears paradigmatic of what we referred to as automation bias or automation complacency, a well-known phenomenon in the field of AV where users overly trust the system and become less vigilant, diminishing their ability to intervene promptly when required.¹³¹
- *Inadequate transition demand design* – This leads to a second problem: the timing and clarity of transition demands. An AV system that fails to detect a hazard until the last possible moment – such as the pedestrian in this case – must not only issue an alert but also ensure the user has sufficient time to regain situational awareness and act effectively.¹³² If this period is excessively

¹³¹ Simon Chesterman, *We, the Robots? Regulating Artificial Intelligence and the Limits of the Law* (CUP, 2021), 35. See also Raja Parasuraman and Dietrich H. Manzey, ‘Complacency and Bias in Human Use of Automation: An Attentional Integration’ (2010), 52(3) *Human Factors The Journal of the Human Factors and Ergonomics Society*, 382.

¹³² While the AI Act is silent on the adequate design of transition demands, national legal frameworks seem to explicitly address these issues. For instance, Section 7 of the UK Automated Vehicles Act imposes specific obligations on AV providers to ensure that: transition demands allow the user-in-charge sufficient time to prepare for and assume control;

long, it undermines the purpose of trading control back to the human driver, while an overly short time frame risks making the demand for human intervention unreasonable or even impossible to fulfill.¹³³ The 0.2-second transition period in the Herzberg case is insufficient by any standard to allow meaningful intervention. Even under optimal conditions, requiring the user-in-charge to respond effectively in mere seconds or even milliseconds might be no more than an illusion. Delayed or ambiguous warnings substantially nullify the capacity of the user to properly and reasonably react to avoid an imminent harm.

The concerns raised by automation bias and the lack of *de facto* control underscore the delicacy of reprise clauses. Poorly formulated reprise clauses risk rendering immunity provisions functionally void by imposing unrealistic or superhuman expectations on users. For example, the German Road Traffic Act does not clearly define what ‘remaining aware’ entails, even as it requires users to retake control also in the absence of an explicit transition demand. This vagueness seems to imply that an especially high standard of attentiveness is required as part of the reasonable care necessary for driving a Level 3 AV, and that anything short of it is negligent. Such a standard is not totally unthinkable. Psychology research does not imply that it is *impossible* for the average person to remain constantly alert while not driving, just that it requires extraordinary attention and effort.¹³⁴

Under this heightened standard of care, Vazquez would probably be deemed to fall short of her duty of care and be held criminally liable, insisting that she should

alerts are clear, perceptible, and actionable, enabling the user to understand when and how to intervene; the transition period balances the urgency of human intervention with the practicalities of human cognitive and physical limitations. Yet, the UK legislator also notes that they are not in the position to specify how long this period should be, leaving then a major gap in the proposed regulation. Ideally, this should be determined through actual data obtained from empiric testing.

¹³³ Schellekens (n 82), 245. See also Selbst (Ch III n 113), 1346.

¹³⁴ Selbst (Ch III n 113), 1348.

have kept situational awareness and that should have been aware that the conditions for using the self-driving feature were not there anymore.

This high standard of care, however, may place undue expectations on users-in-charge, contravening the principle that individuals should not be required to pay attention while self-driving features are engaged. Moreover, the reasonable driver must know when the AV is malfunctioning in order to effectively take over – which is not always the case in complex human-machine interactions. If negligence law requires a higher standard of care than the average person can keep up, it could not only undermine the practical utility of immunity clauses but also compromise the fairness of liability allocations, turning users into mere legal crumple zones.

Given these considerations, it is safe to assume that if the Herzberg incident occurred in Europe under existing regulatory frameworks, the user-in-charge would likely not bear the bulk of the criminal responsibility if the transition demand was unreasonable or impossible to act upon. The 0.2-second timeframe would likely absolve Vasquez of negligence, as no reasonable person could react effectively under such conditions.

Instead, the criminal liability focus would likely shift to Uber in its capacity as the provider and developer of the AV system. The liability of the corporation (or of its directors/officers, in those jurisdictions where corporate criminal liability is not recognized) would be assessed for:

- *Negligent design* – Does the system’s inability to detect the pedestrian and to issue a timely, actionable transition demand indicate design flaws and failure to meet safety standards required for AVs under existing EU regulatory frameworks?
- *Negligent testing* – Were the AV’s limitations properly tested and mitigated under conditions representative of real-world scenarios, such as a pedestrian crossing unexpectedly?

- *Failure to ensure effective human oversight* – Were user-in-charge’s responsibilities as well as the system’s operational limitations adequately communicated? and was the training provided to them in handling AVs sufficient?

If Uber failed to meet these obligations to develop a comprehensive risk management system and implement adequate oversight safeguards, it could face regulatory fines, civil liability, or even criminal charges where permitted by law.

Given the information provided in the comprehensive NTSB investigation report, there appears to be a basis which would support the criminal liability of Uber or of its directors and high-level managers for the death of Elaine Herzberg. In fact, the AV’s inability to correctly classify jaywalkers as pedestrians highlights a fundamental flaw in its system design and testing. This flaw, combined with evidence from the NTSB investigation that Uber was aware of deficiencies in its AV’s object classification and safety protocols,¹³⁵ underscores at the very minimum a *reckless disregard* for adequate safety practices or, that is the same, tolerance for practices that jeopardized public safety – factors that directly contributed to the Uber vehicle colliding with Ms. Herzberg.¹³⁶ Additionally, specific criminal provisions addressing endangerment could be introduced to penalize failures in ensuring proper oversight policies for AVs. For instance, such provisions could target acts such as deploying AVs on public roads without adequately trained backup drivers capable of responding effectively in emergency scenarios, as well as failures to account for well-documented phenomena such as automation complacency in backup drivers. Incorporating such provisions could also align with principles found in European legislation, i.e. Article 14(1) of the AI Act, which mandates that ‘(h)igh-risk AI systems shall be designed and developed in such a way, including with appropriate human-

¹³⁵ U.S. NTSB (n 93), 40.

¹³⁶ Stamp (n 128), 32.

machine interface tools, that they can be effectively overseen by natural persons during the period in which they are in use’.

Ultimately, the risk of criminal liability may incentivize AV providers to force a handoff to a human backup driver in precisely such emergency situations because doing so increases the likelihood that the user-in-charge, rather than the provider itself, will bear the brunt of liability. The Herzberg case underscores the necessity of refining legal and technical frameworks to address the complexities of trading control between humans and machines. By ensuring reasonable transition demands and accounting for human cognitive limits, the law can avoid inequitable attribution of liability and prevent scenarios where users-in-charge are unfairly scapegoated for system failures.

- Hypothesis II: use of self-driving features in a dangerous state

As already mentioned, under national legal frameworks such as the UK Automated Vehicles Act and the German Road Code, users-in-charge enjoy immunity from driving offences, provided that they have engaged the self-driving features correctly and adhered to relevant operating conditions.

Now, consider the case of a user-in-charge activating self-driving features while the vehicle or the surrounding conditions are in a dangerous state – for example, engaging the system despite severe weather conditions, hazardous road environments, or explicit operational design domain (ODD) restrictions. The user disregards clear warnings or preconditions for safe use, relying on AV capabilities. As a result, the system fails to detect a pedestrian crossing the road due to heavy rain.

Certainly, this case is fairly straightforward. The concept of immunity for users-in-charge must be balanced with their obligation to engage self-driving features responsibly. If it had been obvious to a competent and reasonable user-in-charge, at the moment when the self-driving feature in question was engaged, that the current state of the vehicle would make it dangerous to use it in the way in which it is being

used, they could be held criminally liable for recklessly endangering public safety, especially if their actions lead to harm.¹³⁷

However, liability would depend on whether the warnings were clear, specific, and unambiguous, and whether the dangerous state was apparent and avoidable by a reasonable user. AV providers have a duty to ensure that their systems are equipped to detect and communicate operational limits effectively. Under Article 14 of the AI Act, they are required to implement clear warnings and prevent users from activating self-driving features in unsafe conditions. For instance, if an AV allows activation in conditions where it cannot function reliably (e.g., snow-covered roads where lane markings are invisible) without issuing adequate warnings or implementing technical safeguards (e.g., preventing activation in extreme weather), this could indicate a failure in design or risk management and providers may share liability for harm caused. Providers are expected to foresee such misuse and mitigate risks through design safeguards and clear user instructions.

- Hypothesis III: failure to engage self-driving feature resulting in accident

Finally, it is useful to examine the hypothesis where a user-in-charge fails to engage the self-driving features of an AV when doing so could have prevented an accident. For example, imagine an SAE Level 3 vehicle capable of safely navigating specific scenarios autonomously, but the user-in-charge neglects to activate the self-driving mode while driving in those conditions, overestimating human capabilities in handling the situation. An accident occurs as a result of their error, which the AV could have avoided.

As AVs surpass human capabilities in handling safety-critical conditions, failing to utilize such technology when it is demonstrably safer will probably raise questions of negligence in the near future. Indeed, it will not be long before AV is safer than the average driver, and then safer than any human driver at handling safety-

¹³⁷ UK Automated Vehicles Act, Ch 2(54).

critical conditions. The principle of harm avoidance suggests that individuals have a duty to minimize foreseeable risks when reasonable precautions are available. If AVs provide a safer alternative to human operation, failing to engage their features might breach this duty. Legal scholars argue that the traditional ‘reasonable person’ standard could evolve to embrace AI, where human conduct is judged against what a ‘reasonable person using AI’ would have done under similar circumstances.¹³⁸ This means that if the user could reasonably foresee that their manual operation posed a greater risk than engaging the AV, their decision may constitute negligence.

For instance, if an AV can stop within 40 meters while a human driver requires 50 meters, failing to activate the AV could constitute negligence, particularly in scenarios where a fatal accident occurs. Today, in such a case, a defendant may not be liable for striking a child suddenly running in front of the car if a reasonable driver would not have been able to stop. But that person would soon be liable under the exact same circumstances if an AV would have prevented the injury. Courts might rely on expert testimony and technical data to determine whether the AV’s use could have prevented harm. Users might also face an expectation to engage AV features in situations where their performance is known to be superior, such as in poor visibility or dense traffic. Courts might consider situational factors, such as whether the user had sufficient time or knowledge to activate the AV.¹³⁹

Current regulatory frameworks do not impose an explicit duty to use self-driving features, and it is unlikely that the law will prohibit people from driving even when AVs become substantially safer than people.¹⁴⁰ This underscores an inherent tension between encouraging AV engagement for safety and preserving individual autonomy to drive manually. Instead of imposing a ban on human driving, legislations may evolve to permit manual driving while holding drivers to a higher standard of

¹³⁸ Abbott (Ch II n 94), 66-70, arguing that in future human drivers will be held to a ‘reasonable person using AI’ standard.

¹³⁹ *Id.*, 67.

¹⁴⁰ France’s Article L319-3 specifies that driving functions must be appropriately delegated to the automated system, suggesting that users might be at fault for failing to use such features when required.

care based on AI capabilities. Therefore, liability could arise under general principles of criminal negligence, if a user's decision to manually operate the vehicle is deemed unreasonable compared to the safer alternative of engaging the AV. A reasonable person would not have driven; a reasonable person would have chosen to automate.¹⁴¹

Providers also bear a responsibility to ensure that AV systems clearly communicate their capabilities and encourage users to activate safety-critical features when necessary. If a provider fails to design systems that adequately notify users of the advantages of activating autonomous features or permits risky manual operation, they might share liability for resulting harm. Practical challenges arise in defining the threshold at which AV use becomes a legal expectation, such as determining whether liability should attach when AVs are marginally safer than humans or when their performance is significantly better but not infallible. Transparency and ease of use are critical; users must understand the operational benefits and limitations of AV features, and engaging these features must be intuitive.

¹⁴¹ *Id.* 68.

SECTION 4

Human-machine interaction #3: human-*outside*-the-loop

4.1. Current legal framework

The last form of human-machine interaction that we want to address in our analysis is the ‘human-outside-the-loop’ (or human-out-of-the-loop) model, where decision-making authority is fully transferred or delegated to the AI system with minimal or no human intervention.

In this paradigm, intelligent systems operate autonomously, executing tasks and making decisions without a human overseer monitoring or approving each action in real time. Unlike models that rely heavily on human decision-making, the HOTL model signifies a transformative shift, empowering machines to make autonomous decisions in environments where time, complexity, and scalability make constant human oversight is impractical, inefficient, or even impossible.

While this may sound futuristic, the seeds of this paradigm have been quietly blooming for years. For instance, SAE Level 4 driverless vehicles are designed to perform all driving tasks under specific conditions without requiring human involvement. When operating within a geofenced area, these systems do not require a human driver to intervene, as the system can independently navigate the route, manage traffic scenarios, and respond to emergencies based on pre-programmed algorithms and real-time sensor data.¹⁴² Another prominent example is algorithmic trading systems in financial markets. These systems execute split-second trading decisions based on pre-set strategies and real-time market data, without human traders being involved in each individual transaction. Because time-to-decision must be minimal when executing transactions, the human component had to be removed from the loop. Again, large language models like ChatGPT can generate human-like

¹⁴² Contissa and Sever (n 84), 3.

text, answer complex questions, and assist with creative or technical tasks autonomously without requiring human decision-making for each interaction.

While this model of human-machine interaction allows for highly efficient, scalable, and often faster decision-making, it also presents the greatest departure from traditional decision-making models and is perhaps the most challenging from a legal perspective. As they become more prevalent, HOTL systems poses unique legal challenges for regulatory frameworks traditionally built around human agency and direct control, like the AI Act.

The fundamental shift lies in the removal of immediate human involvement, a cornerstone of traditional liability and regulatory frameworks, requiring a rethinking of how we regulate, manage, and control AI systems in the future. In this respect, the principle of human oversight under Article 14 of the AI Act becomes a central point of inquiry when applied to HOTL systems. This provision requires mechanisms to ensure that high-risk AI systems can be effectively overseen by humans, with the aim of mitigating risks to fundamental rights, safety, and societal interests. Human oversight might then entail implementing robust monitoring mechanisms, requiring detailed logging of decision-making processes, or mandating fail-safes that enable humans to evaluate and intervene retrospectively. For traditional or even partially autonomous systems, this principle is straightforward: a human operator supervises the AI, intervenes as necessary, and bears responsibility for ensuring its proper functioning. However, the HOTL model challenges this framework.

The extent to which Article 14 can apply to HOTL systems ultimately depends on how the concept of oversight is interpreted and operationalized in contexts where direct human intervention is absent. In such scenarios, oversight might shift from active engagement during operations to responsibilities in the post-deployment phases of the system lifecycle. A key challenge in this discussion is determining who, if anyone, qualifies as the ‘user’ of an HOTL system and whether they can be held criminally liable for actions or harms resulting from a system operating entirely beyond their immediate control.

The absence of immediate human involvement in the system's real-time decisions complicates traditional notions of liability. Traditional doctrines of negligent liability depend on attributing blame to an individual who either directly causes harm or negligently fails to prevent it. However, when autonomous decision-making is fully delegated to an AI system, this framework struggles to address situations where the operator's role is limited to *ex-post* monitoring. Such detachment raises profound legal questions: can responsibility be meaningfully assigned to someone whose oversight is retrospective, rather than real-time? If not, how should liability be distributed among the system's developers, deployers, and other stakeholders?

4.2. Legal duties of 'not-in-charge' operators

In HOTL models, the human role is largely relegated to *ex-post* monitoring. Once the system is operational, the 'user' undergoes a fundamental shift in legal qualification, moving away from direct control to a mere 'supervisory' function.

The EU High-Level Expert Group on AI frames this concept under the broader umbrella of the 'human-in-command' principle. This framework assigns responsibility to a subject tasked with overseeing the AI system's post-deployment activity and retaining the authority to decide whether, when, and how the system should be deployed in a given context.¹⁴³ While this definition provides some clarity, its practical application in HOTL scenarios remains ambiguous.

Some indications can be found in the AI Act. In particular, Article 26, outlining the responsibilities of those who deploy high-risk AI systems, stipulated that

'Deployers shall monitor the operation of the high-risk AI system and, where relevant, inform providers in accordance with Article 72. Where deployers have reason to consider that the use of the high-risk AI system in accordance with the instructions may result in that AI system presenting a risk within the meaning

¹⁴³ HLEG (n 4), 5.

of Article 79(1), they shall, without undue delay, inform the provider or distributor and the relevant market surveillance authority, and shall suspend the use of that system.¹⁴⁴

Accordingly, Article 72 AIA requires deployers of high-risk AI systems to establish a *post-market monitoring system*. This system should collect and analyze data on the performance of these AI systems throughout their lifetime, ensuring they continue to comply with regulations.

To explain, it is useful to take once again autonomous driving as example, given that there are already legislations at national level addressing responsibility of human users. In particular, many EU member states are creating a legal basis for developing SAE level 4 driverless vehicles (e.g., automated trucks, taxis, and buses) for regular use on public roads. Currently, each country has the freedom to authorize test drives for vehicles with autonomous driving functions to SAE Level 4 on public roads in accordance with the Vienna Convention and the Geneva Convention on Road Traffic.

Across the European Union, a legal framework for fully automated vehicles with autonomous driving functions was finalized in July 2022 through an adaption of the existing type-approval regulation, which was expanded to include fully automated vehicles. The revised version will create a binding regulatory framework that includes fully automated driving in regular operation. The new regulation no longer requires the mandatory use of a safety driver for fully automated Level 4 trucks and buses.

In Germany, national legislation has acted early. In 2021 and 2022, the German government became the first country in the world to create the legal framework for Level 4 automated driving for cars and trucks. On July 28, 2022, two pieces of legislation came into effect: the Autonomous Driving Act and the Ordinance Regulating the Operation of Motor Vehicles with Automated and Autonomous Driving Functions and Amending Road Traffic Regulations

¹⁴⁴ AI Act, Article 26(5).

(AFGBV).¹⁴⁵ These allow autonomous driving functions in fully automated passenger and freight transport vehicles on defined routes. The presence of a safety driver in the vehicle is no longer mandatory. But who is, exactly, expected to supervise Level 4 vehicles? And what is their role and duties?

Importantly, the German road traffic rules on AVs distinguish between level 3 and level 4 users. While Level 3 vehicle users are referred to as ‘drivers’ (ie, users-in-charge) under § 1a(4) of the Road Traffic Act, as is for conventional vehicles, level 4 vehicle users are termed ‘technical supervisors’ under § 1d(3) of the Road Traffic Act. The obligations of the driver and technical supervisor also differ pursuant to § 1b(2) and § 1f(2) of the Road Traffic Act. As we said, drivers are obliged to take control of the vehicle when the vehicle requests or when they notice or should notice an obvious situation where the intended conditions of the automated driving functions are no longer met. On the contrary, with level 4 automation, the problem encountered at level 3, that a driver must take control at some point does not occur. On a vehicle deploying level 4 features, a whole journey can be completed without any kind of intervention by a human on board.

In this context, the technical supervisor is intended as a licensed person (natural or legal persons, coinciding for example with the company providing a passenger service.) which oversees vehicles possessing Level 4 self-driving features. The technical supervisor is a *remote operator* who need to have ‘oversight’ of the vehicle any time the self-driving feature is engaged on a road or in another public place: they are expected to constantly monitor the vehicle’s performance and, if necessary, deactivate the vehicle when requested by the autonomous system, but they are not expected to monitor the driving environment.

Similarly, the UK Automated Vehicles Act of 2024 introduced a new regulated legal subject called ‘no-user-in-charge operator’ (‘NUiC operator’) who is

¹⁴⁵ The Eighth Act Amending the Road Traffic Act (*Achtes Gesetz zur Änderungen des Straßenverkehrsgesetzes*) represents German legislators’ first steps to regulate highly and fully automated driving functions: <https://www.gesetze-im-internet.de/stvg/>

not in the vehicle or in a position to exercise control at any point in the journey.¹⁴⁶ The licensed NUiC operator will be responsible for the detection of, and response to, problems arising during journeys when the NUiC feature is engaged or there is no individual in the vehicle who is exercising control of it.¹⁴⁷ Oversight duties would include both remote assistance (for example, if the AV detects an object in its lane which is too large to avoid and stops, remote assistance could imply providing instructions to the vehicle on how to deal with the obstruction) and fleet operations (for example, dealing with law-enforcement agencies or paying tolls).

At this point, it is safe to claim that the obligations of the level 4 technical supervisor/NUiC are less than those of the level 3 vehicle user-in-charge. It is also clear that we cannot escape placing some responsibility on such subjects, particularly when their failure to comply with their duty of care results in fatal accidents. Unlike users-in-charge, NUiC operators are not the addressee of statutory immunity from criminal offences such as the one proscribed for the user-in-charge in the UK Act for those situations where self-driving features are engaged.

4.3. Variations on a theme: failure scenarios and liability profiles in SAE level 4

- Hypothesis I: NUiC failure to deactivate AV presenting clear signs of malfunction

First and foremost, certain offences for harmful events regarding the ‘use’ of the vehicle might apply also to a NUiC operator based on the violation of their duty of vigilance and intervention. This hypothesis considers the liability implications when a NUiC operator fails to deactivate an SAE Level 4 AV that is displaying evident signs of malfunction. The failure can occur under two primary conditions: (1) the AV issues an alert indicating a critical malfunction and requests deactivation, or (2) the

¹⁴⁶ UK Automated Vehicles Act, Ch 1(4)(3). The Act provides for a new licensing regime and register for NUiC operators.

¹⁴⁷ *ibid.*, Ch 2(5)(a).

NUIC operator independently observes signs of failure but does not take the appropriate action to deactivate the vehicle.

In the first scenario, the AV actively issues an alert indicating a critical malfunction or requests immediate intervention. The operator's liability depends on two primary factors: notice and capacity to act. Here, criminal liability would center on whether the NUIC had clear and reasonable notice of the AV's malfunction through an explicit alert or warning. Regulatory frameworks, such as those envisaged under the AI Act and existing national legislations, often require that AVs provide timely, clear, and actionable notifications to ensure operators can intervene effectively. A failure of the NUIC to respond appropriately to such alerts may expose to negligence-based liability, provided it can be demonstrated that the NUIC had the capacity to deactivate the vehicle in time and that the inaction directly caused or failed to prevent harm. On the contrary, if the AV's design left the operator without sufficient opportunity to respond, imposing criminal negligence would violate fundamental principles of culpability.

In the second scenario, the AV does not issue an explicit alert, but the NUIC operator could reasonably be expected to detect and act on the anomaly based on performance monitoring. This scenario introduces complexities tied to the standard of care expected from NUICs. The legal question here is whether the NUIC met the standard of care expected of someone in their position. A reasonable operator, even in a supervisory role, is expected to exhibit attentiveness proportional to the risks involved. If the anomaly was glaring (e.g., erratic behaviour, failure to respond to inputs), and the operator failed to deactivate the system or take preventive action, criminal negligence may be invoked. The prosecution would need to show that the operator's omission was a substantial deviation from what a reasonable person would do under similar circumstances. If the signs of failure were evident and the NUIC failed to intervene, the burden may rest on the operator for failing to uphold a duty of reasonable vigilance.

However, if the AV lacked mechanisms to clearly communicate critical issues, liability might shift to the manufacturer or system provider for failing to ensure

adequate oversight capabilities or user notifications, potentially violating Article 14 of the AI Act. Moreover, if the NUiC received inadequate training or unclear instructions regarding when and how to report issues, liability may shift to the provider or fleet manager.

- Hypothesis II: NUiC failure to inform relevant parties about impending failure

Additionally, the NUiC may face criminal liability charges due to ‘failure to inform’ relevant parties – such as a fleet manager, system provider, or regulatory body – about a detected malfunction or anomaly in the AV that foreseeably cause harm. In particular, Article 26 AIA stipulates that

(2) ‘Deployers shall assign human oversight to natural persons who have the necessary competence, training and authority, as well as the necessary support’.

(5) ‘Where deployers have reason to consider that the use of the high-risk AI system in accordance with the instructions may result in that AI system presenting a risk within the meaning of Article 79(1), they shall, without undue delay, inform the provider or distributor and the relevant market surveillance authority, and shall suspend the use of that system’.

The existence of a legal or contractual duty to report system failures or anomalies is central to attributing liability. This duty to inform is not new in the product liability scenario; the user is put in a position to comply with it as soon as he is provided with instructions by the manufacturer, which should indicate the potential risks that may arise when using the good. If, therefore, this duty of information exists for all products, it is, however, true that in the case of high-risk intelligent systems, unlike for other products, the duty of information is certainly more stringent because of their greater potential for harm.

In this vein, new criminal provisions could punish the non-disclosure or misrepresentation of information to the regulator (*motu proprio* or when responding

to regulators' requests) – aggravated in cases where the misrepresentation or non-disclosure leads to death or serious injury of a subject.¹⁴⁸

Of course, if the NUiC received inadequate training or unclear instructions regarding when and how to report issues, liability may shift to the provider or fleet manager. Providers have a critical role in ensuring that reporting protocols are clear, actionable, and well-communicated to operators. Article 14 of the AI Act requires providers to implement mechanisms for effective human oversight, which includes ensuring that operators are adequately trained to recognize and report anomalies.¹⁴⁹

The crucial importance of the NUiC's duty to inform lies in its role as a pivotal link in enabling corrective actions by providers. Article 20(1) of the AI Act stipulates that if a provider possesses information or has reason to believe that a high-risk AI system they have placed on the market or put into service is non-compliant with the Regulation, they must immediately take corrective measures. These measures include bringing the system into conformity, withdrawing it, disabling it, or recalling it, as appropriate, and informing the competent market surveillance authorities. Failure to fulfill these obligations exposes providers to regulatory sanctions and, in some cases, civil or criminal liability if non-compliance contributes to harm.

This explains why the NUiC's timely communication of anomalies or failures is indispensable in this process: without it, providers may remain unaware of critical issues, delaying corrective actions and undermining the oversight framework envisioned by the AI Act. This is reason enough to argue in favor of criminalizing failures in information-sharing, which serve to safeguard the legal interest in ensuring the correct functioning of competent authorities and effective oversight mechanisms.

¹⁴⁸ The UK Act defines the NUiC's duty to inform as 'duty of candour'. The Act contemplates a range of information-sharing duties. The Act also sets out penalties for non-compliance with a warrant or a requirement to provide information. Ch 3(14) and (20).

¹⁴⁹ See AI Act, Article 26(2) and recital 48 indicating that a certain level of know-how is expected from those humans tasked by the user to perform the oversight.

- Hypothesis III: NUiC engaging AV outside its operational design domain

Finally, the easiest hypothesis to analyze involves the scenario where the NUiC engages the AV outside its operational design domain (ODD). Automated driving systems are designed to handle traffic situations they encounter within their ODD, which delineates the specific conditions under which the system is permitted to operate – such as on a motorway during daytime and in clear weather.¹⁵⁰

Operating an AV outside its defined ODD can lead to hazardous situations as the system is not equipped to handle conditions beyond its intended design. In such cases, the NUiC's decision to engage or allow the AV to operate under these circumstances may constitute a breach of their duty of care. This breach can give rise to criminal liability if their conduct results in harm that a reasonable operator would have foreseen and avoided.

Criminal liability may be more clearly attributable when an AV explicitly warns the NUiC that conditions exceed its ODD, and the NUiC disregards this warning. This represents a deliberate deviation from established safety protocols, making it easier to argue gross negligence or even recklessness. Conversely, if the AV fails to issue an alert but a reasonable operator in the NUiC's position should have recognized that the vehicle was outside its ODD based on observable conditions (e.g., severe weather or off-road terrain), liability could also arise based on a failure to exercise appropriate vigilance.

¹⁵⁰ By contrast, Level 5 systems, which are not yet commercially available, are envisioned to operate under any conditions and without any ODD limitations.

SECTION 5

Provisional conclusions

Chapter IV explored the critical responsibility of AI users/deployers under the AI Act, specifically the requirement for ‘human oversight’, which admittedly carries significant implications in the realm of criminal liability.

Although the AI Act does not explicitly address criminal liability of deployers, it seems to place a significant emphasis on the role and duties of the human overseer. In particular, the concept of *meaningful human control*, as outlined in Article 14 AIA to address the problem of responsibility gaps in high-risk AI systems, is becoming a central concept when discussing human responsibility for AI, especially in non-forgiving scenarios in which fundamental human rights and safety are at stake.¹⁵¹ The core idea is that humans should ultimately remain in control of, and thus morally and legally responsible for, the behaviour and output of AI systems. As a result, ‘if an autonomous system causes harm to human beings, having a human in the loop provides trust that somebody would bear the consequence of such mistakes’.¹⁵²

Nevertheless, meaningful human control has also received the critique to be a rudimentary, ill-defined concept that ignores operational contexts and does not provide concrete guidelines.¹⁵³ Many authors have pointed out that the blanket requirement for human oversight over any operation of an AI system, while seemingly reassuring, ultimately undermines the very effectiveness of such control.¹⁵⁴ At its core, this approach suffers from a circular logic: it delegates certain decision-making responsibilities to a machine, effectively removing human involvement ‘out

¹⁵¹ Santoni de Sio and Van den Hoven (n 14), 15.

¹⁵² European Parliamentary Research Service, ‘The Ethics of Artificial Intelligence: Issues and Initiatives’ (2020), 35.

¹⁵³ Santoni de Sio and Van den Hoven (n 14), 103–115.

¹⁵⁴ Green and Kak (n 36).

of the door’, but then reintroduces human oversight ‘from the window’ to ensure safety, without thoroughly evaluating whether meaningful human intervention is even feasible in the first place.¹⁵⁵ In other words, a human is required to supervise a system that was created to overcome her and make up for her shortcomings. This false sense of security not only risks diverting attention from more pressing concerns – namely, the inherent risks posed by certain AI applications that may remain harmful regardless of oversight mechanisms, or the potential liability shield for providers¹⁵⁶ – but also contradicts the very purpose of automation, as it does not actually reduce human effort in any way, but merely shifts the duty of attentiveness from the direct execution of a hazardous activity to an equally pervasive and continuous obligation to monitor the actions of another entity.¹⁵⁷ The risk is to create a (human) scapegoat or ‘human crumple zones’;¹⁵⁸ the contrast with common principles of criminal law (the principle of culpability *in primis*) is self-evident. Since *ad impossibilia nemo tenetur*, more refined regimes or interpretations of the law are necessary to avoid unequitable attribution of responsibility or scapegoating.¹⁵⁹

Against this backdrop, this chapter argued that if human control is to serve as a genuine safeguard, it must be grounded in a realistic assessment of when and how human intervention can meaningfully mitigate risk. Simply requiring human oversight without defining its scope and limitations risks creating a *legal fiction* – one where liability is shifted onto users or operators without ensuring they have the actual

¹⁵⁵ Beck (Ch I n 21), 141.

¹⁵⁶ According to some authors, overburdening users would substantially shield providers from responsibility, reducing their incentives to implement robust safety measures and design safeguards. This dynamic allows providers to market AI as surpassing human capabilities while deflecting responsibility onto users who are often ill-prepared to mitigate the associated risks. See Green and Kak (n 36).

¹⁵⁷ Cappellini (Ch III n 3), 13-14.

¹⁵⁸ Elish (n 39), who uses the term ‘moral crumple zone’ to refer to the danger that responsibility for an action may be misattributed to a human actor who had limited control over the behaviour of the system. ‘Just as the crumple zone in a car is designed to absorb the force of impact in a crash, the human in a highly complex and automated system may become simply a component – accidentally or intentionally – that bears the brunt of the moral and legal responsibilities when the overall system malfunctions’.

¹⁵⁹ Giannini and Kwik (Ch I n 21), 45.

capacity to prevent harm.¹⁶⁰ This, in turn, could lead to a dilution of responsibility, where both AI providers and regulators rely on human oversight as a regulatory ‘quick fix’ rather than addressing the root causes of AI-caused harm.

Therefore, this brought us back to a well-known challenge: defining a clear area of *acceptable risk*. The core issue in this chapter was determining if and to what extent users can reasonably rely on the proper functioning of an AI system and, consequently, if and to what extent they should be exempt from criminal liability. We addressed a critical question: where should the threshold be drawn between permissible reliance and unreasonable neglect?

This chapter aimed to contribute to closing the gap between the theory of meaningful human control, as proposed by the EU legislator, and the practice of human-machine interactions by aligning the analysis with the broader discussion on scenario-based assessment previously examined in relation to AI providers criminal liability. Just as the scenario-based framework for providers helps define acceptable risk thresholds and ensures that legal duties and liability implications are assessed in a structured and proportionate manner, a similar logic could be applied to users and human oversight. This would help move beyond vague, blanket obligations and instead establish a more nuanced approach that considers the specific ways in which risk is generated through different human-machine interaction scenario.

A scenario-based approach for users could help differentiate between situations where oversight failures genuinely contribute to harm and those where intervention would have been impractical or even counterproductive. For instance, in high-automation systems where human operators are unlikely to maintain constant vigilance, imposing liability based on unrealistic expectations of intervention would be both unfair and ineffective. Instead, liability should be assessed based on whether the user’s role within a given scenario allowed for meaningful risk mitigation.

¹⁶⁰ It might even be the case that instead of an actual oversight on the action of the machine, the human is only capable of ‘rubber-stamping’ the results produced. B Wagner, ‘Human Agency in Decision-Making Systems’ (2019) 11(1) Policy & Internet, 114.

This perspective also reinforces the idea that AI-related liability should not be determined by isolated failures but by evaluating system-wide performance against predefined safety benchmarks. Just as providers are required to ensure that their AI systems meet scientifically and statistically acceptable risk thresholds, users should only be held accountable when their actions – or inactions – demonstrably deviate from reasonable oversight expectations within a given operational context.

Interestingly, our analysis of three different types of human-machine interaction scenarios revealed that some European countries have already introduced sector specific frameworks of user’s criminal liability connected to AI systems, all of them sharing some common characteristics. In particular, they all introduce some new legal subjects and legal constructs in an attempt to frame user responsibility within clear operational boundaries: assuming that we will not be able to reduce the risks of AI harm to zero, they aim to distinguish in this area of risk zones of legality from zones of illegality. Some frameworks manage to draw these lines with greater precision than others, but ambiguity remains a recurring challenge.¹⁶¹

Looking ahead, future policy initiatives should strive to move beyond one-size-fits-all approaches to human oversight and toward a more context-driven, evidence-based understanding of user responsibility – one that acknowledges the multiplicity of risks posed by AI while ensuring that individuals are equipped to manage and mitigate those risks in a meaningful way.

In this perspective, a crucial role must be attributed to the *instructions* provided by the provider, which accompany the product throughout its lifecycle, as well as to the provider’s obligation to design human-machine interactions that allow for *de facto* control, taking into account the cognitive constraints that emerge in high-stakes, technology-dependent environments. Factors such as automation

¹⁶¹ While these national case studies only focus on certain AI applications, we discussed the problem of human oversight at a sufficient level of abstraction to allow reproducibility of our conclusions to any domain where contemporary AI is implemented. In this respect, future discussion will need to focus on the possibility and efficacy of adapting similar regimes to address oversight issues in other domains.

complacency, reduced situational awareness, and the psychological tendency to over-rely on AI must be carefully considered when designing user-machine interactions.

A user who follows the provider's instructions should not be held criminally liable merely for having used an autonomous system as intended. The situation changes, however, if the user operates the system beyond the prescribed limits or if the provider has failed to set clear operational boundaries. In cases where the provider has misled users about the system's capabilities or failed to equip them with the necessary tools to manage AI-related risks, the introduction of specific criminal offences for providers could be considered.

Ultimately, an evidence-based approach is essential – one that draws from empirical data and insights from human-machine interaction theory to ensure a fair distribution of liability. This would help prevent unjust scapegoating of either users or providers while fostering a regulatory framework that aligns with both technological realities and fundamental legal principles.

CHAPTER V

THE PREVENTIVE TURN IN REGULATING AI CRIME: PERSPECTIVES *DE* *JURE CONDITO* AND *DE JURE CONDENDO*

SECTION 1

The anticipation of criminal law intervention

1.1. The rationales and limits of anticipatory criminal intervention

The role of risk regulation and criminal law in the AI sector is fundamentally built upon the concept of permissible risk, acknowledging that certain levels of risk are inevitable and must be tolerated in contemporary AI-driven society. The way we define the threshold between acceptable and impermissible risk is therefore not merely a regulatory choice but one that directly impacts criminal liability for harm resulting from AI systems.

As this study has extensively demonstrated, the AI Act marks a significant regulatory milestone by establishing a structured approach to defining (un)acceptable AI-related risks. However, as AI systems continue to evolve, the Act's risk-based approach reveals critical limitations, potentially leaving violations of fundamental rights and interests – harms that, in other legal domains, have already been subject to criminalization – without adequate legal consequences. This gap risks denying victims access to criminal-law remedies, effectively undermining justice and protection.

A key issue lies in the AI Act's reliance on a normative conception of risk ('legal risk'), rather than an empirical assessment of actual harm. While risk regulation aims to structure and mitigate unavoidable AI-related dangers, its effectiveness is limited if it does not generally translate into a coherent criminal liability framework. This highlights the urgent need to recalibrate and expand the role of criminal law, ensuring that liability frameworks evolve alongside AI-driven environments and provide 'effective, proportionate, and dissuasive sanctions' for AI crimes'.¹

¹ AI Act, Article 99.

In this context, criminal law emerges as the only regulatory tool truly capable of capturing the inherent wrongfulness of creating or maintaining illicit risks to fundamental legal interests such as life and physical integrity. Unlike risk regulation, which primarily seeks to manage and mitigate dangers, criminal law has the distinct function of marking certain risks as morally and legally intolerable, thereby providing a stronger protective shield against AI-related threats.²

Given the challenges of directly linking AI crime to individual fault, ‘anticipated’ criminal protection could take the form of endangerment-based offenses, where liability arises from the mere creation of a prohibited risk, rather than from the actual occurrence of harm.³ This approach would enable criminal law to intervene earlier even in the absence of concrete harm, based on the presumption that certain behaviours or situations inherently generate an intolerable risk to legally protected interests.

In European legal systems, the primary tool for anticipating criminal protection – shifting liability from actual harm to the mismanagement of a source of risk – is the category of *crime of (abstract) danger*.⁴ As our risk-oriented society has increased the demand for protection, criminal law has evolved to address new and more complex forms of criminality, often driven by technological advancements. This model of anticipatory criminal intervention applies particularly to failures to comply with legal requirements and safety standards in activities that inherently

² Eric Hilgendorf, ‘Autonome Systeme, künstliche Intelligenz und Roboter’ in Stephan Barton *et al.* (eds), *Festschrift für Thomas Fischer* (CH Beck 2018), 110-111.

³ It could be said that there are as many qualifications for crimes as there are doctrinaires. For example, criminal types are qualified according to the content, the subject who performs the conduct, the legal property protected, the structure, and the time, among other factors. Although all the qualifications of criminal types are valid, they are understood to outline the learning of the theory of crime. One of the classifications that best accommodates criminal reality is the one that states that crimes must be classified according to the action and harmfulness. On the one hand, crimes can be divided into (a) result crimes and (b) crimes of mere conduct, while on the other, crimes can be divided in (a) crimes of injury and (b) endangerment crimes. This classification affirms that result crimes produce substantial injury to the legal asset, while those of mere conduct generate danger to the legal asset.

⁴ See Gaetana Morgante, ‘Criminal law and risk management: from tradition to innovation’ (2016) 16(3) *Global Jurist* 315-350, 338ff.

impact fundamental legal goods, such as public safety, health, or the environment.⁵ Under this approach, liability is established not by proving actual endangerment but solely by demonstrating violations of regulatory standards of diligence and precaution designed to mitigate risks.

Such offenses have the advantage of bypassing the challenges of criminal negligence in areas of complex and rapidly evolving technologies. Unlike traditional liability models, which often hinge on proving a breach of duty or a causal link between conduct and harm, abstract danger offenses would operate on the basis of risk creation itself, meaning that violations would give rise to intent-based liability (*dolus*) rather than mere negligence.⁶

However, while the introduction of danger-based offenses is relatively unproblematic when the risks associated with a given technology are well-established, the situation becomes far more complex when dealing with technological tools whose

⁵ This model is exemplified by the crimes of endangerment listed in article 3(2) of the revised EU directive on the protection of the environment through criminal law entered into force on 21 May 2024. Endangerment crimes under the directive generally refer to the infringement of a specific EU law prohibition, like the interdiction to produce, commercialize, use or release ozone-depleting substances and fluorinated greenhouse gases. Alternatively, some conduct crimes involve a failure to comply with applicable legal requirements when carrying out specific activities likely to have significant effects on the environment, such as the recycling of ships or the treatment of hazardous waste. The rationale for criminalizing noncompliance with standards of diligence and precautions lies in the need to catch cases where the legislator has decided that the infringement of sectoral rules would put the environment at intolerable risk – even though risk may not manifest immediate harm. In this way, the focus shifts from proving causation and specific endangerment to monitoring conduct against established regulatory frameworks, thus enhancing the law’s preventive function. Directive (EU) 2024/1203 of the European Parliament and of the Council of 11 April 2024 on the protection of the environment through criminal law and replacing Directives 2008/99/EC and 2009/123/EC, OJ L, 2024/1203. This model is easy to observe in many other areas, such as product and workplace safety, financial, corporate, and tax matters.

⁶ Monika Simmler and Nora Markwalder, ‘Roboterstrafrecht Zur strafrechtlichen Verantwortlichkeit von Robotern und künstlicher Intelligenz’ (2017) 2 Aktuelle juristische Praxis 171-182, 177-178.

risks remain uncertain or difficult to quantify. In such cases, the risk of overextending criminal law based on precautionary logic must be carefully managed.

Some scholars argue that the need to address hypothetical risk scenarios may justify – albeit problematically – the use of such offenses, with the *caveat* that the concept of ‘danger’ in this context remains highly speculative and detached from scientifically verified risk assessments. In such cases, criminal law reaches its outermost limits, stretching the principle of harm to its breaking point. However, this preemptive approach retains some legitimacy, provided that the conjectures about the dangerousness of a given activity are substantial enough to raise ‘serious concerns about potential threats to fundamental legal interests’.⁷

However, the prevailing legal doctrine challenges the legitimacy of abstract endangerment offenses in cases where the alleged danger is not backed by a solid scientific basis for the alleged risks.⁸ Traditionally, such offenses are justified when the risk is empirically demonstrable, such as with hazardous substances or dangerous industrial practices. In contrast, a precautionary criminalization of AI-related risks would imply sanctioning behaviours based on mere speculation, rather than on established evidence of their harmfulness.⁹

⁷ For the Italian doctrine, see Domenico Pulitanò, ‘Colpa ed evoluzione del sapere scientifico’ (2008) 14(5) *Diritto penale e processo* 647-652, 652. Giorgio Marinucci and Emilio Dolcini, *Corso di diritto penale* (Giuffrè, 2001), 567.

⁸ See G. Quintero Olivares, ‘La robótica ante el derecho penal: el vacío de respuesta jurídica a las desviaciones incontroladas’ (2017) *Revista Electrónica de Estudios Penales y de la Seguridad*, 17-19, does not consider it possible to introduce criminal offenses inspired by the precautionary logic. The argument put forward by the author is of a substantive nature, based on the incompatibility between precaution and abstract endangerment offenses. Cfr. Donini (Ch II n 163) 120-121, according to whom, in the case of the precautionary principle, the epistemological basis is reversed, as a presumption of dangerousness or lack of safety is established. Cfr, also Castronuovo (Ch II n 163), 47 ff.

⁹ The legitimacy of crimes of abstract danger has been widely debated across legal systems. Similar concerns have emerged in Belgian and French systems regarding the *infraction formelle de mise en danger*, also referred to as *délit obstacle*. These offences have sparked controversy due to their emphasis on punishing prohibited conduct itself rather than a tangible violation of a legal good. Kutu, *Principes généraux du droit penal belge. Tome II:*

Traditional judicial interpretation suggests that such offences can align with the principles of criminal law only if they are structured as a mediated form of protection for real legal interests.¹⁰ This aligns with the German model of *konkretes Gefährdungsdelikt* (or crimes of concrete endangerment), which requires a demonstrable link between the prohibited conduct and a concrete danger to a protected legal interest. In other words, a crime of concrete endangerment is committed when the criminalized action has as a characteristic the effective possibility of damaging a specific legal good.¹¹ The Italian Penal Code follows suit, opting for a notion of danger as a concrete structural element of the criminal type.¹²

The shift toward concrete endangerment aims to address some of the critical concerns associated with abstract endangerment. Unlike abstract endangerment offences – where criminal liability arises from hypothetical risks – the key advantage of concrete endangerment offences lies in its stronger alignment with fundamental criminal law principles, particularly the principle of harm. By requiring evidence that a situation has progressed beyond mere potentiality to an actualized risk, concrete endangerment provisions offer a clearer and more justifiable basis for intervention.

l'infraction penal (Bruxelles, 2010), 94; G. Schamps, *La mise en danger: un concept fondateur d'un principe general de responsabilit  . Analyse de droit compare* (Bruxelles-Paris, 1998).

¹⁰ Morgante (n 4), 338.

¹¹ In crimes of concrete endangerment, the danger is an element contained in the type. The effective possibility of damaging a specific legal good must be examined from an ex-ante perspective, and it becomes the actual and concrete endangerment of a protected interest. By contrast, a crime of abstract endangerment is committed when, in the criminalized action, it is not necessary to verify the dangerous potentiality of injury, but rather, it is presumed that it can occur if the conduct that has been undertaken is not stopped.

¹² In the Italian legal system, this is the case of public integrity crimes, originally included in the Criminal Code of 1930. Emblematic of this approach is the case of the so-called ‘unnamed disaster’ provision under Article 434 of the Italian Criminal Code. This provision, deliberately drafted in general terms, reflects the inherent difficulty of addressing emerging sources of danger and newly evolving phenomena of harm. Specifically, it punishes with imprisonment from one to five years anyone who, outside the confines of previously regulated disasters (such as air, marine, or rail disasters), commits an act intended to cause a disaster not otherwise covered by existing provisions. By leaving the concept of *disaster* undefined, the provision aims to ensure a future-proof framework that remains adaptable to unpredictable technological and societal developments.

This ensures greater proportionality in criminal sanctions, as liability is attached to conduct that has demonstrably created a real, imminent threat rather than mere speculation.

This raises a critical question: can we legitimately anticipate criminal intervention in AI-related domains without a solid scientific basis for the alleged risks? If AI trustworthiness is framed as a legal good, the challenge will be to define its endangerment in terms that are legally and empirically sound, ensuring that criminal law remains a tool for effective risk management rather than a mechanism for precautionary overreach.

1.2. Is it time to consider new endangerment offences in the AI domain?

In the context of AI regulation, crimes of danger could serve as a key legal mechanism for anticipating criminal protection against the risks posed by AI systems, particularly regarding their potential adverse impact on fundamental rights and interests. After all, the categorization of AI risks operated by the European legislator, though inherently limited – since it defines a legally relevant ‘danger’ of fundamental rights and interests violations rather than identifying risk in relation to the actual occurrence of harm – nonetheless plays a crucial role in shaping the objective parameters for the potential introduction of new endangerment offenses.

In this light, the concept of ‘AI trustworthiness’ – understood within the EU regulatory framework as the assurance that AI systems do not systematically generate unlawful risks to fundamental rights and interests – could be elevated to the status of a new *legally protected interest*, meaning that criminal law would explicitly safeguard society’s trust in the integrity and reliability of AI systems as an instrumental public good, warranting criminal protection to foster responsible innovation and prevent harm to primary legal interests such as life, health, or property. This approach would shift the focus of criminal intervention from reactive measures – addressing harm

after it occurs – to proactive prevention, emphasizing the responsibility of AI operators to ensure that AI systems are inherently safe and trustworthy.

It follows that new crimes of danger could be introduced to criminalize: (i) the act of producing or releasing certain AI systems in areas where their use is deemed excessively risky, insufficiently beneficial, or incapable of ensuring the necessary protection of fundamental human rights and interests;¹³ (ii) the failure by the provider to design, deploy, or maintain AI systems within the legally established area of acceptable risk (e.g., mismanagement of AI risks or failure to implement adequate oversight mechanisms); and (iii) the improper use or maintenance of AI systems by the user (e.g., deactivation, failure to activate, or failure to update the safety measures with which the AI system was originally equipped) regardless of whether harm actually occurs.¹⁴

For instance, if an AI system is designed or deployed in a way that foreseeably and unreasonably endangers human life or public safety – such as an autonomous vehicle system that fails to adequately detect pedestrians or a medical diagnostic AI that consistently overlooks critical health conditions – the responsible parties could be held liable for knowingly or recklessly creating such a risk, even if no actual injury or death has yet occurred. This would apply the principle of *abstract endangerment*

¹³ For example, criminal offences could be established that sanction infringements of specific EU law prohibitions, such as the interdiction against producing, commercializing, using, or releasing certain prohibited AI systems under Article 4 of the AI Act. This approach draws inspiration from the field of biolaw, where certain jurisdictions have introduced strict prohibitions and criminal sanctions for specific high-risk activities, such as human cloning or the unauthorized use of genetic data. These prohibitions are designed to protect fundamental rights and societal values, and they serve as a model for how the law can preemptively address emerging technologies that pose significant ethical and legal challenges. Translating this model to the field of AI would involve similarly identifying and criminalizing activities that fundamentally undermine trust, safety, and human rights. Cfr. for example the German *Embryo Protection Act* ('*Embryonenschutzgesetz*', EschG).

¹⁴ See *ex multis* Consulich (Ch I n 21); Lagioia and Sartor (Ch III n 155), 459; Hilgendorf (n 2), 110-111; Athina Sachoulidou, 'AI Systems and criminal Liability. A Call for Action' (2024) 11(1) Oslo Law Review 1-10.

to AI trustworthiness to indirectly protect the legal interest of *human life and bodily integrity*, which are among the most fundamental and universally recognized rights.¹⁵

It could be argued at length whether such a shift away from an event-based approach toward a danger-based paradigm is commendable. In a liberal criminal law system, the event traditionally occupies a central role as a cornerstone for attributing liability. When the significance of the event diminishes, subsides, or loses its foundational role – becoming merely a condition for liability to punishment, or even disappearing entirely from the conditions for liability – it fundamentally alters the principles underlying criminal liability. Indeed, criminalizing hypothetical risks detached from empirical assessments of actual harm could push the boundaries of offensivity too far, potentially leading to over-regulation in the form of over-criminalization, an outcome that would be hardly compatible with the ancillary nature of criminal law and the protection of fundamental rights in this realm.¹⁶

That said, what remains certain is that, in some sectors, particularly those involving advanced technologies like AI, the harmful event often eludes the scope of authority of the human agent and, more often than not, is substantially independent from the implementation of safeguards.¹⁷ Given the unique and systemic risks associated with AI – including opacity, automation bias, and self-learning capabilities – the idea of criminal law as an anticipatory safeguard remains a compelling option – but only under strict conditions: legal intervention must be scientifically grounded, proportionate, and precisely delineated to prevent an unchecked expansion of liability.

Several benefits could be expected out of this criminal intervention. These measures aim to foster a culture of compliance, incentivizing organizations to prioritize safety, take their regulatory duties seriously, and reinforcing the notion that AI innovation must align with societal and ethical norms. Additionally, this may alleviate the judiciary's burden of proving causation and *mens rea* in complex AI-

¹⁵ Consulich (Ch I n 21), 291.

¹⁶ See in general Manes (Ch III n 69).

¹⁷ Consulich (Ch I n 21), 303.

related harm cases. However, implementing such offences also presents significant challenges from a criminal law theory perspective.

First, from a structural standpoint, defining the specific conduct required or prohibited under such a regime would be inherently problematic. These norms would essentially operate as blanket criminal provisions, relying on external legal obligations to define the scope and content of the criminal command. While such reliance allows criminal law to ensure compliance with the evolving regulatory framework, it also introduces a degree of dependency on non-criminal standards and precautions, assuming their adequacy and clarity in delineating the boundaries of permissible conduct. This assumption, however, may not always hold true, particularly in the context of rapidly evolving and inherently complex technologies like AI systems.

This is exemplified by the nature of existing AI regulation, which often prioritizes flexibility and procedural-administrative requirements over rigid prescriptive rules aimed at preventing specific outcomes. For example, the AI Act requirements do not exhaustively define the factual assumptions underpinning prohibited or mandated conduct. Instead, they rely heavily on harmonized technical standards developed by standardization bodies to provide clarity and operational detail.

While such standards are intended to complement and integrate criminal provisions, their inherent immaturity, ongoing development, and generality risks introducing ambiguities into the criminal framework. The lack of specificity in these standards can leave operators without concrete guidance on permissible risk-taking and the precise boundaries of their responsibilities. This ambiguity may lead to uncertainty about whether their actions meet the prescribed standards, increasing the risk of unintentional non-compliance and, potentially, unjust sanctions.

Additionally, the incorporation of external, non-legal sources into criminal provisions raises questions about the principle of *nullum crimen sine lege certa* (principle of strict legality), a cornerstone of criminal law that requires legal norms to be sufficiently precise to allow individuals and entities to understand and predict what

behaviour is punishable and thus ensure foreseeability. When criminal liability is contingent on compliance with external and potentially evolving standards, individuals may struggle to anticipate the legal consequences of their actions. This could not only erode foreseeability in criminal law, challenging the legitimacy of sanctions and increasing the likelihood of arbitrariness in enforcement, but also risks exposing operators to disproportionate risks and potentially discouraging the adoption and innovation of AI technologies in the face of unclear regulatory expectations.

Therefore, these challenges raise fundamental questions: how can we ensure that such offenses remain within the limits of legality and proportionality? If AI-related criminalization is based on risk rather than concrete harm, what standards should be used to determine which risks are ‘unacceptable’ and warrant criminal sanction? Answering these questions will be critical to ensuring that criminal intervention in AI remains a tool of last resort, grounded in scientific and legal certainty rather than precautionary overreach.

For instance, *de jure condito*, we could refer, among others, to the sanctions outlined in Italy’s *Consumer Code* (Legislative Decree 206/2005) in the context of mass-produced technologies. Notably, Article 112, paragraph 2 of the Consumer Code is particularly pertinent to our discussion, stipulating that ‘unless the act constitutes a more serious offence, the producer who places *dangerous products* on the market shall be punished by imprisonment of up to one year and a fine of between 10,000 and 50,000 euros’ (emphasis added). When products are placed on the market in large quantities, any defect, non-conformity or alteration in their condition constitutes a risk to the health, safety or life not only of an individual but of an indeterminate community.

Although such a provision may not be directly applicable to the field of AI,¹⁸ it could serve as a model to hold providers liable if they market AI systems that fail

¹⁸ Notably, it would not probably be accurate to categorize AI systems as ‘products’ under Article 3, paragraph 1, letter e) of the Consumer Code, which defines a product as ‘any item

to meet the standards of a trustworthiness. For example, if an AI system is marketed as safe and reliable but lacks essential safeguards or performs poorly in critical tasks, the provider could be held liable for releasing a ‘dangerous’ or ‘defective’ product. However, the main challenge lies in defining what makes an AI system ‘dangerous’ or ‘defective’ in a way that is fair, scientifically sound, and legally precise.

To address this, the concept of a dangerous AI should be tied to predefined performance standards within specific operational contexts. Instead of criminalizing vague or hypothetical risks, liability would be based on whether an AI system fails to meet clearly defined performance thresholds in real-world scenarios. This ensures that the legal framework focuses on demonstrable deficiencies that create an intolerable level of risk, rather than penalizing abstract or speculative dangers.

For example, if it is established that the accident rate must stay below a specified threshold of x% relative to AI usage, then the expectations for the provider become clear. If the provider discovers that the accident rate from the system’s deployment in a certain operational context exceeds this threshold, despite adhering to the mandated precautionary measures, and fails to implement further risk mitigation strategies – such as issuing warnings to consumers, recalling or withdrawing the product – they could be held liable when the provider is aware of the existence of the performance thresholds that must be upheld for preventive purposes but fails to take necessary action.¹⁹

In this perspective, *de jure condendo*, national legislators are already defining new offences of endangerment related to AI safety. Notably, the Italian government has recently become the first to legislate specifically on AI-related criminal provisions.

intended for consumers, even as part of a service provision, or likely, under reasonably foreseeable conditions, to be used by the consumer, regardless of whether it was intended for them, supplied or made available for a fee or free of charge in the context of a commercial activity [...]’. The application of this definition to AI systems seems feasible only in scenarios where the AI algorithm is embedded within a good meant for consumer use, such as an autonomous driving system within a vehicle. It appears there is little scope for extending this regulation to so-called digital products, which refer to software that exists independently of any hardware.

¹⁹ Fragasso (Ch II n 62), 38.

On 23 April 2024, the Italian Council of Ministers approved a draft law (Disegno di Legge, or DDL) titled ‘Principles regarding artificial intelligence’ aiming to introduce domestic provisions that complement and expand upon the European AI Act.²⁰

Among the most intriguing aspects of this draft law are its criminal provisions, which envisage a range of offences committed through or in relation to the use of AI systems. Specifically, Article 22(5)(b) of the draft foresees the establishment of one or more autonomous criminal offences, covering both intentional and negligent conduct, to address failures in adopting or adapting safety measures concerning the production, circulation, and professional use of AI systems.

Ultimately, the core idea here is that the *prudent organization of risky activities* is essential to preventing harm caused by AI technology and to constructing criminal offenses that serve as a safeguard within the broader regulatory framework. This means that the legal system should focus on ensuring that those who develop, deploy, or manage AI systems take proactive steps to organize their activities in a way that minimizes risks and prevents harm. Criminal liability, in this context, would act as a final layer of safeguard, in line with the principle of *extrema ratio*, ensuring that failures to meet these organizational standards are met with appropriate legal consequences.

At the same time, non-criminal measures could play a significant role in this framework by providing more immediate and flexible responses, particularly given the likely limited severity of penalties that might be imposed for endangerment offenses. For instance, administrative measures – such as bans, fines, or mandatory oversight – could be more effective than a minor criminal penalty in ensuring compliance and effective public monitoring.

²⁰ Disegno di legge n. 1146, ‘Disposizioni e delega al Governo in materia di intelligenza artificiale’, 20 May 2024, currently being discussed in commission, at <https://www.senato.it/leg/19/BGT/Schede/Ddliter/58262.htm>.

1.3. The new frontiers of an integrated penalty system to face AI risks: back to criminal law as a last resort

Given the complexity of AI-related risks and the limitations of both purely regulatory and purely punitive approaches, criminal law should remain a measure of last resort, deployed only when other safeguards prove insufficient. In this regard, according to some legal scholars, an injunction-based model could offer a more proportionate and preventive mechanism, ensuring coordination between criminal sanctions and extensive administrative oversight.²¹

Under this model, authorities would be empowered to issue binding injunctions against the development, deployment, or continued use of AI systems that fail to meet established safety thresholds – such as implementing extra precautions, subjecting intelligent systems to additional training or testing phases, or, in extreme cases, ordering their deactivation. Criminal liability would then apply only in cases of deliberate defiance – where an entity knowingly disregards these prohibitions, either by refusing to comply with corrective measures or by continuing to operate AI systems deemed excessively dangerous.

Additionally, obligations could extend to encompass the duty to share relevant know-how with supervisory authorities, particularly for the assessment and mitigation of potential risks that have not yet materialized as adverse events. This duty might also include reporting any damages caused by AI systems and the frequency rates of such incidents, providing authorities with the necessary information to enhance oversight and improve regulatory mechanisms.

From a theoretical perspective, these criminal offences are functional to the effectiveness of the administrative control of the thresholds of danger.²² In this

²¹ Piergallini (Ch I n 39) 1771-1773.

²² In Italian legal system the injunction model was historically inspired to the punitive pattern of art. 650 of penal code. Piergallini (Ch II n 155), 526. ‘While it may partially conflict with the principle of legal reserve—since the sublegal measure does not merely serve as a technical

framework, the legal interest of AI trustworthiness receives only indirect protection, as its preservation is entrusted to public authorities responsible for enforcing compliance.

Although traditionally met with skepticism – given its potential to reduce criminal intervention to a mere sanctioning appendix of sector-specific administrative regulations – this protective technique demonstrates practical and systemic coherence. Indeed, while the imposition of detailed technical requirements from above may seem at odds with fundamental principles of legality and offensivity, this tension is partially mitigated by mechanisms that enhance democratic participation in administrative decision-making. By fostering transparency and inclusiveness, the model ensures that technical judgments are not imposed in a purely authoritarian manner (command-and-control), but instead emerge from deliberative processes that secure broader legitimacy and consent (cooperative compliance).²³

This approach reflects the subsidiarity principle of criminal law, reinforcing the idea that penal intervention should be reserved for the most serious cases of non-compliance, while regulatory enforcement, administrative sanctions, and compliance obligations handle the bulk of risk mitigation. By structuring AI liability around graduated enforcement mechanisms, the injunction model balances legal certainty with technological flexibility, ensuring that criminal law remains a targeted tool for addressing egregious AI-related misconduct rather than a blunt instrument of overcriminalization.

specification but effectively defines the substantive content of the offense, making non-compliance a direct violation—the injunction model represents a necessary trade-off for safeguarding intermediate legal interests, which are deemed essential for the protection of fundamental rights and overarching legal goods’.

²³ See in general Morgante (n 4) 341-343.

SECTION 2

From individual to corporate liability for AI crime

2.1. The organizational solution: towards an algorithmic corporate liability?

Section 1 of this chapter emphasized the importance of anticipating criminal intervention to tackle AI crime. However, we must now turn to a more fundamental question: who should be held criminally liable for failures in AI risk management?

In view of the traditionally individualistic nature of criminal law, the analysis has so far focused on the *individual operator* – whether the provider or the user – as the primary catalyst of liability for AI harm. The individual-centric approach emphasizes the errors and omissions of individuals, presuming that risks of harm arises because people fail to pay sufficient attention to their tasks or because they fail to uphold existing technical standards and procedures. This approach follows a linear

causal model, where individual actions are scrutinized in isolation, while the broader organizational and systemic context remains in the background. As a result, remedial efforts tend to focus on frontline individuals, leading to an outcome that centres on blame attribution. If an individual is deemed culpable, the response is often punitive – removing or sanctioning the ‘bad apple’ without addressing systemic factors that may have contributed to the harm.

While this approach may be effective when clear individual fault is present, it fails to capture the significant degree of technical complexity (of actors and procedures of the organization and management of the risk itself) inherent to a risk management system, where the conditions that enable harm often arise from systemic and structural failures rather than isolated personal mistakes. In this sense, the prudent management of a source of risk and the prevention of harm related to dangerous but useful activities immediately evokes the importance of the *collective* (rather than individual) factor.²⁴

This emphasis on the ‘organizational dimension’ of technological risks carries significant implications for criminal law. Every organization, beyond its formal institutional role, inevitably develops patterns of deviance from legal standards and normative expectations.²⁵ This organizational deviance – whether intentional, structural, or systemic – becomes particularly problematic in cases where harmful events stem from structural deficiencies rather than individual misconduct – such as technological disasters, environmental accidents, unsafe labor conditions or, for what is of interest here, the widespread deployment of hazardous AI systems. In this

²⁴ This shift is not unique to AI. In the criminal law of risk, liability has increasingly expanded from individual fault to models that capture the collective and systemic dimensions of harmful activities. In this perspective, prudent risk management and harm prevention in high-risk technological activities inherently require the involvement of collective entities rather than just individuals. Morgante (n 4) 326 ff.

²⁵ Piergallini (Ch II n 155), 317ff. See also Giorgio Marinucci, ‘Innovazioni tecnologiche e scoperte scientifiche: costi e tempi di adeguamento delle regole di diligenza’ (2005) *Rivista Italiana di Diritto e Procedura Penale*, 29.

context, the issue emerges as to *who* should be sanctioned (whether criminally or not).

Within organized and complex contexts, however, it is almost useless to try to isolate the role and, consequently, the share of responsibility of the individual. As the American sociologist Diane Vaughan says, ‘in the complex organizations individuals do not make mistakes, but the mistakes are socially organized and systematically produced’.²⁶ While it may be true that an incident is triggered by the error of an individual, it is equally true that such errors often occur within flawed organizational systems characterized by latent vulnerabilities, which remain dormant until activated by human error. In many cases, the conditions for human error are inadvertently pre-established by organizational choices and then ‘inherited’ by individuals.²⁷ For that reason, human error becomes the starting point of analysis, not its conclusion.

Therefore, in cases of AI-related harm, the key question is not merely who should be punished but how criminal liability should be structured to ensure that organizations themselves bear responsibility for failures in risk governance. At stake is the development of a liability framework that does more than just assign blame – it

²⁶ Vaughan, *Controlling Unlawful Organizational Behaviour: Social Structure and Corporate Misconduct* (Chicago University Press, 1983).

²⁷ The shift from individual to organizational perspective reflects the general recognition that human error in complex and interconnected organizational contexts is ‘normal’ – and often even inevitable. The theory of ‘organizational error’ has reshaped the understanding of incidental events, redirecting attention from individual fault to the risk culture, systemic properties of an organization, and its risk management frameworks. Cfr. Perrow (Ch II n 112) argues that accidents in highly complex systems are not merely the result of individual errors or component failures but stem from the unpredictable and often catastrophic interactions between different system components. In this vein, Perrow defines such incidents as ‘normal’, not because they occur with high frequency, but because they are an inherent byproduct of the normal functioning of highly complex systems. Perrow’s insights underline the inherent limitations of relying solely on individual accountability in managing risks within complex technological systems, highlighting the need for systemic approaches to safety and liability.

must incentivize organizations to adopt risk-aware practices and proactively mitigate AI risks.

In this context, the classic criminal law, as such inextricably linked to the individual and his responsibility, is largely unsuitable. The complexity of AI systems, their reliance on multidisciplinary teams, and the integration of diverse processes across design, training, and deployment make it not only useful but also unfair to isolate the share of responsibility of individual human actors.²⁸

One promising avenue, in cases of AI harm, is to hold open the possibility of *corporate liability*. As corporations replace employees with algorithms, corporate liability becomes the *only* means of redress, targeting organizational mismanagement of AI risks and ensuring that entities are held accountable for systemic failures, even in the absence of direct harm: if we cannot change human nature, we might instead focus on transforming the organizational conditions in which individuals operate. This suggests that liability for harm caused by AI or other technological systems should not rest solely on the actions of individuals, but rather on whether the organization has put in place effective safeguards, organizational practices, and oversight to prevent foreseeable risks.²⁹

In this regard, the key question is whether the legal person (i.e., the corporation behind the AI) can *de jure condito* (or should *de jure condendo*) be held criminally liable in place or in addition to individual human actors for what is usually referred to as ‘algorithmic corporate misconduct’.³⁰ This Section will first explore the

²⁸ In this regard, it is stressed, on the one hand, the ineffectiveness, in terms of general prevention, much of the traditional crimes as the punitive sanction; on the other, the potential fallacy of hindsight, certainly not very sensitive to the differences between the cognitive situation of the agent inside the organization before a fact occurs and that of the observer from outside the organization after an occurred fact. Morgante (n 4), 327-328.

²⁹ Piergallini (Ch II n 155), 317ff.

³⁰ On the problem of algorithmic corporate misconduct see Mihailis E Diamantis, ‘The Extended Corporate Mind: When Corporations Use AI to Break the Law’ (2020) 98 N C L Rev 895ff, 898. See also Federico Mazzacuva, ‘The Impact of AI on Corporate Criminal Liability: Algorithmic Misconduct in the Prism of Derivative and Holistic Theories’ (2021) 1 RIDP 143ff

possibility that an organization may be held accountable for AI-related harms by looking at existing national regulatory frameworks in continental Europe, and then investigating possible techniques to make corporations criminally liable in this field as well as potential counter arguments to such a regulatory aim.³¹

2.2. Corporate criminal liability models in a comparative perspective

The rise of AI technologies is not the first instance where our society of risk challenges the foundational principles of traditional criminal law. Importantly, the evolving dynamics of organizational risk and the emergence of corporate crime have already driven significant shifts in criminal law, breaking away from individualistic dogmas such as the *societas delinquere et puniri non potest* doctrine. In many jurisdictions, corporations have been recognized as independent subjects of crime, capable of being held accountable for offenses stemming from systemic failures or structural inadequacies.

As AI systems become integral to organizational operations – and are poised to surpass employees as the primary contributors to corporate wrongdoing – the relevance of corporate liability grows increasingly critical. Yet, holding legal entities criminally liable for AI-related harm inevitably reopens long-standing debates in criminal law theory. Questions about whether corporations can truly qualify as legal persons for the purposes of criminal accountability – given that they do not act autonomously, lack moral agency, and cannot be culpable in the same way individuals

³¹ It is worth emphasizing that the issue herein discussed does not refer to cases where employees purposely, knowingly, or recklessly design AI systems to break the law; rather to the case where the employee's misconduct is 'removed' from the picture, so that law has to handle a purely algorithmic corporate misconduct. In these cases, criminal liability can be ascribed to the persons who intentionally use the tools in question for such purposes and, where appropriate, also to the legal person under the ordinary rules of the applicable corporate criminal liability law. On this topic see also Ivan Salvadori, 'Agenti artificiali, opacità tecnologica e distribuzione della responsabilità penale' (2021) 1 *Rivista italiana di diritto e procedura penale* 83ff.

are – remain contentious.³² This debate has historically drawn a categorical divide between common law and civil law traditions, as well as among various legal systems across continental Europe – and it is doubtful that unanimity will emerge in response to AI harm.³³

Nonetheless, this divergence does not exempt legal scholars and policymakers from addressing the applicability of various models of corporate liability – ranging from those based on strict liability or vicarious liability, to those focused on organisational fault or failure to prevent specific offences. On the contrary, the challenges posed by AI underscore the urgency of revisiting and refining these frameworks, ensuring they remain responsive to the realities of modern technological and organizational landscapes.

Before addressing the implications of corporate liability for AI harm, some preliminary points must be clarified. Regardless of jurisdiction, concepts such as *mens rea*, the type and purpose of punishment, and its consequences must be redefined when applied to legal persons. This redefinition must align these concepts with the processes of decision-making and action that arise from the internal dynamics of complex corporate structures.

In particular, it can be argued that the main issues that emerge in this context are a) the relation between the crime and the legal person, which involves determining how and when the actions of individuals within the organization can be attributed to the corporation itself; b) the juridical nature of the related responsibility, concerning the foundational question of whether corporate liability should be seen as direct (the corporation as the primary offender) or derivative (the corporation held liable based on the actions of its agents); and c) the role of the criminal conduct of a human being, as many jurisdictions require a human actor's wrongful act to serve as the basis for corporate liability, raising questions about whether this requirement remains

³² See for all Carlo Enrico Paliero, 'La colpa di organizzazione tra responsabilità collettiva e responsabilità individuale' (2018) 31(1-2) Riv. trim. dir. pen. econ. 175-219.

³³ M Dubber and T Hörnle, *Criminal Law: A Comparative Approach* (OUP 2014) 329-42.

appropriate in the context of autonomous or semi-autonomous AI systems (predicate offence).

Each of these issues is complex and contentious, with different countries and legal traditions adopting varied approaches. These variations are shaped by factors such as the prevalence and gravity of corporate crime, constitutional principles, and the legislature's sensitivity to this specific type of criminal activity. In this context, an almost unanimous interpretation has emerged, recognizing the coexistence of two distinct paradigms of attribution within corporate liability frameworks.³⁴

The first follows a *derivative* model of responsibility. Here, the criminal act of a human agent, carried out in the interest or to the advantage of the corporation, is attributed to the corporation through the principle of organic identification. This paradigm essentially treats the corporation as an extension of the individual offender, 'deriving' its liability directly from their actions. Accordingly, this theory does not require proof of a corporate state of mind, but only the material element of the wrongdoing (*actus reus*), while corporate *mens rea* is defined in terms of employee mental states.³⁵

The second paradigm underscores the autonomy of corporate liability by shifting the focus from individual wrongdoing to the corporation's failure to

³⁴ Carlo Piergallini, 'La maggiore età della responsabilità dell'ente: nodi ermeneutici e pulsioni di riforma' (2021) Arch. pen., web, 1, 3ff. More generally on the different models of corporate criminal liability, see James Gobert, 'Corporate criminality: four models of fault' (1994) 14 Legal Stud 393ff.

³⁵ More specifically, this is typical of the US and English law, where corporate liability is based on *respondeat superior* or vicarious liability. According to this principle, the corporation shall be held liable when an offence has been committed by one of its employees while performing collective activities to obtain a general, rather than an individual, benefit. For more details on these models, see Nicola Selvaggi, 'Criminal Liability of Corporations and Compliance Programs in the U.S. System' in Antonio Fiorella (ed.), *Corporate Criminal Liability and Compliance Programs*, vol. I, *Liability 'Ex Crimine' of Legal Entities in Members States* (2012), 601 ff; Celia Wells, 'Corporate Criminal Liability in England and Wales: Past, Present and Future' in Mark Pieth, Radha Ivory (eds.), *Corporate Criminal Liability. Emergence, Convergence and Risk* (Springer 2011).

implement adequate preventive measures.³⁶ When an organisational structure is objectively negligent in taking the necessary precautions to prevent the commission of offences, the harmful event is attributable to the entity based on ‘organisational fault’ and criminal sanctions may therefore be imposed on it.³⁷ In this context, the corporation’s *mens rea* is reflected in its organization and management systems (so-called compliance programs), which serve as concrete indicators of the corporation’s structure and the effectiveness (or shortcomings) of its risk management and compliance efforts.³⁸ Compliance programs play a pivotal role within the discipline of this paradigm of corporate liability: if it is proven that the governing body has adopted and effectively implemented, before the act was committed, adequate organizational and management measures to prevent crimes of the type that occurred, and that the

³⁶ This approach to ascribing criminal liability to corporations is known as ‘holistic theory’. This expression refers to the fact that the criminal offence can be considered as the product of a collective decision (*uti universi*), which is different from the sum of the individual decision-makers within the organization (*uti singuli*). For a comparative study, see for instance Enrica Villani, ‘Compliance Programs and Organisational fault in Europe’ in Antonio Fiorella (ed.), *Corporate Criminal Liability and Compliance Programs*, vol. II, *Towards a Common Model In The European Union*, (2012), 249ff. Cf. also Coffee Jr., ‘No soul to damn, no body to kick: an unscandalized inquiry into the problem of corporate punishment’ (1981) Mich. law rev., 386.

³⁷ The organizational fault model is a form of culpability specific to collective entities and constitutes the basic paradigm of the Italian ‘*ex crimine* administrative liability of legal persons’ introduced by Legislative Decree No. 231/2001 (Decree 231). It is worth remembering that, even if the responsibility is formally defined by law as ‘administrative’, it can be considered at least as para-criminal liability, due to the punitive weight of the sanctions provided by the Decree 231. See Paliero (n 32), 175ff; Giorgio Marinucci, ‘La responsabilità penale delle persone giuridiche. Uno schizzo storico-dogmatico’ (2007) Riv. it. dir. pen. proc. 464 ff.

³⁸ These models, though exhibiting notable differences, draw heavily from the compliance program framework of the American legal system. Essentially, they serve as the tangible embodiment of organizational duties, providing a structured mechanism for ensuring adherence to legal standards. In practice, they also pave the way for a form of technocratic oversight within corporate operations, effectively shaping their internal organization and processes to prioritize legal compliance above other operational considerations. Paliero (n 32), 172.

individual person who committed the crime has fraudulently evaded the model, then the corporation will not respond for the predicate offence of its employees.³⁹

However, an additional condition for attributing liability to the corporation is the establishment of a clear link between an organizational deficiency and the predicate offence.⁴⁰ This connection between the responsibility of the corporation and the individual's misconduct still frames corporate liability as 'derived' or 'dependent'.⁴¹ In other words, while organizational fault represents a distinct form of corporate culpability, it is not entirely independent of individual liability. Corporate liability still depends on the commission of a predicate offence by an individual within the entity. In fact, the basic idea underlying this model is that corporations can be held responsible for failing to put in place compliance programs or adequate procedures to prevent the individual criminal offence from occurring.

Ultimately, the bottom-line of all these theories is that corporations can only be held accountable if a human agent within the organization commits a criminal offense that satisfies all its essential elements (*actus reus* and *mens rea*). Corporate liability models remain rooted in individual action, making them less equipped to address situations where organizational structures and processes, rather than any single individual's conduct, contribute to harmful outcomes.

A more suitable model for addressing preventive needs would shift the focus to the *original fault of the organization* itself, moving beyond the actions or intent of individual agents. Under this model, liability could be attributed to the corporation for systemic failures, even when the actions of individual employees are lawful within the scope of their roles. The impossibility both in practice and on a theoretical level

³⁹ Vincenzo Mongillo, *La responsabilità penale tra individuo ed ente collettivo* (Giappichelli 2018), 193, 205.

⁴⁰ Additionally, according to the Italian Decree 231, in order to hold a corporation liable it is necessary, first of all, that one of the crimes listed by the Decree 231 has been committed 'in the interest or for the benefit' of the corporation itself by a person placed in the corporate hierarchy, from the lowest-level operations to the highest. Article 5, paragraph 1 of the Decree 231/2001.

⁴¹ Art. 6 co. 1 Legislative Decree 231/2001. Domenico Pulitanò, 'Responsabilità amministrativa per i reati delle persone giuridiche' (2002) 6 Enc. dir., agg., 963.

of making a human being liable (either intentionally, recklessly or negligently) for predicate crimes means that legal persons cannot also be held liable. For instance, if the organization's structure, processes, or culture systematically allows risks to materialize, this structural failure would constitute the foundation for liability, independent of individual culpability.

This debate is particularly relevant in the context of corporations increasingly relying on AI systems to make decisions traditionally handled by humans. AI's opacity and autonomy introduce new challenges for attributing responsibility under existing models.

2.3. The problems with holding corporations liable for AI harm

To explore how legislative models might evolve to hold corporations liable for AI-caused harm, it is essential to assess whether criminal policy reforms are necessary and whether such changes would be both appropriate and fair. Effectively sanctioning corporations when their AI systems cause harm or facilitate criminal conduct requires addressing three significant challenges:

- i. the difficulty of identifying culpable employees for the underlying offence within the corporation;
- ii. the challenge of proving that the offence was committed 'in the interest or for the benefit' of the corporation by someone within its hierarchy;
- iii. and the limitations of systems with 'closed-catalogue' predicate offences, such as those in Italy, which do not currently encompass scenarios where AI use results in severe outcomes like death or serious injuries.

The first issue, already mentioned in the previous paragraph, highlights a core limitation of traditional corporate liability frameworks: they often depend on the identification of a specific individual whose wrongful actions can be linked to the corporation. However, with AI, this task becomes increasingly challenging. AI systems operate in highly complex and decentralized processes where responsibilities are distributed across multiple individuals and teams. For instance, one team may be responsible for algorithm design, another for training the AI, and yet another for its deployment and monitoring. This diffusion of roles creates substantial obstacles in pinpointing a single individual whose actions—or omissions—can be directly tied to the harm caused. Moreover, many AI systems function as ‘black boxes’, making it difficult to trace the causal chain from human input to the AI’s harmful output.

This gap undermines the foundation of traditional corporate liability models, which often require a direct link between the conduct of an individual and the corporation. In an AI context, harm may result from systemic issues such as flawed training data, inadequate testing, or insufficient post-deployment oversight – failures that reflect organizational shortcomings rather than the fault of a single employee. As a result, current legal frameworks struggle to address situations where no individual can be deemed directly at fault, but the harm is nonetheless a foreseeable consequence of corporate actions or omissions.

This is certainly true of models that emphasize *derivative* liability, such as the identification principle or vicarious liability. Here, the key difficulty lies in drawing a clear connection between an individual employee’s culpable mental state and the corporation’s *mens rea* regarding the offence committed. This linkage is especially problematic in AI-related contexts, where harm may result from the autonomous or emergent behaviours of the system, making it difficult to attribute a specific employee’s intent or negligence to the corporation’s criminal responsibility.

In contrast, organizational fault models, which prioritize the corporation’s systemic and structural failures over individual culpability, might seem to offer a more fitting approach. These models shift the emphasis to the entity’s overarching responsibility for harm prevention, focusing on the lack, deficiency, or inadequacy of

risk management practices, internal controls, and compliance programs. By assessing whether the corporation has implemented effective organizational and prevention measures the basis of assessment of the specific risks deriving from the activities entrusted to and, in any case, carried out by AI systems in their interest or to their advantage, organizational liability could theoretically establish a basis for holding corporations accountable, even in the absence of direct individual fault.

While this could be a viable solution, some *caveats* are needed. Indeed, even these models remain fundamentally tied to the requirement of proving the commission of a ‘predicate offence’ in all its essential elements, both material and subjective, or at the very least the objective components of human conduct and the resulting harm. This enduring connection to traditional notions of criminal liability limits the potential of organizational models to address the unique challenges posed by autonomous AI systems, where identifying a clear causal link between human actions and AI-driven harm remains often elusive.

Additionally, the legal person can be held responsible only for those AI-related crimes that are committed *in its interest* or from which the legal person *derives an advantage*. This requirement introduces a significant challenge when applied to AI-related offences, given the inherent incongruence in attributing intent or purpose to an AI system.

AI systems, as they currently exist, operate without subjective intentionality or volition. Consequently, the event leading to harm is typically an unintended byproduct of the AI’s operation, rather than a deliberate act by the system to benefit the corporation. Indeed, even assuming that AI can be considered as a ‘person’ (a hypothesis that presupposes the controversial recognition of AI legal personhood), the existence of such a requirement is not automatic with reference to algorithmic misconduct, unless we demonstrate either of two conditions:

- i) *Intentional use of AI for unlawful gains* < The corporation must have deliberately employed the AI system to achieve illegal profits. For instance, this might occur if the corporation knowingly designed or

deployed the AI in a manner that exploits regulatory loopholes, ignores compliance requirements, or engages in deceptive practices to maximize financial returns.

- ii) *Cost-saving advantages* < Alternatively, the use of untrustworthy AI in business operations could itself represent a form of corporate benefit, such as cost reductions in compliance processes. If these cost-saving measures directly or indirectly contribute to the harmful outcome, this could establish the requisite link between the corporation's interest and the AI-related offence.⁴²

Finally, it must be pointed out that – at least in certain jurisdictions, like the Italian legal system – the legal person is not called to respond, not even potentially, for every crime committed in its interest, but only for those crimes which are strictly identified by the law. In those national legal systems that base corporate liability to a closed list of offences, the extension of such list to the criminal offences (particularly negligent crimes of event, such as homicide or physical injury) which can be committed through, by, or against AI systems is recommended *de lege ferenda*.⁴³

2.4. Potential innovative solutions to corporate liability for AI harm

As we delve into the evolving legal landscape, it is imperative to reflect concisely yet thoughtfully on the legislative techniques that could be employed in the foreseeable future to hold corporations liable for offences arising from the development and use

⁴² Such an interpretation has been adopted by Italian courts with reference to the cases in which work accidents are caused by inadequate health and security measures adopted by the employer: see for instance: see Cass., Sez. Un., sent. 24 April 2014, n. 38343 and, more recently, Cass., Sez. IV, sent. 8 January 2021, n. 32899.

⁴³ Piergallini (Ch II n 155), 1756.

of AI systems. This involves not only considering the technical and procedural frameworks that might enable such liability but also critically evaluating whether these proposed criminal policy strategies would be both equitable and effective in addressing the unique challenges posed by AI-driven harms.

In our opinion, three punitive strategies can be theoretically considered and developed to address the vexed question of attribution of corporate mental states for AI harm:

- i. A new model of ‘autonomous’ corporate liability, not based on the liability of the individual natural person, but on the failure to adopt compliance measures to prevent the AI-caused harmful events of the type of the specific one that occurred.⁴⁴
- ii. Criminal sanctions directed against the corporation that are entirely unrelated to the occurrence of harm or criminal offences, based on the pure violation of the compliance requirements and obligations that need to be fulfilled in connection with the introduction of such AI systems on the market and their post-market monitoring.
- iii. Criminal sanctions directed against the AI system as such (of course with the prior introduction of AI legal personhood), considering the possibility of triggering the ‘ordinary’ corporate criminal liability rules of the current various jurisdictions if such AI crimes are committed in the context of a corporation.

⁴⁴ For a general overview on this subject see Vincenzo Mongillo, ‘Corporate criminal liability for AI-related crimes: possible legal techniques and obstacles’ in Lorenzo Picotti and Beatrice Panattoni (eds.), *Traditional Criminal Law Categories and AI: Crisis or Palingenesis?* (RIDP - International Review of Penal Law - 94 Issue 1 2024) 77ff. See also Council of Europe’s European Committee on Crime Problems (CDPC), ‘Feasibility study on a future Council of Europe instrument on artificial intelligence and criminal law’ (2020).

The first option allows the legal person to be held liable even in cases where a natural person directly responsible for the harm is not individually punishable due to particular conditions or circumstances or remains unidentified or unidentifiable. In such cases, it would suffice to establish that an objectively typical and unlawful act was committed in the interest or to the advantage of the organization.

Some scholars in the United States have explored the possibility of extending traditional corporate vicarious liability models to cases of *corporate algorithm misconduct*. This perspective suggests that AI wrongdoing could theoretically be treated as a corporate wrong, akin to the misconduct of human employees.⁴⁵ A notable model in this context, often referred to as *system intentionality*, posits that corporations express their states of mind through their systems of conduct, including policies, practices, and increasingly, AI systems.

The premise of this model is straightforward and intuitive. Since corporations lack natural minds, memories, or the continuity of human actors – given that employees come and go – they ‘think’ through their systems, including AI. Thus, assessing these systems could offer insights into corporate intentions and knowledge, which could then be used to establish a deficient corporate mental state. By evaluating AI systems, courts could theoretically infer corporate ‘intention’ or ‘knowledge’ relevant to determining liability.⁴⁶

Building on this notion, Diamantis has proposed the idea of treating algorithmic conduct as corporate conduct, thereby activating what he calls the *extended corporate mind* theory.⁴⁷ According to this theory, the mental states of corporations – traditionally tied to the activities and intentions of individual employees – can extend to include functionally integrated corporate systems, such as AI. This functionalist view asserts that corporate ‘minds’ are not restricted to human

⁴⁵ The reference is here to Diamantis (n 30), 893.

⁴⁶ Here, it is useful to remember that the primary doctrine for attributing mental states to corporations—*respondeat superior*—defines corporate mental states in terms of employee mental states.

⁴⁷ Mihail Diamantis, ‘Algorithms Acting Badly: A Solution from Corporate Law’ (2021) 89(4) *George Washington Law Review* 801.

operators but encompass other systems through which corporations act or think. Therefore, under an extended interpretation of vicarious liability or *respondeat superior* doctrines, legal frameworks could hold corporations accountable for the ‘wrongdoing’ of their algorithms.⁴⁸

While innovative, this approach raises several critical challenges and concerns. First, it implicitly treats algorithms as quasi-employees, a conceptual leap that could destabilize existing liability frameworks. More controversially, it suggests the possibility of attributing guilt, blame, or *mens rea* to algorithms, treating them as though they possess intentionality or moral agency. This assumption conflicts with the prevailing understanding of AI. The general consensus remains that AI systems lack free will, consciousness, or subjective intentionality – qualities traditionally regarded as prerequisites for criminal guilt. Without these attributes, the idea of holding an algorithm culpable or equating its actions to corporate *mens rea* stretches the boundaries of legal theory and could lead to troubling implications.

By contrast, certain models of organizational fault already possess regulatory tools to establish an *exclusive* and *autonomous* corporate liability independently of the identification of the individual offender. Notably, the Italian legal system may already be predisposed to such a development, as the provisions of Article 8 of Legislative Decree 231/2001 encapsulate an early form of *anonymous corporate guilt*. Under this framework, a corporation can be held liable even when the natural person responsible for the crime within the corporate context has not been identified or is not imputable.⁴⁹ The basic principle underlying Article 8 is that corporate liability arises from the entity’s failure to prevent the commission of the crime, rather than from the actions of a specific individual. This shifts the focus from individual culpability to systemic deficiencies within the organization, such as lapses in risk

⁴⁸ *ivi*, 895ff.

⁴⁹ ‘1. La responsabilità dell’ente sussiste anche quando: a) l’autore del reato non è stato identificato o non è imputabile’ (art. 8 d.lgs. 231/2001).

management, inadequate compliance programs, or violations of due diligence standards.⁵⁰

This raises an important question: if harm caused by an AI system is attributable to a breach of the organization's duty of care – such as the failure to adhere to industry standards, inadequate oversight of AI deployment, or insufficient post-market monitoring – but the individual perpetrators of the misconduct cannot be identified, could such a scenario give rise to corporate liability under such a model (in Italy but also in other EU countries recognizing corporate criminal liability)?

Critically, the extension of *anonymous guilt* models to AI-related cases is not without challenges. First, the complexity and opacity of AI systems often make it difficult to draw a clear causal link between organizational deficiencies and specific harm. Second, the principle of proportionality must be preserved to ensure that corporations are not penalized excessively for failures that might involve residual risks inherent in AI technologies. Finally, to avoid sliding into forms of strict liability based purely on risk or circular reasoning – where the mere failure to identify an individual offender automatically proves an organizational deficiency – it is crucial to maintain the requirement that the harm must still arise from culpable conduct on the part of the entity. In other words, the corporation's liability cannot rest solely on the fact that no human perpetrator has been identified. Instead, it must be demonstrated that there exists a functional correlation between the organizational deficiency (e.g., lapses in oversight, risk management, or compliance frameworks) and the predicate offence itself.

Taking into account all these issues, option 3 might offer a more balanced and pragmatic solution. This approach would entail the creation of *crimes of danger*, which would hold organizations accountable not for the actual harm caused by AI

⁵⁰ Federico Consulich, 'Il principio di autonomia della responsabilità dell'ente. Prospettive di riforma dell'art. 8' (2018) 4 Rivista 231, 197 ff; Id, 'Il nastro di Möbius. Intelligenza artificiale e imputazione penale nelle nuove forme di abuso del mercato' (2018) 71(2) Banca, borsa e titoli di credito 195-234, 226.

systems, but for violations of legal compliance measures and standards designed to ensure the production and deployment of *trustworthy* AI systems.

CHAPTER VI

CONCLUDING REMARKS: AI CRIME AND THE FUTURE OF CRIMINAL RESPONSIBILITY

1. Drawing clear lines? A theoretical conundrum turned legal necessity

We are only beginning to grasp the profound transformations AI is bringing to our lives. What remains certain is that the rapid proliferation of increasingly unpredictable – yet remarkably accurate and efficient – AI applications across diverse sectors presents unprecedented challenges for legal systems. These systems must now regulate technologies they cannot fully comprehend, racing to keep pace with developments that defy traditional jurisprudential categories.

Nowhere is this tension more acute than in criminal law, whose foundational principles of personal culpability for human acts appear fundamentally incompatible with offenses emerging from autonomous AI decision-making. When harmful outcomes result solely from an AI system's independent operations, the very notion of *reproachability* – central to criminal liability – becomes problematic.

This study has revealed the core challenge: establishing clear boundaries between *justified* responsibility gaps (where society reasonably accepts technological risks) and *unjustified* ones (where harms stem from negligent or reckless practices). This line-drawing exercise is not merely an academic puzzle but an urgent legal imperative, as ambiguous accountability frameworks risk either stifling innovation or permitting dangerous impunity.

The conceptual tool used to balance these extremes and establish the limits of criminal liability for 'hard AI crime' is that of *permissible risk*, which serves to balance innovation with safety and liability concerns. The adoption of the AI Act marks a major step in this direction, directly impacting on decisions related to the definition of acceptable risk in AI development and deployment. However, its methodology warrants critical examination. Rather than assessing risk through empirical evidence of actual harm or probabilistic danger, the Act operates through an *ex-ante* classification system that prioritizes legal categorization over concrete threat evaluation. This creates a regulatory paradigm focused on compliance with predetermined risk thresholds rather than responsiveness to demonstrated hazards.

This approach leads to the central operational dilemma: how to translate these abstract regulatory classifications and principles into practical, actionable legal standards capable of effectively governing AI systems while establishing a coherent foundation for criminal liability. The challenge is compounded by several factors.

First, the epistemic challenge: unlike conventional products with established safety expectations, AI systems operate in novel contexts without clear benchmarks for ‘reasonable’ performance. This makes it difficult to determine when a developer has crossed the line from acceptable risk-taking to culpable negligence. Should the threshold focus on the thoroughness of testing protocols? The statistical probability of harmful outcomes? The cost-benefit ratio of potential harms versus benefits?

Second, the oversight paradox: while human oversight requirements appear as an obvious solution, they often prove illusory in practice. For complex, adaptive AI systems, human operators may lack either the technical capacity or the decision-making timeframe to provide meaningful supervision. This creates a dangerous façade of control where none substantively exists.

Within this framework, criminal law scholarship is coalescing around two principal approaches to preserving a role for criminal law in the protection of legal interests endangered by the development and use of AI systems.

The first adopts what might be called a bright-line methodology, seeking to establish clear boundaries between zones of liability and immunity through evidence-based standards. This approach draws from empirical studies of human-machine interaction to create predictable accountability frameworks that prevent the scapegoating of individual developers and users while maintaining proportional consequences for genuine negligence. Its strength lies in providing legal certainty, though critics argue it may oversimplify the complex sociotechnical nature of AI systems.

The second approach emphasizes *ex ante* criminal protections, recognizing that AI’s inherent unpredictability undermines traditional mechanisms for attributing harmful outcomes. This perspective advocates for criminalizing negligent

development practices before harms materialize – that is, *the act of releasing insufficiently safeguarded AI systems into the market*. As some have aptly noted, ‘if the categories of criminal law cannot be stretched to cover the unforeseeable, we must focus instead on penalizing, with measured sanctions, the disvalue of conduct that reflects non-compliance with preventive protocols in the use of advanced technologies’.¹ From this perspective, the legal protection against AI-related risks would reaffirm criminal law’s role as a measure of last resort. It would serve as a safety net, complementing the tools offered by other areas of law (e.g., administrative law, civil liability) which are better equipped to safeguard fundamental rights in the context of socially useful risk.

These competing approaches, despite their differences, collectively highlight AI’s unique value as a stress test for traditional legal categories, as well as require a regulatory approach that would include new rules addressing criminal liability. The challenges posed by autonomous systems force us to re-examine and potentially revitalize underutilized legal mechanisms, from enhanced corporate liability frameworks to crimes of endangerment. If we opt for such an approach, the next step will be to decide if such rules addressing criminal liability should be adopted at national or EU level.

2. Tackling AI crime at EU level: harmonized liability rules on the road ahead?

As some have aptly noted, ‘the EU’s cautious approach to imposing criminal liability related to AI systems comes as no surprise. On the one hand, criminal law is seen as a last resort in both the Member States and the EU, and on the other hand, the EU can only use criminal enforcement mechanisms under specific circumstances’.² Nevertheless, we argue that the EU level presents a theoretically desirable avenue for adopting harmonized criminal liability rules on AI for several compelling reasons.

¹ Consulich (Ch I n 21), 27ff.

² Gless and Ligeti (Ch I n 196), 35.

First, the majority of Member States are not significant producers of AI systems, nor are they systematically involved in their development.³ This reality underscores a critical knowledge gap at the national level when it comes to understanding the technical and operational intricacies of AI technologies. Such a gap admittedly poses a substantial challenge for Member States seeking to craft effective and precise criminal provisions tailored to the risks and complexities associated with AI systems. In this context, the absence of technical expertise and practical familiarity with AI development could result in poorly drafted criminal laws that either fail to address the specificities of AI-related harm or impose unreasonable burdens on stakeholders. This lack of know-how could also lead to significant variations in the scope and application of criminal sanctions for the violation of the relevant EU obligations across jurisdictions, undermining the uniformity and predictability essential for addressing transnational AI risks.

On a similar note, AI systems are inherently cross-border: they rely on data from multiple Member States, and companies from various Member States contribute to designing, developing, and deploying these systems within the common market – whether as manufacturers, providers, distributors, or authorized representatives. This cross-border nature necessitated the EU's adoption of harmonized rules for the production and implementation of AI systems, as their functionality would otherwise be severely limited. By extension, harm related to human-machine interactions also exhibits a cross-border dimension, making it impractical for individual Member States to address them effectively in isolation. Therefore, a common stand on definitions of duties of care, distributions of responsibilities, and possible exceptions from criminal liability appears both logical and necessary.

This would also align with the principle of subsidiarity, as it recognizes that certain challenges, particularly those involving highly specialized knowledge or cross-border implications, are better addressed at the supranational level. Among other things, harmonizing criminal offences would contribute to safeguarding legal certainty

³ *Id.*, 51.

for both the actors involved in the AI industry, on the one hand, and the victims of accidents involving AI systems, on the other, as liability would hinge on measurable and clearly defined obligations, rather than uncertain outcomes. In fact, a lack of uniform implementation across Member States could undermine the internal market's level playing field by putting operators who strictly comply with EU requirements at a disadvantage and create legal uncertainty.⁴ For example, without a unified approach, different jurisdictions may designate varying thresholds for when a human operator is expected to intervene and take back control of a malfunctioning AI system. In some jurisdictions providing for a narrow criminal framework, intervention might be required only when the system emits a warning signal or a takeover request, reflecting a choice for liability exemption for some risks. In other jurisdictions adopting a broader framework, the operator might be expected to act every time they recognize signs of failure or malfunction, which could depend on subjective assessments of what constitutes an imminent risk. This lack of uniformity creates significant challenges, especially for multinational organizations deploying AI systems across borders.

However, desirability is one thing, practical feasibility is another. Even if theoretically desirable, resorting to the mechanisms of harmonization of substantive criminal law as set out in Article 83 TFEU exposes us to their inherent limitations, whether one seeks to meet the criterion of serious, cross-border crime (para 1) in order to justify the expansion of the list of the so-called euro-crimes⁵ or resorts to the solution of 'functional' substantive criminal law (para 2).⁶

⁴ *ibid.*

⁵ According to the principle of conferral enshrined in art 5(2) TFEU, the EU may only adopt criminal law measures in relation to specific serious crimes with a cross-border dimension listed in Article 83(1).

⁶ According to art 5(3) TFEU also states that it can only exercise such competence if EU action is comparatively more efficient than national regulation. See Petter Asp, *The Substantive Criminal Law Competence of the EU* (Jure 2013) 134-35; Valsamis Mitsilegas, *EU Criminal Law* (2nd ed, Hart Publishing 2022), 47ff.

The latter solution (which appears the most plausible one)⁷ presupposes the existence of a Union policy in an area already subject to harmonization measures. In this case, establishing minimum rules on offences related to the field of AI safety should be deemed *essential* for ensuring the *effective implementation* of the common AI policy. It is doubtful whether criminal law measures are essential to achieve the effective implementation of the AI Union policy.⁸ While a cautious approach (that complies with the principle of subsidiarity and proportionality) would require a serious enforcement deficit that cannot be adequately addressed through alternative means (such as administrative sanctions or a robust civil liability regime) to the threat of punishment,⁹ some authors argue that the importance of the legal goods at stake (eg, individual and public safety) and the gravity of the harm that can arise out of

⁷ The basic precondition for criminal law intervention in the field of AI under art 83(1) TFEU is the amendment of the list of serious crimes with cross-border dimension therein. Crimes connected to most AI sectors are not – yet – listed in art 83(1), so a Council decision would be necessary to extend the ‘Euro crimes’ to these crimes. In favour of this option, Igor Vuletic and Tunjica Petrašević, ‘Is It Time to Consider EU Criminal Law Rules on Robotics?’ (2020) Croatian Yearbook of European Law and Policy. Offences related to automation could fulfil the two cumulative criteria that trigger the EU’s legal harmonization competence (seriousness and cross-border dimension), but the ultimate decision to expand the scope of Euro crimes is a policy choice made by the Member States. Whereas since the adoption of the Lisbon Treaty such expansion was not discussed at EU level, more recently the case of violation of EU restrictive measures lead to the use of art 83(1) as a legal basis, thus showing its appetite to expand the list of Euro crimes – even in cases where theoretically the legal basis of art 83(2) TFEU could be employed. Cfr. Leonardo Romanò and Lorenzo Bernardini, ‘Enforcing Restrictive Measures Against Russia Through Criminal Law A Truly Effectiveness-enhancing Choice?’ (2024) Edinburgh Student Law review.

⁸ For a narrow interpretation, see Jacob Oberg, *Limits to EU powers* (Hart Publishing 2017) 314-6; for a broader interpretation, see Asp (n 9), 134-35; Mitsilegas (n 9) 118.

⁹ According to the Commission’s Communication *Towards an EU Criminal Policy: Ensuring the effective implementation of EU policies through criminal law*, when assessing the essentiality of criminalization, factors that should be considered include, inter alia: (a) the seriousness and character of the breach of law; (b) the need to stress strong disapproval in order to ensure deterrence; and the efficiency of the sanction system being enforced as well as the extent to which, and (c) the reasons why, existing sanctions do not achieve the desired level of enforcement. Commission, Communication from the Commission to the European Parliament, the Council, the European Economic and Social Committee and the Committee of the Regions, Towards an EU Criminal Policy: Ensuring the effective implementation of EU policies through criminal law, COM(2011) 573 final.

certain AI applications could militate in favour of establishing criminal law measures even before testing the effectiveness of administrative sanctions or of a civil liability regime.¹⁰

Nevertheless, a significant obstacle to the adoption of harmonized criminal law rules on offences related to AI safety may stem from subsidiarity concerns raised by Member States. The EU's reluctance to initiate discussions on criminal law harmonization in this field could be attributed to the inherent link between AI-related harm and traditional crimes against life and physical integrity, such as homicide and bodily injury – areas typically falling within the jealous purview of Member States.¹¹ Additionally, a one-size-fits-all approach, even if loosely framed, might appear overly rigid to domestic lawmakers, especially considering that this step may presuppose interventions in the area of the general part of criminal law, the provisions of which are not harmonized and which, given the cultural differences of the European legal orders, should preserve its unique character to avoid conflicts with fundamental tenets of national criminal justice systems.¹² Member States may view the imposition of harmonized rules as an infringement on their autonomy to address these matters in a manner consistent with their unique legal and cultural contexts.

Ultimately, whether user criminal liability in the field of AI safety will be governed by common European minimum standards or remain subject to independent domestic rules is a complex and unresolved issue. Establishing such rules will not be an easy exercise for the EU, especially since negligence continues to

¹⁰ Gless and Ligeti (Ch I n 196), 53, who argue that the effective implementation of the common AI policy would be jeopardized if the EU were to establish criminal liability only after administrative sanctions and/or civil liability were found to be an insufficient deterrent.

¹¹ Jannemieke W. Ouwerkerk, 'Old wine in a new bottle: Shaping the foundations of EU criminal law through the concept of legal interests (Rechtsgüter)' (2022) 27(4-6) European Law Journal 1, 9, who comments that 'The acceptance of life and physical integrity as legal interests that are worthy of protection through EU criminal law would, after all, raise the challenging question whether classical crimes against life and physical integrity could also be held as types of crime in which the EU potentially could be given competences'.

¹² Art 83(3) TFEU.

be considered a grey area at the European level and such rules may not align with the national legal traditions of Member States.

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